Local Langlands for $\mathbf{GL}_2(\mathbf{F})$

— ***EYNTKA*** **Series** —

Nathanael Chwojko-Srawley

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Contents

0.0.1 Inspiration Very heavily inspired by [1]

Introduction

The Langlands program sets out to understand the absolute Galois group $\mathrm{Gal}(\overline{F}/F)$ of a field. The pursuit has spawned a web of connections linking number theory (Galois representations), harmonic analysis (automorphic forms), and geometry (cohomology of algebraic varieties). At its heart lies a dictionary, the global *Langlands correspondence*, which translates arithmetic data into analytic data and back.

While the global theory is vast and often technically overwhelming, the *Local Langlands Correspondence* provides the essential vocabulary for this dictionary. It asserts that for a local field F (like $\mathbb{Q}_p$), the representation theory of the groups $\mathrm{GL}_n(F)$ for $n \in \mathbb{N}_{>0}$ is mirrored by the representation theory of the absolute Galois group of F.

The Goal The goal of this book is to provide a complete, self-contained, and accessible proof of the Local Langlands Correspondence for the case $n = 2$.

This book follows *The Local Langlands Conjecture for GL(2)* by Colin J. Bushnell and Guy Henniart, and is written in the *EYNTKA* (Everything You Need To Know About) style. The aim is to expose the narrative arc of the subject, to build the background needed for the n-dimensional case, and to prepare the reader for the connection between elliptic curves and modular forms.

The Roadmap

1. *The analytic side* (chapter 1 to chapter 4). The theory of *smooth representations* is established first. Unlike the representation theory of finite groups or of Lie groups, the vector spaces here are infinite dimensional while the groups are totally disconnected. The “easy” representations, the principal series, are classified by parabolic induction. The “hard” ones, the cuspidals, are then constructed explicitly by *compact induction* from open subgroups, and the tame ones are parameterized by *admissible pairs*.

2. *The invariants* (chapter 5). Matching a representation on one side against one on the other

requires quantities that exist in both worlds. These are the local L-function and the local constant, or ε-factor. The chapter closes with the *converse theorem*: if the L- and ε-factors of two representations agree after enough twisting, the representations are isomorphic.

3. *The arithmetic side* (chapter 6). The Galois group is refined to the *Weil group* $\mathcal{W}_F$ and enriched to *Deligne representations*. The enrichment is what accounts for the special (Steinberg) representations on the analytic side.

4. *The correspondence* (chapter 7). The two sides are brought together. The principal series match reducible Weil representations. For the cuspidals, admissible pairs and a rectifying character produce a bijection preserving the arithmetic invariants.

5. *An appendix* (appendix A) proves that irreducible cuspidal representations are projective in the category of smooth representations with a fixed central character, which is the precise form of the "atomicity" the main text appeals to.

0.0.2 Limitations of the Book Three things are deliberately out of scope. The correspondence in residual characteristic $p = 2$ is not covered: the dyadic case is substantially harder and would need several chapters of its own. The Jacquet-Langlands correspondence, treated in [1], is not covered either. Neither is the passage from the local statement to the global one, though chapter 7 indicates where that passage begins.

Prerequisites

The reader is assumed to be comfortable with:

- the basic theory of local fields (valuations, rings of integers, p-adic numbers), as covered in [10],
- Galois theory, as covered in [7],
- the representation theory of finite groups (characters, induction, Schur's lemma), as covered in [2],
- topological groups, as covered in [6].

Familiarity with profinite groups is useful but not assumed. Section 1.1 develops what is needed.

0.0.3 Prerequisites and Place in the Series This volume builds on *Class Field Theory* [10], *Algebraic Groups and Representations* [8], and *Harmonic Analysis* [13]. In turn, it is a prerequisite for *Smooth Representations of p-adic Groups*, which generalizes the worked $\mathrm{GL}_2(F)$ case treated here to arbitrary p-adic reductive groups. That volume is planned rather than written, so it is named here in prose rather than cited.
The reader has access to every completed volume of the series but is not assumed to have read all of them. Prerequisites are cited where invoked.

1

Introduction to Smooth Representations

The Langlands program is, at its core, a bridge between number theory and harmonic analysis. On the number theoretic side, we have Galois groups $\mathrm{Gal}(\overline{F}/F)$, which encode the arithmetic structure of fields. On the analytic side, we have groups of matrices like $\mathrm{GL}_n(F)$. To build this bridge, we must first understand the ground on which the analytic side stands. The goal of this chapter is to establish the foundational language for the "analytic" side of the correspondence: the theory of *smooth representations* of locally profinite groups. Unlike the representation theory of finite groups or Lie groups, the topology of p-adic fields requires a unique approach. We cannot simply look at continuous representations on Hilbert spaces. Instead we must look at representations that respect the totally disconnected topology of $\mathbb{Q}_p$.

We begin in Section 1.1 by defining locally profinite groups and exploring the topology of F and $\mathrm{GL}_n(F)$. In Section 1.2, we define smooth representations (actions that are locally constant) and show that they satisfy an analogue of Schur's Lemma. In Section 1.3, we develop the necessary integration theory (Haar measure) to analyze these groups. Finally, in Section 1.4, we introduce the Hecke algebra $\mathcal{H}(G)$, an associative algebra that allows us to translate problems of smooth group representations into the language of modules over a ring, mirroring the relationship between finite groups and the group ring $\mathbb{C}[G]$.

1.1 Locally Profinite Groups

All finite groups will by assumption have the discrete topology. For a deeper review of topological groups, see [6, topological groups].

The field $\mathbb{Q}_p$ is a *locally profinite groups* with its topology endowed by the metric $|-|_p$. This is a substantially different topology to $\mathbb{C}$, and hence a moment must be taken to understand better its structure.

Definition 1.1.1: P-adic Topology

Let $\mathbb{Q}_p$ be a p-adic field for a (finite) prime p and let $\mathbb{Z}_p$ be the corresponding ring of integers. Define the neighbourhood basis:

$$\mathbb{Z}_p \supseteq p\mathbb{Z}_p \supseteq p^2\mathbb{Z}_p \supseteq \cdots$$

Giving $\mathbb{Z}_p$ a p-adic topology which can be easily extended to $\mathbb{Q}_p$. As each $p^k\mathbb{Z}_p$ is a closed ball, this matches the metric topology induced by $|-|_p$:

$$B_r(a) = \{x \in \mathbb{Q}_p : |x-a|_p < r\}$$

which extends to $\mathbb{Q}_p$ by defining:

$$\left|\frac{m}{n}\right|_p = \frac{|m|_p}{|n|_p}$$

Some basic properties of this field are established first.

Lemma 1.1.2: P-adic Field Totally Disconnected

The field $\mathbb{Q}_p$ with the topology induced by the metric $|-|_p$ is totally disconnected. More generally any space X whose topology is induced by an ultra-metric is totally disconnected.

***Proof*:**
Let $a, b \in \mathbb{Q}_p$. As $a \neq b$, there must exist a d such that:

$$0 < |a-b|_p = p^{-k} = d$$

Let:

$$B_d(a) = \{x \in \mathbb{Q}_p : |x-a| < d\}$$

By construction, $a \in B_d(a)$ and $b \notin B_d(a)$.

Let us show that

$$B_d(a)^c = \{x \in \mathbb{Q}_p : |x-a| \geq d\}$$

is also open. Take $x \in B_d(a)^c$ so that $D = |a-x|_p \geq d$ and consider $B_d(x)$ and take $y \in B_d(x)$ so that $\delta = |x-y|_p \leq d$. We want to show that $y \in B_d(a)^c$ so $|y-a|_p \geq d$. Then by the isosceles property of the ultra-metric, for any three points a, x, y two of $d(a,x)$, $d(a,y)$ or $d(x,y)$ must be to the larger one. As $|a-x|_p \geq d$, $|y-a| \geq d$, and so $y \in B_d(a)^c$, showing that $B_d(x) \subseteq B_d(a)^c$ showing that it is *also* open.

Thus, $\mathbb{Q}_p$ is totally disconnected, as we sought to show.

Lemma 1.1.3: P-adic Integers Open Subgroup

The subgroup $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is an open subgroup with the topology induced by the valuation. Hence, it is also closed

This differs strongly from $\mathbb{C}$: if $H \subseteq \mathbb{C}$ is an open subgroup then as noted earlier it must also be closed, hence connected, and hence $H = \mathbb{C}$. As is $\mathbb{Q}_p$ is totally disconnected, it is possible to have clopen subsets that are not the entire set.

***Proof*:**
First, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$. Take $a \in \mathbb{Z}_p$ (so that $|a| \leq 1$) and consider:

$$B_1(a) = \{x \in \mathbb{Q}_p : |x - a|_p < 1\}$$

I claim $x \in \mathbb{Z}_p$. Indeed:

$$|x|_p = |x + (-a + a)|_p \leq \max(|x - a|_p, |a|_p) = \max(t, 1) = 1$$

where $t < 1$ as $|x - a|_p < 1$. But then $x \in \mathbb{Z}_p$, and hence $B_1(a) \subseteq \mathbb{Z}_p$.

Thus, as $\mathbb{Q}_p$ is a topological group, $\mathbb{Z}_p$ is also closed.

Other groups built from p-adic fields occur throughout, for example finite extension of $\mathbb{Q}_p$, the ring of integers of $L_{\mathfrak{p}}/\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$, $\mathbb{C}_p$, $M_n(\mathbb{Q}_p)$, $M_n(\mathcal{O}_{L/\mathbb{Q}_p})$, $M_n(\mathfrak{m}_{\mathfrak{p}})$, various matrix subgroups of this groups (GL_n, SL_n, PGL_n, Sp_n, O_n, SU_n, etc.) and so forth. Thus, it is worth going over a general theory that encapsulates the topology of these spaces. Recall first some basic properties of topological groups:

- The topology of G is translation invariant: If $U \subseteq G$ is open, then Ux and xU is open for all $x \in G$
- If $H \leqslant G$, G/H has the quotient topology, and if $H \trianglelefteq G$ then G/H is a topological group.
- If K_1, K_2 are compact subset of G, so is $K_1 K_2$
- subgroups in relation to open/closed:
  - if $H \leqslant G$ is open, it must be closed ($\bigcup_{g \in G \setminus H} gH$ is the union of open sets and the compliment of H). More strictly, if H is open if and only if it contains an open neighbourhood around the identity (if $e \in U \subseteq H$ with open U, then $hU \subseteq H$ and $H = \bigcup_{h \in H} hU$). If G is connected, it has no non-trivial proper open subgroups. Totally disconnected groups are the ones that matter here.
  - If $H \leqslant G$ is an open subgroup, then the quotient space G/H and $H \backslash G$ each have the discrete topology.
  - A closed subgroup $H \leqslant G$ need not be open (ex. $\mu_n \subseteq \mathbb{C}$). However if $H \leqslant G$ is closed and finite index, it is open (the finite union of closed is closed). If G is first-countable, H is closed if it contains all its limit points (more generally, it it contains all the limits of nets).
  - A subgroup need not be open or closed (ex. $\mathbb{Q} \subseteq \mathbb{R}$).

  - if H is a subgroup of G, so it $\overline{H}$ (the closure under limits/nets).
  - IF $G_0 \subseteq G$ is the connected component of the identity, then G_0 is closed, G_0 is the intersection of all normal subgroups of G, and G/G_0 is the largest totally disconnected quotient.

- symmetric neighbourhood:
  - For every open neighborhood of e, say for example U, there is a symmetric neighborhood (if $A \subseteq G$ with $A = A^{-1}$ is called *symmetric*) V of e such that $V \subseteq U$
  - For every neighborhood U of e, there is a neighborhood V of e with $VV \subseteq U$

- Density: the groups considered here often have dense subsets or subgroups. Sometimes the sequence definition of density will not be equivalent to the definition of density, hence a reminder for those who have internalized the sequence definition to the point that they forgot the general topological definition: A set $S \subseteq X$ is *dense* if it intersects every nonempty open subset of X: $U \cap S \neq \emptyset$ for all open $U \subseteq X$.

Lemma 1.1.4: Characterizing Hausdorffness

Let G be a topological group, $e \in G$ the identity, and $H = \bigcap_{\substack{U \subseteq G \text{ open} \\ e \in U}} U$ Then:

1. H is a subgroup
2. H is the closure of e, i.e. $\overline{e} = H$
3. G/H is Hausdorff
4. G is Hausdorff if and only if $H = \{e\}$

***Proof*:**

1. If $a, b \in H$ and U is an open neighbourhood of e, continuity of $(x, y) \mapsto xy^{-1}$ at (e, e) gives an open neighbourhood V of e with $VV^{-1} \subseteq U$. Since $a, b \in V$ we get $ab^{-1} \in U$. As U was arbitrary, $ab^{-1} \in H$.
2. An element x lies in $\overline{e}$ if and only if every open neighbourhood of x contains e, which by translation invariance holds if and only if every open neighbourhood of e contains x. That is the defining condition for H.
3. By (2), H is closed, so G/H is a topological group in which the class of the identity is closed. A topological group whose identity is closed is Hausdorff: the diagonal is the preimage of the closed set $\{\overline{e}\}$ under the continuous map $(x, y) \mapsto xy^{-1}$.
4. If G is Hausdorff then points are closed, so $\overline{e} = \{e\}$ and $H = \{e\}$ by (2). Conversely if $H = \{e\}$ then $G/H = G$ is Hausdorff by (3), as we sought to show.

Hausdorffness is assumed throughout unless specified otherwise.

Definition 1.1.5: Profinite Group

Take a cofiltered diagram of finite groups, each carrying the discrete topology. A *profinite group* is a topological group isomorphic to the limit of such a system, with the topology induced from the product. Given any topological group G, its *profinite completion* is

$$\widehat{G} = \varprojlim_{N} G/N$$

where N ranges over the open normal subgroups of finite index, ordered by containment.

Any group G can be given the profinite topology by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then $\varphi : G \to \widehat{G}$ is a continuous morphism and G is dense in $\widehat{G}$.

Example 1.1.6: Profinite Groups

The following are the "trivial" examples:

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The following is the "usual" example and is better suited for the intuition of profinite groups. Let $G = \mathbb{Z}$. Then

$$\widehat{\mathbb{Z}} = \varprojlim_{n} \mathbb{Z}/n\mathbb{Z} \cong \prod_{p} \mathbb{Z}_p$$

where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$. The main idea is that for any element $n \in \mathbb{Z}$, in chain of normal subgroups (say $p^k\mathbb{Z} \subseteq p^{k+1}\mathbb{Z} \subseteq \cdots$) the element mod $p^k\mathbb{Z}$ eventually stabilize. For example, take $p = 7$ and $n = 129'654 = 2 \cdot 3^3 \cdot 7^4$. Then the value is zero for the until $\mathbb{Z}/7^5\mathbb{Z}$ where it becomes $12'005$ mod 7^n for $n \in \{5, 6\}$ and then $129'654$ forever onward. $\mathbb{Z}_p$ allows for never-stabilizing chains (mirroring how stable sequences of rational numbers converge to rational numbers).

For Langlands, the important non-abelian example is the profinite completion of galois Galois groups, namely $\mathrm{Gal}_K(\overline{K})$. See [7] for the Galois theory and [5] for the profinite refinement.

More generally, profinite groups can be seen as the object that captures all the finite behaviour "up to infinity" of a system. In the $\mathbb{Z}_p$ example, it appeared when taking a solution mod each p^n and for galois it comes from amalgamating all symmetries of roots of some infinite family of (finite) polynomial. Another place this shall show up is in the definition of the étale fundamental group. See *Étale Cohomology* [4].

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both.

Lemma 1.1.7: Profinite Completion Injective

Let G be a topological group and $\widehat{G}$ its profinite completion. Then $\varphi\colon G \to \widehat{G}$ is injective if and only if the intersection of all open normal subgroups of finite index in G is trivial.

***Proof*:**
The map φ sends g to the compatible family $(gN)_N$, so $g \in \ker\varphi$ precisely when $gN = N$ for every open normal subgroup N of finite index, that is, precisely when g lies in every such N. Therefore $\ker\varphi = \bigcap_N N$, and φ is injective if and only if that intersection is trivial, as we sought to show.

The completion comes equipped with a natural universal property:

Theorem 1.1.8: Universal Property of Profinite Completion

let $\varphi : G \to H$ be continuous homomorphism with H a profinite group. Then there is a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\varphi"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

***Proof*:**
Uniqueness is the density statement: G has dense image in $\widehat{G}$, H is Hausdorff, and two continuous maps into a Hausdorff space agreeing on a dense subset agree everywhere.

For existence, write $H = \varprojlim_i H_i$ with each H_i finite and discrete, and let $p_i\colon H \to H_i$ be the projections. Each composite $p_i \circ \varphi\colon G \to H_i$ is a continuous homomorphism into a finite discrete group, so its kernel is an open normal subgroup N_i of finite index. Hence $p_i \circ \varphi$ factors through G/N_i, and therefore through $\widehat{G} = \varprojlim_N G/N$. These factorizations are compatible with the transition maps of the system defining H, because the corresponding composites with φ are, and G is dense. They therefore assemble into a continuous homomorphism $\widehat{G} \to H$ making the triangle commute, completing the proof.

A striking result is that all profinite groups can be characterized by a simple topological property:

Theorem 1.1.9: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
Let G be profinite. Then it is a closed subgroup of a direct product of finite groups, hence compact and totally disconnected.

Conversely, suppose G is compact and totally disconnected. The key point is that such a group has a basis of open normal subgroups at the identity. In a compact totally disconnected space the connected component of a point equals its quasi-component, so the clopen sets separate points. Hence every neighbourhood of e contains a compact open set C with $e \in C$. Put $U = \{g \in G : gC = C\}$, the stabilizer of C under left translation. It is a subgroup, it is open because C is compact and open, and $U \subseteq C$ since $e \in C$. Intersecting the finitely many conjugates of U, which are finite in number because U is open in a compact group and so of finite index, produces an open *normal* subgroup of finite index inside the original neighbourhood.

Writing $\mathcal{N}$ for the set of these subgroups, the canonical map $G \to \varprojlim_{N \in \mathcal{N}} G/N$ is injective because $\bigcap_{N \in \mathcal{N}} N = \{e\}$ by the previous paragraph, and surjective by compactness: a compatible family determines a family of non-empty closed cosets with the finite intersection property, whose intersection is therefore non-empty. Each G/N is finite and discrete, so G is the limit of a cofiltered system of finite groups, which is what definition 1.1.5 requires, completing the proof.

Corollary 1.1.10: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion. More precisely,

$$G \cong \varprojlim_{U} G/U$$

where U ranges over the *open* normal subgroups of G, ordered by containment.
Moreover G is isomorphic to its profinite completion *as an abstract group* (one says G is *strongly complete*) if and only if every subgroup of finite index in G is open.

***Proof*:**
For the first assertion, G is compact by theorem 1.1.9, so every open subgroup has finite index and the two indexing systems in the definition of $\widehat{G}$ and in the display coincide. The canonical map $G \to \varprojlim_U G/U$ is injective by lemma 1.1.7, since a compact totally disconnected group has a basis of open normal subgroups at the identity whose intersection is $\{e\}$. It is surjective because G is compact and the maps $G \to G/U$ are surjective, so a compatible family is realized by an element of the intersection of a family of non-empty closed sets with the finite intersection property. A continuous bijection from a compact space to a Hausdorff one is a homeomorphism.

For the second, the profinite completion as an abstract group is the limit over *all* finite-index normal subgroups, whereas G itself is the limit over the *open* ones. The two agree exactly when every finite-index subgroup is open, which is the stated criterion, completing the proof.

Proposition 1.1.11: Dense Subsets of Profinite Group

Let $S \subseteq G$ be a subset of a profinite group. Then S is *dense* if and only if the natural map $S \to G/U$ is surjective for every open normal subgroup $U \subseteq G$.

***Proof*:**
By the proof of corollary 1.1.10, the open normal subgroups form a basis of neighbourhoods of the identity, so the cosets gU with U open normal form a basis for the topology of G.

Suppose S is dense and let U be open normal. Each coset gU is a non-empty open set, so it meets S. An element of $S \cap gU$ maps to gU, so $S \to G/U$ is surjective.

Conversely, suppose every such map is surjective, and let $V \subseteq G$ be non-empty and open. Pick $g \in V$. Since the cosets of open normal subgroups form a basis, there is an open normal U with $gU \subseteq V$. By hypothesis some $s \in S$ satisfies $sU = gU$, hence $s \in gU \subseteq V$. So S meets every non-empty open set and is dense, as we sought to show.

Definition 1.1.12: Locally Profinite Group

A topological group G is *locally profinite* if it is Hausdorff and every open neighbourhood of the identity contains a compact open subgroup of G.

Notice how there is no mention of an inverse system. The definition implies G is totally disconnected, and hence if it is also compact then a profinite group is recovered. Every *discrete* group is locally profinite, the trivial subgroup being compact and open. Closed subgroups of a locally profinite group are locally profinite, and so are quotients by closed normal subgroups. A locally profinite group is locally compact, and conversely any locally compact totally disconnected group is locally profinite.

The canonical example for these notes is the non-Archimedean local field $(F, +)$, as proven in [10]. The multiplicative group $(F^\times, \times)$ is locally profinite as well, though it is *not* profinite: it is not compact, since the valuation surjects it onto $\mathbb{Z}$. What is profinite is the unit group $U_F = \mathfrak{o}^\times$, and the subgroups

$$U_F^n = 1 + \mathfrak{p}^n$$

form a basis of compact open neighbourhoods of 1 in $F^\times$, which is what local profiniteness asks for.

Being locally profinite is preserved by finite products, so F^n is locally profinite and hence so is $(M_n(F), +) \cong F^{n^2}$. Neither is profinite, for the same reason F is not: they are not compact. The group $\mathrm{GL}_n(F) \subseteq M_n(F)$ is an open topological subgroup under multiplication, and it is locally profinite because $\mathrm{GL}_n(\mathfrak{o})$ is profinite. That subgroup is compact, being homeomorphic to a closed subset of $\mathfrak{o}^{n^2}$, a product of compact groups. It is open because $\mathfrak{o} = \{x \in F : |x|_\mathfrak{p} \leq 1\}$ is open. This was all done in coordinates. More generally the same applies to $\mathrm{End}_F(V)$ and $\mathrm{Aut}_F(V)$ for a finite-dimensional F-vector space V.

Characters on locally profinite groups come next.

Theorem 1.1.13: Continuous Character

let $\varphi : G \to \mathbb{C}^\times$ be a group homomorphism where G is locally profinite. Then the following are equivalent:

1. φ is continuous
2. the kernel of φ is open

If φ satisfies one of these conditions and G is the union of its compact open subgroups (such as $\mathbb{Q}_p$), then $\mathrm{im}\varphi \subseteq S^1 \subseteq \mathbb{C}^\times$.

***Proof*:**
If the kernel is open, then if $U \subseteq \mathbb{C}^\times$ and $K = \ker\varphi$,

$$\varphi^{-1}(U) = \bigcup_{g \in \varphi^{-1}(U)} gK$$

showing φ is continuous. Conversely, Let φ be continuous and U an open neighbourhood of 1. Then $\varphi^{-1}(U)$ is open, and thus by local-profiniteness contains a compact open subgroup $K \subseteq G$. As this applies to any U, we can pick U sufficiently small so that it contains no non-trivial subgroup of $\mathbb{C}^{\times}$. As the image of group is a group and $K \subseteq \varphi^{-1}(U)$, $K \subseteq \ker\varphi$. As K is open, $\ker\varphi$ contains an open subgroup and hence by topological group theory $\ker\varphi$ is open.

Next, S^1 is the unique maximal compact subgroup of $\mathbb{C}^{\times}$. If K' is a compact subgroup of G, $\varphi(K') \subseteq S^1$, giving the 2nd result as we sought to show.

This condition has interesting algebraic consequences, for example take the map $\mathbb{Z} \to S^1$ by sending 1 to $e^{2\pi i\theta}$ where θ is irrational. As $\mathbb{Z}$ is dense in $\mathbb{Z}_p$, this map extends to a map $\mathbb{Z}_p \to S^1$. It's kernel is $\{0\}$, i.e. it is not open, and so this map is not continuous with the profinite topology on $\mathbb{Z}_p$.

Corollary 1.1.14: Continuous Maps To General Linear Group

Let $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ be a group homomorphism where G is locally profinite. Then φ is continuous if and only if the kernel of φ is open.

If moreover G is the union of its compact open subgroups, then φ takes values in the unitary group $U(n)$, by the same argument as in theorem 1.1.13: the image of each compact open subgroup is a compact subgroup of $\mathrm{GL}_n(\mathbb{C})$, and every compact subgroup is conjugate into $U(n)$.

***Proof*:**
Note that $\mathrm{GL}_n(\mathbb{C})$ is a lie group and that there exists an open neighbourhood of I_n that contains no non-trivial subgroups, the argument follows.

Definition 1.1.15: Continuous Character

A *character* for a locally profinite group G is a continuous homomorphism

$$\varphi : G \to \mathbb{C}^{\times}$$

Write 1_G, or just 1, for the trivial character (i.e. the constant character).

If $\varphi(G) \subseteq S^1$, φ is called a *unitary character*.

The characters of a local field organize themselves by how deeply ramified they are, and that filtration is what the next section builds on. Write $\widehat{F}$ for the set of all characters of F. As F is a local field, it is the union of its compact open subgroup $\mathfrak{p}^n$ for $n \in \mathbb{N}$, hence all characters are unitary. Then if $\varphi \neq 1$ is a character, there exists $d \in \mathbb{Z}$ such that $\mathfrak{p}^d \subseteq \ker(\varphi)$; the least such number is defined as the level of φ .Fixing a d, the set of characters of level $\leq d$ is a subgroup. This defines a duality which extends to the multiplicative set, in other words this is indeed the right notion of character!

1.2 Smooth Representations of Locally Profinite Groups

All representations shall be complex unless specified otherwise. When working with $\mathbb{Q}_p$ representations and more generally representations of locally profinite groups, it is important to take into account the topology, or else we will get representations that give no valuable information:

Example 1.2.1: "Useless" Representation

Take the algebraic closure of $\mathbb{Q}_p$, $\overline{\mathbb{Q}}_p$. By field theory, it embeds into $\mathbb{C}$ (using the axiom of choice). Thus there is an injective ring homomorphism $\mathbb{Q}_p \to \mathbb{C}$ and hence a (valid) representation:

$$\pi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathrm{GL}_n(\mathbb{C})$$

But this has disregarded all topological (and hence arithmetic) information of $\mathbb{Q}_p$.

The first step to fix this would be to only consider continuous representations as is done in the representation of topological groups. Often when G is not compact, the vector space V is infinite dimensional with which is either a Banach or Hilbert space and $\mathrm{GL}(V)$ is the set of all invertible bounded linear operators. This thread is explored further in [11].

When restricting to locally profinite groups, there is a useful stronger notion called *smooth*. It must be that this notion of smoothness differs from smoothness for manifolds, since the maps in question are $\mathbb{Q}_p \to \mathbb{C}$ which don't share the local topology ($\mathbb{Q}_p$ is also totally disconnected, so there is no way to put a reasonable atlas on it). Instead, the intuition comes from how smoothness (or differentiability) means that zooming in far enough gives the same linear behaviour, or a constant one in suitable coordinates. In terms of locally profinite groups, as they are the union of open compact subgroups, a representation being constant on these groups abstractly shares this same property. This loose analogy turns out to be very practical, hence:

Definition 1.2.2: Smooth Map For Locally Profinite Groups

Let G be a locally profinite group. Then a map $f : G \to \mathbb{C}^n$ is *smooth* if it is locally constant, namely for any $x \in G$ there exists an open neighbourhood U containing x such that $f(x) = f(y)$ for all $x, y \in U$. The collection of all smooth functions $f : G \to \mathbb{C}$ is denoted $C^\infty(G)$.
If the function also has compact support, the space is denoted $C_c^\infty(G)$ and called the *smooth compact support space* or *Schwartz-Bruhat Space.*

1.2.3 Precision in Vocabulary Via Harmonic Analysis There is a more precise comparison when considering the Fourier transform with distributions, so the following assumes [13] for the following.

1. On $\mathbb{R}$: For the Fourier transform, the space of "ideal" functions is the Schwartz space, consisting of functions that are both smooth (C^∞) and rapidly decaying. The Fourier transform is a perfect map on this space: the transform of a smooth, rapidly decaying function is another smooth, rapidly decaying function.

2. On $\mathbb{Q}_p$: There is an analogous theory of Fourier analysis, covered in chapter 5. The "ideal" space of functions (called the Schwartz-Bruhat space) consists of functions that are locally constant and have compact support, namely since such functions under the Fourier transform map back to $C_c^\infty(G)$. Note that in the real case the image of real smooth function with compact support is the Paley-Wiener space which are holomorphic function with a certain decay property, something which does not arise in p-adic analysis. Thus, as Fourier relates smoothness to decay, in p-adic analysis being a compact support function "is" smooth from the Fourier perspective.

Proposition 1.2.4: Equivalence Notion of Smoothness

Let G be a locally profinite group and $f : G \to \mathbb{C}^n$ a map. Then f is smooth if and only if for *any* subset $U \subseteq \mathbb{C}^n$ (which is *not* necessarily open), $f^{-1}(U) \subseteq G$ is open.

Thus, if f is smooth, it is continuous with $\mathbb{C}^n$ having the discrete topology.

***Proof*:**
Suppose f is smooth and let $U \subseteq \mathbb{C}^n$ be any subset. If $x \in f^{-1}(U)$, local constancy gives an open neighbourhood V of x on which f is constant, necessarily equal to $f(x) \in U$. So $V \subseteq f^{-1}(U)$, and $f^{-1}(U)$ is open.

Conversely, suppose every preimage is open and let $x \in G$. Applying the hypothesis to the singleton $U = \{f(x)\}$ shows $f^{-1}(f(x))$ is an open set containing x on which f is constant, so f is locally constant, as we sought to show.

The key in the notion of smoothness being useful is that on $\mathbb{Q}_p$ there are non-trivial locally constant functions:

Example 1.2.5: Smooth Functions On Locally Profinite Groups

1. The p-adic valuation $|-|_p : \mathbb{Q}_p^\times \to \mathbb{R}$ is smooth. To see this, take any $x \in \mathbb{Q}_p$ so that $|x|_p = p^{-n}$ for some $n \in \mathbb{Z}$ giving the range. Then the pre-image of p^{-n} is $p^n\mathbb{Z}_p^\times$ which is open.

   Note that this is not smooth on $\mathbb{Q}_p$ because the pre-image of 0 is 0, i.e. there is no open neighbourhood of 0 on which the absolute value is constant. This can even be stretched to the absolute value on $\mathbb{R}$, when restricting to $\mathbb{R}^\times$ it is smooth, but naturally it is not smooth at 0.

2. The map $\mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{R}$ given by $g \mapsto |\det(g)|_p$ is smooth.

Often, a smooth function should be thought of as being well-defined modulo cosets of an open subgroup.

Identifying $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ gives the notion of a smooth representation:

Definition 1.2.6: Smooth Representation

Let G be a locally profinite group. Then a *complex representation* (or just *representation*) of G is a pair (π, V) where V is a complex vector space and $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ a group homomorphism.

A representation (π, V) is called *smooth* if for every $x \in V$, there is a compact open subgroup $K_x \subseteq G$ such that for all $k \in K_x$:

$$\pi(k) \cdot x = x$$

Equivalently, if V^K denotes the space of $\pi(K)$-fixed vector in V,

$$V = \bigcup_K V^K$$

where K ranges over all compact open subgroups of G.

A smooth representation is called *admissible* if V^K is finite dimensional for all compact open $K \subseteq G$.

Proposition 1.2.7: Smooth Representation Alternative

A representation $\pi : G \to \mathrm{Aut}_{\mathbb{C}}(V)$ is smooth if

$$\mathrm{Stab}_G(v) = \{g \in G : \pi(g) \cdot v = v\}$$

is an open subgroup of G for all $v \in V$

***Proof*:**
Suppose $\mathrm{Stab}_G(v)$ is open for every $v \in V$. Fix v. By definition 1.1.12 the open neighbourhood $\mathrm{Stab}_G(v)$ of the identity contains a compact open subgroup K_v of G, and every element of K_v fixes v by construction. That is exactly the condition in definition 1.2.6, so π is smooth.

The converse also holds: if π is smooth then $\mathrm{Stab}_G(v) \supseteq K_v$ for a compact open K_v, and a subgroup containing an open neighbourhood of the identity is open, as we sought to show.

If (π, V) is a smooth representation of G, then any G-stable subspace of V is another smooth representation, and similarly if $U \subseteq V$ is a G-subspace, the induced representation of G on V/U is smooth. A representation (π, V) is *irreducible* if if $V \neq 0$ and V has no nontrivial G-stable subspaces. A G-homomorphism between two representations (π_1, V_1), (π_2, V_2) is a map that commutes with the group action induce by π_i.

Example 1.2.8: Smooth Representations

1. If $(\mathbb{C}, \chi)$ is a one-dimensional representation, by theorem 1.1.13 it is smooth if and only if it is continuous. Hence in one dimensions these two concepts are the same. Indeed, if $\chi(g) \cdot v = v$, as $\chi(g)$ is a complex number and v is nonzero:

$$\chi(g) = 1$$

   Hence, χ is smooth if and only if the kernel is open (and as G is locally profinite it contains a compact subgroup), which by theorem 1.1.13 is true if and only if χ is continuous.

   By corollary 1.1.14, any finite dimensional continuous representation is smooth.

2. Let $G = \mathbb{Q}_p^*$; let us look for maps $\psi : \mathbb{Q}_p^* \to \mathbb{C}^*$ which as $\mathbb{Q}_p$ is the union of compact open sets, theorem 1.1.13 shows every such map lands in S^1. As $\mathbb{Q}_p^* \cong \mathbb{Z} \times \mathbb{Z}_p^*$ (since $x = p^n u$ for $n \in \mathbb{Z}$ and $u \in \mathbb{Z}_p^*$; namely $n = \nu_p(x)$), and ψ is a group homomorphism, the map is determined by where it maps $(1, 0)$ and $\mathbb{Z}_p^*$. Let us look at the two parts seperatly:

$$\psi((c, u)) = \psi((c, 1)) \cdot \psi((0, u)) = \psi_1(c) \cdot \psi_2(u)$$

   Thus, for any choice of $s \in \mathbb{C}$, let:

$$\psi_1(\nu_p(x)) = p^{-s\nu_p(x)}$$

   Next the units. Note that this includes $1 \in \mathbb{Z}_p^*$, and that the kernel must be open, and as $\varphi(1) = 1$, that means that an open neighbourhood around $1 \in \mathbb{Z}_p^*$ maps to $1 \in \mathbb{C}^*$. The open neighborhoods around 1 are:

$$U_k = 1 + p^k\mathbb{Z}_p = \left\{u \in \mathbb{Z}_p^* : u \equiv 1 \bmod (p^k)\right\}$$

   Thus, for some $k \in \mathbb{N}$ $\psi_2(U_k) = 1$ By the isomorphism theorem the cosets will be:

$$\mathbb{Z}_p * / U_K \cong (\mathbb{Z}/p^k\mathbb{Z})^*$$

   But now ψ_2 reduces to $(\mathbb{Z}/p^k\mathbb{Z})^* \to \mathbb{C}^*$ which is just the familiar roots of unity maps. Note how this is "ramified". For this reason, ψ_1 is often called the *unramified component* while ψ_2 is called the *ramified component*.

   Combining these, with $\psi_1(x) = |x|_p^s$ and $\psi_2(x) = \psi(x)$:

$$\chi(x) = |x|_p^s \psi(x)$$

   These classify all continuous/smooth characters of $\mathbb{Q}_p^*$.

3. Let us next consider the next dimension up: $G = \mathrm{GL}_n(\mathbb{Q}_p)$ an $V = \mathbb{C}$, that is smooth characters on $\mathrm{GL}_n(\mathbb{Q}_p)$. Then As $\mathbb{C}^*$ is commutative, $\chi : \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^*$ must be trivial on the commutator. It is an exercise in linear algebra to show that $[G, G] = \mathrm{SL}_n(\mathbb{Q}_p)$[a]. As the identity has determinant 1, $I \in \mathrm{SL}_n(\mathbb{Q}_p)$ and as χ is a group homomorphism it must map to 1. Thus $\chi(\mathrm{SL}_n(\mathbb{Q}_p)) = 1$.

   Next, for any $A, B \in \mathrm{GL}_n(\mathbb{Q}_p)$ such that $\det(A) = \det(B)$ so that $AB^{-1} \in \mathrm{SL}_n(\mathbb{Q}_p)$, necessarily $\chi(A) = \chi(B)$, as χ is a homomorphism. But then, χ is only dependent on the determinant of a matrix.

To construct a character $\chi\colon \mathrm{GL}_n(\mathbb{Q}_p) \to \mathbb{C}^*$, choose a 1-dimensional character $\psi : \mathbb{Q}_p^* \to \mathbb{C}^*$ and compose it with the determinant map $A \mapsto \det(A)$ so that:

$$\chi(A) = \psi(\det(A))$$

It can be shown that all characters are of this form.

4. Take $G = \mathbb{Q}_p$ and $V = \mathbb{C}$, so the objects are the characters $\chi\colon \mathbb{Q}_p \to \mathbb{C}^*$. These are all formed by the *standard character*. For any $x \in \mathbb{Q}_p$ the p-adic representation is of the form

$$x = \sum_{n}^{\infty} a_i p^i \qquad n \in \mathbb{Z},\ a_i \in \{0, 1, ..., p-1\}$$

The p-adic fractional part is the sum of the negatively indexed terms. Write $\{x\}_p$ for it. Define:

$$\psi_p : \mathbb{Q}_p \to \mathbb{C}^* \qquad \psi_p(x) = e^{2\pi i \{x\}_p}$$

Notice the kernel of this map is $\mathbb{Z}_p$, and hence it is continuous/smooth.

5. Characters on $\mathfrak{m}_p$ arise by restricting characters on $\mathbb{Q}_p$. Show that $\hat{\mathfrak{m}}_p \cong \mathbb{Q}_p/\mathbb{Z}_p$.

6. For a continuous non-smooth example, let $G = \mathbb{Z}_p$ and $V = C(\mathbb{Z}_p, \mathbb{C})$ be the $\mathbb{C}$-vector space of continuous characters with the uniform-convergence topology. Then take the left-regular representation (ρ, V):

$$(\rho(g) \cdot f)(x) = f(x - g)$$

The reader can show this is continuous. Let us look at the stabilizer of the vector $g \mapsto |g|_p$. Note this map is certainly continuous as the norm-map is continuous, and thus $|-|_p \in C(\mathbb{Z}_p, \mathbb{C})$. Let us consider the stabilizer of $|-|_p$ that is all, $g \in \mathbb{Z}_p$ such that :

$$g \cdot |\bullet|_p = |\bullet - g|_p = |\bullet|_p$$

Let $g \in \mathrm{Stab}_{\mathbb{Z}_p}(|-|_p)$ so that:

$$|g - x|_p = |x|_p \forall x \in \mathbb{Z}_p$$

If $x \neq g$, then it easy to see that equality *fails*, hence the only stabilizer is 0. But then $\mathrm{Stab}_{\mathbb{Z}_p}(|-|_p) = \{0\}$ which is *not* open.

7. (principal series representation) Let:

$$B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q}_p)\} \right\}$$

Let us define a character on B by choosing two characters $\chi_1, \chi_2 : \mathbb{Q}_p^* \to \mathbb{C}^*$ given by:

$$\varphi\left(\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\right) = \chi_1(a)\chi_2(c)$$

Now, define the complex vector space V to be smooth complex function $f : \mathrm{GL}_2(\mathbb{Q}_p) \to \mathbb{C}$ such that:

$$f(bg) = \psi(b)f(g) \qquad b \in B,\ g \in \mathrm{GL}_2(\mathbb{Q}_p)$$

Then define the action to be:

$$(\rho(h) \cdot f)(g) = f(gh)$$

Then this is an infinite dimensional smooth representation. For a choice of χ_1, χ_2 denote it $I(\chi_1, \chi_2)$. It is almost always irreducible, but there shall be there are non-reducible representations given the right choices of χ_1, χ_2, see example 1.2.13.

By Schur's Lemma, the abelian groups have no higher dimensional irreducible representations. For higher finite dimensional smooth representations of the non-abelian groups, theorem 2.4.16 shall show that every one of them factors through the determinant, even though these groups are not abelian. This shows that to capture some of their more non-abelian structure of these groups, it will be important to look at (well-behaved) infinite dimensional representations, hence the need to add the smooth condition.

Note that by Lie-Kolchin theorem, (complex) irreducible representations of solvable connected groups are always 1-dimensional. Looking $\mathrm{GL}_b(F)$ where F is nonarchemedian in the Zariski topology, it is irreducible and hence connected and hence if it *were* solvable this would apply. It is however *not* the case that it is solvable.

[a] this is true as long as the field F is not the field with two elements

Proposition 1.2.9: Generalized Maschke's Theorem

Let G be compact (thus profinite) and (π, V) an irreducible representation of G. Then V must be finite-dimensional.

***Proof*:**
If $0 \neq v \in V$, then $v \in V^K$ for an open subgroup $K \subseteq G$. As G is profinite, $[G : K]$ is finite and so:

$$\{\pi(g) \cdot v : g \in G/K\}$$

spans V. We can then take the smaller $K' = \bigcup_{g \in G/K} gKg^{-1}$ which is an open normal subgroup of G, still of finite index, and the action is invariant on K'. Thus, V can be seen as an irreducible representation of the finite discrete group G/K', and hence is certainly finite dimensional.

As there are many hierarchies of types of representations in representation theory (semi-stable, crystalline, etc), it is worth naming the category described thus far:

Definition 1.2.10: Smooth Representation Category

Let G be a locally profinite group. Then $\mathrm{Rep}_{\mathrm{smooth},K}(G)$ is the smooth representation category whose objects are smooth representations (V, π) and morphisms are G-equivariant maps. The representations are always smooth, so the notation is abbreviated to $\mathrm{Rep}_K(G)$ instead of $\mathrm{Rep}_{\mathrm{smooth},K}(G)$. The field K is dropped if it is clear from context. By default in this book it shall be $\mathbb{C}$.

By a similar argument as the usual category for representations of finite groups, $\mathrm{Rep}(G)$ is an abelian category. For finite groups, representations can be thought of as $\mathbb{C}[G]$-modules. With the smoothness condition imposed, there are more elements in the $\mathbb{C}[G]$-module V than smooth representations. The remedy is to pass to *Hecke algebras*, as discussed in section 1.4.

The standard decomposition result from finite group representation theory upgrades as follows:

Proposition 1.2.11: Semisimple Equivalence

Let G be a locally profinite group, and let (π, V) be a smooth representation of G. The following conditions are equivalent:

1. V is the sum of its irreducible G-subspaces;
2. V is the direct sum of a family of irreducible G-subspaces;
3. any G-subspace of V has a G-complement in V.

***Proof*:**
(1) $\Rightarrow$ (2). Take the set of irreducible G-subspaces U_i of V

$$\{U_i : i \in I\}$$

such that $V = \sum_{i\in I} U_i$. Consider the set $\mathcal{I}$ of subsets J of I such that the sum $\sum_{i\in J} U_i$ is direct. The set $\mathcal{I}$ is non-empty; Indeed, for an induction argument suppose we have a totally ordered set $\{J_a : a \in A\}$ of elements of $\mathcal{I}$. Put $J = \bigcup_{a\in A} J_a$. If the sum $\sum_{j\in J} U_j$ is not direct, there is a finite subset S of J for which $\sum_{j\in S} U_j$ is not direct. Since S must be contained in some J_a, this is impossible. Therefore $J \in \mathcal{I}$. We can now apply Zorn's Lemma to get a maximal element J_0 of $\mathcal{I}$. For this set, we have

$$V = \bigoplus_{a\in J_0} U_i,$$

as required for (2).

(2) $\Rightarrow$ (3). In (3), let W be a G-subspace of V. By (2), we can assume that $V = \bigoplus_{i\in I} U_i$, for a family (U_i) of irreducible G-subspaces of V. We consider the set $\mathcal{J}$ of subsets J of I for which $W \cap \sum_{i\in J} U_i = 0$. Again, the set $\mathcal{J}$ is nonempty and inductively ordered by inclusion. If J is a maximal element of $\mathcal{J}$, the sum $X = W + \bigoplus_{j\in J} U_j$ is direct. If $X \neq V$, there is $i \in I$ with $U_i \not\subset X$, so the sum $X + U_i$ is direct, and $J \cup \{i\} \in \mathcal{J}$, contrary to hypothesis. Thus (2) $\Rightarrow$ (3).

(3) $\Rightarrow$ (1) Let V_0 be the sum of all irreducible G-subspaces of V and write $V = V_0 \oplus W$, for some G-subspace W of V. Assume for a contradiction that $W \neq 0$. By its definition, the space W has no irreducible G-subspace. However, there is a non-zero G-subspace W_1 of W which is finitely generated over G. By Zorn's Lemma, W_1 has a maximal G-subspace W_0, and then W_1/W_0 is irreducible. We have $V = V_0 \oplus W_0 \oplus U$, for some G-subspace U of V, and hence a G-projection $V \to U$. The image of W_1 in U is an irreducible G-subspace of U, hence an irreducible subspace of V which is not contained in V_0. This is nonsense, so $V = V_0$ and (3) $\Rightarrow$ (1).

Definition 1.2.12: Semisimple Representation

A representation (π, V) is G-*semisimple* if it satisfies one of the equivalent conditions of proposition 1.2.11.

Unlike finite groups, locally profinite groups will have (many) non-semisimple complex representations:

Example 1.2.13: Reducible non-semisimple representation

The details will be left for the reader. Let $\chi_1(x) = |x|_p$ and $\chi_2(x) = |x|_p^{-1}$ and take the principal series representation $I(\chi_1, \chi_2)$. This representation has a unique 1-dimensional subrepresentation which is a character of $\mathrm{GL}_2(\mathbb{Q}_p)$. Denote it χ. The quotient of $I(\chi_1, \chi_2)$ by this representation is called the *Steinberg representation* and is denoted St. This fit into a short exact sequence:

$$0 \to \chi \to I(\chi_1, \chi_2) \to \mathrm{St} \to 0$$

However, this sequence *does not split*. Thus, it is *reducible* but *indecomposable*, hence *not semisimple*.

The compact opens subgroups shall fair better:

Lemma 1.2.14: Compact Open Semistable Representation

Let G be a locally profinite group, and let K be a compact open subgroup of G. Let (π, V) be a smooth representation of G. The space V is the sum of its irreducible K-subspaces:

$$V = \sum_{\substack{W \subseteq V \\ \text{irred. } K\text{-subspace}}} W$$

Such a V is called K*-semisimple*.

***Proof*:**

Let $v \in V$. As in example 1.2.8, v is fixed by an open normal subgroup K' of K, and it generates a finite-dimensional K-space W on which K' acts trivially. Thus W is effectively a finite-dimensional representation of the finite group K/K' and so is the sum of its irreducible K-subspaces. Since $v \in V$ was chosen at random, the lemma follows.

Thus, the following is a meaningful definition:

Definition 1.2.15: Isotypic Component

Let G be a locally profinite group and $K \subseteq G$ a compact open subgroup. Let $\widehat{K}$ denote the set of equivalence classes of irreducible representations. Let $\rho \in \widehat{K}$ and (π, V) be a smooth representation of G. Then:

$$V^\rho = \sum_{\substack{W \subseteq V \\ K\text{-subspace} \\ \text{of class } \rho}} W$$

is called the *ρ-isotypic component* of V.

By definition, V^K is the isotypic subspace for the class of trivial representation 1 of K.

Proposition 1.2.16: K-isotypic Decomposition

Let (π, V) be a smooth representation of G and let K be a compact open subgroup of G.

1. The space V is the direct sum of its K-isotypic components:
$$V = \bigoplus_{\rho \in \widehat{K}} V^\rho$$
2. Let (σ, W) be a smooth representation of G. For any G-homomorphism $f : V \to W$ and $\rho \in \widehat{K}$,
$$f(V^\rho) \subset W^\rho \text{ and } W^\rho \cap f(V) = f(V^\rho)$$

***Proof*:**
By proposition 1.2.11:
$$V = \bigoplus_{i \in I} U_i,$$
for a family of irreducible K-subspaces U_i of V. We let $U(\rho)$ be the sum of those U_i of class ρ. We then have
$$V = \bigoplus_{\rho \in \widehat{K}} U(\rho)$$
If W is an irreducible K-subspace of V of class ρ, then $W \subset U(\rho)$. Otherwise there would be a non-zero K-homomorphism $W \to U_i$, for some U_i of some class $\tau \neq \rho$. We deduce that $V^\rho = U(\rho)$ and (1) follows.

In (2), the image of V^ρ is a sum of irreducible K-subspaces of W, all of class ρ and therefore contained in W^ρ. Moreover, $f(V)$ is the sum of the images $f(V^\tau)$, $\tau \in \widehat{K}$, and $f(V^\tau) \subset W^\tau$. Since the sum of the W^τ is direct, $f(V)$ is the direct sum of the $f(V^\tau)$ and the second assertion follows.

Part (2) of the proposition is used most often in the following context:

Corollary 1.2.17: Exactness reduced to compact subgroup

Let $a : U \to V$, $b : V \to W$ be G-homomorphisms between smooth representations U, V, W of G. The sequence
$$U \xrightarrow{a} V \xrightarrow{b} W$$
is exact if and only if
$$U^K \xrightarrow{a} V^K \xrightarrow{b} W^K$$
is exact, for every compact open subgroup K of G.

***Proof*:**
Both directions rest on proposition 1.2.16: taking K-fixed vectors is the projection onto the isotypic component of the trivial character of K, and by that proposition V is the direct sum of its K-isotypic components, so $V \mapsto V^K$ is an exact functor on smooth representations.

If the original sequence is exact, applying that exact functor gives exactness of the sequence of K-fixed vectors for every compact open K.

Conversely, suppose every K-fixed sequence is exact. Smoothness gives $V = \bigcup_K V^K$, and likewise for U and W. If $v \in \ker b$, choose K fixing v. Then $v \in \ker(b|_{V^K})$, which by hypothesis equals the image of U^K, so $v \in a(U)$. Hence $\ker b \subseteq \operatorname{im} a$, and the reverse inclusion is obtained the same way, as we sought to show.

For a subgroup H of G, define

$$V(H) = \text{the linear span of } \{v - \pi(h)v : v \in V, h \in H\}$$

In particular, $V(H)$ is an H-subspace of V.

Corollary 1.2.18: Splitting Using Compact Subgroups

Let G be a locally profinite group, and let (π, V) be a smooth representation of G. Let K be a compact open subgroup of G. Then

$$V(K) = \bigoplus_{\substack{\rho \in \hat{K} \\ \rho \neq 1}} V^\rho, \quad V = V^K \oplus V(K),$$

and $V(K)$ is the unique K-complement of V^K in V.

***Proof*:**
The sum $W = \bigoplus V^\rho$, with ρ not trivial, is a K-complement of V^K in V. There is therefore a K-surjection $V \to V^K$ with kernel W. Clearly, $V(K)$ is contained in the kernel of any K-homomorphism $V \to V^K$; we conclude that W contains $V(K)$. On the other hand, if U is an irreducible K-space of class $\rho \neq 1$, then $U(K) = U$, so $V^\rho \subset V(K)$.

The next concept is induced representation, and how representations are constructed from it. Recall that in the finite case a subgroup $H \subseteq G$ which has a representation (W, σ) will have a natural induced representation $(\mathbb{C}[G]_{\mathbb{C}[H]} W, \operatorname{Ind}_H^G \sigma)$ where the space $\mathbb{C}[G]_{\mathbb{C}[H]} W$ can be represented as H-equivariant functions and Ind_H^G acts on the set of these functions. The following generalizes this:

Definition 1.2.19: Induced Representation

Let G be a locally profinite group, and let $H \subseteq G$ be a closed subgroup of G so that H is also locally profinite. Let (σ, W) be a smooth representation of H.

Define the space X of functions $f : G \to W$ which satisfy:

1. equivariant: $f(hg) = \sigma(h)f(g)$, $h \in H$, $g \in G$
2. smoothness: there is a compact open subgroup K of G (depending on f) such that $f(gx) = f(g)$, for $g \in G$, $x \in K$ (so f is smooth i.e. continuous with W having the discrete topology)

Then define a homomorphism $\Sigma : G \to \mathrm{Aut}_{\mathbb{C}}(X)$ by

$$\Sigma(g)f : x \longmapsto f(xg), \quad g, x \in G$$

The pair (Σ, X) provides a smooth representation of G and is said to be *smoothly induced by* σ. It is usually denoted

$$(\Sigma, X) = \mathrm{Ind}_H^G \sigma$$

Proposition 1.2.20: Induced Representation is A Functor

The map $\sigma \mapsto \mathrm{Ind}_H^G \sigma$ gives a functor $\mathrm{Rep}(H) \to \mathrm{Rep}(G)$.

***Proof*:**
Let $u \colon (\sigma, W) \to (\sigma', W')$ be an H-homomorphism. Post-composition sends a function $f \colon G \to W$ to $u \circ f \colon G \to W'$. This preserves the two defining conditions of definition 1.2.19: equivariance because $u(\sigma(h)w) = \sigma'(h)u(w)$, and smoothness because $u \circ f$ is fixed by any compact open subgroup fixing f. It also commutes with the action $\Sigma(g)f \colon x \mapsto f(xg)$, so it is a G-homomorphism $\mathrm{Ind}_H^G \sigma \to \mathrm{Ind}_H^G \sigma'$. Functoriality is immediate, since post-composition respects identities and composites, completing the proof.

There is a canonical H-homomorphism

$$\alpha_\sigma : \mathrm{Ind}_H^G \sigma \longrightarrow W, \quad f \longmapsto f(1)$$

The pair $(\mathrm{Ind}_H^G, \alpha)$ has the following fundamental property:

Theorem 1.2.21: Frobenius Reciprocity

Let H be a closed subgroup of a locally profinite group G. For a smooth representation (σ, W) of H and a smooth representation (π, V) of G, the canonical map

$$\begin{gathered} \mathrm{Hom}_G(\pi, \mathrm{Ind}_H^G \sigma) \longrightarrow \mathrm{Hom}_H(\pi, \sigma) \\ \varphi \mapsto \alpha_\sigma \circ \varphi, \end{gathered}$$

is an isomorphism that is functorial in both variables π, σ.

***Proof*:**
Let $f : V \to W$ be an H-homomorphism. We define a G-homomorphism $f_* : V \to \mathrm{Ind}\sigma$ by letting $f_*(v)$ be the function $g \mapsto f(\pi(g)v)$. The map $f \mapsto f_*$ is then the inverse of the above map.

The following shows how this is a very useful functor:

Proposition 1.2.22: Induced Representation is Additive and Exact

The functor $\mathrm{Ind}_H^G : \mathrm{Rep}(H) \to \mathrm{Rep}(G)$ is additive and exact.

***Proof*:**
For a smooth representation (σ, W) of H, temporarily let $I(\sigma)$ denote the space of functions $G \to W$ satisfying the first condition $f(hg) = \sigma(h)f(g)$ of the definition above. Thus I is a functor to the category of abstract representations of G; it is clearly additive and exact, while $\mathrm{Ind}_H^G(\sigma) = I(\sigma)^\infty$. Thus Ind_H^G is additive, and left-exactness follows because a left-exact sequence of H-spaces induces one on equivariant function spaces.

To prove it is right-exact, let (σ, W), (τ, U) be smooth representations of H and let $f : W \to U$ be an H-surjection. Take $\varphi \in I(\tau)^\infty$, and choose a compact open subgroup K of G which fixes φ. The support of φ is a union of cosets HgK, and the value $\varphi(g) \in U$ must be fixed by $\tau(H \cap gKg^{-1})$. By corollary 1.2.18 (applied to the group H and the trivial representation of its compact open subgroup $H \cap gKg^{-1}$), there exists $w_g \in W$, fixed by $\sigma(H \cap gKg^{-1})$, such that $f(w_g) = \varphi(g)$. We define a function $\Phi : G \to W$ to have the same support as φ and $\Phi(hgk) = \sigma(h)w_g$, for each $g \in H\backslash \mathrm{supp}\varphi/K$. The function Φ is fixed by K, and hence lies in $I(\sigma)^\infty$. Its image in $I(\tau)^\infty$ is φ, as required.

Example 1.2.23: Induced Representation

1. here
2. here

As the compact open subgroups are the key subgroups to consider for this book, there is the following useful variation of induced representation.

Definition 1.2.24: Compact Induction

Given (σ, W) for a representation of closed $H \subseteq G$ and similarly defined X as in definition 1.2.19, define:

$$X_c = \{f \in X : \text{compactly supported modulo } H\}$$

that is the image of $\mathrm{supp}(f)$ in $H \setminus G$ is compact (equivalently, $\mathrm{supp}(f) \subseteq HC$ for some compact set $C \subseteq G$). The induced functor is denoted $\text{c-}\mathrm{Ind}_H^G\sigma$.

Proposition 1.2.25: Compact Induction is Well-defined

The space X_c along with c-$\mathrm{Ind}_H^G \sigma$ is a smooth representation.

***Proof*:**
The space X_c is a subspace of X: the supports of $f_1 + f_2$ and of λf are contained in the union of the supports, and a union of two sets compact modulo H is again compact modulo H.

It is stable under the action. For $g \in G$ the function $\Sigma(g)f \colon x \mapsto f(xg)$ has support $(\mathrm{supp} f)g^{-1}$, whose image in $H\backslash G$ is the translate by g^{-1} of the image of $\mathrm{supp} f$. Right translation is a homeomorphism of $H\backslash G$, so the translate is again compact.

Smoothness is inherited from X: each $f \in X_c$ is already fixed by a compact open subgroup of G, because it satisfies the second condition of definition 1.2.19. Hence $(\text{c-}\mathrm{Ind}_H^G \sigma, X_c)$ is a smooth representation of G, as we sought to show.

Proposition 1.2.26: Compact Induction Repreis Additive and Exact

The functor c-$\mathrm{Ind}_H^G \colon \mathrm{Rep}(H) \to \mathrm{Rep}(G)$ is additive and exact.

***Proof*:**
Additivity is clear, since post-composition with a direct sum decomposition splits X_c accordingly.

For exactness, let $0 \to W' \to W \to W'' \to 0$ be exact in $\mathrm{Rep}(H)$. Left exactness is inherited from Ind_H^G (proposition 1.2.22), because c-$\mathrm{Ind}_H^G \sigma$ is a subspace of $\mathrm{Ind}_H^G \sigma$ and the support condition is preserved by an injection.

Right exactness is where the compact support helps rather than hinders. Let $\varphi \in \text{c-}\mathrm{Ind}_H^G \sigma''$ and choose a compact open K fixing it. Its support is a finite union of double cosets HgK, since the image of the support in $H\backslash G$ is compact and covered by the open sets HgK. On each such coset pick, using surjectivity of $W \to W''$ and corollary 1.2.18 applied to $H \cap gKg^{-1}$, a vector $w_g \in W$ fixed by $\sigma(H \cap gKg^{-1})$ mapping to $\varphi(g)$. Defining $\Phi(hgk) = \sigma(h)w_g$ on that coset and 0 elsewhere gives a well-defined element of c-$\mathrm{Ind}_H^G \sigma$ with the same, hence compact modulo H, support, lifting φ. So c-Ind_H^G is exact, completing the proof.

Example 1.2.27: Compact Induction

1. here
2. here

There is a natural G-embedding:

$$\text{c-}\mathrm{Ind}_H^G(\sigma) \to \mathrm{Ind}_H^G(\sigma) \qquad \text{(i.e. a natural transformation c-}\mathrm{Ind}_H^G \Rightarrow \mathrm{Ind}_H^G)$$

This map is an isomorphism if and only if $H \setminus G$ is compact. [1] says that there are examples of such a map being an isomorphism even when $H \setminus G$ is not compact with the right choice of G, H, σ.

The map is mostly interesting when $H \subseteq G$ is open in which case there is a natural H-homomorphism:

$$\alpha_\sigma^c : W \to \text{c-Ind}(\sigma) \qquad w \mapsto f_w$$

Frobenius reciprocity for *compact* induction is not developed here. The adjunction it would supply is useful downstream: it is what converts an orbit description of $C_c^\infty(M_2(F))$ into a computable Hom space, and its absence is why lemma 5.2.8 is quoted rather than proved. See [1, §2] for the statement.

1.2.28 Hypothesis From Now On For any compact open $K \subseteq G$, G/K is countable.

If G/K is countable, so is G/K' for any other compact open. Indeed $K \cap K'$ is compact open and a finite index of K thus

$$\frac{G}{K \cap K'} \to G/K$$

has finite fibers, thus $G/K \cap K'$ and G/K' are both countable. All groups that will be considered in this book are have this property, but a few further results bellow don't require them. The main point is so that:

Lemma 1.2.29: Irreducible Smooth Representation At Most Countable

Let (π, V) be a smooth irreducible representation of a locally profinite group G. Then $\dim_{\mathbb{C}}(V)$ is countable

***Proof*:**
Let $0 \neq v \in V$ and choose $K \subseteq G$ such that $v \in V^K$. As V is irreducible,

$$\{\pi(g) \cdot v : g \in G/K\}$$

spans V, and this set is countable completing the proof.

This also allows for the following:

Theorem 1.2.30: Schur's Lemma For Smooth Representations

Let (π, V) be a smooth irreducible representation of a locally profinite group G. Then

$$\text{End}_{\mathbb{C}}(V) = \mathbb{C}$$

***Proof*:**
Let $0 \neq \varphi \in \text{End}_{\mathbb{C}}(V)$. As the image and kernel of φ must be G-subspaces, by irreducibility it must be bijective (which is enough in this category for invertible). Thus, $\text{End}_{\mathbb{C}}(V)$ is at least a division $\mathbb{C}$-algebra.

Next, $\text{End}_{\mathbb{C}}(V)$ must have countable dimension for if $0 \neq v \in V$, the G-translate of v spans V, and so $\varphi \in \text{End}_{\mathbb{C}}(V)$ is uniquely determined by the value of $\varphi(v)$. Now, if $\varphi \in \text{End}_{\mathbb{C}}(V)$ where $\varphi \tilde{\in} \mathbb{C}$, φ would be transcendental over $\mathbb{C}$ and hence the subset

$$\left\{(\varphi - a)^{-1} : a \in \mathbb{C}\right\}$$

would be linearly independent, and hence $\mathrm{End}_{\mathbb{C}}(V)$ would have uncountable $\mathbb{C}$-dimension, a contradiction. Thus, $\mathrm{End}_{\mathbb{C}}(V) \cong \mathbb{C}$, as we sought to show.

Note that this is an if and only if when working with finite groups. If G is compact, this becomes an if and only if. This is false in general ([1] cites section 9.10).

Corollary 1.2.31: Central Character

Let (π, V) be an irreducible smooth representation of G and $Z = Z(G)$ is the center of G. Then Z acts on V via the character

$$\omega_\pi : Z \to \mathbb{C}^* \qquad \text{i.e.} \qquad \pi(z) \cdot v = \omega_\pi(z) \cdot v$$

for all $v \in V$ and $z \in Z$

***Proof*:**
As $z \in G$, the left-multiplication map is an endomorphism, hence $\pi(z) \in \mathrm{End}_G(V) \cong G$. Thus $\pi(z)$ is some scalar multiple of id_V, hence $\pi(z) = \omega_\pi(z)$. To shows it's a character, note that if K is a compact open subgroup of G such that $V^K \neq 0$, Then ω_π is trivial om the compact open subgroup $K \cap Z$, showing it is indeed a smooth representation, and hence ω_π is a character of Z.

Definition 1.2.32: Central Character

The function $Z \to \mathbb{C}^*$ defined in corollary 1.2.31 is called a *central character*.

Definition 1.2.33: Smooth Representation Admitting Central Character

Let (π, V) be a smooth representation of G and χ a central character of $Z(G)$. One says that (π, V) *admits χ as a central character* if:

$$\pi(z) \cdot v = \chi(z) \cdot v \qquad \forall v \in V,\ z \in V$$

Proposition 1.2.34: Irreducible Smooth Representation of Abelian Groups

Let G be an abelian locally profinite group. Then an smooth irreducible representation is one-dimensional

***Proof*:**
By corollary 1.2.31, any representation π will act like the character at every element $g \in G$ since $Z(G) = G$. Now, if W is any one-dimensional subspace, as π acts by scaling, W is a G-subspace. Hence, any irreducible subspace must be one-dimensional, completing the proof.

Proposition 1.2.35: KZ Space

Let (π, V) be a smooth representation of G, admitting χ as a central character. Let K be an open subgroup of G such that KZ/Z is compact. Then:

1. Let $v \in V$. The KZ-space spanned by v is of finite dimension, and is a sum of irreducible KZ-spaces.

2. As representation of KZ, the space V is semisimple.

***Proof*:**
The vector v is fixed by a compact open subgroup K_0 of K. The set KZ/K_0Z is finite, so the space W spanned by $\pi(KZ)v$ has finite dimension. Then W is K_0Z-semisimple. Since v was chosen arbitrarily, we get 2 as we sought to show.

The final elementary concept to adders is the dual space and how a representation (π, V) relates to it:

The full linear dual of a smooth representation is almost never smooth: a functional built out of infinitely many coordinates is fixed by no compact open subgroup. The remedy is the same one that produced the definition of a smooth representation in the first place, namely to keep only the vectors that are fixed by something open.

Definition 1.2.36: Smooth Dual (Contragradient)

Let (π, V) be a smooth representation of a locally profinite group G, and let V^* denote the full linear dual of V, carrying the action

$$\langle \pi^*(g)\lambda, v\rangle = \langle \lambda, \pi(g^{-1})v\rangle \qquad g \in G,\ \lambda \in V^*,\ v \in V$$

The *smooth dual* (or *contragredient*) of (π, V) is the subspace of smooth vectors

$$\check{V} = \bigcup_K (V^*)^K$$

where K runs over the compact open subgroups of G, equipped with the restriction $\check{\pi}$ of π^*. It is a smooth representation of G by construction.

The following lemma is used in the discussion of smooth duals below.

Lemma 1.2.37: Semisimple With Finite-index and Induction

Let G be a locally profinite group, and let H be an open subgroup of G of finite index.

1. If (π, V) is a smooth representation of G, then V is G-semisimple if and only if it is H-semisimple.

2. Let (σ, W) be a semisimple smooth representation of H. The induced representation $\mathrm{Ind}_H^G \sigma$ is G-semisimple.

***Proof*:**
Suppose that V is H-semisimple, and let U be a G-subspace of V. By hypothesis, there is an H-subspace W of V such that $V = U \oplus W$. Let $f : V \to U$ be the H-projection with kernel W. Consider the map

$$f^G : v \mapsto (G : H)^{-1} \sum_{g \in G/H} \pi(g) f(\pi(g)^{-1} v), \quad v \in V$$

The definition is independent of the choice of coset representatives and it follows that f^G is a G-projection $V \to U$. We then have $V = U \oplus \operatorname{Ker} f^G$ and $\operatorname{Ker} f^G$ is a G-subspace of V. Thus V is G-semisimple by proposition 1.2.11.

Conversely, suppose that V is G-semisimple. Thus V is a direct sum of irreducible G-subspaces by proposition 1.2.11, and it is enough to treat the case where V is irreducible over G. As representation of H, the space V is finitely generated and so admits an irreducible H-quotient U. Suppose for the moment that H is a normal subgroup of G. By Frobenius Reciprocity (theorem 1.2.21), the H-map $V \to U$ gives a non-trivial, hence injective, G-map $V \to \operatorname{Ind}_H^G U$. As representation of H, the induced representation $\operatorname{Ind}_H^G U = \oplus_{g \in G/H} gU$ is a direct sum of G-conjugates of U. These are all irreducible over H, so $\operatorname{Ind} U$ is H-semisimple. Proposition 1.2.11 then implies that $V \subset \operatorname{Ind}_H^G U$ is H-semisimple.

In general, we set $H_0 = \bigcap_{g \in G/H} gHg^{-1}$. This is an open normal subgroup of G of finite index. We have just shown that the G-space V is H_0-semisimple; the first part of the proof shows it is H-semisimple.

This completes the proof of (1), and (2) follows readily from the same arguments.

The smooth dual is well behaved exactly on the admissible representations, where it becomes an involution. The mechanism is that passing to K-fixed vectors turns it into ordinary linear duality of finite-dimensional spaces, where reflexivity is automatic.

Proposition 1.2.38: Smooth Dual

Let (π, V) be a smooth representation of a locally profinite group G.

1. For every compact open subgroup K of G, restriction of functionals induces an isomorphism $\check{V}^K \cong (V^K)^*$.
2. If π is admissible, so is $\check{\pi}$, and the canonical map $V \to \check{\check{V}}$ is an isomorphism of representations.
3. If π is admissible, then π is irreducible if and only if $\check{\pi}$ is irreducible.

***Proof*:**

1. By corollary 1.2.18 we have $V = V^K \oplus V(K)$, where $V(K)$ is spanned by the vectors $v - \pi(k)v$. A functional $\lambda \in V^*$ is fixed by K precisely when $\lambda(v - \pi(k)v) = 0$ for all v and all $k \in K$, that is, precisely when λ annihilates $V(K)$. Such a λ is determined by its restriction to V^K, and conversely any functional on V^K extends by zero on $V(K)$ to a K-fixed element of V^*, which lies in $\check{V}$ by definition. The two constructions are mutually

inverse.

2. If π is admissible then V^K is finite dimensional, so $\check{V}^K \cong (V^K)^*$ is too by (1), and $\check{\pi}$ is admissible.

   For $v \in V$ let ev_v be evaluation at v. If $v \in V^K$ then ev_v is K-fixed, so the canonical map lands in the smooth dual of $\check{V}$ and is G-equivariant. Taking K-fixed vectors, it becomes the canonical map $V^K \to (\check{V}^K)^* \cong ((V^K)^*)^*$ of a finite-dimensional space into its double dual, which is an isomorphism. Since every vector on either side is fixed by some compact open subgroup, and V^K increases to V as K shrinks, the map is an isomorphism.

3. Suppose π is reducible, say U is a G-subspace with $0 \neq U \neq V$, and set

   $$U^\perp = \left\{\lambda \in \check{V} : \lambda|_U = 0\right\}$$

   a G-subspace of $\check{V}$. It is non-zero: choose $v \in V \setminus U$ and K fixing v, so that $v \in V^K$ while $v \notin U^K = U \cap V^K$. A functional on V^K vanishing on U^K but not at v then gives, via (1), a non-zero element of $U^\perp$. It is proper: choosing $0 \neq u \in U$ and a compact open K fixing u, a functional on V^K non-zero at u gives an element of $\check{V}$ outside $U^\perp$. Hence $\check{\pi}$ is reducible.

   Conversely, if $\check{\pi}$ is reducible then the same argument applied to $\check{\pi}$, which is admissible by (2), shows $\check{(\check{\pi})}$ is reducible. By (2) again that representation is π, completing the proof.

1.3 Measures and Duality

The space $\mathbb{C}[G]$ is $\mathbb{C}$-algebra isomorphic to $(\mathrm{Hom}_{\mathbf{Top}}(G, \mathbb{C}), +, *)$ where $*$ is convolution. This requires summing over each $g \in G$. In this book, G is a.e. infinite, and so we replace summation with an appropriate measure on G.

Let G be a locally profinite group, $C_c^\infty(G)$ the space of functions $f : G \to \mathbb{C}$ which are locally constant and of compact support (i.e. smooth and of compact support). Let $f \in C_c^\infty(G)$. Then compact support and smoothness (locally constant), here exists open subgroup $K_1, K_2 \subseteq G$ such that

$$f(k_1 g) = f(g) = f(g k_2) \qquad \forall g \in G,\ k_i \in K_i$$

By letting $K = K_1 \cap K_2$, f must be a finite linear combination of characteristic functions of double cosets KgK. Now, $G \curvearrowright C_c^\infty(G)$ by left translation and right translation:

$$\begin{cases} \lambda : g \mapsto (\lambda_g f : x \mapsto f(g^{-1}x)) \\ \rho : g \mapsto (\rho_g f : x \mapsto f(xg) \end{cases} \qquad x, g \in G,\ f \in C_c^\infty(G)$$

Note that the left action is the one investing since the action is on functions. To see this, apply the operator λ_{g_2} to the function f. This produces a new function:

$$h = \lambda_{g_2} f$$

The definition of this new function h is:

$$h(y) = (\lambda_{g_2} f)(y) = f(g_2 y)$$

Now we apply the operator λ_{g_1} to h; call the new function k:

$$k = \lambda_{g_1} h = \lambda_{g_1}(\lambda_{g_2} f) \qquad \text{so} \qquad k(z) = (\lambda_{g_1} h)(z) = h(g_1 z)$$

therefore:

$$k(z) = f(g_2(g_1 z)) = f(g_2 g_1 z)$$

showing why the left input needs to be the one inverted. The reader should check that $(C_c^\infty(G), \lambda)$ and $(C_c^\infty(G), \rho)$ are both smooth G-representation.

Definition 1.3.1: Haar Integral

A *right Haar integral* on G is a non-zero linear functional:

$$I : C_c^\infty(G) \to \mathbb{C}$$

such that

1. $I(\rho_g f) = I(f)$ for all $g \in G,\ f \in C_c^\infty(G)$
2. For all $f \in C_c^\infty(G)$ where $f \geq 0$, $I(f) \geq 0$.

And similarly for the left Haar integral.

Proposition 1.3.2: Existence of Haar Integral

There exists right Haar integral $I : C_c^\infty(G) \to \mathbb{C}$. A linear functional $I' : C_c^\infty(G) \to \mathbb{C}$ is a right Haar integral if and only if $I' = cI$ for some $c > 0$.

The same then applies to the left Haar integral.

***Proof*:**

Let K be a compact open subgroup of G. Denote by ${}^K C_c^\infty(G)$ the space of functions in $C_c^\infty(G)$ fixed by $\lambda(K)$. Then viewing ${}^K C_c^\infty(G)$ as G-space via right translation, it is then identical to the representation of G compactly induced from the trivial representation 1_K of K:

$${}^K C_c^\infty(G) = c\text{-}\mathrm{Ind}_K^G 1_K$$

Now, viewing $\mathbb{C}$ as the trivial G-space (and as a consequence K-space), by compact Frobenius reciprocity:

$$\dim_{\mathbb{C}} \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \mathrm{Hom}_K(\mathbb{C}, \mathbb{C}) = 1$$

Next, let's show there exists a non-zero element $I_K \in \mathrm{Hom}_G({}^K C_c^\infty(G), \mathbb{C})$ such that $I_K(f) \geq 0$ whenever $f \geq 0$, and if h_K is the characteristic function of K, then $I_K(h_K) > 0$. Indeed, for $g \in G$, let f_g denote the characteristic function of Kg. The set of functions f_g, $g \in K\backslash G$, then forms a $\mathbb{C}$-basis of the space ${}^K C_c^\infty(G)$. The functional $I_K : f_g \mapsto 1$ has the required properties, noting that $h_K = f_1$.

Now, choose a descending sequence $\{K_n\}_{n \geq 1}$ of compact open subgroups K_n of G such that

$\bigcap_n K_n = 1$. Then:

$$C_c^\infty(G) = \bigcup_{n\geq 1} {}^{K_n}C_c^\infty(G)$$

For each $n \geq 1$, there is a unique right G-invariant functional I_n on ${}^{K_n}C_c^\infty(G)$ which maps the characteristic function of K_n to $(K_1 : K_n)^{-1}$. We have $I_{n+1}|_{{}^{K_n}C_c^\infty(G)} = I_n$, and so the family $\{I_n\}$ gives a functional on $C_c^\infty(G)$ of the required kind. The uniqueness statement is immediate, completing the proof.

In the above, think $K = \mathbb{Z}_p$ so that $f_g \to 1$ so $\mu(\mathbb{Z}_p) = 1$. The rest of the compact sets are given by the translation invariance and taking the index $[H : K]$ when $K \subseteq H$. This gives $\mu(a + p^n\mathbb{Z}_p) = \mu(p^n\mathbb{Z}_p) = p^{-n}$. This extends to a left Haar integral on $\mathbb{Q}_p$. For a finite extension $F/\mathbb{Q}_p$, simply replace p with $\mathfrak{p}$ in the above. For $\mathrm{GL}_k(\mathbb{Q}_p)$, recall the chain of compact open subsets:

$$K_m = \{g \in K_0 \big| g \equiv I \pmod{p^m}\} = I + p^m M_k(\mathbb{Z}_p)$$

where $K_0 = \mathrm{GL}_k(\mathbb{Z}_p)$ has measure 1, and you do the same process.

Corollary 1.3.3: Left-Right Haar Integral

For $f \in C_c^\infty(G)$, define $\check{f} \in C_c^\infty(G)$ by $\check{f}(g) = f(g^{-1})$, $g \in G$. The functional

$$I' : C_c^\infty(G) \longrightarrow \mathbb{C},$$

$$I'(f) = I(\check{f}),$$

is a left Haar integral on G. Moreover, any left Haar integral on G is of the form cI', for some $c > 0$.

***Proof*:**
Inversion $g \mapsto g^{-1}$ is a homeomorphism of G carrying compact sets to compact sets and locally constant functions to locally constant functions, so $f \mapsto \check{f}$ is a linear automorphism of $C_c^\infty(G)$, and I' is a non-zero positive linear functional.

For left invariance, let $x \in G$ and write $\lambda_x f \colon g \mapsto f(x^{-1}g)$. Then $(\lambda_x f)^\vee(g) = f(x^{-1}g^{-1}) = \check{f}(gx)$, which is the right translate of $\check{f}$ by x. Since I is a right Haar integral it is invariant under right translation, so $I'(\lambda_x f) = I(\check{f}) = I'(f)$. Thus I' is left invariant.

Uniqueness up to a positive scalar is the uniqueness clause of proposition 1.3.2 applied on the left, as we sought to show.

This defines a measure called the *Haar measure*. Let I be a left Haar integral on G and $\emptyset \neq S \subseteq G$ a compact open subset of a locally profinite group.

$$\mu_G(S) = I(\Gamma_S)$$

where Γ_S is the characteristic function of S. Then $\mu_G(S) > 0$ and μ_G is left-translation invariant:

$$\mu_G(gS) = \mu_G(S) \qquad \forall g \in G$$

The reader should check that this is indeed a measure (see [13] for a review of measure theory)

Definition 1.3.4: Haar Measure

The measure μ_G defined above is called the *Haar measure*.

The function I is often written as:

$$I(f) = \int_G f(g) d\mu_G(g) \qquad f \in C_c^\infty(G)$$

with the usual abbreviation of:

$$\int_S f(x) d\mu_G(x) := \int_G \Gamma_S(x) f(x) d\mu_G(x)$$

Definition 1.3.5: Unimodular Group

Let G be a locally profinite group (more generally a topological group with a Haar measure defined on it). Then G is *unimodular* if and only if the left Haar integral and is a right Haar integral.

Example 1.3.6: Left and Right Haar Measures

1. put the standard ones and interesting ones here, including the ones alluded earlier.

This is well-defined when working with functions with compact support. However, there are times when one have non-compact induction $\mathrm{Ind}_H^G \sigma$ where $H \backslash G$ is not compact (in this case it is often countable). To that end, extend the domain of integration of the Haar measure. Consider $f : G \to \mathbb{C}$ that is invariant under left translation by a compact subgroup $K \subseteq G$, and let μ_G be a left Haar measure on G. Then if the series:

$$\sum_{g \in K \backslash G} \int_{Kg} |f(x)| d\mu_G(x)$$

converges, so does the version without absolute values, so the following is well defined:

$$\int_G f(x) \mu_G(x) = \sum_{g \in K \backslash G} \int_{Kg} f(x) d\mu_G(x)$$

Notice the cosets are right cosets, because the integral is left-invariant, and hence would have been a sum of constants and hence necessarily diverge. Furthermore, as μ_G is translation invariant this series is independent of the choice of compact open subgroup, since $K = K_1 \cap K_2$ may be taken, which has finite index with both K_1 and K_2.

Characters are often constructed using tensor products, which requires extending a Haar measure. Let G_1, G_2 be locally profinite groups, and set $G = G_1 \times G_2$. Then G is locally profinite. An element $\sum_{1 \le i \le r} f_i^1 \otimes f_i^2$ of the tensor product $C_c^\infty(G_1) \otimes C_c^\infty(G_2)$ gives a function on G by

$$\Phi(g_1, g_2) = \sum_i f_i^1(g_1) f_i^2(g_2)$$

Then $\Phi \in C_c^\infty(G)$. Importantly, this gives an isomorphism

$$C_c^\infty(G_1) \otimes C_c^\infty(G_2) \cong C_c^\infty(G)$$

Let μ_j be a left Haar measure on G_j, $j = 1, 2$. Then there is a unique left Haar measure μ_G on G such that

$$\int_G f_1 \otimes f_2(g)\, d\mu_G(g) = \int_{G_1} f_1(g_1)\, d\mu_1(g_1) \int_{G_2} f_2(g_2)\, d\mu_2(g_2),$$

for $f_i \in C_c^\infty(G_i)$. One writes $\mu_G = \mu_1 \otimes \mu_2$.

For a general $f \in C_c^\infty(G)$, the function

$$f_1(g_1) = \int_{G_2} f(g_1, g_2)\, d\mu_2(g_2)$$

lies in $C_c^\infty(G_1)$, and

$$\int_G f(g)\, d\mu_G(g) = \int_{G_1} f_1(g_1)\, d\mu_1(g_1),$$

and symmetrically: if f is of the form $f_1 \otimes f_2$, this is obvious, and such functions span $C_c^\infty(G)$.

Irreducible representations bring in smooth functions of the form $f\colon G \to V$ that need to be integrated. Let V be a complex vector space, and consider the space $C_c^\infty(G; V)$ of locally constant, compactly supported functions $f : G \to V$. This space is isomorphic to $C_c^\infty(G) \otimes V$ in the obvious way: a tensor $\sum_i f_i \otimes v_i$ gives the function

$$g \mapsto \sum_i f_i(g) v_i$$

If μ_G is a left Haar measure on G, there is a unique linear map $I_V : C_c^\infty(G; V) \to V$ such that

$$I_V(f \otimes v) = \int_G f(g) d\mu_G(g) \cdot v$$

and so define:

$$I_V(\varphi) := \int_G \varphi(g) d\mu_G(g), \quad \varphi \in C_c^\infty(G; V)$$

This has the same invariance properties as the Haar integral on scalar-valued functions.

There are cases where the left and right Haar measures do not agree (recall example 1.3.6). By the uniqueness of the left and right Haar measure up to a constant, one can cleverly equate the left/right Haar integration up to a constant. Let μ_G be a left Haar measure on G. For $g \in G$, consider the functional

$$\begin{aligned} I_g : C_c^\infty(G) &\to \mathbb{C}, \\ f &\mapsto \int_G f(xg)\, d\mu_G(x) \end{aligned}$$

Then I_g is a left Haar integral on G, so there is a unique $\delta_G(g) \in \mathbb{R}_+^\times$ such that

$$\delta_G(g) \int_G f(xg)\, d\mu_G(x) = \int_G f(x)\, d\mu_G(x),$$

for all $f \in C_c^\infty(G)$. The function δ_G is a homomorphism $G \to \mathbb{R}_+^\times$. It is trivial if G is abelian. More generally, δ_G is trivial if and only if G is unimodular. If f is the characteristic function of a compact open subgroup K of G, then δ_G is trivial on K and hence a character of G.

Definition 1.3.7: Haar Module of G

The function δ_G as define above is the *module* of G.

Note that if G is compact, then $\delta_G = 1$ and G is unimodular. In the general case, any character $G \to \mathbb{R}_+^*$ is trivial on compact subgroups, since $\mathbb{R}_+^*$ has only the trivial compact subgroup.

Using the module, the functional

$$f \mapsto \int_G \delta_G(x)^{-1} f(x)\, d\mu_G(x), \quad f \in C_c^\infty(G),$$

is a right Haar integral on G. The mnemonic

$$d\mu_G(xg) = \delta_G(g) d\mu_G(x)$$

would help on how the left/right measures “commute”.

Next, many important irreducible representations shall be G-subspaces of simpler induced representations, very often by taking a character on a closed subgroup and inducing it. Let H be a closed subgroup of G, with module δ_H. Let $\theta : H \to \mathbb{C}^\times$ be a character of H. Consider the space of functions $f : G \to \mathbb{C}$ which are G-smooth under right translation, compactly supported modulo H, and satisfy the “induction” homogeneity property:

$$f(hg) = \theta(h)f(g), \quad h \in H, g \in G$$

Definition 1.3.8: Induction Space of Smooth Functions

Label this space

$$C_c^\infty(H\backslash G, \theta)$$

and view it as a smooth G-space via right translation ρ.

Note that $C_c^\infty(H\backslash G, \theta) = \text{c-Ind}_H^G \theta$.

Theorem 1.3.9: Compact Induction on Haar Measure

Let $\theta : H \to \mathbb{C}^\times$ be a character of H. The following are equivalent:

1. There exists a non-zero linear functional $I_\theta : C_c^\infty(H\backslash G, \theta) \to \mathbb{C}$ such that $I_\theta(\rho_g f) = I_\theta(f)$, for all $g \in G$.
2. $\theta\delta_H = \delta_G|_H$.

When these conditions hold, the functional I_θ is uniquely determined up to constant factor.

***Proof*:**

(1) $\Rightarrow$ (2) Let μ_G, μ_H be left Haar measures on G, H respectively. Define a G-map

$$C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta) \qquad f \mapsto \tilde{f} \qquad \text{where} \qquad \tilde{f}(g) = \int_H \theta\delta_H(h)^{-1} f(hg)\, d\mu_H(h)$$

This map satisfies

$$\widetilde{\lambda_k f} = \delta_H\theta(k)^{-1}\tilde{f},$$

for $k \in H$ and $f \in C_c^\infty(G)$. If this map is surjective, we shall be able to factor I_ω through this map, allowing to link the two moduli.

To that end, If K is a compact open subgroup of G, the space $C_c^\infty(G)^K$ by a simple generalization of proposition 2.2.6 this space is spanned by the characteristic functions of cosets $gK, g \in G/K$. On the other hand, each coset HgK supports, at most, a one-dimensional space of functions in $C_c^\infty(H\backslash G, \theta)^K$. These subspaces span $C_c^\infty(H\backslash G, \theta)^K$, so the map $f \mapsto \tilde{f}$ is surjective on K-fixed functions, proving surjectivity.

Suppose that the space $C_c^\infty(H\backslash G, \theta)$ admits a functional I_θ of the required kind. The map $f \mapsto I_\theta(\tilde{f})$ is then a non-trivial G-homomorphism

$$(C_c^\infty(G), \rho) \to \mathbb{C}$$

However, the space $\operatorname{Hom}_G(C_c^\infty(G), \mathbb{C})$ has dimension 1 by Frobenius Reciprocity:

$$\dim_{\mathbb{C}} \operatorname{Hom}_G({}^K C_c^\infty(G), \mathbb{C}) = \dim_{\mathbb{C}} \operatorname{Hom}_K(\mathbb{C}, \mathbb{C}) = 1$$

and is spanned by a right Haar integral. Therefore, the function I_θ exists if and only if the right Haar integral $C_c^\infty(G) \to \mathbb{C}$ factors through the quotient map $C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta)$. When it does hold, I_θ is determined up to a constant factor.

Now, the kernel of the map $f \mapsto \tilde{f}$ contains all functions $\lambda_h f - \delta_H\theta(h)^{-1} f$, for $h \in H$ and $f \in C_c^\infty(G)$. Applying the right Haar integral on G to such functions, we get

$$\int_G (\lambda_h f(g) - \delta_H\theta(h^{-1}) f(g))\delta_G(g)^{-1}\, d\mu_G(g)$$
$$= \left(\delta_G(h^{-1}) - \delta_H\theta(h^{-1})\right) \int_G f(g)\delta_G(g)^{-1}\, d\mu_G(g)$$

This vanishes identically if and only if $\delta_G(h^{-1}) = \delta_H\theta(h^{-1})$, for all $h \in H$. Thus (1) $\Rightarrow$ (2).

(2) $\Rightarrow$ (1) Take a function $f \in C_c^\infty(G)$ such that $\tilde{f} = 0$. The function f is fixed by a compact open subgroup K, and it is enough to treat the case where $\operatorname{supp} f \subset HgK$, for some $g \in K$. Thus f is a finite linear combination of the characteristic function of cosets $h_i gK$, $h_i \in H$. The condition $\tilde{f} = 0$ then amounts to

$$\mu_H(H \cap gKg^{-1}) \sum_i \theta\delta_H(h_i)^{-1} f(h_i g) = 0,$$

since $\theta\delta_H$ is trivial on the compact subgroup $H \cap gKg^{-1}$ of H. On the other hand,

$$\int_G f(x)\delta_G(x)^{-1} d\mu_G(x) = \mu_G(K)\delta_G(g)^{-1} \sum_i \delta_G(h_i)^{-1} f(h_i g)$$

If (2) holds, this will vanish, giving a well-defined non-zero functional I_θ, as we sought to show.

When the conditions of the proposition hold, the character θ takes only positive real values. Let $f \in C_c^\infty(G)$ satisfy $f(g) \geq 0$, for all $g \in G$; then $\tilde{f}(g) \geq 0$ for all g. Consequently:

Corollary 1.3.10: Existence of Compact Induction on Haar Measure

Suppose that the conditions of theorem 1.3.9 hold. There is then a non-zero linear functional I_θ on $C_c^\infty(H\backslash G, \theta)$ such that:

1. $I_\theta(\rho_g f) = I_\theta(f)$, for $f \in C_c^\infty(H\backslash G, \theta), g \in G$;
2. $I_\theta(f) \geq 0$, for $f \in C_c^\infty(H\backslash G, \theta), f \geq 0$.

These conditions determine I_θ uniquely, up to a positive constant factor, and the notation is:

$$I_\theta(f) = \int_{H\backslash G} f(g)\, d\mu_{H\backslash G}(g), \quad f \in C_c^\infty(H\backslash G, \theta)$$

***Proof*:**
Existence of a non-zero G-invariant functional, and its uniqueness up to a scalar, are exactly the content of theorem 1.3.9 under the stated condition $\theta\delta_H = \delta_G|_H$. Only positivity and the sign of the scalar remain.

The averaging map $\pi\colon C_c^\infty(G) \to C_c^\infty(H\backslash G, \theta)$ given by $\pi(F)(g) = \int_H F(hg)\theta(h)^{-1}\, d\mu_H(h)$ is surjective and carries non-negative functions to non-negative functions. Composing a Haar integral on G with a section of π shows that I_θ may be chosen with $I_\theta(\pi(F)) = \int_G F\, d\mu_G$, which is non-negative whenever F is. Since every non-negative element of $C_c^\infty(H\backslash G, \theta)$ is $\pi(F)$ for some non-negative F, this I_θ satisfies (2).

Finally, if I' is another functional satisfying (1) then $I' = cI_\theta$ for some $c \in \mathbb{C}^\times$ by the uniqueness clause, and evaluating both at a non-zero non-negative f shows $c > 0$ precisely when I' also satisfies (2). So the pair of conditions pins I_θ down up to a positive constant, as we sought to show.

Definition 1.3.11: Semi-invariant Measure

The measure $\mu_{H\backslash G}$ is called a positive *semi-invariant measure* on $H\backslash G$.

Since $\theta = \delta_H^{-1}\delta_G|H$ is uniquely determined, there is no real need to refer to it again, and so it shall be taken as implicit.

This allows for taking the dual of a compact induced smooth representation:

Theorem 1.3.12: Duality Theorem

Let $\dot{\mu}$ be a positive semi-invariant measure on $H\backslash G$. Let (σ, W) be a smooth representation of H. Then there is a natural isomorphism:

$$(c\text{-}\mathrm{Ind}_H^G \sigma)^\vee \cong \mathrm{Ind}_H^G \delta_{H\backslash G} \otimes \check{\sigma},$$

depending only on the choice of $\dot{\mu}$.

***Proof*:**
We view $\delta_{H\backslash G} \otimes \check{\sigma}$ as acting on the same space $\check{W}$ as $\check{\sigma}$. Let $(\check{w}, w) \mapsto \langle \check{w}, w \rangle$ be the evaluation pairing $\check{W} \times W \to \mathbb{C}$. Let $\varphi \in \text{c-Ind}\,\sigma$, $\Phi \in \text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma}$, and consider the function

$$f : g \mapsto \langle \Phi(g), \varphi(g) \rangle, \quad g \in G$$

This lies in $C_c^\infty(H\backslash G, \delta_{H\backslash G})$, so we have a G-invariant pairing

$$\begin{gathered} \text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma} \times \text{c-Ind}\,\sigma \to \mathbb{C}, \\ (\Phi, \varphi) \mapsto \int_{H\backslash G} \langle \Phi(g), \varphi(g) \rangle \, d\dot{\mu}(g) \end{gathered}$$

This induces a G-homomorphism $\text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma} \to (\text{c-Ind}\,\sigma)^\vee$. It is clearly natural in σ; we show it is an isomorphism.

To do this, we take a compact open subgroup K of G and describe the space $(\text{c-Ind}\,\sigma)^K$ of K-fixed functions in $\text{c-Ind}\,\sigma$. The support of such a function, f say, is a finite union of cosets HgK. The value $f(g) \in W$ is then fixed by the compact open subgroup $H \cap gKg^{-1}$ of H. We choose a set $\mathcal{G}$ of representatives for the coset space $H\backslash G/K$. For each $g \in \mathcal{G}$, we choose a basis $\mathcal{W}_g$ of the space $W^{H\cap gKg^{-1}}$. For each $g \in \mathcal{G}$ and each $w \in \mathcal{W}_g$, there is a unique function $f_{g,w} \in (\text{c-Ind}\,\sigma)^K$, supported on HgK, such that $f_{g,w}(g) = w$. The set

$$\{f_{g,w} : g \in \mathcal{G}, w \in \mathcal{W}_g\}$$

is then a basis of $(\text{c-Ind}\,\sigma)^K$. That is, the space $(\text{c-Ind}\,\sigma)^K$ consists of the finite linear combinations of functions $f_{g,w}$, $g \in \mathcal{G}$, $w \in \mathcal{W}_g$.

Let $\mathcal{W}_g^*$ denote the basis of $\check{W}_g = \text{Hom}_{\mathbb{C}}(W^{H\cap gKg^{-1}}, \mathbb{C})$ dual to $\mathcal{W}_g$. For $g \in \mathcal{G}$, $\check{w} \in \mathcal{W}_g^*$, we define $f_{g,\check{w}} \in (\text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma})^K$ in the same way as before. We then have that the space $(\text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma})^K$ consists of all functions f such that, for any $g \in \mathcal{G}$, the restriction $f|_{HgK}$ is a finite linear combination of functions $f_{g,\check{w}}$, $\check{w} \in \mathcal{W}_g^*$.

For $g_1, g_2 \in \mathcal{G}$, $w \in \mathcal{W}_{g_1}$, $\check{w} \in \mathcal{W}_{g_2}^*$, we have

$$\langle f_{g_2,\check{w}}, f_{g_1,w} \rangle = \begin{cases} \dot{\mu}(Hg_1K)\langle \check{w}, w \rangle & \text{if } Hg_1K = Hg_2K, \\ 0 & \text{otherwise,} \end{cases}$$

for some $\dot{\mu}(Hg_1K) > 0$. The pairing thus identifies $(\text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma})^K$ with the linear dual of $(\text{c-Ind}\,\sigma)^K$, and hence $\text{Ind}\,\delta_{H\backslash G} \otimes \check{\sigma}$ with $(\text{c-Ind}\,\sigma)^\vee$, as required.

1.4 Hecke Algebra

Linear representations of finite groups are in bijective correspondence to the modules over the group algebra $\mathbb{C}[G]$. A similar conditions holds for smooth representations of locally profinite groups given the suitable generalization of $\mathbb{C}[G]$ to Hecke algebras.

Though not necessary, to avoid technicalities that don't add insight, throughout this entire section the locally profinite group G will be assumed to be unimodular.

Fix a Haar measure μ on G. Then for any $f_1, f_2 \in C_c^\infty(G)$, define the convolution:

$$f_1 * f_2(g) = \int_G f_1(x) f_2(x^{-1}g) d\mu(x)$$

The integrand $(x, g) \mapsto f_1(x)f_2(x^{-1}g)$ lies in $C_c^\infty(G \times G)$, and integrating a locally constant compactly supported function over the compact set $\mathrm{supp} f_1$ leaves a locally constant function whose support lies in $\mathrm{supp} f_1 \cdot \mathrm{supp} f_2$, itself compact; hence $f_1 * f_2 \in C_c^\infty(G)$. This operation is associative: Here's the equation in an align* environment:

$$\begin{aligned} f_1 * (f_2 * f_3)(g) &= \int \int f_1(x) f_2(y) f_3(y^{-1}x^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(y) d\mu(x) \\ &= \int \int f_1(x) f_2(x^{-1}y) f_3(y^{-1}g) d\mu(x) d\mu(y) \\ &= (f_1 * f_2) * f_3(g) \end{aligned}$$

Definition 1.4.1: Hecke Algebra

Let G be a unimodular locally profinite group. Then:

$$\mathcal{H}(G) = (C_c^\infty(G), *)$$

is is an associative $\mathbb{C}$-algebra called the *Hecke algebra* of G.

This is an example of a *non-unital* algebra if G is not finite. If G is commutative, so is $\mathcal{H}(G)$. Note that if μ, ν are two choices of Haar measures giving rise to $\mathcal{H}_\mu(G)$ and $\mathcal{H}_\nu(G)$ on $C_c^\infty(G)$, then there exists a constant $c > 0$ such that $\nu = c\mu$ and a map $f \mapsto c^{-1}f$, and hence

$$\mathcal{H}_\mu(G) \cong \mathcal{H}_\nu(G)$$

Concretely, if $\nu = c\mu$ then $f_1 *_\nu f_2 = c\,(f_1 *_\mu f_2)$, so $f \mapsto cf$ is an isomorphism $\mathcal{H}_\nu(G) \to \mathcal{H}_\mu(G)$. The Hecke algebra is therefore well defined up to canonical isomorphism, and the choice of Haar measure will not be tracked.

What makes the algebra worth introducing is that it converts smooth representations into modules, exactly as $\mathbb{C}[G]$ does for a finite group. The unit is the only casualty: $\mathcal{H}(G)$ has none, and the substitute is a family of idempotents.

Definition 1.4.2: Hecke Idempotents

For a compact open subgroup $K \subseteq G$ let $e_K = \mu(K)^{-1}\mathbf{1}_K \in \mathcal{H}(G)$, where $\mathbf{1}_K$ is the characteristic function of K. A module M over $\mathcal{H}(G)$ is called *smooth* if $\mathcal{H}(G) * M = M$.

Each e_K satisfies $e_K * e_K = e_K$, and $e_K * f = f$ precisely when f is invariant under left translation by K; since every $f \in C_c^\infty(G)$ is invariant under some such K, the e_K form an approximate identity.

Theorem 1.4.3: Smooth Representations as Hecke Modules

Let G be a unimodular locally profinite group. The assignments

$$\pi(f)v = \int_G f(g)\,\pi(g)v\,d\mu(g), \qquad \pi(g)(f * m) = (\lambda_g f) * m$$

where $\lambda_g f\colon x \mapsto f(g^{-1}x)$, are mutually inverse and define an equivalence between the category of smooth representations of G and the category of smooth $\mathcal{H}(G)$-modules.

***Proof*:**
From representations to modules. Let (π, V) be smooth and $f \in C_c^\infty(G)$. The integrand $g \mapsto f(g)\pi(g)v$ is locally constant with compact support: f is, and $g \mapsto \pi(g)v$ is locally constant because v has open stabilizer by proposition 1.2.7. So the integral is a finite sum and $\pi(f)v$ is defined. A change of variables gives $\pi(f_1 * f_2) = \pi(f_1)\pi(f_2)$, so V is an $\mathcal{H}(G)$-module. It is smooth: taking K with $v \in V^K$ gives $\pi(e_K)v = v$, so $v \in \mathcal{H}(G) * V$.

From modules to representations. Let M be a smooth $\mathcal{H}(G)$-module. Every $m \in M$ is a finite sum $\sum_i f_i * m_i$, and setting $\pi(g)(f * m) = (\lambda_g f) * m$ extends by linearity. This is well defined: if $\sum_i f_i * m_i = 0$ then $\sum_i (\lambda_g f_i) * m_i = 0$, because λ_g is an automorphism of the left $\mathcal{H}(G)$-module $\mathcal{H}(G)$ and unimodularity makes λ_g compatible with convolution on the right. Since $\lambda_{gh} = \lambda_g \lambda_h$, this is a homomorphism $G \to \mathrm{Aut}_{\mathbb{C}}(M)$.

It is smooth. Given $m = f * m'$, pick a compact open K under which f is left invariant; then $\lambda_k f = f$ for $k \in K$, so $\pi(k)m = m$ and the stabilizer of m is open. Applying proposition 1.2.7 again, (π, M) is a smooth representation.

The two constructions are inverse. Starting from (π, V) and returning, the recovered action sends g to the operator determined by $\pi(\lambda_g f) = \pi(g)\pi(f)$, which is $\pi(g)$ on $\mathcal{H}(G) * V = V$. Starting from M and returning, the recovered module structure is $f * m$ again, since $\int_G f(g)\pi(g)\,d\mu(g)$ applied to $h * m'$ reproduces $(f * h) * m'$ by the associativity computed above. Both assignments carry G-maps to $\mathcal{H}(G)$-maps and conversely, so they give an equivalence of categories, completing the proof.

This is the promised generalization: the classification of smooth representations of G becomes a question about modules over a single associative algebra, and the compact open subgroups enter through the idempotents e_K, whose images $e_K * M = M^K$ are the finite-dimensional pieces an admissible representation is assembled from.

2

Smooth Representations of GL2: General Structure and Principal Series

Having established the general framework of smooth representations, we now narrow our focus to the group $G = \mathrm{GL}_2(F)$ for a non-Archimedean field F. This group can be thought of as the group being able to represent the "2D" information of the (local) absolute galois group of F. The goal of this chapter is to classify the "continuous spectrum" of representations of G, known as the *Principal Series*. These are the representations that can be constructed by parabolic induction from characters of the diagonal torus. In doing so, the "discrete spectrum" (the cuspidal representations) is identified by what it is not.

We begin by looking at the representation theory of $\mathrm{GL}_2(\mathbb{F}_q)$ for finite fields, which serves as a crucial finite model for the p-adic case. We then analyze the geometry of $\mathrm{GL}_2(F)$ through its various decompositions (Iwasawa, Cartan, Bruhat). A key tool in our analysis will be the *Mirabolic subgroup* M (Section 2.3), whose representation theory is simple enough to serve as a lever for understanding G. Finally, using the Jacquet module (Section 2.4), we provide a complete classification of the principal series, determining when they are irreducible and how they decompose when reducible (leading to the Steinberg representation).

2.1 Important Background: Representation of $\mathrm{GL}_2(\mathbb{F}_p)$

When analyzing $\mathrm{GL}_2(\mathbb{Q}_p)$ and related groups, we shall draw inspiration of this special case. We shall first single out the following subgroups:

Definition 2.1.1: Matrix Subgroups

Let F be (an arbitrary) field. Then define the groups:

1. (Borel subgroup) $B = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
2. (unipotent subgroup) $N = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong (F, +)$
3. (torus subgroup) $T = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathrm{GL}_2(F) \right\}$
4. (center) $Z = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in \mathrm{GL}_2(F) \right\} \cong F^*$

Note the Borel subgroup is not normal and its normalizer is G.

Lemma 2.1.2: Bruhat Decomposition

Let $G = \mathrm{GL}_2(F)$. Then its *Bruhat decomposition* is:

$$G = B \cup BwB$$

where

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Hence, $\{1, w\}$ is a set of representative of the coset space $B\backslash G/B$.

Furthermore, $B = NT = TN$, hence $B \cong N \rtimes T$

***Proof*:**
Let $M \in GL_2(F)$ so that $\det(M) = ad - bc \neq 0$ if $c = 0$, $ad \neq 0$, so $a, d \neq 0$. Then $M \in B$.

If $c \neq 0$, choose two arbitrary $B_1, B_2 \in B$, calculate $B_1 w B_2$, use the fact that $\det(B_1), \det(B_2) \neq 0$ to get an equation for each entry of M. Note that the bottom-left entry *must* be nonzero and hence $B \cap BwB = \emptyset$, proving the decomposition into disjoint cosets.

For the $B = NT = TN$, notice that:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix} \in TN$$

It can also be factored as:

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} 1 & b/d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \in NT$$

Showing they are a semidirect products completing the proof.

By constructing, any representative of the coset BwB has a nonzero entry in the $(2,1)$-position. As $B = TN = NT$ and w normalizes T ($wTw = T$):

$$BwB = NwB = BwN$$

Furthermore, the map:

$$B \times N \to BwN \qquad (b, n) \mapsto bwn$$

is *bijective.*

Next, For any $M \in \mathrm{GL}_2(F)$, define $\mathrm{ch}_M(t) = \det(tI - M) = t^2 - \mathrm{tr}(M)t + \det(M)$, that is $\mathrm{ch}_M(t)$ is the characteristic polynomial of M, i.e. a monic quadratic F-polynomial and $ch_M(0) = \det(M) \neq 0$.

Proposition 2.1.3: Character Polynomial For $\mathrm{GL}_2(F)$

Let F be an arbitrary field, $M \in \mathrm{GL}_2(F)$, and $f(t) = \mathrm{ch}_M(t) = \det(tI - M)$. Then:

1. If $f(t)$ is irreducible, the F-algebra $F[M] \subseteq M_2(F)$ is a field with $\mathrm{GL}_2(F)$-centralizer $F[M]^*$. Given $f(t) = t^2 + at + b$, then M is $\mathrm{GL}_2(F)$-conjugate to:

$$\begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$$

   Furthermore, an element $N \in \mathrm{GL}_2(F)$ is GL_2-conjugate go M if and only if $\mathrm{ch}_N(t) = f(t)$.

2. If $f(t)$ has distinct roots $a, b \in F^*$, then M is $\mathrm{GL}_2(F)$-conjugate to:

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

3. If $f(t)$ has a repeated root $a \in F^*$, Then M is $\mathrm{GL}_2(F)$-conjugate to either:

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \qquad \text{or} \qquad \begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$$

   In the first case, M lies in the center of $\mathrm{GL}_2(F)$, and in the 2nd the $\mathrm{GL}_2(F)$-centralizer of M is $F[M]^*$.

***Proof*:**
We shall do one and leave the rest as an exercise.

1. As $f(t)$ is irreducible, by Cayley-Hamilton $F[M] \cong F[t]/(f(t))$ where the characteristic polynomial $f(t)$ is also the minimal polynomial. For the conjugacy result, as M is irreducible it has no eigenvalues and so $\{v, Mv\}$ forms a basis for any $v \in F^2$. Finding the matrix representation of the linear transformation corresponding to M shall give the desired result

2. exercise

3. exercise

Now let $F = \mathbb{F}_q$ be a field with $q = p^l$ elements for some $l \in \mathbb{N}$, and let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Let find the irreducible complex representations of G. First off:

Lemma 2.1.4: Order and Irreducible Representations of G

Let $G = \mathrm{GL}_2(\mathbb{F}_q)$. Then:

$$|G| = (q^2 - 1)(q^2 - q)$$

Furthermore, G has $q^2 - 1$ conjugacy classes.

Thus, there are $q^2 - 1$ irreducible representations.

***Proof*:**
Let the first column be any nonzero vector, giving $q^2 - 1$ possibilities. The second column cannot be a scalar multiple of the first. Given a choice v_1 of the first column, the second cannot be kv_1 for all $k \in \mathbb{F}_q$. This gives $q^2 - q$ possibilities. These choices are linearly independent, thus:

$$|G| = (q^2 - 1)(q^2 - q)$$

For the conjugacy classes we repeatedly use proposition 2.1.3. By the first case, as $\mathrm{GL}_2(\mathbb{F}_q)$ has exactly $(q^2 - q)/2$ irreducible monic polynomials of degree 2, there are $(q^2 - q)/2$ conjugacy classes. The second case will give $(q - 1)(q - 2)/2$ cases, and the third $2(q - 1)$ cases. Summing these up gives the desires result.

Thus, there are $q^2 - 1$ irreducible representations. As G is finite, character theory applies, so some characters are constructed next. First, note that $N \cong (\mathbb{F}_q, +)$ via $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$.

Fix a nontrivial character of $(\mathbb{F}_q, +)$ χ (ex. $\chi(x) = e^{2\pi i(x)/p}$). Then $(a\chi)(x) = \chi(ax)$ ranges over all characters of $(\mathbb{F}_q, +)$. This gives a natural action of conjugation by T (or B) on N. Namely the action to T on N produces an action on the characters:

$$(t\chi)(n) = \chi(t^{-1}nt)$$

when interpreting n as a matrix in N via the isomorphism $N \cong (F_q, +)$ given above. Certainly all non-trivial characters are in the orbit of T, hence the conjugacy classes split into the single trivial character and all other ones.

2.1.5 Non-trivial Characters of N Therefore, all non-linear character characters on N are similar up to conjugation.

The trivial one is important in a moment for finding the first irreducible representation.

Let χ_1, χ_2 be characters of $\mathbb{F}_q^*$, and define the character:

$$\chi = \chi_1 \otimes \chi_2 : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_1(a)\chi_2(b)$$

This is a character of T, and may be regarded as a character on B trivial on N via the quotient:

$$B \to B/N \cong T$$

This process is given a name:

Definition 2.1.6: Inflation

Let $H \trianglelefteq G$ be a normal subgroup and (π, V) a representation of G/H. Then the *inflation* of π from H to G is a new $\tilde{\pi} : G \to V$ such that $\tilde{\pi}(gH) = \pi(\overline{g})$; that is, it is trivial on the cosets of G/H.

After inflating from T to B, form the induced representation of $\mathrm{GL}_2(\mathbb{F}_q) = G$

$$\mathrm{Ind}_B^G \chi$$

and consider its irreducible components:

Lemma 2.1.7: Irreducible Components of G

Let π be an irreducible representation of G. Then the following are equivalent:

($\Rightarrow$) π is equivalent to a G-subspace of $\mathrm{Ind}_B^G \chi$ for a character χ of T seen as a character on B trivial on N.

($\Rightarrow$) π contains the trivial character of N.

***Proof*:**
First, π contains the trivial character of N if and only if it is contains an irreducible representation σ of B containing to the trivial character of N. Indeed, restrict G to B, $\mathrm{Res}_B^G \pi$ and then restrict further to N

$$\mathrm{Res}_N^B(\mathrm{Res}_B^G \pi)$$

By Maschke's theorem breakdown $\mathrm{Res}_B^G \pi$ into irreducibles. Call this irreducible representation of B σ. As Res is additive over direr sums, we see the above if and only if holds. Now as we just proved above this is equivalent to σ being an inflation of a character T. By Frobenius reciprocity:

$$\mathrm{Hom}_G(\mathrm{Ind}_B^G \chi, \pi) \cong \mathrm{Hom}_B(\chi, \mathrm{Res}_B^G \pi)$$

As π is irreducible we can apply the same ideas as Schur's Lemma to finish the proof.

So if π contains the trivial character of N, it is irreducible. By the lemma, analyze the induced representations $\mathrm{Ind}_B^G \chi$. To start, define the character of T:

$$\chi^w : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mapsto \chi_2(a)\chi_1(b)$$

where w is the permutation in G swapping columns. Abusing notation, this is a character if B that is trivial on N.

Proposition 2.1.8: Analyzing Inflation of Torus Character

Let χ, ξ be character of T viewed as characters of B trivial on N. Then:

1. The space $\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi))$ is trivial, unless $\chi = \xi$ and $\chi = \xi^w$.
2. The space
$$\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\chi)) \quad \text{and} \quad \mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\chi^w))$$
have the same dimension: it is 2 if $\chi = \chi^w$ and 1 otherwise.

***Proof*:**
We use the canonical isomorphism of Frobenius Reciprocity
$$\mathrm{Hom}_G(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi)) \cong \mathrm{Hom}_B(\mathrm{Res}_B^G(\mathrm{Ind}_B^G(\chi)), \xi) \tag{2.1}$$
The restriction-induction formula of Mackey gives
$$\mathrm{Res}_B^G \mathrm{Ind}_B^G(\chi) \cong \sum_{y \in B \backslash G / B} \mathrm{Ind}_{B \cap B_y}^B \left(\mathrm{Res}^{B_y \cap B}(\chi^y)\right), \tag{2.2}$$
where χ^y denotes the character $b \mapsto \chi(yby^{-1})$ of the group $B^y = y^{-1}By$. By Bruhat decomposition we only have to consider the cases $y = 1$ and $y = w$. The term in (2.2) corresponding to $y = 1$ is just χ, and so contributes a factor $\mathrm{Hom}_T(\chi, \xi)$ to (2.1). The term corresponding to w contributes
$$\mathrm{Hom}_B(\mathrm{Ind}_T^B(\chi^w), \xi) \cong \mathrm{Hom}_T(\chi^w, \xi)$$
Altogether, (2.1) reduces to
$$\mathrm{Hom}_B(\mathrm{Ind}_B^G(\chi), \mathrm{Ind}_B^G(\xi)) \cong \mathrm{Hom}_T(\chi, \xi) \oplus \mathrm{Hom}_T(\chi^w, \xi),$$
and the result follow from Schur's lemma, completing the proof.

Corollary 2.1.9: Irreducible Representations Given Torus Character

Let χ be a character of T viewed as a character of B trivial on N. Then:

1. The representation $\mathrm{Ind}_B^G \chi$ is irreducible if and only if $\chi \neq \chi^w$
2. If $\chi = \chi^w$, then $\mathrm{Ind}_B^G \chi$ has length 2 with distinct component factors

***Proof*:**
The preceding proposition computes $\mathrm{End}_G(\mathrm{Ind}_B^G \chi) \cong \mathrm{Hom}_T(\chi, \chi) \oplus \mathrm{Hom}_T(\chi^w, \chi)$. The first summand is always one-dimensional, and the second is one-dimensional if $\chi = \chi^w$ and zero otherwise, since two characters of T admit a non-zero intertwiner only when they are equal.

If $\chi \neq \chi^w$ the endomorphism algebra is therefore $\mathbb{C}$. A smooth representation of finite length with scalar endomorphism algebra is irreducible, since a proper non-zero subrepresentation would supply an idempotent outside $\mathbb{C}$; finite length here comes from corollary 2.4.11. Hence $\mathrm{Ind}_B^G\chi$ is irreducible, which is (1).

If $\chi = \chi^w$ the endomorphism algebra is two-dimensional, so $\mathrm{Ind}_B^G\chi$ is reducible and decomposes with at least two constituents; by corollary 2.4.11 the length is at most three, and the two projections coming from the two-dimensional endomorphism algebra split it into exactly two distinct factors, giving (2) and completing the proof.

Example 2.1.10: Induced Character of Trivial Character

Consider the trivial character 1_B of B. The trivial character 1_G occurs in $\mathrm{Ind}_B^G 1_B$, so

$$\mathrm{Ind}_B^G 1_B = 1_G \oplus \mathrm{St}_G,$$

for a unique irreducible representation St_G, called the Steinberg representation. Clearly, $\dim \mathrm{St}_G = q$ since $[G : B] = q + 1$ and $\dim 1_G = 1$.

These irreducible representations are given a name:

Definition 2.1.11: Principle Series Representations

The representation Ind_B^G of a character χ of T is called the *principal series*. The term principle series is often extended to include the irreducible components, namely all irreducible representations of G containing the trivial character of N.

Corollary 2.1.12: Counting Principal Series

Up to isomorphism, the group $\mathrm{GL}_2(F)$ has precisely $\frac{1}{2}(q^2 + q) - 1$ irreducible representations which contain the trivial character of N.

***Proof*:**

This is a counting argument: if $\chi = \chi^w$, there are $2(q-1)$ such characters (2 for the length 2). If they are not equal there are $\frac{1}{2}\cdot[(q-1)^2-(q-1)]$ (where we half since $\mathrm{Ind}_B^G\chi_1\otimes\chi_2) = \mathrm{Ind}_B^G\chi_2\otimes\chi_1)$). Adding them gives the desired result.

Notice this number is less than $q^2 - 1$, meaning not all irreducible representations are of this form. In particular there are $\frac{1}{2}(q^2 - q)$ such irreducible representations. An irreducible representation need not be trivial on N; such representations are given a name:

Definition 2.1.13: Cuspidal Representation

An irreducible representation of $\mathrm{GL}_2(F)$ not containing the trivial character of N is called a *cuspidal representation.*

A cuspidal representation σ must contain some non-trivial character of N. As it is non-trivial on N, any two non-trivial characters of N are B-conjugate (see remark 2.1.5), so σ contains all non-trivial characters of N. This adds complexity, in particular the cuspidal representations cannot be constructed directly by induction.

To to that end, let $L/\mathbb{F}_q$ be a quadratic extension so that it has q^2 elements and L^* has q^2-1 elements. Let $F = \mathbb{F}_q$ to simply notation. The characters θ of L^* correspond to the (q^2-1) primitive roots of unity, of which there are $q^2-1-1 = q^2-2$. A character is called regular if it is *not* invariant under the Frobenius automorphism, that is $\theta(x^q) \neq \theta(x)$. A character $\theta_j(g^k) = \omega^{jk}$ is *not regular* if j divides $q+1$, as can be quickly verified. There are $(q^2-1)/(q+1) = q-1$ non-regular characters, and hence $q(q-1)$ regular characters. Choose a basis of L so that $L \cong_F F$ and $\mathrm{GL}_2(F)$ is associated with $\mathrm{Aut}_F(L)$. There is a natural action L^* on L, which corresponds to a subgroup of $\mathrm{GL}_2(F) \cong \mathrm{Aut}_F(L)$. Identify:

$$L^* \stackrel{\simeq}{=} E \subseteq \mathrm{GL}_2(F)$$

Importantly, note that the elements of G with irreducible characteristic polynomial is conjugate to an element of E. For, if $0 \neq \lambda \in L$, then the corresponding matrix in $\mathrm{GL}_2(F)$ must have λ as its eigenvalue, and as it must be irreducible the other root must be the galois conjugate λ^q, but this shows the correspondence.

Now, let θ be a regular character of E and ψ a non-trivial character of N. Define θ_ψ of ZN as:

$$\theta_\psi : \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} u \mapsto \theta(a)\psi(u)$$

Note that $\mathrm{Ind}_{ZN}^G \theta_\psi$ is independent of choice of ψ up to equivalence (see remark 2.1.5).

Theorem 2.1.14: Cuspidal Representation of Finite Matrix Group

Let θ be a regular character of E and ψ a non-trivial character of N.

1. The virtual representation
$$\pi_\theta = \mathrm{Ind}_{ZN}^G \theta_\psi - \mathrm{Ind}_E^G \theta$$
is an irreducible cuspidal representation of G of dimension $q-1$.
2. Let θ_1, θ_2 be regular characters of E; then $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.
3. Every irreducible cuspidal representation of G is of the form π_θ, for some regular character θ of E.

***Proof*:**

1. As we are working with finite groups, representations are characterized by characters. The character of the virtual representation π_θ is given by the difference of the characters of the two induced representations:
$$\mathrm{tr}\pi_\theta(g) = \mathrm{tr}(\mathrm{Ind}_{ZN}^G \theta\psi)(g) - \mathrm{tr}(\mathrm{Ind}_E^G \theta)(g)$$
We use the general formula for the character of an induced representation, namely if (σ, W) is a representation of a subgroup $H \leq G$, then the character of $\Pi = \mathrm{Ind}_H^G \sigma$ is:
$$\mathrm{tr}(\Pi(g)) = \frac{1}{|H|} \sum_{x \in G \text{ s.t. } x^{-1}gx \in H} \sigma(x^{-1}gx)$$
We compute this for each part and for each type of conjugacy class in $G = \mathrm{GL}_2(k)$. First take $\Pi_1 = \mathrm{Ind}_{ZN}^G \theta\psi$ so $H = ZN = B$ with $|H| = (q-1)q$. The character is $\sigma_1 = \theta\psi$.

- Let $g = z \in Z$. Since z is central, $x^{-1}zx = z$ for all $x \in G$. The element z is in ZN, so the condition is always met.

$$\mathrm{tr}(\Pi_1(z)) = \frac{1}{(q-1)q}\sum_{x\in G}\theta\psi(z) = \frac{|G|}{(q-1)q}\theta(z) = \frac{(q^2-1)(q^2-q)}{(q-1)q}\theta(z) = (q^2-1)\theta(z)$$

- Let $g = zu$ with $z \in Z, u \in N, u \neq 1$. An element with a repeated eigenvalue is only conjugate to other elements with repeated eigenvalues. The conjugates of g that lie in ZN are only those of the form $z'u'$. A more detailed analysis of the character sum shows that the sum over the unipotent parts averages to -1 when paired against a non-trivial character ψ. This yields:

$$\mathrm{tr}(\Pi_1(zu)) = -\theta(z)$$

- Let $g = y \in E \setminus Z$. The characteristic polynomial of y is irreducible over k. Every element of ZN has its eigenvalues in k, so its characteristic polynomial is reducible. Thus, no conjugate of y can be in ZN. The sum in the character formula is empty.

$$\mathrm{tr}(\Pi_1(y)) = 0$$

Next, consider $\Pi_2 = \mathrm{Ind}_E^G\theta$. Here $H = E^\times$ is a subgroup of order q^2-1. The character is $\sigma_2 = \theta$.

- Let $g = z \in Z$. Since $z \in F^\times \subset E^\times$, the condition $x^{-1}zx \in E^\times$ is always met. Next, consider $\Pi_2 = \mathrm{Ind}_E^G\theta$ Here $H = E^\times$, a subgroup of order q^2-1. The character is $\sigma_2 = \theta$.

$$\mathrm{tr}(\Pi_2(z)) = \frac{1}{q^2-1}\sum_{x\in G}\theta(z) = \frac{|G|}{q^2-1}\theta(z) = (q^2-q)\theta(z)$$

- Let $g = zu$ with $u \neq 1$. The element g has a repeated eigenvalue. No conjugate of g can have an irreducible characteristic polynomial, so no conjugate of g lies in $E^\times \setminus Z$. Conjugation into Z is impossible for $u \neq 1$. The sum is empty.

$$\mathrm{tr}(\Pi_2(zu)) = 0$$

- Let $g = y \in E\setminus Z$. The centralizer of y in G is $E^\times$. The conjugacy class of y intersects $E^\times$ in exactly two elements: y and its Galois conjugate y^q. The formula for the character of an induced representation on an element $h \in H$ simplifies in this case to $\sum_i \theta(g_ihg_i^{-1})$ where g_i are representatives for $H\backslash G/H$. For GL_2, this gives:

$$\mathrm{tr}(\Pi_2(y)) = \theta(y) + \theta(y^q) = \theta(y) + \theta^q(y)$$

We now subtract the character values to get:

- For $g = z \in Z$:

$$\mathrm{tr}\pi_\theta(z) = (q^2-1)\theta(z) - (q^2-q)\theta(z) = (q-1)\theta(z)$$

- For $g = zu$, $u \neq 1$:

$$\mathrm{tr}\pi_\theta(zu) = -\theta(z) - 0 = -\theta(z)$$

- For $g = y \in E \setminus Z$:

$$\mathrm{tr}\pi_\theta(y) = 0 - (\theta(y) + \theta^q(y)) = -(\theta(y) + \theta^q(y))$$

This completes the derivation of the character:

Conjugacy Class Representative g	**Character Value** $\mathrm{tr}\pi_\theta(g)$
$z \in Z$	$(q-1)\theta(z)$
zu where $z \in Z, u \in N, u \neq 1$	$-\theta(z)$
$y \in E \setminus Z$	$-(\theta(y) + \theta^q(y))$
g not conjugate to an element of $ZN \cup E$	0

Now, we can take the inner product of the character with itself to get:

$$\frac{1}{|G|} \sum_{g \in G} |\mathrm{tr}\pi_\theta(g)|^2 = 1$$

Now, recall that a representation V is irreducible if given the basis of characters and the inner product with itself, we get 1. In this case, we may have gotten a virtual character, that is the coefficient was negative. This is checked not to be the case by computing that

$$\mathrm{tr}\pi_\theta(e) = q - 1 = \dim(V)$$

If it were virtual, this would be negative, showing it is indeed a character and furthermore π_τ is an irreducible representation. Next, as

$$\sum_{u \in N} |\mathrm{tr}\pi_\theta(u)| = 0$$

we see that $\pi_\theta|_N$ does not contain the trivial character of N, and hence π_θ is cuspidal.

2. Two representations are isomorphic if they have the same character. It is easy to see from the character table above that $\pi_{\theta_1} \cong \pi_{\theta_2}$ if and only if $\theta_2 = \theta_1$ or $\theta_2 = \theta_1^q$.

3. As E^* has $q^2 - q$ characters and the group F has $q-1$ we get a total $q^2 - q - (q-1) = q^2 - q$ number of regular characters given of this form, which combining with corollary 2.1.12 gives us the total of $q^2 - 1$ irreducible representations, and hence we found all irreducible representations by lemma 2.1.4, completing the proof.

The book presents this interesting relation between characters and determinants as an exercise that I want to put down because it's pretty cool:

Example 2.1.15: Character and Determinants

Let ψ be a non-trivial character of N, and consider the representation

$$F = \mathrm{Ind}_N^G \psi$$

σ be an irreducible representation of G. Show that:

1. If $\sigma = \varphi \circ \det$, for a character φ of $k^\times$, then σ does not occur in F.
2. Otherwise, σ occurs in F with multiplicity one.

2.2 Linear Groups over Local Fields

Given the additional structure, we have some more interesting decompositions. We have B, N, T, and Z as before as these are define for any field F. Note that $B/N \cong T$ and $B \cong T \ltimes N$ are homeomorphisms.

Proposition 2.2.1: Iwasawa Decompositon

Let B the be Borel subgroup and $K = \mathrm{GL}_2(\mathfrak{o})$. Then:

$$G = BK$$

***Proof*:**
Take $g \in G$; if the $(2,1)$-entry of g is zero, then $g \in B$. Otherwise, post-multiplying by the permutation matrix $w \in K$ if necessary, we can assume $v_F(g_{21}) \geqslant v_F(g_{22})$. We can then post-multiply by a lower triangular matrix in K to achieve $g_{21} = 0$.

Corollary 2.2.2: Compact Quotient

The quotient space

$$B \setminus G$$

is compact

***Proof*:**
$B \setminus G$ is the continuous image of the compact subgroup $K = \mathrm{GL}_2(\mathfrak{o})$.

Proposition 2.2.3: Cartan Decomposition

Let ϖ be a prime element of F. Then the matrices:

$$\begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix} \qquad a, b \in \mathbb{Z},\ a \leq b$$

form a set of representatives for the coset space $K \backslash G / K$.

***Proof*:**
Permuting rows and columns, using the permutation matrix in K, we can arrange for the largest entry of g (in absolute value) to be in the $(1,1)$-place. Multiplying by elementary matrices from K, we can then arrange for g to be diagonal, and unit factors can be absorbed into K. This

gives

$$G = \bigcup_{a \leq b} K \begin{pmatrix} \varpi_0^a & 0 \\ 0 & \varpi_b \end{pmatrix} K$$

We have to prove this union is disjoint. That is, we have to recover the integers a, b from the coset KgK, where

$$g = \begin{pmatrix} \varpi^a & 0 \\ 0 & \varpi^b \end{pmatrix}$$

First, we have $a + b = \nu_F(\det h)$, for any $h \in KgK$. Next, the group index $(K : K \cap hKh^{-1})$ depends only on the coset KhK, and

$$(K : K \cap gKg^{-1}) = 1$$

if $b = a$, or

$$(K : K \cap gKg^{-1}) = (q + 1)q^{b-a-1}$$

if $b > a$, as we sought to show.

Corollary 2.2.4: Countable Quotient

If K is a compact open subgroup of G, the set G/K is countable.

***Proof*:**
As observed under box 1.2.28, it is enough to show that G/K is countable for one choice of K: we take $K = GL_2(\mathfrak{o})$. The space $K \backslash G/K$ is certainly countable, and each double coset KgK contains only finitely many cosets $g'K$.

The Iwahori decomposition comes next. First, the standard Iwahori subgroup of $\mathrm{GL}_2(F)$ is the compact open subgroup $I \leqslant G$:

$$I = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, d \in U_F,\ b \in \mathfrak{o},\ c \in \mathfrak{p} \right\}$$

Let $N' = N^w$ denote the subgroup of lower triangular unipotent matrices of G. Then:

Proposition 2.2.5: Iwahori Decomposition

If I is the Iwahori subgroup, then:

$$I = (I \cap N')(I \cap T)(I \cap N)$$

More precisely, the product map

$$I \cap N' \times I \cap T \times I \cap N \longrightarrow I$$

is bijective, and a homeomorphism, for any ordering of the factors on the left hand side.

***Proof*:**

Write an element of I as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, d \in U_F$, $b \in \mathfrak{o}$ and $c \in \mathfrak{p}$. Since a is a unit, the identity

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ ca^{-1} & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d - ca^{-1}b \end{pmatrix} \begin{pmatrix} 1 & a^{-1}b \\ 0 & 1 \end{pmatrix}$$

exhibits the factorization. The three factors lie in the asserted groups: $ca^{-1} \in \mathfrak{p}$ so the first is in $I \cap N'$; $a \in U_F$ and $d - ca^{-1}b \in U_F$ because $ca^{-1}b \in \mathfrak{p}$ while $d \in U_F$, so the second is in $I \cap T$; and $a^{-1}b \in \mathfrak{o}$ so the third is in $I \cap N$.

Uniqueness is read off the same formula: the three entries ca^{-1}, $(a, d - ca^{-1}b)$ and $a^{-1}b$ are determined by the original matrix, so the product map is injective, and it is surjective by the computation above. Both the map and its inverse are given by polynomial expressions in the entries and in a^{-1}, hence are continuous, so it is a homeomorphism. Reversing the roles of N and N', which amounts to factoring out d instead of a, gives the statement for any ordering of the factors, completing the proof.

This is useful for the following. Let $K = GL_2(\mathfrak{o})$; under the canonical surjection $K \to GL_2(k)$, the image of I is the standard Borel subgroup of $GL_2(k)$. The Bruhat decomposition for $GL_2(k)$ implies

$$K = I \cup IwI$$

Combining this with the Iwasawa decomposition for G gives the more symmetric double-coset decomposition

$$G = BI \cup BwI = B(I \cap N') \cup Bw(I \cap N)$$

The cosets BwI, BI are both open in G.

Furthermore, by letting $L = \mathfrak{o} \oplus \mathfrak{o}, L' = \mathfrak{o} \oplus \pi$, the Iwahori subgroup I is then the common G-stabilizer of the two lattices L, L'.

The decompositions above describe the shape of $\mathrm{GL}_2(F)$; what the rest of this section needs is the measure theory that goes with them. Throughout, F remains non-archimedean.

Proposition 2.2.6: Smooth Functions Over F

The vector-space $C_c^\infty(F)$ is spanned by indicator functions (characteristic functions) of sets $a + \mathfrak{p}^n$ for $a \in F$ and $n \in \mathbb{Z}$

***Proof*:**

Certainly the indicator function of $a + \mathfrak{p}^m$ lies in $C_c^\infty(F)$. Conversely, let $\varphi \in C_c^\infty(F)$. Since φ has compact support, there exists $n \in \mathbb{Z}$ such that $\operatorname{supp} \varphi \subset \mathfrak{p}^n$. Also, φ is fixed under translation by a compact open subgroup of F, hence by $\mathfrak{p}^m$, for some $m \in \mathbb{Z}$. Thus φ is a linear combination of characteristic functions of sets $a + \mathfrak{p}^m$, $a \in \mathfrak{p}^n / \mathfrak{p}^m$.

The Haar integral on the relevant groups is defined as follows. Let μ be the Haar measure on F (automatically left and right as F is abelian), and let Φ_0 denote the indicator function (or characteristic function) on $\mathfrak{o}$. Then as $\mathfrak{o}$ is compact:

$$\int_F \Phi_0(x) d\mu(x) = c_0 \qquad c \in \mathbb{R}_{>0}$$

If Φ_b is the indicator function on $a+\mathfrak{p}^b$ for $a \in F$ and $b \in \mathbb{Z}$, by translation invariance:

$$\int_F \Phi_b(x) d\mu(x) = c_0 q^{-b}$$

Since indicator functions span $C_c^\infty(F)$, this integrates any function.this integral has the following "homogeneity" property: if $\Phi \in C_c^\infty(F)$ and $y \in F^*$, then by the above equation it follows that:

$$\int_F \Phi(xy) d\mu(x) = \|y\|^{-1} \int_F \Phi(x) d\mu(x)$$

where $\|y\| = q^{\nu_F(y)}$. Thus, define the Haar measure μ^* on F^* given by:

$$\mu^*(x) = \frac{\mu(x)}{\|x\|}$$

Then if $\Phi \in C_c^\infty(F^*)$, the function $x \mapsto \|x\|^{-1}\Phi(x)$ lies in $C_c^\infty(F)$ (and vanishes at 0), so:

$$\int_{F^\times} \Phi(x) d\mu^\times(x) = \int_F \Phi(x) \|x\|^{-1} d\mu(x), \qquad \text{for} \qquad \Phi \in C_c^\infty(F^\times)$$

and hence this certainly defines a Haar integral on F^*.

Viewing $M_2(F)$ under addition, its Haar measure is the product of four copies of F, with the measure given by the tensor product of 4 copies of a Haar measure on F. Let $G = \mathrm{GL}_2(F)$. Then:

Proposition 2.2.7: Haar measure on $\mathrm{GL}_2(F)$

Let μ be a Haar measure on $M_2(F)$. For $\Phi \in C_c^\infty(\mathrm{GL}_2(F))$, the function $x \mapsto \Phi(x)\|\det x\|^{-2}$ (vanishing on $M_2(F) \setminus \mathrm{GL}_2(F)$) lies in $C_c^\infty(M_2(F))$. The functional

$$\Phi \mapsto \int_A \Phi(x) \|\det x\|^{-2}\, d\mu(x), \quad \Phi \in C_c^\infty(G),$$

is a left and right Haar integral on $\mathrm{GL}_2(F)$, i.e. $\mathrm{GL}_2(F)$ is unimodular.

***Proof*:**
Let $A = M_2(F)$ and $G = \mathrm{GL}_2(F)$. Take $g \in G$ and consider the functionals

$$\Phi \mapsto \begin{cases} \int_A \Phi(gx)\, d\mu(x), \\ \int_A \Phi(xg)\, d\mu(x), \end{cases} \qquad \Phi \in C_c^\infty(A)$$

Each is a Haar integral on A and differs from the initial one by a positive constant (depending on g). To evaluate this constant, take Φ to be the indicator function of $\mathfrak{m} = M_2(\mathfrak{o})$. In the first instance, the function $x \mapsto \Phi(gx)$ is the characteristic function of the lattice $\mathfrak{m}' = g^{-1}\mathfrak{m}$. Thus

$$\int_A \Phi(gx)\, d\mu(x) = \mu(\mathfrak{m}') = \mu(\mathfrak{m}) \frac{[\mathfrak{m}' : \mathfrak{m} \cap \mathfrak{m}']}{[\mathfrak{m} : \mathfrak{m} \cap \mathfrak{m}']}$$

This quotient of indices depends only on the double coset KgK, $K = \mathrm{GL}_2(\mathfrak{o})$. Taking g in diagonal form the Cartan decomposition (proposition 2.2.3), we get

$$\int_A \Phi(gx)\, d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x)\, d\mu(x)$$

The second instance is treated in the same way to get

$$\int_A \Phi(xg)\, d\mu(x) = \|\det g\|^{-2} \int_A \Phi(x)\, d\mu(x)$$

The proposition then follows from simple manipulations. For example, if $\Phi \in C_c^\infty(G)$,

$$\begin{aligned}\int_A \Phi(xg)\|\det x\|^{-2}\, d\mu(x) &= \|\det g\|^{-2} \int_A \Phi(x)\|\det xg^{-1}\|^{-2}\, d\mu(x)\\ &= \int_A \Phi(x)\|\det x\|^{-2}\, d\mu(x),\end{aligned}$$

as we sought to show.

With a Haar measure on $\mathrm{GL}_2(F)$ in hand, turn to the relevant subgroups. Let N, T, B, Z be as in definition definition 2.1.1. The subgroup N, Z, and T, are taken care of as $N \cong F$, $Z \cong F^*$, and $T \cong F^* \times F^*$, leaving B. As $B \cong T \ltimes N$, define a linear functional on $C_c^\infty(B) := C_c^\infty(T) \otimes C_c^\infty(N)$ by:

$$\Phi \mapsto \int_T \int_N \Phi(tn) d\mu_T(t) d\mu_N(n)$$

where μ_T, μ_N are the Haar measure of T and N respectively induced by the aforementioned isomorphisms. Certainly, this functional is left B-invariant, and hence a left-Haar integral on B. By the above functional, take $\mu_B := \mu_T \otimes \mu_N$. Importantly, this tensor product doesn't commute, taking into account the semidirect product defining B. In terms of μ_B:

$$\Phi \mapsto \int_B \Phi(b) d\mu_B(b)$$

As the tensor product does not commute, μ_B is not unimodular, and hence there is a non-trivial module:

Proposition 2.2.8: Module of Bruhat Group

The module δ_B of the group B is given by

$$\delta_B : tn \mapsto \|t_2/t_1\|, \quad n \in N, t = \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \in T \tag{2.3}$$

***Proof*:**
Setting

$$c = sm, \quad m \in N, \quad s = \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix} \in T,$$

we get

$$\int_B \Phi(bc)\, d\mu_B(b) = \int_T \int_N \Phi(ts\, s^{-1} ns\, m)\, d\mu_T(t) d\mu_N(n)$$

We use the obvious isomorphism $N \to F$ to identify μ_N with a certain Haar measure μ_F on F. For $\varphi \in C_c^\infty(N)$, we then have

$$\int_N \varphi(s^{-1}ns)\, d\mu_N(n) = \int_F \varphi\left(\begin{pmatrix} 1 & s_1^{-1}xs_2 \\ 0 & 1 \end{pmatrix}\right) d\mu_F(x)$$
$$= \|s_1 s_2^{-1}\| \int_N \varphi(n)\, d\mu_N(n)$$

By definition,

$$\int_B \Phi(bc)\, d\mu_B(b) = \delta_B(c)^{-1} \int_B \Phi(b)\, d\mu_B(b),$$

completing the proof.

This defines the desired semi-invariant measure:

Corollary 2.2.9: semi-invariant measure on $\mathrm{GL}_2(\mathfrak{o})$

The space $C_c^\infty(B\backslash G, \delta_B^{-1})$ admits a positive semi-invariant measure $\dot{\mu}$, where δ_B is given by proposition 2.2.8. If $K = \mathrm{GL}_2(\mathfrak{o})$, there is a Haar measure μ_K on K such that

$$\int_{B\backslash G} f(g)\, d\dot{\mu}(g) = \int_K f(k)\, d\mu_K(k),$$

for $f \in C_c^\infty(B\backslash G, \delta_B^{-1})$.

***Proof*:**
The character δ_B is trivial on the compact group $K \cap B$. Restriction of functions is an isomorphism $C_c^\infty(B\backslash G, \delta_B^{-1}) \to C_c^\infty(K \cap B\backslash K, 1)$, where 1 denotes the trivial character of $K \cap B$. The semi-invariant measure $\dot{\mu}$ thus restricts to a semi-invariant measure on $C_c^\infty(K \cap B\backslash K, 1)$, but so does any Haar measure on K.

Note that μ_K is effectively the restriction of a Haar measure μ_G on G. It follows that there is a left Haar measure μ_B on B such that

$$\int_G \varphi(g)\, d\mu_G(g) = \int_K \int_B \varphi(bk)\, d\mu_B(b) d\mu_G(k), \quad \varphi \in C_c^\infty(G) \tag{2.4}$$

showing that the measure and integral "commute" with the decomposition $G = BK$.

2.3 Representation of the Mirabolic Group

A lot of the representation of $\mathrm{GL}_2(F)$ can be simplified by first finding the representation of the following subgroup:

Definition 2.3.1: Mirabolic Subgroup

The group $M \subseteq \mathrm{GL}_2(F) = G$ given by:

$$M = \left\{ \begin{pmatrix} a & x \\ 0 & 1 \end{pmatrix} : a \in F^*,\ x \in F \right\}$$

is called the *mirabolic subgroup* of G. Letting $S = T \cap M \cong F^*$, then $M \cong S \ltimes N$.

My professor once said that mirabolic should be remembered as "miracle-parabolic" group, an apt mnemonic.

Start by analyzing the smooth representations of N, the source of the induction. Let (π, V) be a smooth representation and ϑ a character of N. Let $V(\vartheta)$ be the subspace generated by all $\pi(n)v - \vartheta(n)v$. This is an N subspace, and $V_\vartheta = V/V(\vartheta)$ is well-defined and, by construction, as the maximal N-quotient where the action of N acts via the character ϑ. If ϑ is trivial, then $V(\vartheta) = V(N)$ and so $V_\vartheta = V_N$.

The following lemma generalizes the test in Fourier analysis where $\int f e^{2i\pi k} dx = 0$:

Lemma 2.3.2: Generalized Harmonic Decomposition

Let μ_N be a Haar measure on N and ϑ a character of N.

1. Let (π, V) be a smooth representation of N and $v \in V$. The vector v lies in $V(\vartheta)$ if and only if there is a compact open subgroup N_0 of N such that

$$\int_{N_0} \vartheta(n)^{-1} \pi(n) v \, d\mu_N(n) = 0 \tag{2.5}$$

2. The process $(\pi, V) \mapsto V_\vartheta$ is an exact functor from $\mathrm{Rep}(N)$ to the category of complex vector spaces.

Think of the integral as the generalized projection operator, $\int_{N_0} \vartheta(n)^{-1} \pi(n) v \, d\mu_N(n) = 0$. Multiplying by $\vartheta(n)^{-1}$ is replaces the $e^{-2\pi i k}$.

***Proof*:**

1. First assume that ϑ is trivial on the trivial character. The group $N \cong F$ is the union of an ascending sequence of compact open subgroups. So, if

$$v = \sum_{i=1}^{r} v_i - \pi(n_i) v_i \in V(N),$$

there is a compact open subgroup N_0 of N containing all n_i. Then the relation given by equation 2.5 holds for this choice of N_0.

Conversely, let $v \in V$ and suppose the integral relation holds. Then there is an open normal subgroup N_1 of N_0 such that $v \in V^{N_1}$. The space V^{N_1} carries a representation of the finite group N_0/N_1. Therefore, in the obvious notation, $V^{N_1} = V^{N_1}(N_0/N_1) \oplus V^{N_0}$

and the map

$$w \mapsto \mu_N(N_0)^{-1} \int_{N_0} \pi(n) w \, d\mu_N(n), \quad w \in V^{N_1},$$

is the N_0-projection $V^{N_1} \to V^{N_0}$. This has kernel $V^{N_1}(N_0/N_1) \subset V(N)$. But then this proves (1) for the trivial character of N.

Now let ϑ be an arbitrary character of N, and consider the representation (π', V') of N, where $V' = V$ and $\pi'(n) = \vartheta(n)^{-1}\pi(n)$. Then $V(\vartheta) = V'(N)$ and so (1) follows in general.

2. right-exactness is clear as we are taking the quotient. For the other way, if $f : U \to V$ is injective we want that the induced $\overline{f}$ is injective. If $\overline{f}(v) = 0$, then for injectivity it must be in $V(\vartheta)$. As it is equal to $\overline{0}$ we get:

$$\int_{N_0} \vartheta(n)^{-1}\pi(n) \cdot (f(v)) \, d\mu_N(n) = 0$$

As f is an N-homomorphism, it commutes with the action given by π

$$\int_{N_0} \vartheta(n)^{-1} f(\pi(n) \cdot v) \, d\mu_N(n) = 0$$

and by linearity f comes out, and by injectivity of f we can drop f, and finally by part 1 we get that $v \in V(\vartheta)$ giving injectivity of $\overline{f}$, completing the proof.

Some important remarks. If (π, V) is a smooth representation of N (or of M), then $V(N)$ is an N- (or M-) subspace of V. The exact sequence

$$0 \to V(N) \to V \to V_N \to 0$$

gives an exact sequence

$$0 \to V(N)_N \to V_N \to V_N \to 0$$

in which the map $V_N \to V_N$ is the identity. Therefore

$$V(N)_N = 0 \quad \text{and} \quad V(N)(N) = V(N)$$

If $\vartheta \neq 1$, as N acts trivially on $V/V(N)$, one has $(V/V(N))_\vartheta = 0$ and so the inclusion $V(N) \to V$ induces an isomorphism

$$V(N)_\vartheta \cong V_\vartheta$$

Proposition 2.3.3: Character Invariant Subspace Never Whole Space

Let (π, V) be a smooth representation of N and let $0 \neq v \in V$. Then there exists a character ϑ such that $v \notin V(\vartheta)$.

***Proof*:**
We write

$$N_j = \begin{pmatrix} 1 & \mathfrak{p}^j \\ 0 & 1 \end{pmatrix}, \quad j \in \mathbb{Z}$$

Take $v \in V$, $v \neq 0$. We choose $j_0 \in \mathbb{Z}$ such that N_{j_0} fixes v. For $j \leq j_0$, let V_j denote the

N_j-space generated by v. This is the direct sum of isotypic components $V_j^{\eta_j}$, as η ranges over the characters of N_j trivial on N_{j_0}. For each $j \leq j_0$, there exists η_j such that $V_j^{\eta_j} \neq 0$; by definition, we have

$$\int_{N_j} \eta_j(n)^{-1}\pi(n)v \, dn \neq 0$$

The N_{j-1}-space generated by $V_j^{\eta_j}$ is contained in V_{j-1}, so we may choose η_{j-1} such that $\eta_{j-1}|_{N_j} = \eta_j$. This satisfies the necessary properties to take the limit of these characters, and as N is a limit of the N_j we get that there exists a character ϑ of N such that, for all $j \leq j_0$, we have

$$\int_{N_j} \vartheta(n)^{-1}\pi(n)v \, dn \neq 0$$

Therefore $v \notin V(\vartheta)$, as required.

Corollary 2.3.4: Trivial If Trivial On Characters

Let (π, V) be a smooth representation of N. If $V_\vartheta = 0$ for all characters ϑ of N, then $V = 0$

***Proof*:**
This is the contrapositive of the preceding proposition. If $V \neq 0$, pick $0 \neq v \in V$; that proposition produces a character ϑ of N with $v \notin V(\vartheta)$, so the quotient $V_\vartheta = V/V(\vartheta)$ is non-zero. Hence if $V_\vartheta = 0$ for every character ϑ then $V = 0$, completing the proof.

Now let (π, V) be a smooth representation of M. Then:

Corollary 2.3.5: Trivial Representation of Mirabolic Group

Let (π, V) be a smooth representation of M. Suppose $V_N = 0$ and for some non-trivial character ϑ of N, $V_\vartheta = 0$. Then $V = 0$.

***Proof*:**
Now let (π, V) be a smooth representation of M. The space $V(N)$ is then an M-subspace of V and V_N carries a natural representation of $M/N = S$: On the other hand, $\pi(S)$ permutes transitively the subspaces $V(\vartheta)$, $\vartheta \neq 1$: for $s \in S$, $\pi(s) \cdot V(\vartheta) = V(\vartheta')$, where $\vartheta'(n) = \vartheta(s^{-1}ns)$. The corollary follows immediately.

Now fix a non-trivial character ϑ of N. Consider first $\mathrm{Ind}_N^M \vartheta$ and $\text{c-Ind}_N^M \vartheta$. As $N \trianglelefteq M$ and all non-trivial characters ϑ, ϑ' of N are conjugate, the inducted and compacted inducted representations are M-isomorphic.

Theorem 2.3.6: c-$\mathrm{Ind}_N^M \vartheta$ is Irreducible

Let ϑ be a non-trivial character of N and set $\mathcal{W} = \mathrm{Ind}_N^M \vartheta$, $\mathcal{W}^c = \text{c-}\mathrm{Ind}_N^M \vartheta$. Let $\alpha : \mathcal{W} \to \mathbb{C}$ denote the canonical map $f \mapsto f(1)$.

1. $\mathcal{W}(N) = \mathcal{W}^c(N) = \mathcal{W}^c$ and $(\mathcal{W}/\mathcal{W}^c)(N) = 0$.
2. The map α induces isomorphisms $\mathcal{W}_\vartheta \cong \mathbb{C}$ and $\mathcal{W}^c_\vartheta \cong \mathbb{C}$.

As a consequence, c-$\mathrm{Ind}_N^M \vartheta$ is irreducible over M.

***Proof*:**

1. Let $f \in \mathcal{W}$ and $n \in N$. For $a \in S$, we have $f(an) = \vartheta(ana^{-1})f(a)$. As there is an integer j such that N_j fixes f by proposition 2.3.3, $f(a) = 0$ if $\|\det a\|$ is sufficiently large. On the other hand, $f(an) = f(a)$ if $\|\det a\|$ is sufficiently small, so $nf - f$ vanishes at a if $\|\det a\|$ is sufficiently small. This implies that $nf - f \in \mathcal{W}^c$. Thus $\mathcal{W}(N) \subset \mathcal{W}^c$ and N acts trivially on $\mathcal{W}/\mathcal{W}^c$.

   Next, to prove that $\mathcal{W}^c(N) = \mathcal{W}^c$, let ψ be the character of F defined by
   $$\psi(x) = \vartheta\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\right)$$
   Take $a \in F^\times$, $j \in \mathbb{Z}_{\geq 1}$. Let $f_{a,j} \in \mathcal{W}^c$ be the function such that
   $$f_{a,j} : \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}\begin{pmatrix} au & 0 \\ 0 & 1 \end{pmatrix} \mapsto \psi(x), \quad u \in U_F^j,$$
   and which vanishes elsewhere. By similar augments as done before, the functions $f_{a,j}$ span $\mathcal{W}^c$ over $\mathbb{C}$. We have
   $$\begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} f_{a,j}\left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}\right) = \psi(bz) f_{a,j}\left(\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}\right)$$
   We deduce that $nf_{a,j}$ has the same support as $f_{a,j}$, $n \in N$. We can certainly find $x \in F$ such that the function $u \mapsto \psi(aux)$, $u \in U_F^j$, is constant, equal to c say, with $c \neq 1$. If $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, then $nf_{a,j} - f_{a,j} = (c-1)f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(N)$ and so $\mathcal{W}^c(N) = \mathcal{W}^c$. This implies $\mathcal{W}(N) = \mathcal{W}^c$.

2. The map α induces a surjection $\mathcal{W}_\vartheta \to \mathbb{C}$. On the other hand, since N acts trivially on $\mathcal{W}/\mathcal{W}^c$, the inclusion $\mathcal{W}^c \to \mathcal{W}$ induces an isomorphism $\mathcal{W}^c_\vartheta \cong \mathcal{W}_\vartheta$. To prove (2), therefore, it is enough to show that any $f \in \mathcal{W}^c$ with $f(1) = 0$ belongs to $\mathcal{W}^c(\vartheta)$. A function f, vanishing at 1, is a finite linear combination of functions $f_{a,j}$ with $a \notin U_F^j$, so it is enough to treat such functions. However, as $a \notin U_F^j$, there exists $x \in F$ such that the function $u \mapsto \psi(aux) - \psi(x)$, $u \in U_F^j$, is a non-zero constant. Taking $n = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, the same calculation as before shows that $nf_{a,j} - \vartheta(n)f_{a,j}$ is a non-zero constant multiple of $f_{a,j}$, whence $f_{a,j} \in \mathcal{W}^c(\vartheta)$ and $\mathcal{W}^c(\vartheta) = \mathcal{W}^c$

Finally, for the irreducibility, let V be an M-subspace of $\mathcal{W}^c$. As $\mathcal{W}^c_N = 0$, the spaces $V_N, (\mathcal{W}^c/V)_N$ are both zero. The sequence

$$0 \to V_\vartheta \to \mathcal{W}^c_\vartheta \to (\mathcal{W}^c/V)_\vartheta \to 0$$

is exact. As $\dim \mathcal{W}^c_\vartheta = 1$, we conclude that $\dim V_\vartheta$ is 0 or 1. In the first case, $V = 0$ by corollary 2.3.4. In the second, $(\mathcal{W}^c/V)_\vartheta = 0$ and so $\mathcal{W}^c = V$, showing it is irreducible and completing the proof.

The following from the textbook [1] summarizes the key points:

1. A function $f \in W$ is determined by its restriction to $S \cong F^\times$. The restriction $f|_{F^\times}$ is a smooth function on $F^\times$.
2. A smooth function φ on $F^\times$ is of the form $\varphi = f|_{F^\times}$, for some $f \in W$, if and only if there exists $c > 0$ such that $\varphi(x) = 0$ for all x satisfying $\|x\| > c$.
3. A function $f \in W$ lies in W^c if and only if $f|_{F^\times} \in C_c^\infty(F^\times)$.

2.3.7 Lack of Irreducibility of Dual The above shows the representation $\mathrm{Ind}_N^M \vartheta$ is never irreducible, for any non-trivial character ϑ of N. The Duality Theorem (theorem 1.3.12) implies

$$(\text{c-}\mathrm{Ind}_N^M \vartheta)^\vee \cong \mathrm{Ind}_N^M \check{\vartheta}$$

Thus $\text{c-}\mathrm{Ind}_N^M \vartheta$ provides an example of an irreducible smooth representation with reducible contragredient.

The classification of *all* irreducible representations of the mirabolic group M can now be completed. The following theorem is the key technical tool for showing the list is complete. As before let ϑ be a non-trivial character of N. Let (π, V) be a smooth representation of M. Frobenius Reciprocity gives a canonical isomorphism

$$\mathrm{Hom}_N(V, V_\vartheta) \cong \mathrm{Hom}_M(V, \mathrm{Ind}_N^M V_\vartheta)$$

Let $q : V \to V_\vartheta$ denote the quotient map and let $q_\star$ be the map $V \to \mathrm{Ind}_N^M V_\vartheta$ corresponding to q under this isomorphism: for $v \in V$,

$$q_\star(v) = (m \mapsto q(\pi(m)v))$$

Theorem 2.3.8: Invariant Space equivalent to c-$\mathrm{Ind}_N^M V_\vartheta$

Let (π, V) be a smooth representation of M. The M-homomorphism $q_\star : V \to \mathrm{Ind}_N^M V_\vartheta$ induces an isomorphism $V(N) \cong \text{c-}\mathrm{Ind}_N^M V_\vartheta$.

***Proof*:**
The N-space V_ϑ is a direct sum of copies of ϑ. Therefore $\mathrm{Ind}_N^M V_\vartheta$ is a direct sum of copies of $\mathrm{Ind}_N^M \vartheta$. By theorem 2.3.6

$$(\mathrm{Ind}_N^M V_\vartheta)(N) = \text{c-}\mathrm{Ind}_N^M V_\vartheta = (\text{c-}\mathrm{Ind}_N^M V_\vartheta)(N)$$

The M-homomorphism $q_\star : V \to \mathrm{Ind}_N^M V_\vartheta$ surely maps $V(N)$ to $(\mathrm{Ind}_N^M V_\vartheta)(N) = \text{c-Ind}_N^M V_\vartheta$.

Let $W = \operatorname{Ker} q_\star \cap V(N)$ and $C = \text{c-Ind}_N^M V_\vartheta / q_\star(V(N))$. By lemma 2.3.2 the natural map $W_N \to V(N)_N$ is injective, so $W_N = 0$. Likewise, $(\text{c-Ind}\, V_\vartheta)_N$ is zero, so $C_N = 0$.

Now, the map $q_\star : V \to \text{c-Ind}\, V_\vartheta$ induces a map

$$q_{\star,\vartheta} : V_\vartheta = V(N)_\vartheta \to (\text{c-Ind}\, V_\vartheta)_\vartheta$$

By theorem 2.3.6(2), the canonical N-map $\mathrm{Ind}\, V_\vartheta \to V_\vartheta$ induces an isomorphism $\alpha_\vartheta : (\text{c-Ind}\, V_\vartheta)_\vartheta \to V_\vartheta$. The composite map $\alpha_\vartheta \circ q_{\star,\vartheta} : V_\vartheta \to V_\vartheta$ is the identity. However, the kernel of this map is W_ϑ and its cokernel is isomorphic to C_ϑ. As the spaces $W_N, W_\vartheta, C_N, C_\vartheta$ are all zero, by corollary 2.3.5 both W and C are trivial and $q_\star : V(N) \to \text{c-Ind}\, V_\vartheta$ is an isomorphism, completing the proof.

Theorem 2.3.9: Classification of Irreducible Representation of Mirabolic Group

Let (π, V) be an irreducible smooth representation of M. Either:

1. $\dim V = 1$ and π is the inflation of a character of $M/N \cong F^\times$
2. $\dim V$ is infinite and $\pi \cong \text{c-Ind}_N^M \vartheta$, for any character $\vartheta \neq 1$ of N.

In case (1), $\dim V_N = 1$ and $V_\vartheta = 0$ for $\vartheta \neq 1$. In case (2), $V_N = 0$ and $\dim V_\vartheta = 1$ for all $\vartheta \neq 1$.

***Proof*:**
If $V(N) = 0$, then N acts trivially on V. The group M/N is abelian, so Schur's Lemma 2.6 implies $\dim V = 1$ and we are in case (1).

If $V(N) \neq 0$, then $V(N) = V$ and $V_N = 0$. Therefore $V_\vartheta \neq 0$ for all characters $\vartheta \neq 1$ of N, and so $\dim V$ is infinite. Theorem 2.3.8 implies that $V = V(N)$ is M-isomorphic to $\text{c-Ind}_N^M V_\vartheta$. The N-space V_ϑ is a direct sum of copies of ϑ, so V is a direct sum of copies of $\text{c-Ind}_N^M \vartheta$ and, since it is irreducible, $V \cong \text{c-Ind}_N^M \vartheta$.

2.4 Jacquet Modules and Induced Representations

We now classify the irreducible smooth representations of $\mathrm{GL}_2(F) = G$. In this section, we deal completely with *principal series representations*, that is irreducible representations where $V_N \neq 0$: This is the equivalent of looking to see if π contains the trivial character of N. The terminology comes from functional analysis when taking the spectrum of a topological group with the continuous spectrum associated to the principal (as in main part) series of characters and discrete spectrum's will be later associated to the cuspidal representations. It shall be the case that much of the important representation theory on the Galois side will be in relation to the cuspidal, but that cannot be properly studied without first grasping the whole picture of the representation of $\mathrm{GL}_2(F)$.

Definition 2.4.1: Principal Series Representation

Let (π, V) be an irreducible representations of $\mathrm{GL}_n(F)$ (notice $n \in \mathbb{N}_{>0}$). Then it is called a *principal series* if $V_N \neq 0$.

When $V_N = 0$ these shall be called *cuspidal representations* or *supercuspidal representations*, which could be thought of as the "singular" case (and a singularity in the form of a cusp does appear when looking at the geometric picture of modular forms, for which see *Elliptic Curves* [3]), and are dealt with in chapter 3.

Let (π, V) be a smooth representation of $\mathrm{GL}_2(F)$ and $V(N)$ denote the subspace of V spanned by the vectors $v - \pi(x)v$, for $v \in V$ and $x \in N$. The space $V_N = V/V(N)$ inherits a representation π_N of $B/N = T$, which is smooth. This representation is given a name:

Definition 2.4.2: Jacquet Module

The representation (π_N, V_N) is called the *Jacquet module* of (π, V) at N.
This gives the *Jacquet functor*

$$\begin{aligned}\mathrm{Rep}(G) &\to \mathrm{Rep}(T),\\ (\pi, V) &\mapsto (\pi_N, V_N),\end{aligned}$$

which is exact and additive.

Let (σ, W) be a smooth representation of T. View σ as a smooth representation of B which is trivial on N, and form the smooth induced representation $\mathrm{Ind}_B^G \sigma$. It will often abbreviate $\mathrm{Ind}_B^G \sigma = \mathrm{Ind}\,\sigma$, since B and G are the only groups involved for most of this section. This is often called *parabolic induction.*

If (π, V) is a smooth representation of G, Frobenius Reciprocity gives an isomorphism

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}\,\sigma) \cong \mathrm{Hom}_B(\pi, \sigma)$$

However, σ is trivial on N so any B-homomorphism $\pi \to \sigma$ factors through the quotient map $\pi \to \pi_N$. Thus:

$$\mathrm{Hom}_G(\pi, \mathrm{Ind}\,\sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma)$$

Proposition 2.4.3: Jaquet Module and Principal Series Representation

Let (π, V) be an irreducible smooth representation of G. The following are equivalent:

1. The Jacquet module V_N is non-zero.
2. The representation π is isomorphic to a G-subspace of a representation $\mathrm{Ind}_B^G \chi$, for some character χ of T.

***Proof*:**

(1)⇒(2) We have to show that the Jacquet module V_N admits an irreducible T- (or B-) quotient.
Choose $v \in V, v \neq 0$. Since V is irreducible over G, any element of V is a

finite linear combination of translates $\pi(g)v$ of v, for various $g \in G$. Write $K = \mathrm{GL}_2(\mathfrak{o})$. The vector v is fixed by a subgroup K' of K of finite index; let $\{v_1, v_2, \ldots, v_r\}$ be the distinct elements of the form $\pi(k)v, k \in K$. In particular, $r \leq (K : K')$. Since $G = BK$, the elements $v_1, \ldots, v_r$ generate V over B, and their images generate V_N over T.

Thus V_N is finitely generated as a representation of T. We choose a minimal generating set $\{u_1, \ldots, u_t\}, t \geq 1$, say. A standard Zorn's Lemma argument shows that V_N has a T-subspace U, containing $u_1, \ldots, u_{t-1}$, and maximal for the property $u_t \notin U$. Then U is a maximal T-subspace of V_N and V_N/U is an irreducible representation of T, hence a character .

(2)⇒(1) As $\mathrm{Hom}_G(\pi, \mathrm{Ind}\,\sigma) \cong \mathrm{Hom}_T(\pi_N, \sigma)$. we get

$$\mathrm{Hom}_T(\pi_N, \chi) \cong \mathrm{Hom}_G(\pi, \mathrm{Ind}\,\chi) \neq 0,$$

so $\pi_N \neq 0$.

Next, let μ_N be a Haar measure on N and let $t \in T$. The measure $S \mapsto \mu_N(t^{-1}St)$ is the Haar measure $\delta_B(t)\mu_N$:

$$\int_N f(txt^{-1})\, d\mu_N(x) = \delta_B(t) \int_N f(x)\, d\mu_N(x), \quad f \in C_c^\infty(N)$$

As before, let w denote the permutation matrix

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

If σ is a smooth representation of T, form the representation $\sigma^w : t \mapsto \sigma(wtw^{-1})$, and view it as a representation of B which is trivial on N.

Let α_σ denote the canonical B-map $\mathrm{Ind}\,\sigma \to \sigma$ given by $f \mapsto f(1)$. It induces a canonical T-map $(\mathrm{Ind}\,\sigma)_N \to \sigma$, still denoted α_σ.

Theorem 2.4.4: Restriction-induction Lemma

Let (σ, U) be a smooth representation of T and set $(\Sigma, X) = \mathrm{Ind}_B^G \sigma$. There is an exact sequence of representations of T:

$$0 \to \sigma^w \otimes \delta_B^{-1} \to \Sigma_N \xrightarrow{\alpha_\sigma} \sigma \to 0$$

First, a technical lemma:

Lemma 2.4.5: Technical Lemma

Recall that $G = B \cup BwN$. A function $f \in X$ lies in V if and only if $\operatorname{supp} f \subset BwN$, that is if $f \in X$ then $f \in V$ if and only if there is a compact open subgroup N_0 of N (depending on f) such that $\operatorname{supp} f \subset BwN_0$.

***Proof*:**
A function $f \in X$ lies in V if and only if $f(1) = 0$. Since f is G-smooth, such a function f vanishes on a set BN_0', where N_0' is some compact open subgroup of N'. The identity (for $x \neq 0$)

$$\begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} w \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \in Bw \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix}$$

implies that $\operatorname{supp} f \subset BwN_0$, for some compact open subgroup N_0 of N, as required.

***Proof*:**
(of theorem 2.4.4)

By definition, X is the space of G-smooth functions $f : G \to U$ such that $f(bg) = \sigma(b)f(g)$, $b \in B, g \in G$. The canonical map $\alpha_\sigma : X \to U$ amounts to restriction of functions to B. Set $V = \operatorname{Ker} \alpha_\sigma$. Thus V provides a smooth representation of B and there is an exact sequence

$$0 \to V_N \to X_N \to U \to 0$$

We have to identify the T-representation V_N.

Let $f \in V$; in view of the preceding lemma, we can define a function $f_N : T \to U$ by

$$f_N(x) = \int_N f(xwn)\, dn = \sigma(x) f_N(1), \quad x \in T$$

By lemma 2.3.2, the kernel of the map $f \mapsto f_N$ is $V(N)$, and so $f \mapsto f_N(1)$ gives a bijective map $V_N \to U$. Taking $t \in T$ and $f \in V$, we have

$$\begin{aligned} (tf)_N(x) &= \int_N f(xwnt)\, dn \\ &= \delta_B(t^{-1}) \int_N f(xt^w wn)\, dn = \delta_B^{-1}(t)(t^w f_N)(x) \end{aligned}$$

Thus $f \mapsto f_N(1)$ is a B-homomorphism $V \to \sigma^w \otimes \delta_B^{-1}$ inducing a T-isomorphism $V_N \cong \sigma^w \otimes \delta_B^{-1}$.

Proposition 2.4.6: Non-cuspidal Irreducible Representation Are Admissible

Let (π, V) be an irreducible smooth representation of $\mathrm{GL}_2(F)$ which is not cuspidal. The representation π is admissible.

***Proof*:**
By definition, $V_N \neq 0$. By proposition 2.4.3, π is equivalent to a sub-representation of $\operatorname{Ind}_B^G \chi$, for some character χ of T. It is enough, therefore, to prove that $\operatorname{Ind}\chi$ is admissible.

We fix a compact open subgroup K of G; shrinking it if necessary, we may assume that $K \subset K_0 = \mathrm{GL}_2(\mathfrak{o})$. The space X^K of K-fixed points in $\operatorname{Ind}\chi$ consists of the functions $f : G \to \mathbb{C}$ satisfying

$$f(bgk) = \chi(b) f(g), \quad b \in B, g \in G, k \in K \tag{2.6}$$

We have $G = BK_0$, so the set $B\backslash G/K$ is finite, and each double coset BgK supports, at most, a one-dimensional space of functions satisfying equation 2.6. Thus X^K is finite-dimensional, as we sought to show.

Note that irreducible *cuspidal* representations of $\mathrm{GL}_2(F)$ are likewise admissible, and hence all irreducible representations of $\mathrm{GL}_2(F)$ are admissible; the cuspidal case is proposition 3.1.2, and the two together give corollary 3.1.5. A more granular classification of the principal series follows. First recall from example 1.2.8 that smooth characters on $\mathrm{GL}_n(F)$ factored via the determinant and a character of F^*. Define the following:

Definition 2.4.7: Twist On Representation

Let (π, V) is a smooth representation of G and φ a character of F^*. A *twist* $\varphi\pi$ is a smooth representation $(\varphi\pi, V)$ of G given by:

$$\varphi\pi(g) = \varphi(\det g)\pi(g), \quad g \in G$$

Similarly for characters of T: if $\chi = \chi_1 \otimes \chi_2$ is a character of T and φ is a character of $F^\times$, put $\varphi \cdot \chi = \varphi\chi_1 \otimes \varphi\chi_2$. Inflating $\varphi \cdot \chi$ to a representation of B trivial on N gives $\varphi \cdot \chi = (\varphi \circ \det|_B) \otimes \chi$. It follows immediately that

$$\mathrm{Ind}_B^G(\varphi \cdot \chi) \cong \varphi \mathrm{Ind}_B^G \chi$$

This allows convenient adjustments to the character χ without changing the essential structure of the induced representation. With this definition, the structure of representations of the form $\mathrm{Ind}_B^G\chi$. The main step is:

Theorem 2.4.8: Irreducibility Criterion

Let $\chi = \chi_1 \otimes \chi_2$ be a character of T, and set $(\Sigma, X) = \mathrm{Ind}_B^G\chi$.

1. The representation (Σ, X) is reducible if and only if $\chi_1\chi_2^{-1}$ is either the trivial character or the character $x \mapsto \|x\|^2$ of $F^\times$.
2. Suppose that (Σ, X) is reducible. Then:
   (a) the G-composition length of X is 2;
   (b) one composition factor of X has dimension 1, the other is of infinite dimension;
   (c) X has a 1-dimensional G-subspace if and only if $\chi_1\chi_2^{-1} = 1$;
   (d) X has a 1-dimensional G-quotient if and only if $\chi_1\chi_2^{-1}(x) = \|x\|^2$, $x \in F^\times$.

This refines to a classification of the irreducible principal series representations.

Let:

$$V = \{f \in X : f(1) = 0\}$$

This is a B-subspace of X, and there is an exact sequence

$$0 \to V \to X \to C \to 0,$$

where the one-dimensional space $C = X/V$ carries the character χ of T. By the Restriction-Induction Lemma theorem 2.4.4, so $V_N \cong \delta_B^{-1}\chi^w$.

Proposition 2.4.9: Irreducible Kernel

Let W be the kernel $V(N)$ of the canonical map $V \to V_N$. The space W is irreducible over B.

First an important technical lemma:

Lemma 2.4.10: Representation of V

For $f \in V$, define a function $f_N \in C_c^\infty(N)$ by $f_N(n) = f(wn)$, $n \in N$. The map

$$V \longrightarrow C_c^\infty(N),$$
$$f \longmapsto f_N$$

is an N-isomorphism.

***Proof*:**
As in theorem 2.4.4, the support of $f \in V$ is contained in $Bw\mathcal{N}$, for some compact open subgroup $\mathcal{N}$ of N. The assertions follow immediately (observing that the notation f_N here is not the same as that in theorem 2.4.4).

***Proof*:**
(of proposition 2.4.9)

We shall prove that W is irreducible as a representation of the mirabolic group M: recall under lemma 2.3.2 it was shown that $W_N = 0$ and $W = W(N)$. For $\varphi \in C_c^\infty(N)$ and $a \in F^\times$, we define $a\varphi \in C_c^\infty(N)$ by

$$a\varphi \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \chi_2(a)\varphi \begin{pmatrix} 1 & a^{-1}x \\ 0 & 1 \end{pmatrix}$$

This gives an action of $F^\times$ on $C_c^\infty(N)$ which we regard as an action of the group S of matrices $\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$. We combine this with the natural action of N to give $C_c^\infty(N)$ the structure of a smooth representation of M. With this structure, the map in lemma 2.4.10 is an M-isomorphism.

Let ϑ be a non-trivial character of N. The map $f \mapsto \vartheta f$ is a linear automorphism of $V = C_c^\infty(N)$ carrying $V(N)$ to $V(\vartheta)$. Since $V/V(N)$ is one dimensional, so is $\dim V_\vartheta = 1$. However, since N acts trivially on V_N, the inclusion $W \to V$ induces an isomorphism $W_\vartheta \cong V_\vartheta$, hence $W_\vartheta \cong \vartheta$. Then apply theorem 2.3.9 to get $W = W(N) \cong \text{c-Ind}_N^M \vartheta$ which, by corollary 2.3.5 is irreducible.

A direct consequence of the proposition:

Corollary 2.4.11: Length of Induced Representation by Character

As a representation of B or of M, $\mathrm{Ind}_B^G \chi$ has composition length 3. Two of the composition factors have dimension one, and the third is of infinite dimension. In particular, the G-composition length of the representation $\mathrm{Ind}_B^G \chi$ is at most 3.

***Proof*:**
Chain the two exact sequences already in play. The first is $0 \to V \to X \to C \to 0$ with C one dimensional, and the second is $0 \to W \to V \to V_N \to 0$ with $W = V(N)$. By lemma 2.4.10 and the computation in the proof of proposition 2.4.9, V_N is one dimensional, and W is irreducible over B, indeed over M, and infinite dimensional since it is isomorphic to $\text{c-Ind}_N^M \vartheta$.

Splicing the two sequences exhibits X with the composition series $W \subset V \subset X$, whose factors are W, V_N and C: one of infinite dimension and two of dimension one. So the B-length, and equally the M-length, is exactly 3.

A G-subspace is in particular a B-subspace, so any G-composition series refines to a B-one and the G-length is at most 3, completing the proof.

Proposition 2.4.12: Detecting one-dimensional Subrepresentations

The following are equivalent:

($\Rightarrow$) (1) $\chi_1 = \chi_2$

($\Rightarrow$) (2) X has a one-dimensional N-subspace

When these conditions hold,

1. X has a unique one-dimensional N-subspace X_0;
2. X_0 is a G-subspace of X, and it is not contained in V.

***Proof*:**
If (1) holds, by twisting we may as well take $\chi_1 = \chi_2 = 1$. The (non-zero) constant function then spans a one-dimensional G-subspace of X.

Conversely, let $f \in X$ span an N-stable subspace of dimension 1. Thus N acts on f by right translation as a character. The support of f is left-invariant under B, so $\operatorname{supp} f$ is either G or BwN. The second case is impossible since if $f(1) = 0$, by the restriction induction lemma (theorem 2.4.4) the support of f is confined to BwN_0, for some compact open subgroup N_0 of N. Thus, $\operatorname{supp} f = G$ and f vanishes nowhere so that $f(1) \neq 0$, and $f \notin V$. Then the canonical N-map $X \to \mathbb{C} = X/V$ identifies the N-space $\mathbb{C}f$ with the trivial N-space $\mathbb{C}$. It follows that N fixes f under right translation. Take $x \in F^\times$, and consider the identity

$$w \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & x^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -x^{-1} & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$$

Taking x so that $\|x\|$ is sufficiently large, then f is fixed under right translation by $\begin{pmatrix} 1 & 0 \\ x^{-1} & 1 \end{pmatrix}$.

So, as f is fixed by N, we have

$$f(w) = \chi_1(-1)\chi_1^{-1}(x)\chi_2(x)f(1),$$

for all $x \in F^\times$ of sufficiently large absolute value. Thus $\chi_1 = \chi_2 = \varphi$, say, and $f(g) = \varphi(\det g)f(1)$. We have proved (1) $\Leftrightarrow$ (2), and that the one-dimensional N-subspace is uniquely determined.

The proof of theorem 2.4.8 follows.

***Proof*:**
(of theorem 2.4.8)

Assume X is reducible. By corollary 2.4.11 its G-length is 2 or 3, and it has either a finite-dimensional G-subspace or a finite-dimensional G-quotient

Assume the first alternative. Thus X has a one-dimensional N-subspace, and we are in the situation of proposition 2.4.12: X has a one-dimensional G-subspace L, and $\chi_1 = \chi_2 = \varphi$, say. Moreover, G acts on L as the character $\varphi \circ \det$ and $L \cap V = 0$. The quotient $Y = X/L$ is therefore B-isomorphic to V. If X has G-length 3, then Y has G-length 2. However, V has B-length 2 and a unique B-quotient, which is of dimension 1. This gives a G-quotient of Y on which G must act as a character $\varphi' \circ \det$. This would force $\varphi' \otimes \varphi'$ to appear as a factor in the Jacquet module $Y_N \cong \varphi \cdot \delta_B^{-1}$, which it cannot. Thus X has G-length 2 and we are in case (2)(c) of the Irreducibility Criterion (theorem 2.4.8).

In the other alternative, X has a finite-dimensional G-quotient. The representation $\check{X}$ therefore has a finite-dimensional G-subspace, and we are back with the first alternative. By the Duality Theorem (theorem 1.3.12), $\check{X} \cong \operatorname{Ind}_B^G \delta_B^{-1}\check{\chi}$, so we are in the case (2)(d)of the Irreducibility Criterion (theorem 2.4.8).

Thus, in part (1) of the theorem, we have shown that X reducible implies χ has the stated form. The converse is given by proposition 2.4.12 and the dual case. We have also proved statements (a) to (d), completing the proof.

Having fully determined how to get irreducible representations from inductions/inflation of characters, to get a classification of the irreducible non-cuspidal representations of G requires investigating the homomorphisms between induced representations:

Proposition 2.4.13: Homomorphism between induced Representations

Let χ, ξ be characters of T. The space $\operatorname{Hom}_G(\operatorname{Ind}_B^G \chi, \operatorname{Ind}_B^G \xi)$ has dimension 1 if $\xi = \chi$ or $\chi^w\delta_B^{-1}$, zero otherwise.

***Proof*:**
We use Frobenius Reciprocity:

$$\operatorname{Hom}_G(\operatorname{Ind}_B^G \chi, \operatorname{Ind}_B^G \xi) \cong \operatorname{Hom}_T((\operatorname{Ind}\chi)_N, \xi)$$

The Jacquet module $(\operatorname{Ind}\chi)_N$ fits into the exact sequence

$$0 \to \chi^w\delta_B^{-1} \to (\operatorname{Ind}\chi)_N \to \chi \to 0$$

If we assume that $\chi \neq \chi^w \delta_B^{-1}$, then this sequence splits and the result follows immediately. The equation $\chi = \chi^w \delta_B^{-1}$ amounts to $\chi_1(x) = \|x\| \chi_2(x)$, $x \in F^\times$. In this case, $\operatorname{Ind}\chi$ is irreducible and the result again follows.

By way of some examples, consider in more detail the case where $\operatorname{Ind}_B^G \chi$ is reducible. Thus there is a character φ of $F^\times$ such that $\chi = \varphi \cdot 1_T$ or $\chi = \varphi \cdot \delta_B^{-1}$. Twisting does not affect the situation materially, so take $\varphi = 1$. Consider first the case $\operatorname{Ind}_B^G 1_T$. The irreducible G-quotient of $\operatorname{Ind}_B^G 1_T$ is called the *Steinberg representation* of G, and is denoted St_G:

$$0 \to 1_G \to \operatorname{Ind}_B^G 1_T \to \mathrm{St}_G \to 0 \tag{2.7}$$

Its dimension is infinite and $(\mathrm{St}_G)_N \cong \delta_B^{-1}$. Likewise, for a character φ of $F^\times$ there is an exact sequence

$$0 \to \varphi_G \to \operatorname{Ind}_B^G \varphi_T \to \varphi \cdot \mathrm{St}_G \to 0,$$

where $\varphi_G = \varphi \circ \det$ and $\varphi_T = \varphi \otimes \varphi$. (Representations of the form $\varphi \cdot \mathrm{St}_G$ are sometimes called *special.*)

Taking the smooth dual of (2.7) gives an exact sequence

$$0 \to \mathrm{St}_G^\vee \to \operatorname{Ind}_B^G \delta_B^{-1} \to 1_G \to 0 \tag{2.8}$$

The proposition implies

$$\mathrm{St}_G \cong \mathrm{St}_G^\vee \tag{2.9}$$

2.4.14 Remark The proposition also implies that the space $\operatorname{End}_G(\operatorname{Ind} 1_T)$ has dimension 1, while $\operatorname{Ind} 1_T$ is not irreducible. Thus the converse of Schur's Lemma fails in this context, as remarked in theorem 1.2.30.

Observe also the imperfect parallelism between the proposition above and the corresponding result (theorem 2.3.6) for the finite field case.

A new notation is convenient. For a smooth representation σ of T, define

$$\iota_B^G \sigma = \operatorname{Ind}_B^G(\delta_B^{-1/2} \otimes \sigma) \tag{2.10}$$

This provides another exact functor $\operatorname{Rep}(T) \to \operatorname{Rep}(G)$, known as *normalized* or *unitary smooth induction.* It gives rise to more convenient combinatorics, for example:

$$(\iota_B^G \sigma)^\vee \cong \iota_B^G \check{\sigma} \tag{2.11}$$

In this language, the Irreducibility Criterion (theorem 2.4.8) and proposition 2.4.13 say:

Lemma 2.4.15: Normalized Induction Properties

1. Let $\chi = \chi_1 \otimes \chi_2$ be a character of T. The representation $\iota_B^G \chi$ is reducible if and only if $\chi_1 \chi_2^{-1}$ is one of the characters $x \mapsto \|x\|^{\pm 1}$ of $F^\times$ or, equivalently, $\chi = \varphi \cdot \delta_B^{\pm 1/2}$ for some character φ of $F^\times$.
2. Let χ, ξ be characters of T. The space $\operatorname{Hom}_G(\iota_B^G \chi, \iota_B^G \xi)$ is not zero if and only if $\xi = \chi$ or $\xi = \chi^w$.

***Proof*:**
Both parts are the unnormalized statements read through the twist $\eta = \delta_B^{-1/2}\chi$ of (2.10), so the whole content is bookkeeping with δ_B. Since $\delta_B(t) = \|t_2/t_1\|$ by proposition 2.2.8, as a character of $T = F^\times \times F^\times$ we have $\delta_B^{-1/2} = \|-\|^{1/2} \otimes \|-\|^{-1/2}$, so writing $\eta = \eta_1 \otimes \eta_2$,

$$\eta_1 = \chi_1\|-\|^{1/2}, \qquad \eta_2 = \chi_2\|-\|^{-1/2}, \qquad \text{hence} \qquad \eta_1\eta_2^{-1} = \chi_1\chi_2^{-1}\|-\|$$

(1). By theorem 2.4.8(1) the representation $\iota_B^G\chi = \mathrm{Ind}_B^G\eta$ is reducible if and only if $\eta_1\eta_2^{-1}$ is trivial or equal to $\|-\|^2$. By the display, that happens exactly when $\chi_1\chi_2^{-1}$ equals $\|-\|^{-1}$ or $\|-\|$ respectively, which is the stated criterion. For the reformulation, $\chi = \varphi\cdot\delta_B^{\pm1/2}$ says $\chi_1\chi_2^{-1} = \|-\|^{\mp1}$, the same condition.
(2). By proposition 2.4.13 the space $\mathrm{Hom}_G(\mathrm{Ind}_B^G\eta, \mathrm{Ind}_B^G\theta)$ is non-zero exactly when $\theta = \eta$ or $\theta = \eta^w\delta_B^{-1}$, where θ is the twist of ξ. The first case gives $\xi = \chi$. For the second, $\eta^w = \eta_2 \otimes \eta_1$ and $\delta_B^{-1} = \|-\| \otimes \|-\|^{-1}$, so

$$\eta^w\delta_B^{-1} = \left(\chi_2\|-\|^{-1/2}\|-\|\right) \otimes \left(\chi_1\|-\|^{1/2}\|-\|^{-1}\right) = \chi_2\|-\|^{1/2} \otimes \chi_1\|-\|^{-1/2}$$

Comparing with $\theta = \xi_1\|-\|^{1/2} \otimes \xi_2\|-\|^{-1/2}$ gives $\xi_1 = \chi_2$ and $\xi_2 = \chi_1$, that is $\xi = \chi^w$. So the normalized statement has the clean form claimed, with the δ_B twist absorbed, completing the proof.

Gathering the earlier arguments and results:

Theorem 2.4.16: Classification Theorem

The following is a complete list of the isomorphism classes of irreducible, non-cuspidal representations of G:

1. the irreducible induced representations $\iota_B^G\chi$, where $\chi \neq \varphi\cdot\delta_B^{\pm1/2}$ for any character φ of $F^\times$;
2. the one-dimensional representations $\varphi\circ\det$, where φ ranges over the characters of $F^\times$;
3. the special representations $\varphi\cdot\mathrm{St}_G$, where φ ranges over the characters of $F^\times$.

The classes in this list are all distinct except that, in (1), $\iota_B^G\chi \cong \iota_B^G\chi^w$.

***Proof*:**
The examples in (2.7) show that every irreducible, non-cuspidal representation of G appears in this list. The relations between the irreducibly induced ones are given by proposition 2.4.13. The same result also implies that no special representation $\varphi\cdot\mathrm{St}_G$ can be equivalent to $\varphi'\cdot\mathrm{St}_G$, $\varphi' \neq \varphi$, or any irreducibly induced representation.

3

Cuspidal Representations

In the previous chapter, we classified the principal series representations. However, this left the *cuspidal* representations. These are the "atomic" particles of the theory; they cannot be constructed by inducing from parabolic subgroups, and they have no analogue in the representation theory of $\mathrm{GL}_n(\mathbb{C})$. They are the representations that carry the "actual" arithmetic information that is used in the Langlands Correspondence. The goal of this chapter is to define, analyze, and provide a general construction for cuspidal representations. Unlike the principal series, which are naturally constructed via the Bruhat decomposition, cuspidal representations are constructed using the compact open subgroups of G.

We start in Section 3.1 by giving a more workable definition of cuspidal representations via their matrix coefficients (compact support modulo the center). In Section 3.2, we introduce the machinery of *intertwining* and Hecke algebras for compact open subgroups. This leads to the main result of the chapter: a method to construct irreducible cuspidal representations by *compact induction* from representations of open subgroups that contain the center. We conclude by introducing the concept of *Fundamental Strata* (Section 3.3), which serves as the stratifying for these representations, classifying them by their level of ramification.

3.1 Cuspidal Representations and Coefficients

Our exploration of non-principal series representations interestingly enough starts by revisiting the definition of the cuspidal representation. So far, we have defined it negatively: a smooth representation (π, V) of $\mathrm{GL}_2(F)$ is cuspidal if the Jacquet module (π_N, V_N) is trivial, or equivalently, if it is not isomorphic to a composition factor of an induced representation $\mathrm{Ind}_B^G \chi$. This definition, being the "negative" of non-principal series hides the great importance these representations have in the Langlands program. Hence, we start by giving a more practical and algebraic definition.

The key insight from this new viewpoint is that irreducible cuspidal representations have an algebraic property: they are *projective objects* in the appropriate subcategory of $\mathrm{Rep}(G)$, as shall be elaborated on in appendix A.

Let (π, V) be a smooth representation of $G = \mathrm{GL}_2(F)$. From vectors $v \in V$ and $\tilde{v} \in \check{V}$, we get a smooth function on G by

$$\gamma_{\tilde{v} \otimes v} : g \mapsto \langle \tilde{v}, \pi(g)v \rangle$$

Let $C(\pi)$ be the vector space spanned by the functions $\gamma_{\tilde{v}\otimes v}$ for $\tilde{v} \in \check{V}, v \in V$. The functions $f \in C(\pi)$ are called the *(matrix) coefficients* of π. The space $\check{V} \otimes V$ carries a smooth representation of the group $G \times G$, while $G \times G$ acts on the function space $C(\pi)$ by translation: the first factor acts by left translation and the second by right translation. The map $\tilde{v} \otimes v \mapsto \gamma_{\tilde{v}\otimes v}$ is then a surjective $G \times G$-homomorphism $\check{V} \otimes V \to C(\pi)$.

The space $C(\pi)$ is primarily, but not exclusively, of interest in the case where π is irreducible. When π is irreducible, the centre Z of G acts on V via the central character ω_π of π and

$$\gamma(zg) = \omega_\pi(z)\gamma(g), \quad z \in Z, g \in G, \gamma \in C(\pi)$$

The support of a coefficient is therefore invariant under translation by Z.

Definition 3.1.1: γ-Cuspidal Representation

Let (π, V) be an irreducible smooth representation of G; one says that π is *γ-cuspidal* if every $\gamma \in C(\pi)$ is compactly supported modulo Z.

The term "γ-cuspidal" is convenient but temporary: it turns out to be equivalent to the standard definition. Some technical control comes first.

Proposition 3.1.2: Admissibility and γ-Cuspidality

1. If (π, V) is an irreducible γ-cuspidal representation of G, then π is admissible.
2. Let (π, V) be an irreducible admissible representation of G, and suppose that some non-zero coefficient of π is compactly supported modulo Z; then π is γ-cuspidal.

***Proof*:**

1. Suppose for a contradiction that π is not admissible. We choose a compact open subgroup K such that V^K has infinite dimension. This dimension is countable (by lemma 1.2.29). The dimension of $\check{V}^K \cong \mathrm{Hom}_{\mathbb{C}}(V^K, \mathbb{C})$ is therefore uncountable.

   We fix a non-zero $v \in V^K$ and consider the map $\Gamma_v : \check{V}^K \to C(\pi)$ given by $\tilde{v} \mapsto \gamma_{\tilde{v}\otimes v}$. Since the translates gv $(g \in G)$ span V, the map Γ_v is injective. Its image is a space of functions f on G satisfying

   $$f(zkgk') = \omega_\pi(z)f(g), \quad g \in G, z \in Z, k, k' \in K,$$

   and supported on a finite union of cosets $ZKgK$ (due to the γ-cuspidal property). The dimension of $\Gamma_v(\check{V}^K)$ is therefore at most countable, while Γ_v is injective and $\dim \check{V}^K$ is uncountable. This gives the desired contradiction.

2. The smooth dual $(\check{\pi}, \check{V})$ is irreducible and admissible (proposition 1.2.38). We view the space $\check{V} \otimes V$ as a smooth representation of $G \times G$, and hence as a smooth module over

$\mathcal{H}(G \times G) = \mathcal{H}(G) \otimes \mathcal{H}(G)$. If K is a compact open subgroup of G, we have

$$(\check{V} \otimes V)^{K \times K} = (e_K \otimes e_K) * (\check{V} \otimes V) = \check{V}^K \otimes V^K$$

If K is sufficiently small, the spaces $\check{V}^K, V^K$ are finite-dimensional simple modules over $\mathcal{H}(G, K)$. The Jacobson Density Theorem implies that $\check{V}^K \otimes V^K$ is a simple module over $\mathcal{H}(G, K) \otimes \mathcal{H}(G, K) \cong \mathcal{H}(G \times G, K \times K)$. This holds for all sufficiently small K, so $\check{V} \otimes V$ is an irreducible admissible $G \times G$-space.

The surjective $G \times G$-homomorphism $\gamma : \check{V} \otimes V \to C(\pi)$ is therefore an isomorphism and $C(\pi)$ is irreducible over $G \times G$. If $\gamma \in C(\pi)$ is non-zero, any $\gamma' \in C(\pi)$ is a finite linear combination of functions $(g, h)\gamma$ for $(g, h) \in G \times G$. If γ is compactly supported modulo Z, then so is γ'.

The terminology is justified by the equivalence with the standard definition.

Theorem 3.1.3: Equivalence of Cuspidal Definitions

Let (π, V) be an irreducible smooth representation of G. Then π is cuspidal if and only if it is γ-cuspidal.

***Proof*:**
We first assume that π is cuspidal ($V_N = 0$), and show it is γ-cuspidal. Let ϖ be a prime element of F, and put

$$t = \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix}$$

The set T^+ of powers t^n, $n \geq 0$, provides a family of representatives for $ZK\backslash G/K$, where $K = \mathrm{GL}_2(\mathfrak{o})$ (by Cartan Decomposition, proposition 2.2.3).

Lemma 3.1.4: Vanishing of Coefficients

Let $v \in V, \tilde{v} \in \check{V}$. There exists $m \geq 0$ such that $\gamma_{\tilde{v} \otimes v}(t^n) = 0$ for all $n \geq m$.

***Proof*:**
(of Lemma) We choose a compact open subgroup N_1 of N which fixes $\tilde{v}$. Since $V_N = 0$, we have $v \in V(N)$ and by lemma 2.3.2 there is a compact open subgroup N_2 of N such that $\int_{N_2} \pi(x)v\,dx = 0$. We then have $\int_{N_0} \pi(x)v\,dx = 0$ for any compact open subgroup N_0 containing N_2. However, there exists $m \geq 0$ such that $t^a N_2 t^{-a} \subset N_1$ for all $a \geq m$. For such a we have (for certain positive constants k_1, k_2):

$$\begin{aligned}\langle \tilde{v}, \pi(t^a)v\rangle &= k_1 \int_{N_1} \langle \check{\pi}(x^{-1})\tilde{v}, \pi(t^a)v\rangle\,dx \\ &= k_1 \int_{N_1} \langle \check{\pi}(t^{-a})\tilde{v}, \pi(t^{-a}xt^a)v\rangle\,dx \\ &= k_2 \int_{t^{-a}N_1t^a} \langle \check{\pi}(t^{-a})\tilde{v}, \pi(x)v\rangle\,dx = 0,\end{aligned}$$

since $t^{-a}N_1t^a \supset N_2$.

Returning to the proof of the theorem, fix a non-zero coefficient $f = \gamma_{\tilde{v}\otimes v}$. Write $K = \mathrm{GL}_2(\mathfrak{o})$ and let K' be an open normal subgroup of K fixing both $\tilde{v}$ and v. Let $k_1, \ldots, k_r$ be coset representatives for K/K'. Thus, if $g \in G$, there exists $n \geq 0$ such that $ZKgK = ZKt^nK = \bigcup_{i,j} ZK'k_i^{-1}t^nk_jK'$. It follows that

$$\mathrm{supp} f \subset \bigcup_{1\leq i,j\leq r} ZK'(\mathrm{supp} f_{ij} \cap T^+)K',$$

where f_{ij} denotes the coefficient function $x \mapsto f(k_ixk_j^{-1})$. This set is compact modulo Z by the lemma. Thus π is γ-cuspidal.

Converse: Let π be irreducible and γ-cuspidal. By proposition 3.1.2, π is admissible. By proposition 1.2.38, the dual $(\check{\pi}, \check{V})$ is irreducible and admissible. Let $K_n = 1 + \mathfrak{p}^n M_2(\mathfrak{o})$. Take $v \in V$ and choose $n \geq 1$ so that v is fixed by $\pi(K_n)$. For $\tilde{v} \in \check{V}^{K_n}$, the function $g \mapsto \langle \tilde{v}, \pi(g)v\rangle$ is compactly supported modulo Z; we deduce that $\langle \tilde{v}, \pi(t^a)v\rangle = 0$ for all a sufficiently large. Since $\check{V}^{K_n}$ is finite dimensional, there is a constant c such that this holds for all $\tilde{v} \in \check{V}^{K_n}$ and all $a \geq c$. This implies $\pi(e_{K_n})\pi(t^a)v = 0$ for $a \geq c$. Writing $K_n = N_nT_nN_n'$ (Iwahori decomposition), we set $K_n^{(a)} = t^{-a}K_nt^a$. We have, for $a \geq c$:

$$0 = \pi(e_{K_n})\pi(t^a)v = \pi(t^a)\pi(e_{K_n^{(a)}})v = \pi(t^a)\sum_{x\in N_{n-a}/N_n} \pi(x)\pi(e_{K_n^{(a)}\cap K_n})v$$

However, v is fixed by $K_n^{(a)} \cap K_n$, so this reduces to a sum over N which vanishes. We deduce that $v \in V(N)$. This applies to all $v \in V$, so π is cuspidal.

Combining the previous results gives a structural fact for $\mathrm{GL}_2(F)$.

Corollary 3.1.5: Admissibility of Irreducible Representations

Every irreducible smooth representation of $G = \mathrm{GL}_2(F)$ is admissible.

***Proof*:**
If π is not cuspidal, it is admissible by proposition 2.4.6. If π is cuspidal, it is γ-cuspidal by theorem 3.1.3, and thus admissible by proposition 3.1.2.

3.2 Intertwining and Compact Induction

In this section, we describe a general method for constructing irreducible cuspidal representations of $G = \mathrm{GL}_2(F)$ using compact induction from open subgroups. While the formal framework applies to arbitrary locally profinite groups, its power lies in the specific application to the filtrations of compact subgroups in G.

3.2.1 General Context Throughout this section, G denotes a unimodular locally profinite group satisfying the countability hypothesis. Z denotes the centre of G. For a compact open subgroup K, $\widehat{K}$ denotes the set of isomorphism classes of irreducible smooth representations of K.

Definition 3.2.2: Intertwining

For $i = 1, 2$, let K_i be a compact open subgroup of G and let $\rho_i \in \widehat{K}_i$. Let $g \in G$. The element g *intertwines* ρ_1 with ρ_2 if

$$\mathrm{Hom}_{K_1^g \cap K_2}(\rho_1^g, \rho_2) \neq 0,$$

where $K_1^g = g^{-1}K_1 g$ and ρ_1^g denotes the representation $x \mapsto \rho_1(gxg^{-1})$ of K_1^g.

This property depends only on the double coset $K_1 g K_2$. The representations ρ_1 and ρ_2 are said to *intertwine in* G if there exists some $g \in G$ which intertwines them. For a single pair (K, ρ), the element g intertwines ρ if it intertwines ρ with itself.

Definition 3.2.3: Occurrence

Let K be a compact open subgroup of G, and let (π, V) be a smooth representation of G. The representation π is said to *contain* ρ, or ρ *occurs* in π, if

$$\mathrm{Hom}_K(\rho, \pi) \neq 0$$

These definitions extend to the case where K is open and compact modulo Z, provided π admits a central character.

Proposition 3.2.4: Intertwining implies Common Occurrence

For $i = 1, 2$, let K_i be a compact open subgroup of G and let $\rho_i \in \widehat{K}_i$. Let (π, V) be an irreducible smooth representation of G which contains both ρ_1 and ρ_2. Then there exists $g \in G$ which intertwines ρ_1 with ρ_2.

***Proof*:**
For each i, V decomposes into K_i-isotypic components (proposition 1.2.16). The hypothesis is equivalent to $V^{\rho_i} \neq 0$ for $i = 1, 2$. Let e_2 denote the K_2-projection $V \to V^{\rho_2}$. Since π is irreducible and $V^{\rho_1} \neq 0$, the spaces $\pi(g^{-1})V^{\rho_1} = V^{\rho_1^g}$, $g \in G$, span V. Therefore, there exists $g \in G$ such that $e_2 \circ \pi(g^{-1})$ induces a non-zero map $V^{\rho_1} \to V^{\rho_2}$. This is the required element g.

There is a criterion for intertwining involving the Hecke algebra. Recall the idempotent $e_\rho \in \mathcal{H}(G)$ associated to $\rho \in \widehat{K}$ defined by $e_\rho(x) = \frac{\dim \rho}{\mu(K)} \mathrm{tr}\rho(x^{-1})$ for $x \in K$ and 0 otherwise.

Proposition 3.2.5: Hecke Algebra Criterion

Let K be a compact open subgroup of G, let $g \in G$, and $\rho \in \widehat{K}$. The following are equivalent:

1. there exists $f \in e_\rho * \mathcal{H}(G) * e_\rho$ such that $f|_{KgK} \neq 0$;
2. g intertwines ρ.

***Proof*:**
Consider the space $C^\infty(KgK)$ of G-smooth functions on the coset KgK. This carries a smooth representation of $K \times K$ by $(k_1, k_2)f : x \mapsto f(k_1^{-1}xk_2)$. Let H denote the group of pairs $(k, g^{-1}kg) \in K \times K$ for $k \in K \cap gKg^{-1}$. The map $f \mapsto f(g)$ is an H-homomorphism $C^\infty(KgK) \to \mathbb{C}$ (with H acting trivially). By Frobenius Reciprocity, this induces a $K \times K$-isomorphism

$$C^\infty(KgK) \cong \mathrm{Ind}_H^{K \times K}(1_H)$$

Condition (1) amounts to $e_\rho * C^\infty(KgK) * e_\rho \cong V^{\rho \otimes \check{\rho}} \neq 0$. Equivalently, $\mathrm{Hom}_{K \times K}(\rho \otimes \check{\rho}, V) \cong \mathrm{Hom}_H(\rho \otimes \check{\rho}, 1_H) \neq 0$. This last relation is equivalent to the representation $k \mapsto \rho(k) \otimes \check{\rho}(g^{-1}kg)$ of $K \cap gKg^{-1}$ having a fixed vector, which means $\mathrm{Hom}_{K \cap gKg^{-1}}(\rho, \rho^{g^{-1}}) \neq 0$. This is precisely the condition that g intertwines ρ.

To make this condition operational, one associates a specific algebra of functions to the representation ρ. Let K be an open subgroup of G, containing and compact modulo the centre Z. Let (ρ, W) be an irreducible smooth representation of K. Let $\mathcal{H}(G, \rho)$ be the space of functions $f : G \to \mathrm{End}_{\mathbb{C}}(W)$ which are compactly supported modulo Z and satisfy

$$f(k_1 g k_2) = \rho(k_1) f(g) \rho(k_2), \quad k_i \in K, g \in G$$

The support of any $f \in \mathcal{H}(G, \rho)$ is a finite union of double cosets KgK. Under the convolution operation

$$\varphi_1 * \varphi_2(g) = \int_{G/Z} \varphi_1(x) \varphi_2(x^{-1}g)\, d\dot{\mu}(x),$$

$\mathcal{H}(G, \rho)$ becomes an associative $\mathbb{C}$-algebra with identity. It is called the *ρ-spherical Hecke algebra* (or intertwining algebra).

Lemma 3.2.6: Support of Hecke Functions

Let $g \in G$; there exists $\varphi \in \mathcal{H}(G, \rho)$ with support KgK if and only if g intertwines ρ.

***Proof*:**
Let $f \in \mathrm{End}_{\mathbb{C}}(W)$. For a fixed g, the assignment $kgk' \mapsto \rho(k)f\rho(k')$ is well-defined if and only if $f \in \mathrm{Hom}_{K^g \cap K}(\rho^g, \rho)$. Since ρ is semisimple, the spaces $\mathrm{Hom}(\rho^g, \rho)$ and $\mathrm{Hom}(\rho, \rho^g)$ have the same dimension.

This algebra is intimately related to the endomorphisms of the compactly induced representation. Let $(\Sigma, X) = \text{c-Ind}_K^G \rho$. The space X consists of functions $f : G \to W$, compactly supported modulo Z, satisfying $f(kg) = \rho(k)f(g)$. For $\varphi \in \mathcal{H}(G, \rho)$ and $f \in X$, the convolution $\varphi * f$ lies in X. This defines a homomorphism of $\mathbb{C}$-algebras:

$$\mathcal{H}(G, \rho) \longrightarrow \mathrm{End}_G(\text{c-Ind}_K^G \rho)$$

Proposition 3.2.7: Hecke Algebra Isomorphism

The map $\mathcal{H}(G, \rho) \to \mathrm{End}_G(\text{c-Ind}_K^G \rho)$ is an isomorphism of $\mathbb{C}$-algebras.

***Proof*:**
The relation $\mathrm{End}_G(\text{c-Ind}\rho) \cong \mathrm{Hom}_K(\rho, \text{c-Ind}\rho)$ from Frobenius Reciprocity (theorem 1.2.21) is used here. Let $\varphi_0 : w \mapsto \varphi_0^w$ be the canonical map $W \to X$ where φ_0^w is supported on K with value $\rho(k)w$. The isomorphism is composition with φ_0. Let $\varphi \in \mathrm{Hom}_K(W, X)$. Define $\Phi(g) \in \mathrm{End}_{\mathbb{C}}(W)$ by $\Phi(g)w = \varphi(w)(g)$. One verifies that $\Phi \in \mathcal{H}(G, \rho)$ and this map provides the inverse.

These algebraic structures provide the necessary foundation for the central result of this section, which establishes a method for constructing irreducible cuspidal representations.

Theorem 3.2.8: Construction of Cuspidal Representations

Let K be an open subgroup of $G = \mathrm{GL}_2(F)$, containing and compact modulo Z. Let (ρ, W) be an irreducible smooth representation of K. Suppose that an element $g \in G$ intertwines ρ if and only if $g \in K$. Then the compactly induced representation $\text{c-Ind}_K^G \rho$ is irreducible and cuspidal.

***Proof*:**
Write $(\Sigma, X) = \text{c-Ind}_K^G \rho$. First, it is shown that Σ has a non-zero coefficient compactly supported modulo Z. By the Duality Theorem (theorem 1.3.12), $\check{X} \cong \mathrm{Ind}_K^G \check{\rho}$. This contains $\text{c-Ind}_K^G \check{\rho}$ as a subspace. Canonical embeddings are used to identify W and $\check{W}$ with functions in X and $\check{X}$ supported on K. For $w \in W, \check{w} \in \check{W}$, the coefficient $\gamma_{\check{w} \otimes w}$ is non-zero and supported in K.

Consequently, it suffices to prove that X is irreducible. If it is, it is admissible (by corollary 3.1.5) and thus γ-cuspidal (by proposition 3.1.2), hence cuspidal. The centre Z acts on X via the central character ω_ρ. Thus X is the direct sum of its K-isotypic components. Any K-map $W \to X$ has image in X^ρ. Thus:

$$\mathrm{Hom}_K(W, X^\rho) = \mathrm{Hom}_K(W, X) \cong \mathrm{End}_G(X) \cong \mathcal{H}(G, \rho)$$

The condition that g intertwines ρ iff $g \in K$ implies (by lemma 3.2.6) that functions in $\mathcal{H}(G, \rho)$ are supported on K. Thus $\dim \mathcal{H}(G, \rho) = \dim \mathrm{End}_K(\rho) = 1$. Therefore, $\mathrm{End}_G(X) \cong \mathbb{C}$. Since

$X^\rho \cong W$ is irreducible as a K-module and generates X as a G-module, standard arguments imply X is irreducible.

This construction principle can be immediately applied to classify the representations of level zero. Let $K = \mathrm{GL}_2(\mathfrak{o})$ and $K_1 = 1 + \mathfrak{p}M_2(\mathfrak{o})$. Let I_1 denote the preimage in K of the group of upper triangular unipotent matrices in $\mathrm{GL}_2(k)$.

Theorem 3.2.9: Classification of Level Zero Representations

Let (π, V) be an irreducible smooth representation of G, and suppose that π contains the trivial character of K_1. Exactly one of the following holds:

1. π contains a representation λ of K, inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$;
2. π contains the trivial character of I_1.

In the first case, π is cuspidal, and there exists a representation Λ of ZK such that $\Lambda|_K \cong \lambda$ and

$$\pi \cong \text{c-Ind}_{ZK}^G \Lambda$$

***Proof*:**
The group K stabilizes V^{K_1}, so it decomposes into representations of $K/K_1 \cong \mathrm{GL}_2(k)$. Let λ be one such component, inflated from $\tilde{\lambda}$. Either $\tilde{\lambda}$ is cuspidal (in the finite group sense) or it is not. If not, it contains the trivial character of $N(k)$, and hence λ contains the trivial character of I_1.

It must be shown that the cases are mutually exclusive and that the structure in case (1) holds.

Lemma 3.2.10: Intertwining of Level Zero Cuspidals

For $i = 1, 2$, let $\tilde{\rho}_i$ be irreducible representations of $\mathrm{GL}_2(k)$ and ρ_i their inflations to K. Suppose $\tilde{\rho}_1$ is cuspidal.

1. The representations ρ_i intertwine in G if and only if $\tilde{\rho}_1 \cong \tilde{\rho}_2$.
2. An element $g \in G$ intertwines ρ_1 if and only if $g \in ZK$.

***Proof*:**
Let g intertwine ρ_2 with ρ_1. The element g can be taken to be diagonal of the form $\mathrm{diag}(\varpi^a, 1)$ for $a \geq 0$ (Cartan decomposition). If $a \geq 1$, $K_1^g \cap K$ contains the group N_0 of upper triangular unipotent matrices in K. ρ_2^g is trivial on this group. Since $\tilde{\rho}_1$ is cuspidal, it does not contain the trivial character of N_0, so intertwining is impossible for $a \geq 1$. Thus $g \in ZK$.

By the lemma and proposition 3.2.4, the two cases cannot occur together. If $\tilde{\lambda}$ is cuspidal, π contains an extension Λ of λ to ZK. There is a non-trivial map $\text{c-Ind}_{ZK}^G \Lambda \to \pi$. By the Lemma part (2) and theorem 3.2.8, c-IndΛ is irreducible. Thus the map is an isomorphism.

3.3 Chain Orders and Fundamental Strata

This section constructs a family of compact open subgroups of $G = \mathrm{GL}_2(F)$, along with specific characters of these subgroups. This structure provides the "genetic code" for admissible representations.

The construction begins with the lattices themselves. Let $V = F \oplus F$ be the standard 2-dimensional vector space over F. $G = \mathrm{Aut}_F(V)$ and $A = \mathrm{End}_F(V) = M_2(F)$.

Definition 3.3.1: Lattice Chain

An $\mathfrak{o}$-lattice chain in V is a non-empty set $\mathcal{L}$ of $\mathfrak{o}$-lattices in V such that:

1. $\mathcal{L}$ is linearly ordered under inclusion (if $L, L' \in \mathcal{L}$, then $L \subseteq L'$ or $L' \subseteq L$).
2. $\mathcal{L}$ is stable under multiplication by $F^\times$ (if $L \in \mathcal{L}$ and $x \in F^\times$, then $xL \in \mathcal{L}$).

Given a lattice chain $\mathcal{L}$, one can index its elements as $\mathcal{L} = \{L_i\}_{i\in\mathbb{Z}}$ such that $L_{i+1} \subsetneq L_i$. Stability under $F^\times$ implies the existence of an integer $e = e(\mathcal{L})$, called the *period* of $\mathcal{L}$, such that $\varpi L_i = L_{i+e}$ for all i.

Proposition 3.3.2: Classification of Lattice Chains

Let $\mathcal{L} = \{L_i\}$ be an $\mathfrak{o}$-lattice chain in V. Then $e(\mathcal{L})$ is either 1 or 2.

1. If $e(\mathcal{L}) = 1$, there exists $g \in G$ such that $gL_i = \mathfrak{p}^i \oplus \mathfrak{p}^i$ for all $i \in \mathbb{Z}$.
2. If $e(\mathcal{L}) = 2$, there exists $g \in G$ such that $gL_{2i} = \mathfrak{p}^i \oplus \mathfrak{p}^i$ and $gL_{2i+1} = \mathfrak{p}^i \oplus \mathfrak{p}^{i+1}$ for all $i \in \mathbb{Z}$.

***Proof*:**
The quotient $L_0/L_e = L_0/\varpi L_0$ is a vector space over the residue field k of dimension 2. The chain $L_0 \supset L_1 \supset \cdots \supset L_e$ defines a flag of subspaces in k^2. The length of a strictly decreasing flag in dimension 2 is at most 2. Thus $e \in \{1, 2\}$. The basis construction follows from the elementary divisor theorem.

Associated to a lattice chain is a ring of endomorphisms.

Definition 3.3.3: Chain Order

Let $\mathcal{L}$ be a lattice chain. The associated *chain order* (or hereditary order) is:

$$\mathfrak{A}_{\mathcal{L}} = \{x \in A : xL \subseteq L \text{ for all } L \in \mathcal{L}\} = \bigcap_{i\in\mathbb{Z}} \mathrm{End}_{\mathfrak{o}}(L_i)$$

The period $e(\mathcal{L})$ is denoted $e_{\mathfrak{A}}$.

By proposition 3.3.2, there are exactly two conjugacy classes of chain orders in $M_2(F)$:

1. $e_{\mathfrak{A}} = 1$: $\mathfrak{A} \cong M_2(\mathfrak{o})$ (The standard maximal order).

2. $e_{\mathfrak{A}} = 2$: $\mathfrak{A} \cong \mathcal{I} = \begin{pmatrix} \mathfrak{o} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{o} \end{pmatrix}$ (The standard Iwahori order).

Let $\mathfrak{A}$ be a chain order. Its Jacobson radical is denoted by $\mathfrak{P} = \mathrm{rad}(\mathfrak{A})$.

Proposition 3.3.4: Radical of Chain Order

Let $\mathfrak{A}$ be a chain order with period e.

1. $\mathfrak{P} = \{x \in A : xL_i \subseteq L_{i+1} \text{ for all } i\}$.
2. $\mathfrak{P}$ is an invertible two-sided ideal of $\mathfrak{A}$, and $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A}$.
3. There exists $\Pi \in G \cap \mathfrak{A}$ such that $\mathfrak{P} = \Pi\mathfrak{A} = \mathfrak{A}\Pi$. Such an element is called a *prime element* of $\mathfrak{A}$.

***Proof*:**
Write $\mathfrak{P}' = \{x \in A : xL_i \subseteq L_{i+1} \text{ for all } i\}$ for the set described in (1).

(1). $\mathfrak{P}'$ is a two-sided ideal of $\mathfrak{A}$, since composing with an endomorphism preserving the chain keeps the one-step shift. Reduction modulo $\mathfrak{P}'$ embeds $\mathfrak{A}/\mathfrak{P}'$ into $\prod_{i \bmod e} \mathrm{End}_k(L_i/L_{i+1})$, a finite product of matrix algebras over the residue field, which is semisimple; hence $\mathrm{rad}(\mathfrak{A}) \subseteq \mathfrak{P}'$. In the other direction, iterating the defining condition gives $(\mathfrak{P}')^e L_i \subseteq L_{i+e} = \varpi L_i$, so $(\mathfrak{P}')^e \subseteq \mathfrak{p}\mathfrak{A}$ and $\mathfrak{P}'$ is nilpotent modulo the Jacobson radical of $\mathfrak{A}$; a nil ideal lies in the radical, so $\mathfrak{P}' \subseteq \mathrm{rad}(\mathfrak{A})$. Therefore $\mathfrak{P} = \mathrm{rad}(\mathfrak{A}) = \mathfrak{P}'$.

(2). The computation just made gives $\mathfrak{P}^e \subseteq \mathfrak{p}\mathfrak{A}$, and conversely $\mathfrak{p}\mathfrak{A} \subseteq \mathfrak{P}^e$ because $\varpi L_i = L_{i+e}$ realizes multiplication by ϖ as an e-step shift. Hence $\mathfrak{P}^e = \mathfrak{p}\mathfrak{A}$. Invertibility follows from (3): if $\mathfrak{P} = \Pi\mathfrak{A} = \mathfrak{A}\Pi$ with $\Pi \in G$, then $\mathfrak{P}^{-1} = \mathfrak{A}\Pi^{-1}$ is a two-sided $\mathfrak{A}$-module with $\mathfrak{P}\mathfrak{P}^{-1} = \mathfrak{A}$.

(3). By proposition 3.3.2 it is enough to exhibit Π for the two standard orders. If $e_{\mathfrak{A}} = 1$ then $\mathfrak{A} = M_2(\mathfrak{o})$, the chain is $L_i = \mathfrak{p}^i \oplus \mathfrak{p}^i$, and $\Pi = \varpi I$ shifts it by one step. If $e_{\mathfrak{A}} = 2$ then $\mathfrak{A} = \mathcal{I}$ with $L_{2i} = \mathfrak{p}^i \oplus \mathfrak{p}^i$ and $L_{2i+1} = \mathfrak{p}^i \oplus \mathfrak{p}^{i+1}$, and

$$\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$$

carries L_j onto L_{j+1} for every j, as one checks on the two lattice types. In both cases Π is invertible in A and $\Pi\mathfrak{A} = \mathfrak{A}\Pi$ is exactly the set of elements shifting the chain one step, which is $\mathfrak{P}$ by (1), completing the proof.

For the standard orders:

- If $\mathfrak{A} = M_2(\mathfrak{o})$, $\mathfrak{P} = \varpi M_2(\mathfrak{o})$. A prime element is ϖI.
- If $\mathfrak{A} = \mathcal{I}$, $\mathfrak{P} = \begin{pmatrix} \mathfrak{p} & \mathfrak{o} \\ \mathfrak{p} & \mathfrak{p} \end{pmatrix}$. A prime element is $\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$.

This structure defines a filtration of compact open subgroups. Let $U_{\mathfrak{A}} = \mathfrak{A}^\times$ be the group of units. For $n \geq 1$, define:

$$U_{\mathfrak{A}}^n = 1 + \mathfrak{P}^n$$

These are compact open normal subgroups of $U_{\mathfrak{A}}$. For integers $n > m \geq 1$ such that $2m \geq n$, the map $x \mapsto 1 + x$ induces an isomorphism $\mathfrak{P}^m/\mathfrak{P}^n \xrightarrow{\sim} U_{\mathfrak{A}}^m/U_{\mathfrak{A}}^n$.

The normalizer of the order in G is also needed:

$$\mathcal{K}_{\mathfrak{A}} = \{g \in G : g\mathfrak{A}g^{-1} = \mathfrak{A}\}$$

If $\mathfrak{A} = M_2(\mathfrak{o})$, $\mathcal{K}_{\mathfrak{A}} = F^\times \mathrm{GL}_2(\mathfrak{o})$. If $\mathfrak{A} = \mathcal{I}$, $\mathcal{K}_{\mathfrak{A}}$ is generated by $F^\times \mathcal{I}$ and the prime element $\Pi = \begin{pmatrix} 0 & 1 \\ \varpi & 0 \end{pmatrix}$. Note that $\mathcal{K}_{\mathfrak{A}}$ is open, contains Z, and is compact modulo Z.

With the orders and their unit filtrations in place, attention turns to the characters those filtration quotients carry, which is where the arithmetic of a representation is recorded. Fix an additive character $\psi \in \widehat{F}$ of level one (kernel $\mathfrak{p}$). Define a character on the algebra A by $\psi_A(x) = \psi(\mathrm{tr}(x))$. This induces an isomorphism $A \to \widehat{A}$ via $a \mapsto (x \mapsto \psi_A(ax))$. Under this duality, if $\mathfrak{A}$ is a chain order, the dual lattice of $\mathfrak{P}^n$ is $\mathfrak{P}^{1-n}$.

Definition 3.3.5: Stratum

A *stratum* in A is a triple $(\mathfrak{A}, n, \alpha)$ consisting of:

- A chain order $\mathfrak{A}$;
- An integer $n \geq 0$;
- An element $\alpha \in \mathfrak{P}^{-n}$.

Two strata $(\mathfrak{A}, n, \alpha_1)$ and $(\mathfrak{A}, n, \alpha_2)$ are *equivalent* if $\alpha_1 \equiv \alpha_2 \pmod{\mathfrak{P}^{1-n}}$.

If $n \geq 1$, a stratum $(\mathfrak{A}, n, \alpha)$ determines a character ψ_α on the group $U_{\mathfrak{A}}^n$:

$$\psi_\alpha(1 + x) = \psi_A(\alpha x), \quad x \in \mathfrak{P}^n$$

Since $\alpha \in \mathfrak{P}^{-n}$ and $x \in \mathfrak{P}^n$, the product is in $\mathfrak{A}$. However, the character is well-defined on the quotient $U_{\mathfrak{A}}^n/U_{\mathfrak{A}}^{n+1}$ because if $y \in \mathfrak{P}^{n+1}$, $\mathrm{tr}(\alpha y) \in \mathrm{tr}(\mathfrak{P}) = \mathfrak{p}$, on which ψ is trivial. The equivalence class of the stratum corresponds bijectively to the character ψ_α.

Proposition 3.3.6: Intertwining of Strata

Let $(\mathfrak{A}, n, \alpha)$ and $(\mathfrak{A}, n, \beta)$ be strata with $n \geq 1$. Let $g \in G$. The following are equivalent:

1. g intertwines the character ψ_α with ψ_β (on $U_{\mathfrak{A}}^n$).
2. The cosets intersect: $g^{-1}(\alpha + \mathfrak{P}^{1-n})g \cap (\beta + \mathfrak{P}^{1-n}) \neq \emptyset$.

***Proof*:**
Let $K_n = U_{\mathfrak{A}}^n$. The intertwining condition is that $\psi_\alpha(gxg^{-1}) = \psi_\beta(x)$ for $x \in K_n \cap g^{-1}K_n g$. In additive terms, $\psi_A(\alpha gxg^{-1}) = \psi_A(g^{-1}\alpha gx) = \psi_A(\beta x)$ for all $x \in \mathfrak{P}^n \cap g^{-1}\mathfrak{P}^n g$. This means $g^{-1}\alpha g - \beta$ lies in the dual of the lattice $\mathfrak{P}^n \cap g^{-1}\mathfrak{P}^n g$. By lattice duality properties, $(\mathfrak{P}^n \cap g^{-1}\mathfrak{P}^n g)^* = \mathfrak{P}^{1-n} + g^{-1}\mathfrak{P}^{1-n}g$. Thus $g^{-1}\alpha g - \beta = y_1 + g^{-1}y_2 g$ with $y_i \in \mathfrak{P}^{1-n}$, which rearranges to the intersection condition.

Definition 3.3.7: Fundamental Stratum

A stratum $(\mathfrak{A}, n, \alpha)$ is *fundamental* if the coset $\alpha + \mathfrak{P}^{1-n}$ contains no nilpotent element of A.

Proposition 3.3.8: Criterion for Fundamental Strata

A stratum $(\mathfrak{A}, n, \alpha)$ is fundamental if and only if for every $r \geq 1$, $\alpha^r \notin \mathfrak{P}^{1-rn}$.

***Proof*:**
The condition is a statement about the image of α in the graded algebra $\operatorname{gr}\mathfrak{A} = \bigoplus_{j\in\mathbb{Z}} \mathfrak{P}^j/\mathfrak{P}^{j+1}$: the class of α lives in the piece of degree $-n$, its r-th power is the class of α^r in degree $-rn$, and $\alpha^r \in \mathfrak{P}^{1-rn}$ says exactly that this r-th power vanishes. So the stated condition says that the class of α is not nilpotent in $\operatorname{gr}\mathfrak{A}$.

Fundamental implies the condition. Suppose $\alpha^r \in \mathfrak{P}^{1-rn}$ for some $r \geq 1$; we produce a nilpotent element in the coset. Since $A = M_2(F)$, the Cayley-Hamilton theorem lets us work with the characteristic polynomial $T^2 - (\operatorname{tr}\alpha)T + \det\alpha$. The hypothesis forces both coefficients into the ideals $\operatorname{tr}\alpha \in \mathfrak{p}^{\lceil(1-n)/e\rceil}$ and $\det\alpha \in \mathfrak{p}^{\lceil(1-2n)/e\rceil}$ that make the element α_0 with the same reduction but with characteristic polynomial T^2 lie in $\alpha + \mathfrak{P}^{1-n}$. That α_0 is nilpotent, so the coset is not free of nilpotents and the stratum is not fundamental.

The condition implies fundamental. Conversely, if some $\alpha_0 \in \alpha + \mathfrak{P}^{1-n}$ is nilpotent then $\alpha_0^2 = 0$ in $M_2(F)$, and since $\alpha \equiv \alpha_0 \pmod{\mathfrak{P}^{1-n}}$ the product expansion gives $\alpha^2 \in \alpha_0^2 + \mathfrak{P}^{1-2n} = \mathfrak{P}^{1-2n}$. Taking $r = 2$ violates the condition, completing the proof.

Note that in rank two only $r \leq 2$ is ever needed, since a nilpotent 2×2 matrix squares to zero; the statement is phrased for all r because that is the form in which it generalizes.

This concept is linked to the normalized level of a representation.

Definition 3.3.9: Normalized Level

Let π be an irreducible smooth representation of G. The *normalized level* is

$$\ell(\pi) = \min\{n/e_{\mathfrak{A}} : (\mathfrak{A}, n) \text{ such that } \pi \text{ contains the trivial character of } U_{\mathfrak{A}}^{n+1}\}$$

Theorem 3.3.10: Strata and Representation Level

Let π be an irreducible smooth representation. Suppose π contains the character ψ_α associated to a stratum $(\mathfrak{A}, n, \alpha)$ with $n \geq 1$. Then $(\mathfrak{A}, n, \alpha)$ is fundamental if and only if $\ell(\pi) = n/e_{\mathfrak{A}}$.

***Proof*:**
If the stratum is not fundamental, one can approximate α by a nilpotent element, or conjugate the order to show that ψ_α is trivial on a subgroup $U_{\mathfrak{A}'}^{n'}$ where $n'/e_{\mathfrak{A}'} < n/e_{\mathfrak{A}}$. If it is fundamental, intertwining arguments show the level cannot be lowered.

3.4 Classification of Fundamental Strata

We classify fundamental strata $(\mathfrak{A}, n, \alpha)$ based on the arithmetic of the element α. Let $\nu_{\mathfrak{A}}(\alpha) = -n$.

Definition 3.4.1: Types of Strata

Let $(\mathfrak{A}, n, \alpha)$ be a fundamental stratum.

1. **Essentially Scalar:** $e_{\mathfrak{A}} = 1$ and the image of α in $\mathfrak{A}/\mathfrak{P} \cong M_2(k)$ has a repeated non-zero eigenvalue.
2. **Split:** $e_{\mathfrak{A}} = 1$ and the image of α has distinct eigenvalues in k.
3. **Unramified Simple:** $e_{\mathfrak{A}} = 1$ and the characteristic polynomial of the image of α is irreducible over k.
4. **Ramified Simple:** $e_{\mathfrak{A}} = 2$ and n is odd.

The last two cases are collectively called **Simple Strata**. They are characterized by the fact that the algebra $E = F[\alpha]$ is a field, and α is "minimal" in the following sense:

Definition 3.4.2: Minimal Element

An element $\alpha \in G \setminus Z$ is *minimal over F* if $E = F[\alpha]$ is a field, and setting $n = -\nu_E(\alpha)$:

- If E/F is ramified, n is odd.
- If E/F is unramified, $\gcd(n, 2) = 1$ (relative to the valuation of E, this corresponds to n being coprime to the ramification index, which is 1 here, but the definition is formalized via the residue characteristic polynomial).

Proposition 3.4.3: Properties of Simple Strata

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum. Then:

1. α is minimal over F
2. $E^\times \subset \mathcal{K}_{\mathfrak{A}}$
3. $e(E|F) = e_{\mathfrak{A}}$

***Proof*:**
By definition 3.4.1 a simple stratum is of one of two shapes, and in dimension two each can be handled explicitly. Throughout, $E = F[\alpha]$.

The unramified case: $e_{\mathfrak{A}} = 1$ and the characteristic polynomial of the image $\bar{\alpha} \in \mathfrak{A}/\mathfrak{P} \cong M_2(k)$ is irreducible over k. Irreducibility of that polynomial means $k[\bar{\alpha}]$ is the quadratic extension k_2 of k, so $\bar{\alpha}$ generates a field and in particular α satisfies no polynomial over F of degree one modulo $\mathfrak{P}$; hence $E = F[\alpha]$ is a field, quadratic over F. Its residue extension already has degree two, so $f(E|F) = 2$ and therefore $e(E|F) = 1$: the extension is unramified. This gives

(3), since $e_{\mathfrak{A}} = 1$ here. For (1), the condition in definition 3.4.2 in the unramified case asks $\gcd(n, e(E|F)) = \gcd(n, 1) = 1$, which holds automatically, and E is a field as just shown, so α is minimal over F.

The ramified case: $e_{\mathfrak{A}} = 2$ and n is odd. Here $\mathfrak{A}$ is conjugate to the Iwahori order and $\mathfrak{P}^2 = \mathfrak{p}\mathfrak{A}$ by proposition 3.3.4(2). Since $\alpha \in \mathfrak{P}^{-n} \setminus \mathfrak{P}^{1-n}$ with n odd, α has odd valuation with respect to the filtration by powers of $\mathfrak{P}$, so no power α^r with r odd can lie in F: an element of $F^\times$ has even $\mathfrak{P}$-valuation, because ϖ generates $\mathfrak{P}^2$. In particular $\alpha \notin F$, so $E = F[\alpha]$ is a quadratic field extension, and the valuation v_E extending v_F satisfies $v_E(\alpha) = -n$ with n odd, which is possible only if $e(E|F) = 2$. That is (3), and it also gives (1): definition 3.4.2 in the ramified case asks exactly that n be odd.

Property (2), both cases. It suffices to show that every element of $E^\times$ normalizes $\mathfrak{A}$. The order $\mathfrak{A}$ is the intersection $\bigcap_i \mathrm{End}_{\mathfrak{o}}(L_i)$ over the defining lattice chain (definition 3.3.3), so $g\mathfrak{A}g^{-1} = \mathfrak{A}$ holds as soon as g permutes that chain. In the unramified case E/F is unramified, $\mathfrak{o}_E \subset \mathfrak{A}$ because α and hence $\mathfrak{o}[\alpha]$ stabilizes each L_i, and $E^\times = \varpi^{\mathbb{Z}}\mathfrak{o}_E^\times$; units of $\mathfrak{o}_E$ preserve each lattice and ϖ scales the whole chain, so both normalize $\mathfrak{A}$. In the ramified case, let ϖ_E be a prime element of E. Then $v_E(\varpi_E) = 1$, so ϖ_E shifts the chain by one step, that is $\varpi_E L_i = L_{i+1}$, and it is therefore a prime element of $\mathfrak{A}$ in the sense of proposition 3.3.4(3); shifting the chain permutes it, so $\varpi_E \in \mathcal{K}_{\mathfrak{A}}$. Since $E^\times = \varpi_E^{\mathbb{Z}}\mathfrak{o}_E^\times$ and the units again preserve each lattice, $E^\times \subset \mathcal{K}_{\mathfrak{A}}$, as we sought to show.

3.5 The Classification of Cuspidal Representations

We assume π is an irreducible smooth representation of G.

3.5.1 Exhaustion

Proposition 3.5.1: Principal Series Strata

Let χ be a character of T. Let $\Sigma = \mathrm{Ind}_B^G \chi$.

1. If χ is sufficiently ramified, Σ contains a Split fundamental stratum.
2. If $\chi_1\chi_2^{-1}$ is trivial on units but ramified, Σ contains an Essentially Scalar stratum.
3. If χ is unramified, Σ contains the trivial character of the Iwahori $U_{\mathcal{I}}^1$

Proof:
The point in every case is that Σ is realized on functions on G determined by their restriction to K (proposition 2.2.1), so the characters of the filtration subgroups occurring in Σ can be read off the torus.

(3). If χ is unramified it is trivial on $U_T = T \cap K$, so the function supported on the big cell and equal to 1 on $B(I \cap N')$ is fixed by the Iwahori I, using the decomposition $G = BI \cup BwI$ recorded after proposition 2.2.5. A vector fixed by I is in particular fixed by $U_{\mathcal{I}}^1 \subseteq I$, so Σ contains the trivial character of $U_{\mathcal{I}}^1$.

(1) and (2). Restrict Σ to K and use $\Sigma|_K \cong \mathrm{Ind}_{B\cap K}^{K}(\chi|_{B\cap K})$, again by Iwasawa. The characters of $U_{\mathfrak{A}}^n/U_{\mathfrak{A}}^{n+1}$ occurring in this induced representation are, by Frobenius reciprocity (theorem 1.2.21) and Mackey restriction to the subgroup $U_{\mathfrak{A}}^n$, the K-conjugates of the characters obtained by restricting χ along $T\cap U_{\mathfrak{A}}^n \to U_{\mathfrak{A}}^n$. Writing such a character as ψ_α via the duality of definition 3.3.5, the element α can therefore be taken diagonal, $\alpha = \mathrm{diag}(a_1, a_2)$ with a_i determined by the levels of χ_1 and χ_2.

If χ is sufficiently ramified, meaning χ_1 and χ_2 have distinct levels, the two diagonal entries have distinct valuations, so the image of α in $\mathfrak{A}/\mathfrak{P} \cong M_2(k)$ has distinct eigenvalues in k: that is a Split stratum in the sense of definition 3.4.1(2), and it is fundamental because a diagonal matrix with distinct non-zero eigenvalues is not nilpotent modulo $\mathfrak{P}$.

If instead $\chi_1\chi_2^{-1}$ is trivial on units but ramified, the two entries have the same valuation and the same image in k, so α reduces to a scalar with a repeated non-zero eigenvalue: that is an Essentially Scalar stratum, definition 3.4.1(1), completing the proof.

This characterizes the cuspidals.

Theorem 3.5.2: Exhaustion Theorem

Let π be an irreducible smooth representation of G. Assume $\ell(\pi) \le \ell(\chi\pi)$ for all characters χ of $F^\times$ ("minimal level"). The following are equivalent:

1. π is cuspidal.
2. Either:
   (a) $\ell(\pi) = 0$ and π contains a representation of $K = \mathrm{GL}_2(\mathfrak{o})$ inflated from an irreducible *cuspidal* representation of $\mathrm{GL}_2(k)$.
   (b) $\ell(\pi) > 0$ and π contains a *Simple Stratum*.

***Proof*:**
If π is not cuspidal, it is a subquotient of a principal series. By proposition 3.5.1, it must contain a split or essentially scalar stratum, or be level 0 non-cuspidal. Conversely, the intertwining properties of simple strata (below) prevent them from occurring in induced representations from B.

3.5.2 Cuspidal Types

To generate the representations, we extend the simple stratum character. Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum with $E = F[\alpha]$. We define a filtration of open subgroups in $\mathcal{K}_{\mathfrak{A}}$:

$$H_\alpha = U_E^1 U_{\mathfrak{A}}^{\lfloor n/2\rfloor+1}, \qquad J_\alpha = U_E U_{\mathfrak{A}}^{\lfloor n/2\rfloor+1}, \qquad J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2\rfloor}$$

(Note: if n is odd, $J_\alpha = J$). The character ψ_α on $U_{\mathfrak{A}}^{\lfloor n/2\rfloor+1}$ extends naturally to H_α (using $\psi_A(\alpha\cdot)$).

Theorem 3.5.3: Intertwining of Simple Characters

Let $(\mathfrak{A}, n, \alpha)$ be a simple stratum.

1. An element $g \in G$ intertwines the character ψ_α on $U_{\mathfrak{A}}^{\lfloor n/2 \rfloor+1}$ if and only if $g \in J$
2. For any irreducible representation Λ of J containing ψ_α, the induced representation $\text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal

***Proof*:**
Write $m = \lfloor n/2 \rfloor$, so the character in question lives on $U_{\mathfrak{A}}^{m+1}$ and $J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$.

(1), the easy inclusion. If $g \in E^\times$ then g commutes with α, so $\psi_\alpha(gxg^{-1}) = \psi_A(\alpha gxg^{-1}) = \psi_A(g^{-1}\alpha gx) = \psi_A(\alpha x) = \psi_\alpha(x)$, and g intertwines ψ_α with itself. If $g \in U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$ then $g = 1 + y$ with $y \in \mathfrak{P}^{\lfloor (n+1)/2 \rfloor}$, and for $x \in \mathfrak{P}^{m+1}$ the commutator $\alpha(gxg^{-1} - x)$ lies in $\mathfrak{P}^{-n}\mathfrak{P}^{\lfloor (n+1)/2 \rfloor}\mathfrak{P}^{m+1} \subseteq \mathfrak{P}$, on which ψ_A is trivial. So every generator of J intertwines ψ_α, hence so does J.

(1), the converse. Suppose g intertwines ψ_α on $U_{\mathfrak{A}}^{m+1}$. By proposition 3.3.6 applied to the stratum $(\mathfrak{A}, n', \alpha)$ with $n' = n - m$, this says

$$g^{-1}\big(\alpha + \mathfrak{P}^{-m}\big)g \cap \big(\alpha + \mathfrak{P}^{-m}\big) \neq \emptyset$$

so $g^{-1}\alpha g$ and α agree modulo $\mathfrak{P}^{-m}$, that is, to more than half the depth of α. Because α is minimal over F (proposition 3.4.3), the coset $\alpha + \mathfrak{P}^{-m}$ determines $E = F[\alpha]$ and no proper subfield meets it, so this half-depth agreement forces $g^{-1}Eg = E$ and then $g \in E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor} = J$.

This last forcing step is where the argument leaves what has been developed here: turning agreement to half depth into membership in J is the "intertwining implies conjugacy" mechanism for simple strata, and in the shape needed it rests on the full theory of simple characters rather than on the two-by-two computations above. It is quoted; see [1, §13].

(2). Granting (1), the hypothesis of theorem 3.2.8 is met: J is an open subgroup of G containing the centre and compact modulo it (proposition 3.4.3(2) puts $E^\times$ inside $\mathcal{K}_{\mathfrak{A}}$, which is compact modulo Z), Λ is an irreducible smooth representation of J, and an element of G intertwines Λ only if it intertwines the character ψ_α that Λ contains, hence only if it lies in J. That theorem then gives that $\text{c-Ind}_J^G \Lambda$ is irreducible and cuspidal, as we sought to show.

Definition 3.5.4: Cuspidal Type

A *cuspidal type* is a triple $(\mathfrak{A}, J, \Lambda)$ where:

1. $\mathfrak{A}$ is a chain order.
2. J is an open subgroup of $\mathcal{K}_{\mathfrak{A}}$ as defined above (associated to a simple stratum or level 0).
3. Λ is an irreducible representation of J constructed as follows:
   - **Level 0:** $J = F^\times \mathrm{GL}_2(\mathfrak{o})$, Λ is an extension of a cuspidal rep of $\mathrm{GL}_2(k)$.
   - **Positive Level:** $\Lambda|_{U_{\mathfrak{A}}^{\lfloor n/2 \rfloor+1}}$ is a multiple of ψ_α for a simple stratum α.

Theorem 3.5.5: Classification of Cuspidal Representations

The mapping

$$(\mathfrak{A}, J, \Lambda) \longmapsto \pi_\Lambda = \text{c-Ind}_J^G \Lambda$$

induces a bijection between the set of G-conjugacy classes of cuspidal types and the set of equivalence classes of irreducible cuspidal representations of G.

***Proof*:**
Injectivity: Follows from the Intertwining Theorem. Since g intertwines Λ iff $g \in J$, the Hecke algebra $\mathcal{H}(G, \Lambda)$ is 1-dimensional, implying irreducibility. **Surjectivity:** By the Exhaustion Theorem, π contains a simple stratum (or is level 0). If level > 0, $\pi|_{U^{\lfloor n/2 \rfloor+1}}$ contains ψ_α. One then ascends the hierarchy $H_\alpha \subset J_\alpha \subset J$ using Clifford theory (Heisenberg groups appear here for J_α/H_α). This proves π contains some Λ of the required type.

3.6 Representations with Iwahori-Fixed Vectors

We consider representations π such that $V^I \neq 0$, where I is the Iwahori subgroup.

Proposition 3.6.1: Structure of Iwahori-Fixed Representations

Let (π, V) be irreducible. If $V^I \neq 0$, then π is a subquotient of $\mathrm{Ind}_B^G \chi$ where χ is an unramified character of T. Specifically, π is either:

1. An unramified principal series $\mathrm{Ind}_B^G \chi$ (χ regular).
2. An unramified twist of the Steinberg representation $\chi \otimes \mathrm{St}_G$.
3. An unramified character $\chi \circ \det$.

In particular, such representations are never cuspidal.

***Proof*:**
The category of smooth representations generated by their I-fixed vectors is equivalent to the category of modules over the Hecke algebra $\mathcal{H}(G, I) = e_I * \mathcal{H}(G) * e_I$. The algebra $\mathcal{H}(G, I)$ is generated by two elements T_w (supported on IwI) and T_Π (supported on $I\Pi I$). These satisfy the Affine Hecke Algebra relations (e.g., quadratic relation for T_w). The finite dimensional irreducible modules of this algebra are 1 or 2 dimensional. These correspond precisely to the listed representation types via the functor $M \mapsto \mathcal{H}(G) \otimes_{\mathcal{H}(G,I)} M$.

This analysis is crucial for establishing the square-integrability of the Steinberg representation.

Theorem 3.6.2: Square-Integrability of Steinberg

The Steinberg representation St_G is square-integrable modulo the center. That is, for matrix coefficients f,

$$\int_{G/Z} |f(g)|^2 \, d\dot{g} < \infty$$

***Proof*:**
Let $f(g) = \langle \tilde{v}, \pi(g)v \rangle$ where v is I-fixed. Using the Iwahori decomposition and the specific values of the Hecke operators on the unique I-fixed line in St_G, one calculates that $|f(g)|$ behaves like $q^{-\ell(w)/2}$ where $\ell(w)$ is the length of the element in the affine Weyl group. The number of elements of length k grows linearly/polynomially, while the decay is exponential, ensuring convergence.

4

Quadratic Tame Extensions Paramaterizing Tame Cuspidals

We have seen that cuspidal representations are constructed by compact induction, but the construction in the previous chapter was abstract. To establish the Langlands correspondence, we need an explicit parameterization of these representations that can be matched against Galois representations. The goal of this chapter is to provide this explicit parameterization for the "tame" case (when the residual characteristic p is odd). We will show that tame cuspidal representations are in bijection with *Admissible Pairs* $(E/F, \xi)$, consisting of a quadratic extension and a character.

We begin in section 4.1 by defining admissible pairs and analyzing the arithmetic of tame quadratic extensions. In section 4.2, we explicitly construct a cuspidal representation π_ξ for each pair $(E/F, \xi)$ using the machinery of fundamental strata. Finally, in section 4.3, we prove the Parametrization Theorem, showing that this construction yields a bijection between admissible pairs and the set of tame cuspidal representations. This provides the "Group side" of the correspondence that we will later match to the "Galois side."

4.1 Admissible Pairs

We rely on the arithmetic of tame extensions. Throughout, let $\psi \in \widehat{F}$ be a character of level one. As a reminder:

Definition 4.1.1: Tamely Ramified Extension

A finite field extension E/F is *tamely ramified* if $p \nmid e(E|F)$. Equivalently, $\mathrm{tr}_{E/F}(\mathfrak{o}_E) = \mathfrak{o}_F$.

If $p \neq 2$, every quadratic extension is tamely ramified. If $p = 2$, only the unramified quadratic extension is tame.

Lemma 4.1.2: Properties of Tame Extensions

Let E/F be a tamely ramified extension with ramification index e.

1. For $r \in \mathbb{Z}$, $\mathrm{tr}_{E/F}(\mathfrak{p}_E^{1+r}) = \mathfrak{p}_F^{1+\lfloor r/e \rfloor}$.
2. For $m \geq 1$, the norm $N_{E/F}$ induces an isomorphism
$$U_E^{em}/U_E^{em+1} \xrightarrow{\sim} U_F^m/U_F^{m+1}$$
3. The map $N_{E/F} : U_E/U_E^1 \to U_F/U_F^1$ is surjective if E/F is unramified, and has kernel/-cokernel of order $\gcd(e, q-1)$ if totally ramified.

***Proof*:**
Property (1) follows from the fact that the inverse different $\mathfrak{d}_{E/F}^{-1} = \mathfrak{p}_E^{1-e}$. Property (2) follows from the expansion $N(1+x) = 1 + \mathrm{tr}(x) + \ldots$. See [10, the norm and trace on a tamely ramified extension] for the details.

Define $\psi_E = \psi \circ \mathrm{tr}_{E/F}$. By the lemma, ψ_E has level one in E.

Proposition 4.1.3: Norm and Character Levels

Let χ be a character of $F^\times$ of level $m \geq 1$. Let $\chi_E = \chi \circ N_{E/F}$. Then χ_E has level em. If $c \in \mathfrak{p}_F^{-m}$ satisfies $\chi(1+x) = \psi(cx)$ for $x \in \mathfrak{p}_F^{\lfloor m/2 \rfloor + 1}$, then
$$\chi_E(1+y) = \psi_E(cy), \quad y \in \mathfrak{p}_E^{\lfloor em/2 \rfloor + 1}$$

***Proof*:**
For the level, lemma 4.1.2(2) gives an isomorphism $U_E^{em}/U_E^{em+1} \to U_F^m/U_F^{m+1}$ induced by the norm. Since χ has level m it is trivial on U_F^{m+1} and non-trivial on U_F^m, so $\chi_E = \chi \circ N_{E/F}$ is trivial on U_E^{em+1} and non-trivial on U_E^{em}; that is, χ_E has level em.

For the second assertion, expand the norm of a principal unit. For $y \in \mathfrak{p}_E$,
$$N_{E/F}(1+y) = (1+y)(1+y^\sigma) = 1 + \mathrm{tr}_{E/F}(y) + N_{E/F}(y)$$
where σ generates $\mathrm{Gal}(E/F)$. Take $y \in \mathfrak{p}_E^{\lfloor em/2 \rfloor + 1}$. Then $N_{E/F}(y)$ lies in $\mathfrak{p}_F^{m+1}$, because its valuation is at least twice that of y measured in F, and $2(\lfloor em/2 \rfloor + 1) > em$. As χ is trivial on U_F^{m+1}, that term may be discarded:
$$\chi_E(1+y) = \chi\big(1 + \mathrm{tr}_{E/F}(y)\big)$$
By lemma 4.1.2(1), $\mathrm{tr}_{E/F}(y) \in \mathfrak{p}_F^{\lfloor m/2 \rfloor + 1}$, which is exactly the range on which $\chi(1+x) = \psi(cx)$ holds. Since $c \in F$ the trace is F-linear, so
$$\chi_E(1+y) = \psi\big(c\,\mathrm{tr}_{E/F}(y)\big) = \psi\big(\mathrm{tr}_{E/F}(cy)\big) = \psi_E(cy)$$

as we sought to show.

Definition 4.1.4: Admissible Pair

A pair $(E/F, \chi)$ is called *admissible* if:

1. E/F is a tamely ramified quadratic extension.
2. χ is a character of $E^\times$.
3. χ does not factor through the norm $N_{E/F}$ (i.e., $\chi \neq \varphi \circ N_{E/F}$ for any $\varphi \in \widehat{F^\times}$).
4. If $\chi|_{U_E^1}$ factors through $N_{E/F}$, then E/F is unramified.

Admissible pairs are isomorphic if there is an F-isomorphism $j : E \to E'$ such that $\chi = \chi' \circ j$. Let $P_2(F)$ denote the set of isomorphism classes.

Definition 4.1.5: Minimality of Pairs

An admissible pair $(E/F, \chi)$ is *minimal* if $\chi|_{U_E^n}$ does not factor through $N_{E/F}$, where n is the level of χ.

Any admissible pair is isomorphic to a twist $(E/F, \chi' \otimes \varphi_E)$ where $(E/F, \chi')$ is minimal.

Proposition 4.1.6: Minimality Criterion

Let $(E/F, \chi)$ be admissible with χ of level $m \geq 1$. Let $\alpha \in \mathfrak{p}_E^{-m}$ satisfy $\chi(1+x) = \psi_E(\alpha x)$ for $x \in \mathfrak{p}_E^m$. Then $(E/F, \chi)$ is minimal if and only if the element α is *minimal over* F.

***Proof*:**
Both conditions are read off the same congruence, so the argument proves the equivalence in one pass.

The restriction $\chi|_{U_E^m}$ is the character $1 + x \mapsto \psi_E(\alpha x)$ on $\mathfrak{p}_E^m$. It factors through the norm exactly when it agrees with $\varphi \circ N_{E/F}$ for some character φ of $F^\times$; by proposition 4.1.3 such a φ, of level m' with $em' = m$, is realized by an element $c \in F$ through $\varphi_E(1 + x) = \psi_E(cx)$. So $\chi|_{U_E^m}$ factors through the norm if and only if there is $c \in F$ with

$$\psi_E\big((\alpha - c)x\big) = 1 \qquad \text{for all } x \in \mathfrak{p}_E^m$$

Since ψ_E has level one, this says precisely that $\alpha - c \in \mathfrak{p}_E^{1-m}$, that is, that the coset $\alpha + \mathfrak{p}_E^{1-m}$ meets F.

Now compare with definition 3.4.2. An element α with $-\nu_E(\alpha) = m$ fails to be minimal over F exactly when it can be adjusted modulo $\mathfrak{p}_E^{1-m}$ into a proper subfield of $E = F[\alpha]$, which for a quadratic extension means into F itself. The two failure conditions are therefore the same condition, and $(E/F, \chi)$ is minimal if and only if α is minimal over F, as we sought to show.

4.2 Construction of Representations

We associate an irreducible cuspidal representation π_χ to each admissible pair $(E/F, \chi)$.

4.2.1 Level Zero Construction

Let $(E/F, \chi)$ be an admissible pair where χ has level zero. By definition, E/F must be unramified.

Lemma 4.2.1: Admissibility at Level Zero

Let E/F be unramified quadratic and χ a character of level zero. Let $\sigma \in \mathrm{Gal}(E/F)$ be the non-trivial element. The following are equivalent:

1. $(E/F, \chi)$ is admissible.
2. $\chi \neq \chi^\sigma$.
3. $\chi|_{U_E} \neq \chi^\sigma|_{U_E}$.

***Proof*:**
Since E/F is unramified, $E^\times = F^\times U_E$. The kernel of the norm consists of x/x^σ. χ factors through the norm iff $\chi(x/x^\sigma) = 1$ iff $\chi = \chi^\sigma$.

Embed E into $A = M_2(F)$ such that $E^\times$ normalizes the maximal order $\mathfrak{A} = M_2(\mathfrak{o})$. This induces an embedding of residue fields $k_E \to M_2(k)$. The character $\chi|_{U_E}$ inflates a character $\tilde{\chi}$ of $k_E^\times$. Since $\tilde{\chi} \neq \tilde{\chi}^q$, this generates a cuspidal representation $\tilde{\pi}$ of $\mathrm{GL}_2(k)$ (see theorem 2.1.14). Let Λ be the extension of the inflation of $\tilde{\pi}$ to $K_\mathfrak{A} = F^\times \mathrm{GL}_2(\mathfrak{o})$ such that $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$.

Definition 4.2.2: Representation π_χ (Level 0)

For a level zero admissible pair, define $\pi_\chi = \text{c-Ind}_{K_\mathfrak{A}}^G \Lambda$.

Proposition 4.2.3: Bijection at Level Zero

The map $(E/F, \chi) \mapsto \pi_\chi$ induces a bijection between $P_2(F)_0$ and $\mathcal{A}_2^0(F)_0$ (cuspidals of level 0).

***Proof*:**
Both sides are parameterized by the same finite-field data, and the argument is to identify that datum on each side.

Surjectivity. Let π be cuspidal of level zero. By theorem 3.2.9, π contains a representation λ of $K = \mathrm{GL}_2(\mathfrak{o})$ inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$, and $\pi \cong \text{c-Ind}_{ZK}^G \Lambda$ for an extension Λ of λ to $ZK = K_\mathfrak{A}$. By theorem 2.1.14 the cuspidal representations of $\mathrm{GL}_2(k)$ are parameterized by the characters $\tilde{\chi}$ of $k_E^\times$ with $\tilde{\chi} \neq \tilde{\chi}^q$, two such characters giving the same representation exactly when they are exchanged by $\tilde{\chi} \mapsto \tilde{\chi}^q$. Inflating $\tilde{\chi}$ along $U_E \to k_E^\times$

and extending over $F^\times$ so that the central character matches $\Lambda|_{F^\times}$ produces a character χ of $E^\times = F^\times U_E$ of level zero with $\chi \neq \chi^\sigma$, hence an admissible pair by lemma 4.2.1, and by construction $\pi_\chi \cong \pi$.

Injectivity. If $\pi_{\chi_1} \cong \pi_{\chi_2}$ then the two inducing representations of $K_{\mathfrak{A}}$ are conjugate, since by theorem 3.2.9 the type is determined by π up to conjugacy; hence $\tilde{\chi}_1 \in \{\tilde{\chi}_2, \tilde{\chi}_2^q\}$. Conjugation by the non-trivial element of $\mathrm{Gal}(E/F)$ realizes $\tilde{\chi} \mapsto \tilde{\chi}^q$ and is an isomorphism of pairs, so $(E/F, \chi_1) \cong (E/F, \chi_2)$ once the central characters are compared: both restrict to ω_π on $F^\times$, which pins the extension from U_E to $E^\times$. Since E/F is forced to be the unramified quadratic extension by lemma 4.2.1, there is no choice of field to make, completing the proof.

4.2.2 Positive Level Construction

Let $(E/F, \chi)$ be a minimal admissible pair with χ of level $n \geq 1$. Choose $\alpha \in \mathfrak{p}_E^{-n}$ realizing χ as a character (via ψ_E). Embed E in A such that $E^\times \subset K_{\mathfrak{A}}$ for a chain order $\mathfrak{A}$ (where $e_{\mathfrak{A}} = e(E/F)$). The triple $(\mathfrak{A}, n, \alpha)$ is a simple stratum. Define the groups $J = E^\times U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$ and $J^1 = U_E^1 U_{\mathfrak{A}}^{\lfloor (n+1)/2 \rfloor}$. We must construct a representation $\Lambda \in C(\psi_\alpha, \mathfrak{A})$ of J.

Case 1: n is odd. In this case, $J = E^\times U_{\mathfrak{A}}^{(n+1)/2}$. The character ψ_α is defined on $U_{\mathfrak{A}}^{(n+1)/2}$.

Definition 4.2.4: Representation Λ (Odd)

Define the character Λ of J by:

$$\Lambda|_{U_{\mathfrak{A}}^{(n+1)/2}} = \psi_\alpha, \qquad \Lambda|_{E^\times} = \chi$$

These are consistent since χ corresponds to α.

Case 2: n is even. Here $n = 2m$. E/F is unramified. $J = E^\times U_{\mathfrak{A}}^m$. Let $H^1 = U_E^1 U_{\mathfrak{A}}^{m+1}$ and $J^1 = U_E^1 U_{\mathfrak{A}}^m$. Define a character θ on H^1 by $\theta(ux) = \chi(u)\psi_\alpha(x)$ for $u \in U_E^1, x \in U_{\mathfrak{A}}^{m+1}$.

Lemma 4.2.5: Heisenberg Extension

There is a unique irreducible representation η of J^1 such that $\eta|_{H^1}$ contains θ.

1. $\dim \eta = q$.
2. $\eta|_{H^1}$ is a multiple of θ.
3. There is a unique extension $\tilde{\eta}$ of η to the group $\mu_E J^1$ (where μ_E are roots of unity) such that $\mathrm{tr}\tilde{\eta}(\zeta) = -\chi(\zeta)$ for $\zeta \in \mu_E \setminus \mu_F$.

***Proof*:**
Everything here is an instance of the group-theoretic analysis carried out in section 4.5, which is stated for an abstract finite p-group with cyclic centre precisely so that it can be applied at this point; the verification that the present groups satisfy its hypotheses is the closing paragraph of that section. Take $G_0 = J^1/\ker\theta$ with centre $Z_0 = H^1/\ker\theta$, and $A = \mu_E/\mu_F$ acting by conjugation.

The quotient $V = G_0/Z_0 \cong J^1/H^1 \cong \mathfrak{p}_F^m/(\mathfrak{p}_E^m + \mathfrak{p}_F^{m+1})$ is elementary abelian of order q^2, and the commutator pairing $(x, y) \mapsto \theta([x, y])$ is a non-degenerate alternating form on it. Existence and uniqueness of η with $\eta|_{Z_0}$ containing θ is then the Stone-von Neumann statement recalled at the head of section 4.5, and $\dim \eta = |V|^{1/2} = q$, which is (1); that $\eta|_{H^1}$ is a multiple of θ is part of the same statement, giving (2).

For (3), A is cyclic of order $q + 1$, has order prime to p, acts trivially on Z_0, and acts on V with no non-trivial fixed point, since k_E is a maximal subfield of $M_2(k)$ and so is its own centralizer. These are exactly hypotheses (a), (b) and (c), so lemma 4.5.2 applies and supplies a unique extension $\tilde{\eta}$ of η to $\mu_E J^1$ with $\mathrm{tr}\tilde{\eta}(a) = -1$ for $a \neq 1$ in A. Untwisting by χ, which is what identifies A-characters with characters of μ_E trivial on μ_F, turns this into the normalization $\mathrm{tr}\tilde{\eta}(\zeta) = -\chi(\zeta)$ for $\zeta \in \mu_E \setminus \mu_F$, completing the proof.

Corollary 4.2.6: Representation Λ (Even)

There is a unique irreducible representation Λ of J such that:

1. $\Lambda|_{J^1} \cong \eta$
2. $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$
3. For $\zeta \in \mu_E \setminus \mu_F$, $\mathrm{tr}\Lambda(\zeta) = -\chi(\zeta)$

***Proof*:**
Write $J = E^\times U_{\mathfrak{A}}^m$ and $J^1 = U_E^1 U_{\mathfrak{A}}^m$. Since E/F is unramified in the even case, $E^\times = \varpi^{\mathbb{Z}} \mu_E U_E^1$ with ϖ a prime element of F, so

$$J = \varpi^{\mathbb{Z}}\, \mu_E\, J^1$$

and the three groups $\varpi^{\mathbb{Z}}$, μ_E and J^1 generate J with J^1 normal.

Existence. Lemma 4.2.5(3) supplies the extension $\tilde{\eta}$ of η to $\mu_E J^1$ normalized by $\mathrm{tr}\tilde{\eta}(\zeta) = -\chi(\zeta)$ on $\mu_E \setminus \mu_F$. The remaining generator ϖ is central in J, so setting $\Lambda(\varpi) = \chi(\varpi) \cdot \mathrm{id}$ and $\Lambda|_{\mu_E J^1} = \tilde{\eta}$ defines a representation of J provided the two prescriptions agree on $\varpi^{\mathbb{Z}} \cap \mu_E J^1$, which they do because $\tilde{\eta}$ restricted to $\mu_F \subset \mu_E$ is a multiple of χ by construction. This Λ satisfies (1), (2) and (3), and it is irreducible because its restriction to J^1 is the irreducible η.

Uniqueness. Suppose Λ' also satisfies the three conditions. By (1) the restrictions to J^1 agree, so by Clifford theory Λ' and Λ differ by a character of J/J^1. That quotient is generated by the images of ϖ and of μ_E. Condition (2) forces the two to agree on ϖ, and condition (3) forces the traces to agree on every $\zeta \in \mu_E \setminus \mu_F$; since a character of the cyclic group μ_E/μ_F is determined by its values there, the difference character is trivial and $\Lambda' \cong \Lambda$, completing the proof.

Definition 4.2.7: Representation π_χ (Positive Level)

Let Λ be as defined in Case 1 or Case 2. Define $\pi_\chi = \text{c-Ind}_J^G \Lambda$.

Proposition 4.2.8: Independence of Choices

The equivalence class of π_χ depends only on the isomorphism class of $(E/F, \chi)$.

1. It is independent of the choice of α, ψ, and the embedding $E \to A$.
2. If φ is a character of $F^\times$ such that $(E/F, \chi\varphi_E)$ is minimal, then $\pi_{\chi\varphi_E} = \varphi\pi_\chi$.

***Proof*:**
Any two embeddings are conjugate. Changing α changes ψ_α by a conjugation or a unit, which is absorbed by the uniqueness of the Heisenberg representation.

This defines π_χ for any admissible pair, not necessarily minimal, by twisting: $\pi_\chi = \varphi\pi_{\chi'}$ where $\chi = \chi'\varphi_E$.

4.3 The Parametrization Theorem

We denote $\mathcal{A}_2^0(F)$ the set of irreducible cuspidal representations. $\mathcal{A}_2^{\mathrm{nr}}(F)$ is the subset of unramified representations (twists of level 0 types).

Theorem 4.3.1: Tame Parametrization Theorem

The construction $(E/F, \chi) \mapsto \pi_\chi$ induces a bijection:

$$P_2(F) \xrightarrow{\sim} \begin{cases} \mathcal{A}_2^0(F) & \text{if } p \neq 2 \\ \mathcal{A}_2^{\mathrm{nr}}(F) & \text{if } p = 2 \end{cases}$$

Properties:

1. $\ell(\pi_\chi) = \ell(\chi)/e(E|F)$
2. $\omega_{\pi_\chi} = \chi|_{F^\times}$
3. $\pi_{\check{\chi}} = \check{\pi}_\chi$
4. $\pi_{\chi\varphi_E} = \varphi\pi_\chi$

***Proof*:**
The construction and the two propositions of section 4.4 do the work; what remains is to assemble them and to check that twisting lets the minimal case control the general one.

Step 1: the map is well defined. For a minimal pair this is proposition 4.2.8(1): the class of π_χ does not depend on the choice of α, of ψ, or of the embedding $E \to A$. For a general admissible pair, write $\chi = \chi'\varphi_E$ with $(E/F, \chi')$ minimal, as is always possible, and set $\pi_\chi = \varphi\pi_{\chi'}$. If $\chi = \chi''\varphi'_E$ is a second such factorization then $\chi'(\chi'')^{-1} = (\varphi'\varphi^{-1})_E$ is of the form η_E, so $\pi_{\chi'} = \eta\pi_{\chi''}$ by proposition 4.2.8(2), and the two candidate values agree. Isomorphic pairs give equal representations because an F-isomorphism $j\colon E \to E'$ carries one set of choices to another.

Step 2: injectivity. On minimal pairs this is proposition 4.4.1. In general, suppose $\pi_{\chi_1} \cong \pi_{\chi_2}$ with $\chi_i = \chi_i'(\varphi_i)_{E_i}$ and χ_i' minimal. By definition 3.3.9 a representation and its twists attain a smallest level, and by property (1) below that smallest level is attained exactly at the minimal pairs; twisting both sides by φ_1^{-1} therefore reduces to the case where χ_1 and χ_2 are both minimal, where proposition 4.4.1 applies. Undoing the twist gives $(E_1/F, \chi_1) \cong (E_2/F, \chi_2)$.

Step 3: surjectivity. Let π lie in the target set. Twisting by a character of $F^\times$ we may assume $\ell(\pi) \leq \ell(\varphi\pi)$ for every φ, since the level takes non-negative values in $\frac{1}{2}\mathbb{Z}$ and so attains a minimum on the twist orbit. Proposition 4.4.3 then produces a minimal admissible pair $(E/F, \chi)$ with $\pi \cong \pi_\chi$, and undoing the twist realizes the original π as $\pi_{\chi\varphi_E}$ by property (4).

Step 4: the two cases. By theorem 3.5.5 every irreducible cuspidal representation contains a cuspidal type $(\mathfrak{A}, J, \Lambda)$, and by proposition 3.4.3(3) the chain order satisfies $e_{\mathfrak{A}} = e(E|F)$ for the field $E = F[\alpha]$ of its simple stratum. If $p \neq 2$ both quadratic extensions of F are tamely ramified, so every cuspidal type arises from an admissible pair and the target is all of $\mathcal{A}_2^0(F)$. If $p = 2$ only the unramified quadratic extension is tame, so only chain orders with $e_{\mathfrak{A}} = 1$, that is $\mathfrak{A} \cong M_2(\mathfrak{o})$, are reached; by lemma 4.3.2 these are exactly the representations in $\mathcal{A}_2^{\mathrm{nr}}(F)$.

Step 5: the four properties.

1. The representation π_χ contains the simple stratum $(\mathfrak{A}, n, \alpha)$ with n the level of χ, so $\ell(\pi_\chi) = n/e_{\mathfrak{A}}$ by definition 3.3.9, and $e_{\mathfrak{A}} = e(E|F)$ by proposition 3.4.3(3).
2. By construction $\Lambda|_{F^\times}$ is a multiple of $\chi|_{F^\times}$, in both the odd case (definition 4.2.4) and the even case (corollary 4.2.6); the centre of G is $F^\times$ and acts on c-$\mathrm{Ind}_J^G \Lambda$ through that character.
3. Both G and J are unimodular, J being open and compact modulo the centre, so the Duality Theorem (theorem 1.3.12) gives $\check{\pi}_\chi \cong \mathrm{Ind}_J^G \check{\Lambda}$. The inducing datum attached to $\check{\chi}$ is $\check{\Lambda}$, so c-$\mathrm{Ind}_J^G \check{\Lambda} = \pi_{\check{\chi}}$, and the canonical embedding c-$\mathrm{Ind}_J^G \check{\Lambda} \to \mathrm{Ind}_J^G \check{\Lambda}$ is a non-zero map between two irreducible representations: the source is irreducible by theorem 3.2.8 and the target by proposition 1.2.38, since π_χ is irreducible. A non-zero map of irreducibles is an isomorphism.
4. This is proposition 4.2.8(2) for minimal pairs, and the definition of π_χ by twisting in Step 1 for the rest, completing the proof.

Lemma 4.3.2: Unramified Recognition

A cuspidal representation π is unramified (in $\mathcal{A}_2^{\mathrm{nr}}(F)$) if and only if the chain order $\mathfrak{A}$ in its cuspidal type is isomorphic to $M_2(\mathfrak{o})$.

***Proof*:**
By proposition 3.3.2 there are exactly two conjugacy classes of chain order in $M_2(F)$, distinguished by the period: $e_{\mathfrak{A}} = 1$ gives $\mathfrak{A} \cong M_2(\mathfrak{o})$ and $e_{\mathfrak{A}} = 2$ gives the Iwahori order. By proposition 3.4.3(3) the period of the order occurring in a cuspidal type equals the ramification index $e(E|F)$ of the field $E = F[\alpha]$ generated by the stratum. So $\mathfrak{A} \cong M_2(\mathfrak{o})$ if and only if E/F is unramified.

It remains to see that this is what "unramified" means for π. The set $\mathcal{A}_2^{\mathrm{nr}}(F)$ consists of the twists $\varphi\pi_0$ of representations π_0 of level zero. Twisting by a character of $F^\times$ changes neither the chain order nor the field: by proposition 4.2.8(2) it replaces the pair $(E/F, \chi)$ by $(E/F, \chi\varphi_E)$, leaving E untouched. A level-zero representation has E/F unramified by lemma 4.2.1, so every member of $\mathcal{A}_2^{\mathrm{nr}}(F)$ has $\mathfrak{A} \cong M_2(\mathfrak{o})$. Conversely, if $\mathfrak{A} \cong M_2(\mathfrak{o})$ then E/F is unramified, and twisting by a character of $F^\times$ chosen to cancel the restriction of χ to U_E^1 reduces the level of the pair to zero, exhibiting π as a twist of a level-zero representation, as we sought to show.

4.4 Tame Intertwining Properties

We prove the injectivity of the map constructed above.

Proposition 4.4.1: Injectivity of Parametrization

The map $(E/F, \chi) \mapsto \pi_\chi$ is injective on isomorphism classes of minimal pairs.

***Proof*:**
Suppose $\pi_{\chi_1} \cong \pi_{\chi_2}$. Central characters imply $\chi_1|_{F^\times} = \chi_2|_{F^\times}$. The levels must match. The representations contain simple strata $(\mathfrak{A}, n, \alpha_i)$. These strata must intertwine, meaning the characters ψ_{α_i} are conjugate.

Lemma 4.4.2: Conjugacy of Minimal Elements

Let α_1, α_2 be minimal elements generating extensions E_1, E_2. If their characters intertwine, there exists $u \in U_{\mathfrak{A}}$ such that $uE_1u^{-1} = E_2$ and $u\alpha_1u^{-1} \equiv \alpha_2 \pmod{U_{E_2}^1}$.

***Proof*:**
(of the Lemma.) By proposition 3.3.6 the hypothesis says there is $g \in G$ with

$$g^{-1}\big(\alpha_1 + \mathfrak{P}^{1-n}\big)g \cap \big(\alpha_2 + \mathfrak{P}^{1-n}\big) \neq \emptyset$$

so after replacing α_1 by $g^{-1}\alpha_1 g$, which changes neither minimality nor the field up to isomorphism, the two cosets $\alpha_1 + \mathfrak{P}^{1-n}$ and $\alpha_2 + \mathfrak{P}^{1-n}$ coincide.

Both strata are simple, so by proposition 3.4.3 each α_i is minimal over F and $E_i^\times \subset \mathcal{K}_{\mathfrak{A}}$. Minimality means the coset $\alpha_i + \mathfrak{P}^{1-n}$ contains no element of a proper subfield, so the coset determines the characteristic polynomial of α_i modulo $\mathfrak{p}$, and hence determines $E_i = F[\alpha_i]$ up to F-isomorphism. Two equal cosets therefore give $E_1 \cong E_2$, and since all embeddings of a quadratic extension into $M_2(F)$ normalizing a fixed chain order are conjugate under $U_{\mathfrak{A}}$ (this is the same statement used in proposition 4.2.8(1)), there is $u \in U_{\mathfrak{A}}$ with $uE_1u^{-1} = E_2$.

Finally, $u\alpha_1u^{-1}$ and α_2 generate the same field E_2 and lie in the same coset modulo $\mathfrak{P}^{1-n}$, hence agree modulo $\mathfrak{p}_{E_2}^{1-n}$, which for a unit of valuation $-n$ is the statement $u\alpha_1u^{-1} \equiv \alpha_2 \pmod{U_{E_2}^1}$, as we sought to show.

Using this lemma, we align the fields $E_1 \cong E_2$. The equality of representations implies equality of the inducing types Λ_i. For odd level, $\Lambda = \chi$, so $\chi_1 = \chi_2$. For even level, the trace relation on roots of unity recovers the character on the full group $E^\times$.

Proposition 4.4.3: Minimality Converse

Let $\pi \in \mathcal{A}_2^0(F)$ (or $\mathcal{A}_2^{\mathrm{nr}}(F)$ if $p = 2$). Suppose $\ell(\pi) \leq \ell(\varphi\pi)$ for all φ. Then there exists a minimal pair $(E/F, \chi)$ such that $\pi \cong \pi_\chi$.

***Proof*:**
π contains a cuspidal type $(\mathfrak{A}, J, \Lambda)$. If $\mathfrak{A} = \mathcal{I}$, we are in the ramified case (odd level), and $\chi = \Lambda|_{E^\times}$ works. If $\mathfrak{A} = M_2(\mathfrak{o})$ and level > 0, we are in the even level case. We reconstruct χ from $\Lambda|_{J^1}$ (which gives the restriction to units) and the character formula on roots of unity (which gives the rest).

4.5 A Certain Group Extension

We fix a prime number p. Let G be a finite p-group with cyclic centre Z, such that $V = G/Z$ is an elementary abelian p-group. Let θ be a faithful character of Z. The alternating pairing

$$h : V \times V \longrightarrow \mathbb{C}^\times, \quad (x, y) \longmapsto \theta([x, y]),$$

is non-degenerate. By the Stone-von Neumann theorem, there is a unique irreducible representation η of G such that $\eta|_Z$ contains θ.

We introduce a group of automorphisms. Let A be a finite cyclic group of automorphisms of G, of order relatively prime to p. Assume A acts trivially on Z. Since η is the unique representation with central character θ, and A fixes θ, the representation η is stable under A. Thus, η admits an extension to an irreducible representation of the semi-direct product $AG = A \ltimes G$.

Lemma 4.5.1: Extensions of Heisenberg Representations

Let η_0 be some irreducible representation of AG such that $\eta_0|_G \cong \eta$. Let χ range over the irreducible characters of $AG/G \cong A$. The representations $\chi \otimes \eta_0$ are distinct and

$$\mathrm{Ind}_G^{AG} \eta \cong \bigoplus_{\chi \in \widehat{A}} (\chi \otimes \eta_0)$$

***Proof*:**
The determinant of the representation $\chi \otimes \eta_0$ is $\chi^{\dim \eta} \det \eta_0$. Since $\dim \eta = |V|^{1/2}$ is a power of p and $|A|$ is relatively prime to p, the map $\chi \mapsto \chi^{\dim \eta}$ is a permutation of $\widehat{A}$. Thus the determinants are distinct, implying the representations are distinct. By Frobenius Reciprocity, $\mathrm{Ind}_G^{AG} \eta$ contains each $\chi \otimes \eta_0$. Comparing dimensions ($|A| \dim \eta$ vs $\sum 1 \cdot \dim \eta$), we see they exhaust the induced representation.

Specializing to the case relevant to GL_2 imposes the following hypotheses:

(a) $|V| = q^2$, where q is a power of p.

(b) $|A| = q + 1$.

(c) If $a \in A, a \neq 1$, then a acts on V with only the trivial fixed point (regular action).

Lemma 4.5.2: The Trace -1 Extension

Under hypotheses (a) to (c), there is a unique irreducible representation η_1 of AG such that:

1. $\eta_1|_G \cong \eta$.
2. $\mathrm{tr}\eta_1(a) = -1$ for all $a \in A, a \neq 1$.

***Proof*:**
Let η_i, $1 \leq i \leq q+1$, be the distinct extensions of η to AG. Consider the induced representation

$$\xi = \mathrm{Ind}_{AZ}^{AG}(1_A \otimes \theta),$$

where $1_A \otimes \theta$ is the character $(a, z) \mapsto \theta(z)$ of the abelian group $AZ \cong A \times Z$. We decompose ξ. Since $\xi|_G \cong \mathrm{Ind}_Z^G \theta \cong (\dim \eta)\eta = q\eta$, we must have

$$\xi \cong \bigoplus_{i=1}^{q+1} m_i \eta_i, \quad \text{with } \sum m_i = q$$

We compute the character of ξ restricted to AZ.

$$\xi|_{AZ} \cong \bigoplus_{g \in AZ \backslash AG / AZ} \mathrm{Ind}_{AZ \cap g^{-1}AZg}^{AZ}(1_A \otimes \theta)^g$$

The double coset space $AZ\backslash AG/AZ$ is isomorphic to the orbit space V/A. By hypothesis (c), A acts freely on $V \setminus \{0\}$. Thus we have one trivial orbit (size 1) and $q-1$ regular orbits (size $q+1$). For the trivial orbit ($g = 1$), we get $1_A \otimes \theta$. For non-trivial orbits, the intersection of groups is just Z, so we induce from Z to AZ, obtaining $\mathrm{Reg}_A \otimes \theta$.

$$\xi|_{AZ} \cong (1_A \otimes \theta) \oplus (q-1)(\mathrm{Reg}_A \otimes \theta)$$

Calculating the inner product $\langle \xi, \xi \rangle_{AG} = \langle \xi|_{AZ}, 1_A \otimes \theta \rangle_{AZ} = 1 + (q-1) \cdot 1 = q$. Since $\sum m_i = q$ and $\sum m_i^2 = q$, it must be that one m_i is 0 and the rest are 1. Renumber so $m_1 = 0$. Then $\xi = (\bigoplus \eta_i) - \eta_1$. Since $\bigoplus \eta_i \cong \mathrm{Ind}_G^{AG}\eta$, its trace on any $a \in A \setminus \{1\}$ is 0. Therefore, $\mathrm{tr}\xi(a) = -\mathrm{tr}\eta_1(a)$. From the explicit formula above, $\mathrm{tr}\xi(a) = 1 + (q-1)\mathrm{tr}(\mathrm{Reg}_A)(a) = 1 + 0 = 1$. Thus $\mathrm{tr}\eta_1(a) = -1$.

To confirm this applies in the present context, analyze the conjugacy classes.

Lemma 4.5.3: Conjugacy in AG

Let $a \in A, a \neq 1$.

1. Let $g \in G$; the element ag is G-conjugate to an element of the form az, $z \in Z$.
2. For $z \in Z$, the G-conjugacy class and the AG-conjugacy class of az are the same, and this class has q^2 elements.
3. For $z_1, z_2 \in Z$, the elements az_1, az_2 are AG-conjugate if and only if $z_1 = z_2$.

***Proof*:**
Consider the map $\varphi_a : V \to V$ given by $v \mapsto ava^{-1}v^{-1}$. Since A acts without fixed points on V, this map is injective, hence bijective. This implies every ag can be conjugated by G into aZ, proving (1). Since A is abelian, A centralizes az. Thus the AG-conjugacy class is the same as the G-conjugacy class. The size is $|G|/|C_G(az)|$. The centralizer condition $gag^{-1} = a$ modulo Z is equivalent to g being a fixed point of a in V, which implies $g \in Z$. Thus the centralizer is Z, and the class size is $|G|/|Z| = |V| = q^2$. Part (3) follows because the commutator of AG is contained in G, so $AG/G \cong A$ is abelian. Conjugacy preserves the projection to A.

The construction in section 4.2 satisfies these hypotheses. Here $n = 2m$.

- $G = J^1_\alpha / \ker \theta$.
- $Z = H^1_\alpha / \ker \theta$.
- θ is the character defined by ψ_α and χ.

Here $V = G/Z \cong J^1_\alpha / H^1_\alpha \cong \mathfrak{p}^m_F / (\mathfrak{p}^m_E + \mathfrak{p}^{m+1}_F)$. Identify $A/P \cong M_2(k)$. Since E/F is unramified, the image of $\mathfrak{o}_E$ is the field $k_E \subset M_2(k)$. Then $V \cong M_2(k)/k_E$. This has dimension 2 over k (size q^2). The group $A = \mu_E/\mu_F \cong k_E^\times / k^\times$ has order $(q^2-1)/(q-1) = q+1$. The action is by conjugation. A scalar $x \in k_E^\times$ fixes $y \in M_2(k)$ iff they commute. Since k_E is a maximal subfield, the centralizer is k_E. Thus the action on the quotient $M_2(k)/k_E$ is free. All hypotheses are satisfied.

5

Characterizing Representations using Functional Equations

We have classified representations of G, but to relate them to Galois groups, we need invariant objects that exist on both sides of the fence. In particular, we shall fuse the notion of characters from [2] with the functional theory of [10] to create new invariants. The goal of this chapter is to define these invariants, the local L-function $L(\pi, s)$ and the local constant (or ε-factor) $\varepsilon(\pi, s, \psi)$, and to prove that they completely characterize the representation.

We start in Section 5.1 with the thesis of John Tate, establishing the functional equation and ε-factors for $\mathrm{GL}_1(F)$ (characters). We then upgrade this to $\mathrm{GL}_2(F)$ in Section 5.2 using the Godement-Jacquet theory of Zeta integrals. We provide explicit formulas for the ε-factors of cuspidal representations using Gauss sums. The key result of this chapter is the *Converse Theorem* (Section 5.5), which asserts that if two representations have matching L and ε factors for all twists, they must be isomorphic. This theorem is the engine that makes the Langlands correspondence unique.

5.1 Functional Equation for GL(1)

Recall that if $a \in F$ and $\psi \in \widehat{F}$, then $a\psi$ denotes the character $x \mapsto \psi(ax)$. If $\psi \neq 1$, the map $a \mapsto a\psi$ gives an isomorphism $F \cong \widehat{F}$

Definition 5.1.1: Fourier Transform

Let $\psi \in \widehat{F}$, $\psi \neq 1$ to define an isomorphism $F \cong \hat{F}$, and let μ be a Haar measure on F. For $\Phi \in C_c^\infty(F)$, define the *Fourier transform* $\hat{\Phi}$ of Φ (with respect to to μ and ψ) to be:

$$\hat{\Phi}(x) = \int_F \Phi(y)\psi(xy)\,d\mu(y), \quad x \in F$$

The integrand is locally constant, and hence the integral reduces to a finite sum.

Proposition 5.1.2: Fourier Transform Properties

1. For $\Phi \in C_c^\infty(F)$, the function $\hat{\Phi}$ lies in $C_c^\infty(F)$.
2. There is a positive real number $c = c(\psi, \mu)$ such that
$$\hat{\hat{\Phi}}(x) = c\Phi(-x), \quad \Phi \in C_c^\infty(F),\ x \in F$$
3. For given ψ, there is a unique Haar measure μ_ψ for which $c(\psi, \mu_\psi) = 1$. This measure satisfies $\mu_\psi(\mathfrak{o}) = q^{l/2}$, where l is the level of ψ.
4. For $a \in F^\times$, $\mu_{a\psi} = \|a\|^{\frac{1}{2}}\mu_\psi$.

Notice how (1) is completely false in the real analysis case: By the uncertainly principle it is impossible for f and $\hat{f}$ to be compactly supported, but it *is* the case in p-adic analysis case.

***Proof*:**

1-2. Let l be the level of ψ so that $\psi(\mathfrak{p}^{-l}) = 1$ and $\psi(\mathfrak{p}^{-l-1}) \neq 1$, and let Φ_j be the characteristic function of $\mathfrak{p}^j$. For $a \in F$, the character $a\psi|_{\mathfrak{p}^j}$ is trivial if and only if $a \in \mathfrak{p}^{l-j}$. The support of $\hat{\Phi}_j$ is therefore $\mathfrak{p}^{l-j}$ and

$$\hat{\Phi}_j(x) = \mu(\mathfrak{o})q^{-j}, \quad x \in \mathfrak{p}^{l-j}$$

Therefore, (1) and (2) hold for all functions Φ_j, with $c = \mu(\mathfrak{o})^2 q^{-l}$. Let $\Phi \in C_c^\infty(F)$, $a \in F$, and let Ψ denote the function $x \mapsto \Phi(x-a)$. We then have

$$\hat{\Psi}(x) = \int_F \Phi(y-a)\psi(xy)\,d\mu(y) = a\psi(x)\hat{\Phi}(x)$$

The function $a\psi$ is locally constant, so $\hat{\Psi}$ lies in $C_c^\infty(F)$ provided $\hat{\Phi}$ does. Calculating the Fourier transform again, we get

$$\hat{\hat{\Psi}}(x) = \hat{\hat{\Phi}}(a+x)$$

If parts (1) and (2) of the Lemma hold for Φ, they also therefore hold for Ψ with the same value of c. However, they do hold for the functions Φ_j, and the functions $x \mapsto \Phi_j(x-a)$, $a \in F$, $j \in \mathbb{Z}$, span $C_c^\infty(F)$. The first two parts of the proposition therefore hold for all $\Phi \in C_c^\infty(F)$, with $c = \mu(\mathfrak{o})^2 q^{-l}$, as above.

3. For $b > 0$, we have $c(\psi, b\mu) = b^2 c(\psi, \mu)$; put another way, to achieve $c(\psi, \mu) = 1$, we must have $\mu(\mathfrak{o})^2 q^{-l} = 1$.

4. immediate.

The important concepts from the proof above are worth boxing:

Definition 5.1.3: Self-dual Haar measure

The measure μ_ψ is called *the self-dual Haar measure on F, relative to ψ*. Using μ_ψ to compute the Fourier transform:

$$\hat{\Phi}(x) = \int_F \Phi(y)\psi(xy)\, d\mu_\psi(y),$$

Definition 5.1.4: Fourier Inversion Formula

The 2nd point in proposition 5.1.2 is called the *Fourier inversion formula*

$$\hat{\hat{\Phi}}(x) = \Phi(-x), \quad \Phi \in C_c^\infty(F),\ x \in F \tag{5.1}$$

The multiplicative information is codified by the Mellin transform, i.e. the Gelfand Transform with $(\mathbb{R}_{>0}, \cdot)$ [9].

Definition 5.1.5: Multiplicative Haar Measure

Let μ be a Haar measure on the additive group F. Define a Haar measure μ^* on the multiplicative group $F^\times$ by

$$d\mu^*(x) = \frac{d\mu(x)}{\|x\|}$$

It is easy to show this measure is scaling-invariant. Hence, when integrating over F^*, it is invariant to units. To that end, let χ be a character of $F^\times$, let $\Phi \in C_c^\infty(F)$, and choose a prime element ϖ of F.

Definition 5.1.6: Mellin Transform

Let $\Phi \in C_c^\infty(M)$. Then it's *Mellin transform* (with respect to s) is a new function $\mathcal{M}(\Phi, -, s)$ that inputs multiplicative characters and outputs:

$$\mathcal{M}(\Phi, \chi, s) = \int_{F^\times} \Phi(x)\chi(x)\|x\|^s d\mu^*(x)$$

This simplifies considerably. Recall that the multiplicative group is topologically $F^\times \cong \mathbb{Z} \times U_F$, so the field decomposes into "shells" (or annuli) based on valuation. For $m \in \mathbb{Z}$, let e_m denote the characteristic function of the set $\varpi^m U_F = \mathfrak{p}^m \setminus \mathfrak{p}^{m+1}$.

For $\Phi \in C_c^\infty(F)$, the function $\Phi_m = e_m\Phi$ lies in $C_c^\infty(F^\times) \subset C_c^\infty(F)$. It is identically zero for $m \ll 0$, so one may set:

$$z_m = z_m(\Phi, \chi) = \int_{\varpi^m U_F} \Phi(x)\chi(x)\, d\mu^*(x), \quad m \in \mathbb{Z} \tag{5.2}$$

On the shell $\varpi^m U_F$, the norm $\|x\|$ is exactly q^{-m}. Therefore, $\|x\|^s = (q^{-m})^s = (q^{-s})^m$. Thus the value can be pulled out, giving:

$$\mathcal{M}(\Phi, \chi, s) = \sum_{m \in \mathbb{Z}} (q^{-s})^m \underbrace{\left(\int_{\varpi^m U_F} \Phi(x)\chi(x) d\mu^*(x) \right)}_{\text{This is } z_m}$$

The value z_m captures the correlation between the test function Φ and the character χ specifically on the shell where $v_F(x) = m$. These assemble into a generating function, and the q^{-s} vary (as it can depend on the choice of q) by choosing a formal variable X:

Definition 5.1.7: Local Zeta Integral (Formal Series)

Define a formal Laurent series $Z(\Phi, \chi, X) \in \mathbb{C}((X))$ by

$$\mathcal{Z} : C_c^\infty(F) \to \mathbb{C}((X))$$
$$Z(\Phi, \chi, X) = \sum_{m \in \mathbb{Z}} z_m X^m$$

Its image is denoted $\mathcal{Z}(\chi) = \mathcal{Z}(\chi, X) = \{Z(\Phi, \chi, X) : \Phi \in C_c^\infty(F)\}$.

For $a \in F^\times$ and $\Phi \in C_c^\infty(F)$, write $a \cdot \Phi$ for the function $x \mapsto \Phi(a^{-1}x)$. A simple calculation shows:

$$Z(a \cdot \Phi, \chi, X) = \chi(a) X^{v_F(a)} Z(\Phi, \chi, X)$$

Thus $\mathcal{Z}(\chi)$ is a module over the ring $\mathbb{C}[X, X^{-1}]$ of Laurent polynomials.

Recall that on the shell $\varpi^m U_F$, the norm is constant: $\|x\| = q^{-m}$. Formally substituting $X = q^{-s}$ into the series recovers the standard integral notation:

$$Z(\Phi, \chi, q^{-s}) = \sum_{m \in \mathbb{Z}} \int_{\varpi^m U_F} \Phi(x)\chi(x)(q^{-m})^s \, d\mu^*(x) = \int_{F^\times} \Phi(x)\chi(x)\|x\|^s \, d\mu^*(x)$$

The notations $\zeta(\Phi, \chi, s)$ and $Z(\Phi, \chi, q^{-s})$ are used interchangeably. The function z_m acts as the m-th Fourier coefficient of Φ with respect to the valuation dimension of the group $F^\times$.

Proposition 5.1.8: Rationality of Zeta Integrals

Let χ be a character of $F^\times$; then

$$\mathcal{Z}(\chi, X) = P_\chi(X)^{-1}\mathbb{C}[X, X^{-1}],$$

where

$$P_\chi(X) = \begin{cases} 1 - \chi(\varpi)X & \text{if } \chi \text{ is unramified,} \\ 1 & \text{otherwise.} \end{cases}$$

Hence $\mathcal{Z}(\chi)$ is a PID.

***Proof*:**
Suppose first that $\Phi(0) = 0$. Thus $\Phi|_{F^\times}$ lies in $C_c^\infty(F^\times)$ and $Z(\Phi, \chi, X)$ has only finitely many non-zero coefficients, so:

$$Z(\Phi, \chi, X) \in \mathbb{C}[X, X^{-1}]$$

If Φ is the indicator function of a sufficiently small neighbourhood of 1, then $Z(\Phi, \chi, X)$ is a positive constant, and so $1 \in \mathcal{Z}(\chi)$ and

$$\{Z(\Phi, \chi, X) : \Phi \in C_c^\infty(F^\times)\} = \mathbb{C}[X, X^{-1}]$$

Let Φ_0 be the characteristic function of $\mathfrak{o}$. Then:

$$Z(\Phi_0, \chi, X) = \sum_{m \geq 0} \chi(\varpi^m) X^m \int_{U_F} \chi(x)\, d\mu^*(x)$$

The inner integral is $\mu^*(U_F)$ if χ is unramified (trivial on U_F), and is zero otherwise (by orthogonality of characters on the compact group U_F). In other words:

$$\mu^*(U_F)^{-1} Z(\Phi_0, \chi, X) = \begin{cases} (1 - \chi(\varpi)X)^{-1} & \text{if } \chi \text{ is unramified,} \\ 0 & \text{otherwise.} \end{cases}$$

Since $C_c^\infty(F)$ is spanned by Φ_0 and $C_c^\infty(F^\times)$, the result follows.

The polynomial $P_\chi(X)^{-1}$ is given a name:

Definition 5.1.9: Local L-function

The (multiplicative) inverse of the polynomial $P_\chi(X)$ is called the *Local L-function*. It is traditionally is denoted $L(\chi, s)$ when $X = q^{-s}$ so that:

$$L(\chi, s) = \begin{cases} \frac{1}{1 - \chi(\varpi) q^{-s}} & \text{if } \chi \text{ is unramified,} \\ 1 & \text{otherwise.} \end{cases}$$

Corollary 5.1.10: Poles of Unramified L-functions

Let χ_1, χ_2 be unramified characters of $F^\times$. The following are equivalent:

1. The meromorphic functions $L(\chi_1, s)$ and $L(\chi_2, s)$ have a pole in common;
2. The meromorphic functions $L(\chi_1, s)$ and $L(\chi_2, s)$ have the same sets of poles;
3. $\chi_1 = \chi_2$.

***Proof*:**
The implication (3) $\Rightarrow$ (2) is trivial, and (2) $\Rightarrow$ (1) is immediate since the set of poles is non-empty.

We prove (1) $\Rightarrow$ (3). The poles of $L(\chi_i, s)$ are the zeros of the polynomial $1 - \chi_i(\varpi)q^{-s}$. Thus, s is a pole if and only if $q^s = \chi_i(\varpi)$.

Suppose the functions share a pole s_0. Then:

$$\chi_1(\varpi) = q^{s_0} = \chi_2(\varpi)$$

By definition, an unramified character is trivial on the units U_F. Since $F^\times \cong \mathbb{Z} \times U_F$ (generated by ϖ and U_F), an unramified character is completely determined by its value on the prime element ϖ. Therefore $\chi_1 = \chi_2$.

The main result of the local theory is next: the functional equation relating the additive and multiplicative structures. Fix a character $\psi \in \widehat{F}$, $\psi \neq 1$. For $\Phi \in C_c^\infty(F)$, let $\hat{\Phi}$ denote the Fourier transform relative to the self-dual measure μ_ψ.

Theorem 5.1.11: Local Functional Equation (GL(1))

Let χ be a character of $F^\times$. There exists a unique rational function $c(\chi, \psi, X) \in \mathbb{C}(X)$ such that

$$Z(\hat{\Phi}, \check{\chi}, q^{-1}X^{-1}) = c(\chi, \psi, X) Z(\Phi, \chi, X), \tag{5.3}$$

for all $\Phi \in C_c^\infty(F)$. Here $\check{\chi} = \chi^{-1}$.

This equation is often called *Tate's (local) functional equation*

***Proof*:**
Consider the space Λ of linear maps $\lambda : C_c^\infty(F) \to \mathbb{C}(X)$ satisfying the homogeneity condition:

$$\lambda(a \cdot \Phi) = \chi(a) X^{v_F(a)} \lambda(\Phi), \quad \Phi \in C_c^\infty(F), a \in F^\times$$

Then Λ is a vector space over $\mathbb{C}(X)$ and contains the map $\lambda_0 : \Phi \mapsto Z(\Phi, \chi, X)$ which is non-zero by proposition 5.1.8.

For $\Phi \in C_c^\infty(F)$ and $a \in F^\times$, a simple calculation using the properties of the Fourier transform yields:

$$\widehat{a \cdot \Phi}(y) = \|a\| a^{-1} \cdot \hat{\Phi}(y)$$

It follows that the map

$$\lambda_1 : \Phi \mapsto Z(\hat{\Phi}, \check{\chi}, q^{-1}X^{-1})$$

also lies in Λ. If Λ is one-dimensional, we can pick our desired constant.

Lemma 5.1.12: Uniqueness of Functionals

The space Λ has dimension one over $\mathbb{C}(X)$.

***Proof*:**
(of Lemma) Choose $n \geq 0$ such that $U_F^n \subset \operatorname{Ker}\chi$. Let Φ_k denote the characteristic function of U_F^k, $k \geq 1$. We consider the evaluation map $\Lambda \to \mathbb{C}(X)$ given by $\lambda \mapsto \lambda(\Phi_n)$. We show this is injective.

Suppose $\lambda(\Phi_n) = 0$. The defining property of Λ implies $\lambda(a \cdot \Phi_k) = \lambda(\Phi_k)$ for any $a \in U_F^n$ and $k \geq n$. Since the volume of the units scales by indices, we have $\lambda(\Phi_k) = q^{n-k}\lambda(\Phi_n) = 0$ for $k \geq n$.

Next, any $\Phi \in C_c^\infty(F^\times)$ is a finite linear combination of $F^\times$-translates of functions Φ_k for sufficiently large k. Thus, λ vanishes on $C_c^\infty(F^\times)$. For a general $\Phi \in C_c^\infty(F)$, the value $\lambda(\Phi)$ therefore depends only on $\Phi(0)$ (since $\Phi - \Phi(0)\Phi_{\text{small}}$ vanishes at 0). However, $\lambda(a \cdot \Phi) = \chi(a)X^{v(a)}\lambda(\Phi)$ forces $\lambda(\Phi) = 0$ unless λ is identically zero.

Since Λ is one-dimensional and $\lambda_0 \neq 0$, there exists a unique scalar $c(\chi, \psi, X) \in \mathbb{C}(X)$ such that $\lambda_1 = c(\chi, \psi, X)\lambda_0$.

The above is the notation Tate used. The following is the more traditional notation. Let s be a complex variable.

$$\begin{aligned}
\zeta(\Phi, \chi, s) &= Z(\Phi, \chi, q^{-s}), \\
L(\chi, s) &= P_\chi(q^{-s})^{-1}, \\
\gamma(\chi, s, \psi) &= c(\chi, \psi, q^{-s})
\end{aligned}$$

The functional equation then reads:

$$\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$$

This notation is used from now on.

The structure of $\gamma(\chi, s, \psi)$ deserves a closer look. It is given a name:

Definition 5.1.13: Epsilon Factor

The *epsilon factor* (or local constant) $\varepsilon(\chi, s, \psi)$ defined to be the function $\varepsilon(\chi, s, \psi) \in \mathbb{C}(q^{-s})$ satisfying the relation:

$$\gamma(\chi, s, \psi) = \varepsilon(\chi, s, \psi)\frac{L(\check{\chi}, 1-s)}{L(\chi, s)}$$

Theorem 5.1.11 also yields a functional equation for the epsilon factor.

Corollary 5.1.14: Functional Equation of the Local Constant

The function $\varepsilon(\chi, s, \psi)$ satisfies the functional equation

$$\varepsilon(\chi, s, \psi)\varepsilon(\check{\chi}, 1-s, \psi) = \chi(-1)$$

It is of the form

$$\varepsilon(\chi, s, \psi) = q^{(1/2-s)n(\chi,\psi)}\varepsilon(\chi, 1/2, \psi),$$

for some integer $n(\chi, \psi) \in \mathbb{Z}$.

***Proof*:**
The functional equation in theorem 5.1.11 reads

$$\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s) \tag{5.4}$$

Applying this twice, we get

$$\zeta(\hat{\hat{\Phi}}, \chi, s) = \gamma(\check{\chi}, 1-s, \psi)\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\check{\chi}, 1-s, \psi)\gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$$

Using the Fourier inversion formula $\hat{\hat{\Phi}}(x) = \Phi(-x)$, we see that

$$\zeta(\hat{\hat{\Phi}}, \chi, s) = \int_{F^\times} \Phi(-x)\chi(x)\|x\|^s \, d\mu^*(x) = \chi(-1)\zeta(\Phi, \chi, s)$$

Comparing the two expressions, we obtain

$$\gamma(\check{\chi}, 1-s, \psi)\gamma(\chi, s, \psi) = \varepsilon(\check{\chi}, 1-s, \psi)\varepsilon(\chi, s, \psi) = \chi(-1),$$

which proves the first relation.

For the second assertion, we can rewrite (5.4) in the form

$$\frac{\zeta(\hat{\Phi}, \check{\chi}, 1-s)}{L(\check{\chi}, 1-s)} = \varepsilon(\chi, s, \psi)\frac{\zeta(\Phi, \chi, s)}{L(\chi, s)}$$

The fraction on either side lies in $\mathbb{C}[q^s, q^{-s}]$ (by the definition of the L-function as the generator of the zeta ideal). As in proposition 5.1.8, we may choose Φ such that $\zeta(\Phi, \chi, s) = L(\chi, s)$. We deduce that $\varepsilon(\chi, s, \psi) \in \mathbb{C}[q^s, q^{-s}]$. Likewise, $\varepsilon(\check{\chi}, 1-s, \psi) \in \mathbb{C}[q^s, q^{-s}]$.

The functional equation implies that $\varepsilon(\chi, s, \psi)$ is an invertible element of the ring $\mathbb{C}[q^s, q^{-s}]$, and hence equal to aq^{ms}, for some $a \in \mathbb{C}^\times, m \in \mathbb{Z}$. This can be written in the required form, completing the proof.

5.1.1 Local Constants

L-functions characterize unramified characters. If the character is ramified, we loose all information:

Lemma 5.1.15: L-function of Ramified Character

If the character χ of $F^\times$ is ramified, then $L(\chi, s) = 1$.

***Proof*:**
By definition, $L(\chi, s)$ is the generator of the ideal $\mathcal{Z}(\chi)$ of all zeta integrals. To prove the generator is 1, it suffices to exhibit a test function $\Phi \in C_c^\infty(F)$ such that the integral $Z(\Phi, \chi, s)$ is a non-zero constant.

Since χ is smooth but ramified, there exists an integer $n \geq 1$ such that χ is non-trivial on U_F but trivial on the subgroup $U_F^n = 1 + \mathfrak{p}^n$. Let Φ be the characteristic function of this set U_F^n. The support of Φ is contained in the units, so $v_F(x) = 0$ (and $\|x\|^s = 1$) for all x in the support.

Computing the integral:

$$Z(\Phi, \chi, s) = \int_{F^\times} \Phi(x)\chi(x)\|x\|^s \, d\mu^*(x) = \int_{U_F^n} 1 \cdot \chi(x) \cdot 1 \, d\mu^*(x)$$

Since χ is trivial on U_F^n, this reduces to:

$$Z(\Phi, \chi, s) = \int_{U_F^n} d\mu^*(x) = \mu^*(U_F^n)$$

This is a non-zero constant (independent of s). The ideal $\mathcal{Z}(\chi)$ therefore contains a unit of the ring $\mathbb{C}[q^s, q^{-s}]$, which implies that the generator $L(\chi, s)$ is 1.

More information is therefore needed to characterize irreducible representations using functionals. Theorem 5.5.2 will show the ε-factors are exactly the required information. It will be essential to have a table of values for the local constant $\varepsilon(\chi, s, \psi)$. Observe first that the dependence on the additive character ψ is transparent.

Lemma 5.1.16: Dependence on Additive Character

Let $a \in F^\times$; then

$$\varepsilon(\chi, s, a\psi) = \chi(a)\|a\|^{s-1/2}\varepsilon(\chi, s, \psi)$$

***Proof*:**

The L-functions $L(\chi, s)$ and $L(\check{\chi}, 1-s)$ depend only on the character χ and not on the additive character ψ. Therefore, the ratio between $\varepsilon(\chi, s, a\psi)$ and $\varepsilon(\chi, s, \psi)$ is the same as the ratio between the factors $\gamma(\chi, s, a\psi)$ and $\gamma(\chi, s, \psi)$. We calculate the latter using the functional equation.

Let $\psi' = a\psi$. From proposition 5.1.2, the self-dual measure scales as $\mu_{\psi'} = \|a\|^{1/2}\mu_\psi$. The Fourier transform relative to ψ' is given by:

$$\begin{aligned}
\hat{\Phi}_{\psi'}(x) &= \int_F \Phi(y)\psi(axy) \, d\mu_{\psi'}(y) \\
&= \|a\|^{1/2} \int_F \Phi(y)\psi(axy) \, d\mu_\psi(y) \\
&= \|a\|^{1/2}\hat{\Phi}_\psi(ax)
\end{aligned}$$

We substitute this into the zeta integral on the left side of the functional equation:

$$\zeta(\hat{\Phi}_{\psi'}, \check{\chi}, 1-s) = \int_{F^\times} \|a\|^{1/2}\hat{\Phi}_\psi(ax)\check{\chi}(x)\|x\|^{1-s} \, d\mu^*(x)$$

We perform the change of variables $y = ax$. Since μ^* is a Haar measure on $F^\times$, $d\mu^*(x) = d\mu^*(y)$. We observe that $x = a^{-1}y$, so $\check{\chi}(x) = \check{\chi}(a^{-1})\check{\chi}(y) = \chi(a)\check{\chi}(y)$ and $\|x\|^{1-s} = \|a\|^{s-1}\|y\|^{1-s}$. The integral becomes:

$$\begin{aligned}
\zeta(\hat{\Phi}_{\psi'}, \check{\chi}, 1-s) &= \|a\|^{1/2}\chi(a)\|a\|^{s-1} \int_{F^\times} \hat{\Phi}_\psi(y)\check{\chi}(y)\|y\|^{1-s} \, d\mu^*(y) \\
&= \chi(a)\|a\|^{s-1/2}\zeta(\hat{\Phi}_\psi, \check{\chi}, 1-s)
\end{aligned}$$

Applying the functional equation definition $\zeta(\hat{\Phi}, \check{\chi}, 1-s) = \gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$ to both sides, we obtain:

$$\gamma(\chi, s, a\psi)\zeta(\Phi, \chi, s) = \chi(a)\|a\|^{s-1/2}\gamma(\chi, s, \psi)\zeta(\Phi, \chi, s)$$

Since this holds for all test functions Φ, the result follows.

The lemma above reduces the computation of $\varepsilon(\chi, s, \psi)$ for characters ψ of level one. The self-dual Haar measure for such a ψ satisfies

$$\int_{\mathfrak{o}} dx = q^{1/2}$$

For the purposes of calculation, it will be convenient to choose the Haar measure d^*x on $F^\times$ such that

$$\int_{U_F} d^*x = 1$$

Proposition 5.1.17: Epsilon Factor for Unramified Characters

Suppose that χ is unramified and that ψ has level one. If ϖ is a prime element of F, then

$$\varepsilon(\chi, s, \psi) = q^{s-1/2}\chi(\varpi)^{-1}$$

***Proof*:**
If Φ denotes the characteristic function of $\mathfrak{o}$, we get $\zeta(\Phi, \chi, s) = L(\chi, s)$, as in the proof of proposition 5.1.8. On the other side, since ψ has level 1, the Fourier transform is supported on $\mathfrak{p}$ and given by $\hat{\Phi}(x) = q^{1/2}\Phi(\varpi^{-1}x)$, so

$$\begin{aligned}\zeta(\hat{\Phi}, \chi^{-1}, s) &= q^{1/2}\int_{F^\times} \Phi(\varpi^{-1}x)\chi(x)^{-1}\|x\|^s\, d^*x \\ &= q^{1/2-s}\chi(\varpi)^{-1}\int_{F^\times} \Phi(x)\chi(x)^{-1}\|x\|^s\, d^*x \\ &= q^{1/2-s}\chi(\varpi)^{-1}L(\chi^{-1}, s)\end{aligned}$$

It follows that

$$\varepsilon(\chi, s, \psi) = \frac{\zeta(\hat{\Phi}, \chi^{-1}, 1-s)}{L(\chi^{-1}, 1-s)} = q^{s-1/2}\chi(\varpi)^{-1},$$

as required.

Theorem 5.1.18: Epsilon Factor for Ramified Characters

Suppose that χ has level $n \geq 0$ and is *ramified*. Let $\psi \in \widehat{F}$ have level one. Then

$$\varepsilon(\chi, s, \psi) = q^{n(1/2-s)}\left(\sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1}\psi(\alpha x)\right) q^{-(n+1)/2},$$

for any $\alpha \in F^\times$ such that $v_F(\alpha) = -n$.

***Proof*:**
Let Φ be the characteristic function of U_F^{n+1}, so that

$$\zeta(\Phi, \chi, s) = \mu^*(U_F^{n+1})$$

The Fourier transform $\hat{\Phi}$ has support $\mathfrak{p}^{-n}$ and

$$\hat{\Phi}(y) = q^{1/2-n-1}\psi(y), \quad y \in \mathfrak{p}^{-n}$$

Thus

$$\begin{aligned}\zeta(\hat{\Phi}, \chi^{-1}, s) &= q^{1/2-n-1}\int_{\mathfrak{p}^{-n}\setminus\{0\}} \psi(y)\chi(y)^{-1}\|y\|^s\, d^*y \\ &= q^{1/2-n-1}\sum_{m\geq -n} z_m q^{-ms},\end{aligned}$$

where

$$z_m = \int_{\mathfrak{p}^m\setminus\mathfrak{p}^{m+1}} \psi(y)\chi(y)^{-1}\, d^*y$$

We know from corollary 5.1.14 that $\zeta(\hat{\Phi}, \chi^{-1}, s)$ must reduce to a monomial in q^s, so only one of the coefficients z_m is non-zero. The first one is

$$z_{-n} = \int_{U_F} \psi(\alpha u)\chi^{-1}(\alpha u)\, d^*u,$$

for any $\alpha \in F$ with $v_F(\alpha) = -n$. The integrand is constant on cosets of U_F^{n+1} and so the integral reduces to

$$\int_{U_F} \psi(\alpha u)\chi^{-1}(\alpha u)\, d^*u = \mu^*(U_F^{n+1}) \sum_{x\in U_F/U_F^{n+1}} \chi(\alpha x)^{-1}\psi(\alpha x)$$

If the coefficients $m \geq 1-n$, the proof is complete. To that end, take $\beta \in F$, with $v_F(\beta) = m > -n$, and consider the coefficient

$$z_m = \int_{U_F} \psi(\beta u)\chi(\beta u)^{-1}\, d^*u \tag{5.5}$$

Taking $v \in U_F^n$, we have

$$\int_{U_F} \psi(\beta u)\chi(\beta u)^{-1}\, d^*u = \int_{U_F} \psi(\beta uv)\chi(\beta uv)^{-1}\, d^*u$$

Setting $v = 1+y$, the first factor reduces to $\psi(\beta u)\psi(\beta uy) = \psi(\beta u)$, since $\beta uy \in \mathfrak{p}^{m+n} \subset \mathfrak{p} \subset \operatorname{Ker}\psi$ (as $m > -n \implies m+n \geq 1$). Thus

$$\int_{U_F} \psi(\beta u)\chi(\beta u)^{-1}\, d^*u = \chi(v)^{-1}\int_{U_F} \psi(\beta u)\chi(\beta u)^{-1}\, d^*u$$

We may certainly choose $v \in U_F^n$ so that $\chi(v) \neq 1$ (by the definition of level), so the integral (5.5) vanishes, completing the proof.

Consider the sum in parentheses, the "trigonometric sum":

$$\sum_{x \in U_F/U_F^{n+1}} \chi(\alpha x)^{-1}\psi(\alpha x)$$

where the names comes from the classical case the image of a character would lie in the unit circle. This type of sum mixing an additive and multiplicative character is given a name:

Definition 5.1.19: Gauss Sum

Let χ be a ramified character of $F^\times$ of level $n \geq 0$, and let $\psi \in \widehat{F}$ have level one. Let $c \in F$ satisfy $v_F(c) = -n$. The sum

$$\tau(\chi, \psi) = \sum_{x \in U_F/U_F^{n+1}} \check{\chi}(cx)\psi(cx) \tag{5.6}$$

is called the *Gauss sum* of χ relative to ψ.

Another common convention is to denote the Gauss sum as $\tau(\check{\chi}, \psi)$; the above version is slightly more convenient for the present purposes.

The definition is independent of the choice of coset representative x and element c. Re-writing the equation in theorem 5.1.18 gives:

$$\varepsilon(\chi, s, \psi) = q^{n(1/2-s)} \frac{\tau(\chi, \psi)}{q^{(n+1)/2}} \tag{5.7}$$

The functional equation for the local constant (corollary 5.1.14) gives:

$$\tau(\chi, \psi)\tau(\check{\chi}, \psi) = \chi(-1)q^{n+1} \tag{5.8}$$

The sums of characters can often be simplified (think of the proof of summing the powers of roots of unity). Indeed:

Proposition 5.1.20: Simplification of Gauss Sum

Suppose that χ has level $n \geq 1$. Let $c \in F$ satisfy

$$\chi(1+x) = \psi(cx), \quad x \in \mathfrak{p}^{[n/2]+1}$$

Then

$$\tau(\chi, \psi) = q^{[(n+1)/2]} \sum_y \check{\chi}(cy)\psi(cy), \tag{5.9}$$

where y ranges over $U_F^{[(n+1)/2]}/U_F^{[n/2]+1}$.

***Proof*:**

In the defining sum of equation (5.6), we write $x = y(1+z)$, with $y \in U_F/U_F^{[n/2]+1}$ and

$z \in \mathfrak{p}^{[n/2]+1}/\mathfrak{p}^{n+1}$. Then:

$$\check{\chi}(cy(1+z))\psi(cy(1+z)) = \check{\chi}(cy)\psi(cy)\psi(c(y-1)z)$$

Summing:

$$\tau(\chi,\psi) = \sum_y \check{\chi}(cy)\psi(cy) \sum_z \psi(c(y-1)z)$$

Now, the map $z \mapsto \psi(c(y-1)z)$ is a character of $\mathfrak{p}^{[n/2]+1}/\mathfrak{p}^{n+1}$ since $c(y-1) \in \mathfrak{p}^{-n}$. It is the trivial character if and only if $y \equiv 1 \pmod{\mathfrak{p}^{n-[n/2]}}$, i.e. if and only if $y \in U_F^{[(n+1)/2]}$. The inner sum therefore vanishes unless $y \in U_F^{[(n+1)/2]}$ in which case it takes the value $(\mathfrak{p}^{[n/2]+1} : \mathfrak{p}^{n+1}) = q^{[(n+1)/2]}$, as we sought to show.

The formula of in proposition 5.1.20 has an important application for its use in the proof of the Converse Theorem:

Theorem 5.1.21: Stability Theorem

Let θ, χ be characters of $F^\times$, of level $l \geq 0, n \geq 1$ respectively. Suppose that $2l < n$. Let $\psi \in \widehat{F}$, $\psi \neq 1$, and let $c \in F$ satisfy $\chi(1+x) = \psi(cx)$, $x \in \mathfrak{p}^{[n/2]+1}$. Then

$$\varepsilon(\theta\chi, s, \psi) = \theta(c)^{-1}\varepsilon(\chi, s, \psi)$$

***Proof*:**
By lemma 5.1.16, it is enough to treat the case where ψ has level one. The character $\theta\chi$ has level n, and agrees with χ on $U_F^{[(n+1)/2]}$. Applying proposition 5.1.20, we get

$$\tau(\theta\chi,\psi) = q^{[(n+1)/2]} \sum_y \check{\theta}\check{\chi}(cy)\psi(cy),$$

where the sum is taken over $y \in U_F^{[(n+1)/2]}/U_F^{[n/2]+1}$. Since θ is trivial on $U_F^{[(n+1)/2]}$, this reduces to $\theta(c)^{-1}\tau(\chi,\psi)$, as required.

5.2 Functional Equation for GL(2)

The transition from the theory for $\mathrm{GL}_1(F)$ (Tate's Thesis) to $\mathrm{GL}_2(F)$ (Godement-Jacquet theory) is not merely a change in dimension, but a structural upgrade of every component of the functional equation. The fundamental philosophy remains identical: an L-function is constructed by integrating a "test function" against a "representation" over the group. However, to accommodate the non-abelian nature of the group, scalars must be replaced by matrices, and characters by matrix coefficients.

The ambient space for the test functions shifts from the field F to the full matrix algebra $A = M_2(F)$. Just as the multiplicative group $F^\times$ is an open dense subset of F, the general linear group $G = \mathrm{GL}_2(F)$ is an open dense subset of the algebra A. Consequently, the test functions Φ are chosen from the space $C_c^\infty(A)$, the space of locally constant, compactly supported functions on the matrix ring. This space is spanned by characteristic functions of lattices, such as $a + \varpi^j M_2(\mathfrak{o})$.

This change in ambient space necessitates a corresponding change in the Fourier transform. Both theories rely on an additive Fourier transform on the underlying linear space. For GL_1, the duality on F was given by the product xy. For the matrix algebra $M_2(F)$, the natural non-degenerate bilinear form is given by the trace of the product. Let $\psi \in \widehat{F}$ be a non-trivial additive character. The character ψ_A of the additive group $A = M_2(F)$ is defined by $\psi_A(x) = \psi(\mathrm{tr}_A(x)) = \psi(x_{11} + x_{22})$. The Haar measure μ on A is normalized to be self-dual relative to ψ_A.

Definition 5.2.1: Matrix Fourier Transform

For a test function $\Phi \in C_c^\infty(A)$, the Fourier transform $\hat{\Phi}$ is defined by

$$\hat{\Phi}(x) = \int_A \Phi(y)\psi_A(xy)\, d\mu(y)$$

Just as in the one-dimensional case, the map $\Phi \mapsto \hat{\Phi}$ is an isomorphism of $C_c^\infty(A)$ onto itself, and the Fourier Inversion Formula holds: $\hat{\hat{\Phi}}(x) = \Phi(-x)$.

The most significant conceptual shift concerns the object being integrated. For GL_1, the representation is a character $\chi : F^\times \to \mathbb{C}^\times$. Since it is one-dimensional, the character itself is the function integrated against Φ. For GL_2, an irreducible representation (π, V) is generally infinite-dimensional. The operator $\pi(g)$ cannot be integrated directly to yield a scalar value. Instead, the theory utilizes *matrix coefficients*.

Definition 5.2.2: Matrix Coefficient

Let (π, V) be a smooth representation of G. For vectors $v \in V$ and $\tilde{v} \in \check{V}$ (the smooth dual), the function

$$f(g) = \gamma_{\tilde{v}\otimes v}(g) = \langle \tilde{v}, \pi(g)v \rangle$$

is called a *matrix coefficient* of π. Let $C(\pi)$ denote the vector space spanned by all such coefficients.

The space $C(\pi)$ carries a smooth representation of $G \times G$ via the action $(g_1, g_2) \cdot f(x) = f(g_1^{-1} x g_2)$. In the context of Zeta integrals, these coefficients replace the character $\chi(x)$. The Zeta integrals are then defined by integrating the test function against the matrix coefficient, weighted by a power of the determinant.

Definition 5.2.3: Zeta Integral for GL_2

Let $\Phi \in C_c^\infty(A)$, $f \in C(\pi)$, and $s \in \mathbb{C}$. The *Zeta integral* is defined as:

$$\zeta(\Phi, f, s) = \int_G \Phi(g) f(g) \|\det g\|^s \, d^\times g,$$

where $d^\times g$ is a Haar measure on G.

Convergence is the first thing to settle, and it is not automatic: G is not compact modulo the centre, so nothing yet prevents the integral from diverging for every s. The situation is better than it looks. Sorting G by the valuation of the determinant turns the zeta integral into a formal Laurent series whose coefficients are integrals over *compact* sets, and whose negative part is finite.

Throughout, write $X = q^{-s}$, so that $\|\det g\|^s = X^{v_F(\det g)}$, and set

$$G_m = \{g \in G : v_F(\det g) = m\} \qquad m \in \mathbb{Z}$$

Lemma 5.2.4: Zeta Integrals as Laurent Series

Let (π, V) be a smooth representation of G, let $\Phi \in C_c^\infty(A)$ and let $f \in C(\pi)$. For $m \in \mathbb{Z}$ put

$$c_m(\Phi, f) = \int_{G_m} \Phi(g) f(g)\, d^\times g$$

1. The set $\mathrm{supp}\Phi \cap G_m$ is a compact subset of G, and $c_m(\Phi, f)$ is an absolutely convergent integral.
2. There is an integer $N \geq 0$, depending only on $\mathrm{supp}\Phi$, such that $c_m(\Phi, f) = 0$ for all $m < -2N$.

Consequently $Z(\Phi, f, X) = \sum_{m\in\mathbb{Z}} c_m(\Phi, f) X^m$ is a well-defined element of $\mathbb{C}((X))$, and $\zeta(\Phi, f, s)$ is its value at $X = q^{-s}$ at any s for which the series converges.

***Proof*:**

1. The set $\varpi^m U_F$ is compact, hence closed, in F, and $\det\colon A \to F$ is a polynomial map, hence continuous. Therefore $G_m = \det^{-1}(\varpi^m U_F)$ is closed in A, and $\mathrm{supp}\Phi \cap G_m$ is a closed subset of the compact set $\mathrm{supp}\Phi$. It is therefore compact, and it lies in G, which carries the topology induced from A since G is open in A.

   Both Φ and f are locally constant, so Φf takes only finitely many values on this compact set. A Haar measure is finite on compact sets, so the integral converges absolutely.

2. The lattices $\varpi^{-N} M_2(\mathfrak{o})$, $N \geq 0$, form an increasing open cover of A, so compactness gives N with $\mathrm{supp}\Phi \subseteq \varpi^{-N} M_2(\mathfrak{o})$. For g in the support we then have $\varpi^N g \in M_2(\mathfrak{o})$, whence

   $$\varpi^{2N} \det g = \det(\varpi^N g) \in \mathfrak{o}$$

   and therefore $v_F(\det g) \geq -2N$, completing the proof.

The classification results already proved reduce the analysis of $Z(\Phi, f, X)$ to two cases, and the two are different in character. A cuspidal representation has coefficients that vanish outside a set compact modulo the centre (theorem 3.1.3), so only the central direction is unbounded, and that direction is governed by Tate's theory applied to the central character. Every other irreducible representation is built from a character of the torus (theorem 2.4.16), and unfolding along the Iwasawa decomposition reduces it to a *pair* of Tate integrals. To pass from the induced representations to their subquotients, the following is needed.

Lemma 5.2.5: Coefficients of a Subquotient

Let (Σ, X) be an admissible smooth representation of G and let π be a subquotient of Σ. Then $C(\pi) \subseteq C(\Sigma)$.

***Proof*:**
It suffices to treat a subrepresentation and a quotient separately, since a subquotient is a quotient of a subrepresentation.

Quotients. Let $\pi = X/Y$ for a G-subspace Y. A smooth functional on X/Y is precisely a smooth functional on X vanishing on Y, so there is a canonical injection $\iota\colon (X/Y)^\vee \to \check{X}$. Given $\tilde{u} \in (X/Y)^\vee$ and $\bar{v} \in X/Y$, lift $\bar{v}$ to $v \in X$; then

$$\langle \tilde{u}, \pi(g)\bar{v}\rangle = \langle \tilde{u}, \overline{\Sigma(g)v}\rangle = \langle \iota(\tilde{u}), \Sigma(g)v\rangle$$

which exhibits the coefficient as one of Σ.

Subrepresentations. Let $\pi = (\Sigma|_Y, Y)$ for a G-subspace $Y \subseteq X$, and let $v \in Y$, $\tilde{v} \in \check{Y}$. Choose a compact open subgroup K fixing both, so $v \in Y^K$ and $\tilde{v} \in \check{Y}^K$. By corollary 1.2.18 applied to Y and to X,

$$Y = Y^K \oplus Y(K), \qquad X = X^K \oplus X(K)$$

The first of these intersections is immediate. For the second, $Y(K) \subseteq Y \cap X(K)$ by definition; conversely, if $y \in Y \cap X(K)$, write $y = y_1 + y_2$ with $y_1 \in Y^K$ and $y_2 \in Y(K)$, so that $y_1 = y - y_2$ lies in $X(K)$ as well as in X^K, forcing $y_1 = 0$ and $y = y_2 \in Y(K)$.

The space $Y(K)$ is spanned by the vectors $y - \pi(k)y$, and $\tilde{v}$ is K-fixed, so $\tilde{v}(y - \pi(k)y) = 0$: thus $\tilde{v}$ annihilates $Y(K)$ and is determined by its restriction to Y^K. Since Σ is admissible, X^K is finite dimensional; extend $\tilde{v}|_{Y^K}$ to a linear functional λ on X^K and let $\tilde{u} \in \check{X}$ be λ composed with the projection $X \to X^K$ along $X(K)$. Then $\tilde{u}$ is K-fixed, hence smooth, and $\tilde{u}|_Y = \tilde{v}$ because the two agree on Y^K and both annihilate $Y(K)$. Therefore

$$\langle \tilde{v}, \pi(g)v\rangle = \langle \tilde{u}, \Sigma(g)v\rangle$$

for every $g \in G$, as we sought to show.

One point of normalization has to be settled before the L-function is defined, because getting it wrong makes the functional equation below fail to be symmetric. For GL_1 the zeta integral is attached to s directly; for GL_n the object with the clean $s \leftrightarrow 1-s$ symmetry is the integral shifted by $(n-1)/2$, which for $n = 2$ is a shift by $1/2$. The L-function is therefore defined from the shifted family, and this is what makes $\zeta(\Phi, f, s+\frac{1}{2})$, rather than $\zeta(\Phi, f, s)$, the expression appearing throughout the rest of this section.

Proposition 5.2.6: Rationality of Zeta Integrals

Let (π, V) be an irreducible smooth representation of G.

1. The integrals $\zeta(\Phi, f, s)$ represent rational functions of q^{-s}.
2. The $\mathbb{C}[q^s, q^{-s}]$-module generated by the *shifted* integrals $\zeta(\Phi, f, s+\frac{1}{2})$, for $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$, is a principal fractional ideal generated by a function of the form $P(q^{-s})^{-1}$, where $P(t) \in \mathbb{C}[t]$ and $P(0) = 1$.

This generator is defined to be the *L-function* $L(\pi, s)$.

***Proof*:**
Throughout, π is irreducible, hence admissible by corollary 3.1.5 and equipped with a central character ω_π by corollary 1.2.31.

1. By lemma 5.2.4 the expression $Z(\Phi, f, X)$ is a Laurent series with finitely many negative terms, so the content of the claim is that it is the expansion of a rational function. The argument splits according to whether π is cuspidal, and by theorem 3.1.3 together with theorem 2.4.16 these two cases exhaust the irreducible representations of G. In each case a denominator is produced that depends on π alone and not on Φ or f; this uniformity is what part (2) will consume.

   The cuspidal case. By theorem 3.1.3 the representation π is γ-cuspidal, so every $f \in C(\pi)$ is compactly supported modulo Z. Both G (proposition 2.2.7) and $Z \cong F^\times$ are unimodular, so G/Z carries an invariant measure; fix Haar measures with $d^\times g = d^\times a\, d\dot{g}$. Since $\det(ag) = a^2 \det g$ and $f(ag) = \omega_\pi(a) f(g)$ for $a \in F^\times$ identified with $aI \in Z$,

   $$Z(\Phi, f, X) = \int_{G/Z} \left(\int_{F^\times} \Phi(a\dot{g})\omega_\pi(a) X^{2v_F(a)}\, d^\times a \right) f(\dot{g}) X^{v_F(\det \dot{g})}\, d\dot{g}$$

   Fix $\dot{g} \in G$ and set $\Phi_{\dot{g}}(a) = \Phi(a\dot{g})$ for $a \in F$. The map $a \mapsto a\dot{g}$ is an F-linear injection onto the line $F\dot{g}$, which is closed in A, and it is a homeomorphism onto its image; so $\Phi_{\dot{g}}$ is locally constant with compact support, that is $\Phi_{\dot{g}} \in C_c^\infty(F)$. The inner integral is therefore exactly the Tate zeta integral of definition 5.1.7 attached to $\Phi_{\dot{g}}$, the character ω_π and the variable X^2. By proposition 5.1.8 it converges for $\mathrm{Re}(s)$ large and lies in

   $$P_{\omega_\pi}(X^2)^{-1}\, \mathbb{C}[X^2, X^{-2}]$$

   where $P_{\omega_\pi}(t)^{-1} = L(\omega_\pi, s)$ as in definition 5.1.9.

   For the outer integral, f vanishes off a compact subset of G/Z. There is a compact open subgroup K' of K under which Φ is invariant under right translation: pick N_0 and M with $\mathrm{supp}\Phi \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$ and Φ constant on cosets of $\varpi^M M_2(\mathfrak{o})$, and take $K' = K \cap (1 + \varpi^{M+N_0} M_2(\mathfrak{o}))$. For $y \in \varpi^{-N_0} M_2(\mathfrak{o})$ and $k \in K'$ one has $yk - y = y(k-1) \in \varpi^M M_2(\mathfrak{o})$, so $\Phi(yk) = \Phi(y)$; and for y outside that lattice both values vanish, since k is a unit of $M_2(\mathfrak{o})$ and so preserves it. Shrinking K' further so that it also fixes f, the integrand is constant on the cosets of K' meeting the compact support, of which there are finitely many. The outer integral is thus a finite linear combination, and

   $$Z(\Phi, f, X) \in P_{\omega_\pi}(X^2)^{-1}\, \mathbb{C}[X, X^{-1}]$$

   a rational function of X with denominator depending only on ω_π.

   The non-cuspidal case. By theorem 2.4.16 the representation π is either an irreducible $\iota_B^G \chi$, or $\varphi \circ \det$, or $\varphi \cdot \mathrm{St}_G$; in the latter two cases the exact sequence (2.7) exhibits π as a subquotient of $\mathrm{Ind}_B^G \varphi_T$. In every case π is a subquotient of $\Sigma = \mathrm{Ind}_B^G \chi$ for some character $\chi = \chi_1 \otimes \chi_2$ of T, and Σ is admissible. By lemma 5.2.5 it is enough to prove the claim for $f \in C(\Sigma)$.

   Realize Σ on the space X_χ of smooth $\varphi \colon G \to \mathbb{C}$ with $\varphi(bg) = \chi(b)\varphi(g)$. Since $B \backslash G$ is compact (corollary 2.2.2), the Duality Theorem (theorem 1.3.12) gives $\check{X}_\chi \cong \mathrm{Ind}_B^G \delta_B^{-1} \check{\chi}$ with the pairing $\langle \tilde{\varphi}, \varphi \rangle = \int_K \tilde{\varphi}(k)\varphi(k)\, d\mu_K(k)$, so every $f \in C(\Sigma)$ has the form

   $$f(g) = \int_K \tilde{\varphi}(k)\varphi(kg)\, d\mu_K(k)$$

Substituting this and replacing g by $k^{-1}g$, which leaves $d^\times g$ invariant and does not change $v_F(\det g)$ because $\det k \in U_F$, gives

$$Z(\Phi, f, X) = \int_K \tilde{\varphi}(k) \left(\int_G \Phi^k(g)\varphi(g)X^{v_F(\det g)}\, d^\times g \right) d\mu_K(k), \qquad \Phi^k(g) = \Phi(k^{-1}g)$$

and $\Phi^k \in C_c^\infty(A)$. Apply the Iwasawa decomposition $G = BK$ (proposition 2.2.1) and the associated integration formula (corollary 2.2.9). The resulting integration formula (2.4) reads

$$\int_G F(g)\, d\mu_G(g) = \int_K \int_B F(bk')\, d\mu_B(b)\, d\mu_K(k')$$

for a suitable left Haar measure μ_B on B; the normalizing constant relating μ_G to $d^\times g$ is a positive scalar and affects nothing below, since it does not change the $\mathbb{C}[X, X^{-1}]$-module generated.

Writing $b = tn$ with $t = \operatorname{diag}(t_1, t_2) \in T$ and $n = n(x) \in N$, a left Haar measure on $B = T \ltimes N$ is $d\mu_B(tn) = d^\times t\, d\mu_N(n)$: left translation by $t_0 n_0$ sends (t, n) to $(t_0 t,\ (t^{-1} n_0 t)n)$, which preserves $d^\times t$ in the outer variable and is a translation of N for each fixed t. Since χ is trivial on N and $\det(tn) = t_1 t_2$, the inner integral becomes

$$\int_K \varphi(k') \left(\int_T \int_N \Phi^k(t\, n(x)\, k')\chi_1(t_1)\chi_2(t_2)X^{v_F(t_1)+v_F(t_2)}\, d\mu_N(x)\, d^\times t \right) d\mu_K(k')$$

Fix k, k' and put $\Phi'(y) = \Phi(k^{-1}yk')$, again in $C_c^\infty(A)$. In the inner double integral substitute $y = t_1 x$, so that $d\mu_N(x) = \|t_1\|^{-1} d\mu_N(y)$ and

$$\int_N \Phi' \begin{pmatrix} t_1 & t_1 x \\ 0 & t_2 \end{pmatrix} d\mu_N(x) = \|t_1\|^{-1} \widetilde{\Psi}(t_1, t_2), \qquad \widetilde{\Psi}(t_1, t_2) = \int_F \Phi' \begin{pmatrix} t_1 & y \\ 0 & t_2 \end{pmatrix} d\mu_N(y)$$

The change of variable is what removes the apparent divergence: the function $\widetilde{\Psi}$ lies in $C_c^\infty(F \times F)$. Indeed, choose N_0 with $\operatorname{supp}\Phi' \subseteq \varpi^{-N_0} M_2(\mathfrak{o})$; then the integrand vanishes unless $v_F(t_1), v_F(t_2), v_F(y) \geq -N_0$, so the y-integration runs over the *fixed* compact set $\varpi^{-N_0}\mathfrak{o}$ and $\widetilde{\Psi}$ is supported in $\varpi^{-N_0}\mathfrak{o} \times \varpi^{-N_0}\mathfrak{o}$. Local constancy is inherited from Φ': if Φ' is invariant under translation by $\varpi^M M_2(\mathfrak{o})$ and $t_i \equiv t_i' \bmod \varpi^M \mathfrak{o}$, the two matrices differ by an element of that lattice.

The inner double integral is therefore

$$\int_{F^\times} \int_{F^\times} \widetilde{\Psi}(t_1, t_2)\, \chi_1(t_1)\|t_1\|^{-1}\chi_2(t_2)\, X^{v_F(t_1)+v_F(t_2)}\, d^\times t_1\, d^\times t_2$$

Being locally constant with compact support, $\widetilde{\Psi}$ is constant on the cosets of $\varpi^m\mathfrak{o} \times \varpi^m\mathfrak{o}$ for some m and vanishes outside finitely many of them; the argument of proposition 2.2.6, run on $F \times F$, therefore writes it as a finite sum $\sum_j \widetilde{\Psi}_j^{(1)} \otimes \widetilde{\Psi}_j^{(2)}$ of products of elements of $C_c^\infty(F)$, so the double integral is the finite sum

$$\sum_j Z\big(\widetilde{\Psi}_j^{(1)}, \chi_1\| - \|^{-1}, X\big)\, Z\big(\widetilde{\Psi}_j^{(2)}, \chi_2, X\big)$$

of products of Tate zeta integrals. By proposition 5.1.8 each summand lies in

$$P_{\chi_1\|-\|^{-1}}(X)^{-1} P_{\chi_2}(X)^{-1}\, \mathbb{C}[X, X^{-1}]$$

All the integrands above are bounded by their absolute values in the same way, and the Tate integrals converge absolutely for $\mathrm{Re}(s)$ large, so Fubini's theorem justifies the interchanges on that half plane; both sides are then rational, so the identity persists. Finally the k and k' integrations run over the compact group K with locally constant integrands, hence contribute finite linear combinations, and $Z(\Phi, f, X)$ lies in the displayed fractional ideal, whose denominator depends only on χ.

2. The shift is harmless and can be transported away at the start. Substituting $X = q^{-s}$, the shifted integral $\zeta(\Phi, f, s + \frac{1}{2})$ is the unshifted series $Z(\Phi, f, -)$ evaluated at $q^{-1/2}X$. The map $X \mapsto q^{-1/2}X$ is a $\mathbb{C}$-algebra automorphism of $\mathbb{C}[X, X^{-1}]$ and of $\mathbb{C}(X)$, so it carries the module generated by the unshifted integrals isomorphically onto the module generated by the shifted ones, carries generators to generators, and fixes constant terms, hence preserves the normalization $P(0) = 1$. It therefore suffices to prove the assertion for the unshifted family, which is what follows.

   Let $\mathcal{Z}(\pi)$ denote the $\mathbb{C}[X, X^{-1}]$-submodule of $\mathbb{C}(X)$ generated by the $Z(\Phi, f, X)$, for $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$. It is non-zero: choose $f \in C(\pi)$ with $f(1) \neq 0$ and let Φ be the characteristic function of a small enough compact open neighbourhood of 1 in A on which f is constant and which meets only G_0, so that $Z(\Phi, f, X)$ is a non-zero constant.

   By part (1) there is a single polynomial $D \in \mathbb{C}[X]$, depending only on π, with $\mathcal{Z}(\pi) \subseteq D(X)^{-1}\mathbb{C}[X, X^{-1}]$. The ring $\mathbb{C}[X, X^{-1}]$ is a localization of $\mathbb{C}[X]$, hence a principal ideal domain, and $D(X)^{-1}\mathbb{C}[X, X^{-1}]$ is a free module of rank one over it. A submodule of a free rank-one module over a principal ideal domain is again free of rank at most one, so $\mathcal{Z}(\pi)$ is generated by a single non-zero element $F(X) \in \mathbb{C}(X)$.

   It remains to normalize the generator. The units of $\mathbb{C}[X, X^{-1}]$ are exactly the monomials cX^k with $c \in \mathbb{C}^\times$, and multiplying a generator by a unit again gives a generator. Let $c_0 \in \mathbb{C}^\times$ be the constant found above. Since $c_0 \in \mathcal{Z}(\pi) = F\,\mathbb{C}[X, X^{-1}]$, there is $u \in \mathbb{C}[X, X^{-1}]$ with $c_0 = Fu$, so $F = c_0 u^{-1}$. Write $u = X^k \tilde{u}$ with $\tilde{u} \in \mathbb{C}[X]$ and $\tilde{u}(0) \neq 0$; then

$$F = c_0 X^{-k}\, \tilde{u}(X)^{-1}$$

   and since $c_0 X^{-k}$ is a unit, $\tilde{u}(X)^{-1}$ generates $\mathcal{Z}(\pi)$ as well. Put $P = \tilde{u}/\tilde{u}(0)$, which lies in $\mathbb{C}[X]$ and has $P(0) = 1$; it differs from $\tilde{u}$ by the unit $\tilde{u}(0)$, so $P(X)^{-1}$ is again a generator.

   This P is unique. If $P(X)^{-1}$ and $Q(X)^{-1}$ both generate, with $P, Q \in \mathbb{C}[X]$ and $P(0) = Q(0) = 1$, then Q/P is a unit cX^k of $\mathbb{C}[X, X^{-1}]$; both P and Q have non-zero constant term, which forces $k = 0$, and evaluating at $X = 0$ gives $c = 1$. Hence $P = Q$, and $L(\pi, s) = P(q^{-s})^{-1}$ is well defined, completing the proof.

The proof records two facts that the next sections use directly. For π cuspidal the denominator involves only the central character, and for π a subquotient of $\mathrm{Ind}_B^G(\chi_1 \otimes \chi_2)$ it is built from the two GL_1 denominators attached to $\chi_1\|-\|^{-1}$ and χ_2, each read in the shifted variable. The sharper statements, that $L(\pi, s) = 1$ in the first case and that the L-factor is exactly a product of two GL_1 factors in the second, are proved in section 5.3 and section 5.4 respectively.

The functional equation for GL_2 involves a subtle normalization shift compared to GL_1. In Tate's theory, the equation relates s to $1 - s$. For GL_2, the center of symmetry is shifted. Let $\check{\pi}$ denote the contragredient representation. If $f(g) = \langle \tilde{v}, \pi(g)v \rangle$ is a coefficient of π, then $\check{f}(g) = f(g^{-1}) = \langle v, \check{\pi}(g)\tilde{v} \rangle$ is a coefficient of $\check{\pi}$.

Theorem 5.2.7: Functional Equation (Godement-Jacquet)

Let (π, V) be an irreducible smooth representation. There exists a rational function $\gamma(\pi, s, \psi)$ such that:

$$\zeta\left(\hat{\Phi}, \check{f}, \frac{3}{2} - s\right) = \gamma(\pi, s, \psi)\zeta\left(\Phi, f, s + \frac{1}{2}\right) \tag{5.10}$$

for all $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$.

The shift of $+1/2$ on the right and the evaluation at $3/2 - s$ on the left (which is $2 - (s + 1/2)$) ensures that the resulting ε-factor satisfies the standard symmetry $s \leftrightarrow 1 - s$.

The proof runs on the same principle as Tate's (theorem 5.1.11): exhibit both sides of the equation as elements of one space of equivariant functionals, and show that space is a line. What plays the role of $F^\times$ acting on F is the group $G \times G$ acting on the matrix algebra A by $(g, h)\colon x \mapsto g^{-1}xh$, and the relevant functionals take values not in $\mathbb{C}$ but in the field $\mathbb{C}(X)$, carrying the action

$$(g, h) \cdot P(X) = X^{v_F(\det(gh^{-1}))}P(X)$$

Write $\mathbb{C}(X)_\Theta$ for $\mathbb{C}(X)$ with this action, and set

$$\mathcal{B}(\pi) = \mathrm{Hom}_{G\times G}\big(C_c^\infty(A) \otimes C(\pi),\ \mathbb{C}(X)_\Theta\big)$$

where $G \times G$ acts on $C_c^\infty(A)$ as above and on $C(\pi)$ by $(g, h)f\colon x \mapsto f(g^{-1}xh)$.

Lemma 5.2.8: Uniqueness of Invariant Bilinear Forms

Let (π, V) be an irreducible smooth representation of G. Then $\dim_{\mathbb{C}(X)} \mathcal{B}(\pi) \leq 1$.

***Proof Key ideas*:**
The argument is the one place in this chapter that reaches beyond the machinery developed here, so the structure is given and the technical input is cited. Stratify A by matrix rank. The rank-two locus is G itself, open in A, and the complement $A \setminus G$ is closed and consists of the rank-one matrices together with 0; extension by zero and restriction give a short exact sequence of smooth $G \times G$-representations

$$0 \to C_c^\infty(G) \to C_c^\infty(A) \to C_c^\infty(A \setminus G) \to 0$$

Applying $\mathrm{Hom}_{G\times G}(- \otimes C(\pi), \mathbb{C}(X)_\Theta)$ turns this into a left exact sequence, so the claim reduces to two statements: that the open stratum contributes at most a line, and that the closed stratum contributes nothing.

For the open stratum, $C_c^\infty(G) \otimes C(\pi)$ has a one-dimensional space of such functionals because $C(\pi) \cong \check{V} \otimes V$ is irreducible over $G \times G$ for irreducible admissible π, exactly as established for cuspidal π in the proof of proposition 3.1.2, and an equivariant functional on $C_c^\infty(G) \otimes C(\pi)$ is determined by integration against a single matrix coefficient.

For the closed stratum, the rank-one matrices form a single $G \times G$-orbit, since over a field every rank-one 2×2 matrix is $g\, e_{11}\, h$ for suitable $g, h \in G$. Hence $C_c^\infty(A \setminus G)$ is built from a representation compactly induced from the stabilizer of e_{11}, and Frobenius reciprocity for compact induction converts the Hom space into one over that stabilizer. The resulting condition forces X to satisfy a non-trivial polynomial equation, which cannot hold identically in $\mathbb{C}(X)$; so the contribution vanishes.

The step this book does not supply is the last one: Frobenius reciprocity for *compact* induction is deferred in section 1.2 and is what converts the orbit description into a computable Hom space. See [1, §24-§25] for the argument in full.

Granting the lemma, the functional equation is a computation with the two equivariances.

***Proof*:**
(of theorem 5.2.7)

Step 1: the zeta integral is equivariant. Let $\Phi \in C_c^\infty(A)$, $f \in C(\pi)$ and $(g,h) \in G \times G$. Substituting $y = gzh^{-1}$, which preserves $d^\times z$ because G is unimodular (proposition 2.2.7), and using $v_F(\det y) = v_F(\det z) + v_F(\det(gh^{-1}))$,

$$\begin{aligned} Z\big((g,h)\Phi,(g,h)f,X\big) &= \int_G \Phi(g^{-1}yh)f(g^{-1}yh)X^{v_F(\det y)}\,d^\times y \\ &= X^{v_F(\det(gh^{-1}))}\int_G \Phi(z)f(z)X^{v_F(\det z)}\,d^\times z \end{aligned}$$

so $Z(-,-,X)$ lies in $\mathcal{B}(\pi)$. It is non-zero by the argument in part (2) of proposition 5.2.6.

Step 2: the transformed integral has the same equivariance. The map $z \mapsto gzh^{-1}$ is F-linear on the four-dimensional space A with determinant $\det(g)^2\det(h)^{-2}$, so $d\mu(y) = \|\det(gh^{-1})\|^2 d\mu(z)$. Together with $\operatorname{tr}(x\,gzh^{-1}) = \operatorname{tr}\big((h^{-1}xg)z\big)$ this gives

$$\widehat{(g,h)\Phi} = \|\det(gh^{-1})\|^2\,(h,g)\hat{\Phi}$$

and directly from $\check{f}(x) = f(x^{-1})$,

$$\big((g,h)f\big)^\vee(x) = f\big((h^{-1}xg)^{-1}\big) = \big((h,g)\check{f}\big)(x)$$

Note that both operations *swap* the two factors. Define $W(\Phi,f,X) = Z(\hat{\Phi},\check{f},q^{-2}X^{-1})$ and put $m = v_F(\det(gh^{-1}))$. Applying Step 1 to the pair (h,g), whose determinant valuation is $-m$,

$$W\big((g,h)\Phi,(g,h)f,X\big) = q^{-2m}\left(q^{-2}X^{-1}\right)^{-m}W(\Phi,f,X) = q^{-2m}q^{2m}X^m\,W(\Phi,f,X)$$

The two powers of q cancel exactly, leaving the factor X^m. Thus W lies in $\mathcal{B}(\pi)$ as well: the swap introduced by the Fourier transform is undone by the reflection $X \mapsto q^{-2}X^{-1}$, and the normalization of $\hat{\Phi}$ is what makes the cancellation exact.

Step 3: proportionality. By lemma 5.2.8 the space $\mathcal{B}(\pi)$ has dimension at most one over $\mathbb{C}(X)$, and Step 1 shows it is non-zero. Hence $\mathcal{B}(\pi)$ is a line and W is a $\mathbb{C}(X)$-multiple of Z: there is $\gamma \in \mathbb{C}(X)$ with

$$Z\big(\hat{\Phi},\check{f},q^{-2}X^{-1}\big) = \gamma(X)\,Z(\Phi,f,X)$$

for all Φ and f. Writing $X = q^{-s}$ turns $q^{-2}X^{-1}$ into $q^{-(2-s)}$, so this reads $\zeta(\hat{\Phi},\check{f},2-s) = \gamma\,\zeta(\Phi,f,s)$. Replacing s by $s+1/2$, so that $2-s$ becomes $3/2-s$, gives (5.10) with $\gamma(\pi,s,\psi) = \gamma(q^{-s-1/2})$, completing the proof.

Corollary 5.2.9: Epsilon Factor for GL_2

Define the local constant $\varepsilon(\pi, s, \psi)$ by:

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)\frac{L(\check{\pi}, 1-s)}{L(\pi, s)}$$

Then $\varepsilon(\pi, s, \psi)$ is a monomial of the form cq^{-fs} with $c \in \mathbb{C}^\times$ and $f \in \mathbb{Z}$, and satisfies

$$\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$$

***Proof*:**
The argument is the GL_2 form of corollary 5.1.14. Apply the functional equation twice. Fourier inversion gives $\hat{\hat{\Phi}}(x) = \Phi(-x)$, and $\check{f}^\vee = f$, so

$$\zeta(\hat{\hat{\Phi}}, f, s) = \int_G \Phi(-g)f(g)\|\det g\|^s\, d^\times g = \omega_\pi(-1)\zeta(\Phi, f, s)$$

where the substitution $g \mapsto -g$ leaves $\|\det g\|^s$ unchanged because $\det(-g) = (-1)^2 \det g$ in rank two, and $f(-g) = f((-I)g) = \omega_\pi(-1)f(g)$. Comparing with two applications of (5.10) gives

$$\gamma(\pi, s, \psi)\gamma(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$$

and substituting the defining relation for ε on each factor, the two L-ratios cancel, leaving the stated identity for ε.

For the shape of ε, rewrite the functional equation as

$$\frac{\zeta(\hat{\Phi}, \check{f}, \frac{3}{2}-s)}{L(\check{\pi}, 1-s)} = \varepsilon(\pi, s, \psi)\frac{\zeta(\Phi, f, s+\frac{1}{2})}{L(\pi, s)}$$

By proposition 5.2.6 the L-function is a generator of the module of zeta integrals, so both fractions lie in $\mathbb{C}[q^s, q^{-s}]$, and there are choices of Φ and f making the right-hand numerator equal to $L(\pi, s)$. Hence $\varepsilon(\pi, s, \psi) \in \mathbb{C}[q^s, q^{-s}]$, and the same argument applied to $\check{\pi}$ puts $\varepsilon(\check{\pi}, 1-s, \psi)$ there too. Their product is the non-zero constant $\omega_\pi(-1)$, so $\varepsilon(\pi, s, \psi)$ is a unit of $\mathbb{C}[q^s, q^{-s}]$. The units of that ring are exactly the monomials, giving $\varepsilon(\pi, s, \psi) = cq^{-fs}$ with $c \in \mathbb{C}^\times$ and $f \in \mathbb{Z}$, as we sought to show.

Summary

Concept	Tate (GL(1))	Godement-Jacquet (GL(2))
Ambient Space	F	$M_2(F)$
Representation	$\chi(x)$	$f(g) = \langle\tilde{v}, \pi(g)v\rangle$
Dual Rep	χ^{-1}	$\check{f}(g) = f(g^{-1})$
Duality Pairing	xy	$\mathrm{tr}(xy)$
Norm Factor	$\|x\|^s$	$\|\det g\|^s$
Symmetry	$s \longleftrightarrow 1-s$	$s+\frac{1}{2} \longleftrightarrow \frac{3}{2}-s$

5.3 Epsilon Factors of Cuspidal Representations

The invariants built in section 5.2 are now computed on the representations constructed in chapter 3 and chapter 4. One fact governs the whole computation: the L-function of a cuspidal representation is trivial. Its zeta integrals have no poles at all, the functional equation loses its denominators, and everything a cuspidal representation contributes to this chapter is concentrated in a single monomial, the ε-factor.

The cuspidal case of the proof of proposition 5.2.6 already isolates the only place a pole can come from. A cuspidal coefficient vanishes off a set compact modulo Z, so the zeta integral becomes an integral over G/Z whose inner factor is a Tate zeta integral for the central character ω_π in the variable X^2. By proposition 5.1.8 such an integral has a pole only when ω_π is unramified, and then only through the value of the test function at 0. What is left is to decide whether that pole occurs. It does not, and the mechanism is orthogonality: the residue is an integral of a matrix coefficient against an unramified character of the determinant, and such an integral is a G-homomorphism from π to a one-dimensional representation.

Proposition 5.3.1: Triviality of the L-function of a Cuspidal Representation

Let (π, V) be an irreducible cuspidal representation of G. Then

$$L(\pi, s) = 1$$

***Proof*:**
Throughout, π is irreducible, hence admissible by corollary 3.1.5 and equipped with a central character ω_π by corollary 1.2.31; being cuspidal it is γ-cuspidal by theorem 3.1.3, so every $f \in C(\pi)$ is compactly supported modulo Z. Fix $\Phi \in C_c^\infty(A)$ and $f \in C(\pi)$, and normalize the measures as in the cuspidal case of the proof of proposition 5.2.6, so that $d^\times g = d^\times a\, d\dot{g}$ with $d\dot{g}$ the invariant measure on G/Z.

Step 1: the zeta integral is a finite sum of Tate integrals. Put

$$H(g) = \left(\int_{F^\times} \Phi(ag)\, \omega_\pi(a)\, X^{2v_F(a)}\, d^\times a \right) f(g)\, X^{v_F(\det g)}$$

Neither factor is invariant under $g \mapsto bg$ for $b \in F^\times$, but their product is. Indeed, substituting $a \mapsto ab^{-1}$ multiplies the bracketed factor by $\omega_\pi(b)^{-1} X^{-2v_F(b)}$, while $f(bg) = \omega_\pi(b) f(g)$ and $\det(bg) = b^2 \det g$ multiply the second factor by $\omega_\pi(b) X^{2v_F(b)}$. Thus H descends to a function on G/Z and

$$Z(\Phi, f, X) = \int_{G/Z} H(\dot{g})\, d\dot{g}$$

Writing $\Phi_{\dot{g}}(a) = \Phi(a\dot{g})$, the proof of proposition 5.2.6 shows $\Phi_{\dot{g}} \in C_c^\infty(F)$, and the bracketed factor is the Tate zeta integral $Z(\Phi_{\dot{g}}, \omega_\pi, X^2)$ of definition 5.1.7.

Choose, as in that proof, a compact open subgroup K' of K under which Φ is invariant on the right and which fixes f. Then H is constant on the cosets $\dot{g}K'$: the function $\Phi_{\dot{g}}$ is unchanged, f is unchanged, and $\det u' \in U_F$ for $u' \in K'$. Since f vanishes off a set compact modulo Z, the image of $\operatorname{supp} f$ in G/Z meets only finitely many such cosets, say those of $\dot{g}_1, \ldots, \dot{g}_r$, of measures

$c_1, \ldots, c_r > 0$. Hence

$$Z(\Phi, f, X) = \sum_{i=1}^{r} c_i \, Z(\Phi_{\dot{g}_i}, \omega_\pi, X^2) \, f(\dot{g}_i) \, X^{v_F(\det \dot{g}_i)}$$

Step 2: only one denominator is possible. Let $\Phi_{\mathfrak{o}}$ denote the characteristic function of $\mathfrak{o}$. For $\Psi \in C_c^\infty(F)$ the function $\Psi - \Psi(0)\Phi_{\mathfrak{o}}$ vanishes at 0, so by the first paragraph of the proof of proposition 5.1.8 its zeta integral lies in $\mathbb{C}[Y, Y^{-1}]$; and the computation in the same proof gives

$$Z(\Phi_{\mathfrak{o}}, \omega_\pi, Y) = \begin{cases} \kappa \, (1 - \omega_\pi(\varpi) Y)^{-1} & \text{if } \omega_\pi \text{ is unramified,} \\ 0 & \text{otherwise} \end{cases}$$

where $\kappa > 0$ is the volume of U_F in the measure used, a constant independent of Ψ. Consequently $Z(\Psi, \omega_\pi, Y) - \Psi(0) Z(\Phi_{\mathfrak{o}}, \omega_\pi, Y) \in \mathbb{C}[Y, Y^{-1}]$. Apply this with $\Psi = \Phi_{\dot{g}_i}$ and note that $\Phi_{\dot{g}_i}(0) = \Phi(0)$ for every i, since $0 \cdot \dot{g}_i = 0$.

If ω_π is ramified, every summand in Step 1 is a Laurent polynomial in X and Step 4 may be entered at once. Suppose then that ω_π is unramified. Substituting the decomposition into Step 1,

$$Z(\Phi, f, X) = \frac{\kappa \, \Phi(0)}{1 - \omega_\pi(\varpi) X^2} \, D(X) + R(X), \qquad D(X) = \sum_{i=1}^{r} c_i f(\dot{g}_i) X^{v_F(\det \dot{g}_i)}$$

with $R \in \mathbb{C}[X, X^{-1}]$.

Step 3: the numerator vanishes at the poles. Let $X_0 \in \mathbb{C}^\times$ satisfy $\omega_\pi(\varpi) X_0^2 = 1$ and let χ be the unramified character of $F^\times$ with $\chi(\varpi) = X_0$; this determines χ, since $F^\times = \varpi^{\mathbb{Z}} \times U_F$ and an unramified character is trivial on U_F. For $b \in F^\times$,

$$f(bg) \, \chi(\det(bg)) = \omega_\pi(b) \chi(b)^2 f(g) \chi(\det g) = \left(\omega_\pi(\varpi) X_0^2\right)^{v_F(b)} f(g) \chi(\det g) = f(g) \chi(\det g)$$

so $g \mapsto f(g)\chi(\det g)$ descends to G/Z. It is locally constant, vanishes off a set compact modulo Z, and is constant on the cosets $\dot{g}_i K'$ because χ is unramified. Therefore

$$D(X_0) = \int_{G/Z} f(\dot{g}) \, \chi(\det \dot{g}) \, d\dot{g}$$

Fix $\tilde{v} \in \check{V}$ and define $\ell \colon V \to \mathbb{C}$ by

$$\ell(v) = \int_{G/Z} \langle \tilde{v}, \pi(\dot{g}) v \rangle \, \chi(\det \dot{g}) \, d\dot{g}$$

The integrand descends to G/Z by the calculation just made, the integral converges absolutely because π is γ-cuspidal, and ℓ is linear in v. Since $d\dot{g}$ is invariant under right translation by G, the substitution $\dot{g} \mapsto \dot{g} h^{-1}$ gives

$$\ell(\pi(h) v) = \int_{G/Z} \langle \tilde{v}, \pi(\dot{g} h) v \rangle \chi(\det \dot{g}) \, d\dot{g} = \chi(\det h)^{-1} \ell(v)$$

for every $h \in G$, so ℓ lies in $\mathrm{Hom}_G(\pi, \chi^{-1} \circ \det)$, the target being the one-dimensional smooth representation $g \mapsto \chi(\det g)^{-1}$ of G. Suppose $\ell \neq 0$. Then $\ker \ell$ is a G-subspace of V different from V, hence zero by irreducibility, and $\pi \cong \chi^{-1} \circ \det$. But N lies in the kernel of det, so N acts trivially on $\chi^{-1} \circ \det$ and the Jacquet module of $\chi^{-1} \circ \det$ is $\chi^{-1} \circ \det$ itself, which is non-zero. (Since χ is unramified one may argue instead from proposition 3.6.1: the representation $\chi^{-1} \circ \det$ is trivial on K, so it has non-zero Iwahori-fixed vectors, and no such representation is cuspidal.) This contradicts the cuspidality of π, so $\ell = 0$.

Every $f \in C(\pi)$ is a finite linear combination of coefficients $\gamma_{\tilde{v}\otimes v}$, and $f \mapsto \int_{G/Z} f(\dot{g})\chi(\det \dot{g})\, d\dot{g}$ is linear, so $D(X_0) = 0$.

Step 4: conclusion. The polynomial $1 - \omega_\pi(\varpi)X^2$ has two distinct non-zero roots, and Step 3 applies to each. Writing $D(X) = X^{-M}\widetilde{D}(X)$ with $\widetilde{D} \in \mathbb{C}[X]$, the polynomial $\widetilde{D}$ vanishes at both roots and is therefore divisible by the quadratic; hence $D(X) = (1 - \omega_\pi(\varpi)X^2)E(X)$ with $E \in \mathbb{C}[X, X^{-1}]$ and

$$Z(\Phi, f, X) = \kappa\, \Phi(0)E(X) + R(X) \in \mathbb{C}[X, X^{-1}]$$

The same conclusion holds when ω_π is ramified, by Step 2.

Thus $\mathcal{Z}(\pi) \subseteq \mathbb{C}[X, X^{-1}]$. By part (2) of proposition 5.2.6 the module $\mathcal{Z}(\pi)$ is generated by $P(X)^{-1}$ for a unique $P \in \mathbb{C}[X]$ with $P(0) = 1$; in particular $P(X)^{-1}$ lies in $\mathbb{C}[X, X^{-1}]$, so P is a unit of $\mathbb{C}[X, X^{-1}]$ and therefore a monomial cX^n. A polynomial of that shape with $P(0) = 1$ is $P = 1$, so $L(\pi, s) = P(q^{-s})^{-1} = 1$, as we sought to show.

Compactness of the support modulo Z controls nothing along the line $F\dot{g}$: for a of large valuation the matrix $a\dot{g}$ tends to 0 in A, where Φ need not vanish. The tail is a geometric series, not a finite sum, and it contributes exactly the factor $L(\omega_\pi, 2s)$; what kills it is orthogonality against the one-dimensional representations $\chi^{-1} \circ \det$, not any statement about supports.

The L-function is therefore blind to cuspidal representations. It returns the same value, 1, on every one of them, and cannot separate one cuspidal representation from another, nor a cuspidal representation from any of its twists. Whatever arithmetic a cuspidal representation carries has to be read off the ε-factor instead, which is what the rest of this section computes.

With the L-factors gone, the functional equation simplifies. The ratio $L(\check{\pi}, 1 - s)/L(\pi, s)$ that separates γ from ε in corollary 5.2.9 becomes a ratio of two 1's, provided the contragredient of a cuspidal representation is again cuspidal.

Corollary 5.3.2: Gamma and Epsilon Agree for Cuspidal Representations

Let (π, V) be an irreducible cuspidal representation of G. Then $\check{\pi}$ is irreducible and cuspidal, $L(\pi, s) = L(\check{\pi}, s) = 1$,

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)$$

and

$$\varepsilon(\pi, s, \psi)\, \varepsilon(\check{\pi}, 1 - s, \psi) = \omega_\pi(-1)$$

***Proof*:**
The representation π is admissible by corollary 3.1.5, so by proposition 1.2.38 the smooth dual $\check{\pi}$ is admissible, irreducible, and canonically identified with the smooth dual of its own smooth dual. Write e_{K_0} for the projector of V onto V^{K_0} attached to a compact open subgroup K_0, a finite average because π is smooth; it satisfies $\langle \tilde{v}, e_{K_0}v\rangle = \langle e_{K_0}\tilde{v}, v\rangle$, both sides being the average

of $\langle \tilde{v}, \pi(x)v \rangle$ over $x \in K_0$.

A coefficient of $\check{\pi}$ is a finite linear combination of functions $g \mapsto \langle w, \check{\pi}(g)\tilde{v} \rangle$ with $\tilde{v} \in \check{V}$ and w in the smooth dual of $\check{\pi}$. Writing w as the image of $v \in V$ under the isomorphism just established, such a function is $\langle \tilde{v}, \pi(g^{-1})v \rangle = \check{f}(g)$, where $f = \gamma_{\tilde{v} \otimes v}$; this is the identity recorded in section 5.2, read in the reverse direction. Hence $C(\check{\pi}) = \left\{ \check{f} : f \in C(\pi) \right\}$.

Inversion is a homeomorphism of G carrying Z-stable sets to Z-stable sets, so $\check{f}$ is compactly supported modulo Z whenever f is. By theorem 3.1.3 the representation π is γ-cuspidal, so every coefficient of $\check{\pi}$ is compactly supported modulo Z; thus $\check{\pi}$ is γ-cuspidal, and cuspidal by the same theorem.

Proposition 5.3.1 now applies to π and to $\check{\pi}$, giving $L(\pi, s) = L(\check{\pi}, s) = 1$ and hence $L(\check{\pi}, 1-s) = 1$. The defining relation of corollary 5.2.9 reads

$$\gamma(\pi, s, \psi) = \varepsilon(\pi, s, \psi)\frac{L(\check{\pi}, 1-s)}{L(\pi, s)} = \varepsilon(\pi, s, \psi)$$

and the last identity of the statement is the one already proved in corollary 5.2.9, completing the proof.

The distinction between γ and ε was introduced to strip poles out of the functional equation; for a cuspidal representation there were none to strip. Every statement below can therefore be read with either symbol, and the functional equation of theorem 5.2.7 says, for cuspidal π, that a single non-zero complex monomial converts one Laurent polynomial into another.

That monomial still depends on a choice, namely the additive character ψ used to build the Fourier transform of definition 5.2.1. At GL_1 that dependence was transparent (lemma 5.1.16), and transparency is what allowed the explicit formulas of section 5.1 to be stated for a single normalization of ψ. The same holds here, with the determinant taking over the role that the identity map of F played before.

Lemma 5.3.3: Dependence of the Local Constant on the Additive Character

Let (π, V) be an irreducible smooth representation of G and let $a \in F^\times$. Then

$$\varepsilon(\pi, s, a\psi) = \omega_\pi(a)\, \|a\|^{2(s-1/2)}\, \varepsilon(\pi, s, \psi)$$

***Proof*:**
The L-functions $L(\pi, s)$ and $L(\check{\pi}, 1-s)$ do not involve ψ, so the ratio of the two ε-factors equals the ratio of the two γ-factors, and it suffices to compute the latter. Write $\psi' = a\psi$, so that $\psi'_A(x) = \psi(a\mathrm{tr}x) = \psi_A(ax)$.

Step 1: the self-dual measure on A scales by $\|a\|^2$. For a lattice L in A let $L^* = \{x \in A : \psi_A(xL) = 1\}$ be its dual and Φ_L its characteristic function. Orthogonality on the compact group L gives $\hat{\Phi}_L = \mu(L)\Phi_{L^*}$, and applying the transform again gives $\hat{\hat{\Phi}}_L = \mu(L)\mu(L^*)\Phi_L$, using $L^{**} = L$. The self-dual measure is therefore characterized by $\mu(L)\mu(L^*) = 1$ for every lattice L. Replacing ψ_A by ψ'_A replaces L^* by $a^{-1}L^*$, and A has dimension four over F, so $\mu_{\psi_A}(a^{-1}L^*) = \|a\|^{-4}\mu_{\psi_A}(L^*)$. Writing $\mu_{\psi'_A} = c\,\mu_{\psi_A}$ with $c > 0$ and imposing self-duality yields $c^2\|a\|^{-4} = 1$, that is $c = \|a\|^2$.

Step 2: the Fourier transform. Denoting by $\hat{\Phi}$ the transform relative to ψ and by $\hat{\Phi}^{\psi'}$ the transform relative to ψ',

$$\hat{\Phi}^{\psi'}(x) = \int_A \Phi(y)\psi'_A(xy)\, d\mu_{\psi'_A}(y) = \|a\|^2 \int_A \Phi(y)\psi_A\big((ax)y\big)\, d\mu_{\psi_A}(y) = \|a\|^2\, \hat{\Phi}(ax)$$

Step 3: change of variable in the zeta integral. Substituting $h = ag$, where a is identified with the scalar matrix $aI \in Z$, one has $\det g = a^{-2}\det h$, so $\|\det g\|^{3/2-s} = \|a\|^{2s-3}\|\det h\|^{3/2-s}$; also $\check{f}(g) = f(ah^{-1}) = \omega_\pi(a)\check{f}(h)$, and $d^\times g = d^\times h$ since $d^\times g$ is a Haar measure. Hence

$$\begin{aligned}
\zeta\big(\hat{\Phi}^{\psi'}, \check{f}, \tfrac{3}{2} - s\big) &= \|a\|^2 \int_G \hat{\Phi}(ag)\check{f}(g)\|\det g\|^{3/2-s}\, d^\times g \\
&= \|a\|^2\, \|a\|^{2s-3}\, \omega_\pi(a) \int_G \hat{\Phi}(h)\check{f}(h)\|\det h\|^{3/2-s}\, d^\times h \\
&= \omega_\pi(a)\, \|a\|^{2s-1}\, \zeta\big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\big)
\end{aligned}$$

The zeta integral $\zeta(\Phi, f, s + \frac{1}{2})$ standing on the right-hand side of (5.10) does not involve ψ, so comparing the functional equation for ψ' with the functional equation for ψ gives

$$\gamma(\pi, s, a\psi)\, \zeta\big(\Phi, f, s + \tfrac{1}{2}\big) = \omega_\pi(a)\|a\|^{2s-1}\gamma(\pi, s, \psi)\, \zeta\big(\Phi, f, s + \tfrac{1}{2}\big)$$

This holds for all Φ and all f, and the zeta integrals are not identically zero, so $\gamma(\pi, s, a\psi) = \omega_\pi(a)\|a\|^{2s-1}\gamma(\pi, s, \psi)$. Since $\|a\|^{2s-1} = \|a\|^{2(s-1/2)}$, this completes the proof.

Two features distinguish the formula from its GL_1 counterpart lemma 5.1.16, and both come from the same source. The scalar matrix aI has determinant a^2, so the change of variable $g \mapsto (aI)g$ moves $\|\det g\|^s$ by $\|a\|^{2s}$ rather than by $\|a\|^s$, which doubles the exponent; and it moves the coefficient f by the central character, which is why ω_π stands where χ stood before.

The formula is consistent with corollary 5.2.9. The contragredient has central character ω_π^{-1}, since $\check{\pi}(zI)\tilde{v} = \tilde{v} \circ \pi(z^{-1}I) = \omega_\pi(z)^{-1}\tilde{v}$, so replacing ψ by $a\psi$ multiplies the left-hand side of the identity $\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1 - s, \psi) = \omega_\pi(-1)$ by

$$\omega_\pi(a)\|a\|^{2(s-1/2)} \cdot \omega_\pi(a)^{-1}\|a\|^{2((1-s)-1/2)} = \|a\|^{2(s-1/2)+2(1/2-s)} = 1$$

The identity survives the change of additive character, as it must, since its right-hand side never mentions ψ. This cancellation forces the exponent to be a multiple of $s - \frac{1}{2}$; that the multiple is 2 rather than 1 is precisely the contribution of the determinant of a rank two matrix.

Corollary 5.2.9 says that $\varepsilon(\pi, s, \psi)$ is a monomial, and its two ingredients play different roles. The complex constant is rigid but hard to compute, while the exponent is a discrete invariant that can be read off the degree of the monomial. Fixing a normalization of ψ turns that exponent into an invariant of π alone.

Definition 5.3.4: Epsilon Exponent

Let (π, V) be an irreducible smooth representation of G and let $\psi \in \widehat{F}$ have level one. By corollary 5.2.9 there are unique $c \in \mathbb{C}^\times$ and $d \in \mathbb{Z}$ with $\varepsilon(\pi, s, \psi) = c\, q^{-ds}$. The integer

$$n(\pi, \psi) = d$$

is called the *epsilon exponent* of π relative to ψ, and equivalently

$$\varepsilon(\pi, s, \psi) = \varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)\, q^{(1/2-s)n(\pi,\psi)}$$

The second display is a rewriting of the first and uses nothing beyond the monomial form: evaluating $c\, q^{-ds}$ at $s = \frac{1}{2}$ gives $c = q^{d/2}\varepsilon(\pi, \frac{1}{2}, \psi)$, and substituting back produces the stated shape. The shape is therefore forced by corollary 5.2.9 rather than assumed, and uniqueness of c and d is immediate, since $cq^{-ds} = c'q^{-d's}$ for all s gives $d = d'$ and then $c = c'$.

Restricting to ψ of level one is what makes $n(\pi, \psi)$ well defined. A character $a\psi$ is trivial on $\mathfrak{p}^m$ exactly when ψ is trivial on $\mathfrak{p}^{m+v_F(a)}$, so the level of $a\psi$ is the level of ψ minus $v_F(a)$. The map $a \mapsto a\psi$ is a bijection of F onto $\widehat{F}$, so two characters of level one differ by an element of U_F, and by lemma 5.3.3 the two ε-factors differ by the constant $\omega_\pi(a)$ alone. The exponent is unchanged, so $n(\pi, \psi)$ does not depend on which character of level one was chosen.

The choice of normalization changes the answer. Against a character ψ' of level l, the same recipe returns the exponent $n(\pi, \psi) + 2(1 - l)$, again by lemma 5.3.3, since $\psi' = a\psi$ forces $v_F(a) = 1 - l$. In particular a character of level zero returns $n(\pi, \psi) + 2$; that is the normalization in which an unramified character of $F^\times$ has ε-factor exactly 1, by proposition 5.1.17 together with lemma 5.1.16. Level one is retained here because every explicit formula of section 5.1 is stated for it.

The exponent is carried with its additive character, and the notation records a real dependence: the integer depends on ψ as well as on π, and lemma 5.3.3 says exactly how. Fixing ψ of level one, it measures how much ramification π carries relative to that choice; it is insensitive to unramified twisting, as the computation below shows, and it vanishes on the cuspidal representations of level zero. The same convention, with the same notation, is used on the Galois side in proposition 6.3.3, which is what allows the two sides to be compared in chapter 7. Note that the classical *conductor* exponent is the value taken against an additive character of level zero, so it differs from $n(\pi, \psi)$ by a shift; nothing below depends on which of the two is used, provided one of them is used consistently. For a cuspidal representation of positive level it grows with the level of the simple stratum contained in π; the closed shape of the answer in terms of the cuspidal type of definition 3.5.4 needs the Kirillov model, which this book does not build; it is quoted rather than proved (see [1, §25]).

Example 5.3.5: Epsilon Factor of a Level Zero Cuspidal Representation

Let ψ have level one and let (π, V) be a cuspidal representation of level zero. Level zero is automatically minimal, so theorem 3.5.2 places π in case (2)(a): it contains a representation λ of K inflated from an irreducible cuspidal representation $\tilde{\lambda}$ of $\mathrm{GL}_2(k)$. That is case (1) of theorem 3.2.9, which supplies a representation Λ of ZK with $\Lambda|_K \cong \lambda$ and

$$\pi \cong \text{c-Ind}_{ZK}^{G}\Lambda$$

Write $K_1 = 1 + \mathfrak{p}M_2(\mathfrak{o})$, so that λ is trivial on K_1 and $K/K_1 \cong \mathrm{GL}_2(k)$. Let W be the space of λ, and fix $w \in W$ and $\tilde{w} \in \check{W}$ with $\langle \tilde{w}, w\rangle \neq 0$.

1. *A coefficient and a test function.* The proof of theorem 3.2.8 identifies W and $\check{W}$ with the functions in the induced space supported on ZK, and the resulting coefficient of π is

$$f(g) = \begin{cases} \langle \tilde{w}, \Lambda(g) w\rangle & g \in ZK \\ 0 & g \notin ZK \end{cases}$$

Since K is open and compact in A, the function Φ defined by $\Phi(u) = \langle \tilde{w}, \lambda(u^{-1})w\rangle$ for $u \in K$ and $\Phi = 0$ off K lies in $C_c^\infty(A)$, and it is inflated from $\mathrm{GL}_2(k)$ because λ is.

2. *The zeta integral is a non-zero constant.* The supports meet in K, where $\|\det u\| = 1$, so

$$\zeta(\Phi, f, s) = \int_K \langle \tilde{w}, \lambda(u^{-1})w\rangle\langle \tilde{w}, \lambda(u)w\rangle\, d^\times u = \mathrm{vol}^\times(K_1) \sum_{y \in \mathrm{GL}_2(k)} \langle \tilde{w}, \tilde{\lambda}(y^{-1})w\rangle\langle \tilde{w}, \tilde{\lambda}(y)w\rangle$$

Schur orthogonality on the finite group $\mathrm{GL}_2(k)$ evaluates the sum as $|\mathrm{GL}_2(k)|\,(\dim\tilde{\lambda})^{-1}\langle \tilde{w}, w\rangle^2$, so $\zeta(\Phi, f, s) = c_0$ is a non-zero constant, independent of s.

3. *The Fourier transform on ZK.* The self-dual measure μ on A relative to ψ_A gives $\mu(M_2(\mathfrak{o})) = q^2$ and $\mu(M_2(\mathfrak{p})) = q^{-2}$: the dual of $M_2(\mathfrak{o})$ is $M_2(\mathfrak{p})$, since $\mathrm{tr}(x\, M_2(\mathfrak{o}))$ is the $\mathfrak{o}$-module generated by the entries of x and ψ is trivial exactly on $\mathfrak{p}$, and $\mu(L)\mu(L^*) = 1$. The multiplicative cosets of K_1 in K are additive cosets of $M_2(\mathfrak{p})$, because $yK_1 = y + yM_2(\mathfrak{p}) = y + M_2(\mathfrak{p})$ for $y \in K$. Decomposing K accordingly into the sets $y_j + M_2(\mathfrak{p})$, $1 \le j \le |\mathrm{GL}_2(k)|$, on each of which Φ is constant,

$$\hat{\Phi}(x) = \sum_j \Phi(y_j)\, \psi\big(\mathrm{tr}(xy_j)\big) \int_{M_2(\mathfrak{p})} \psi\big(\mathrm{tr}(xm)\big)\, d\mu(m)$$

The character $m \mapsto \psi(\mathrm{tr}(xm))$ of $M_2(\mathfrak{p})$ is trivial precisely when $x \in M_2(\mathfrak{o})$, so the last integral is q^{-2} for such x and 0 otherwise. Now take $x = \varpi^n u$ with $u \in K$ and $n \in \mathbb{Z}$. Since u has an entry in U_F, the matrix $\varpi^n u$ lies in $M_2(\mathfrak{o})$ if and only if $n \ge 0$, and:

   - for $n \le -1$, $\hat{\Phi}(\varpi^n u) = 0$;
   - for $n \ge 1$, $\mathrm{tr}(\varpi^n u y_j) \in \mathfrak{p} \subseteq \ker\psi$, so $\hat{\Phi}(\varpi^n u) = q^{-2}\sum_j \Phi(y_j) = 0$ by orthogonality of the coefficient of the non-trivial irreducible representation $\tilde{\lambda}$ against the trivial character;
   - for $n = 0$, $\hat{\Phi}(u) = q^{-2}\sum_{y \in \mathrm{GL}_2(k)} \langle \tilde{w}, \tilde{\lambda}(y^{-1})w\rangle\, \eta\big(\mathrm{tr}(\bar{u}y)\big)$, where η is the non-trivial character of $k = \mathfrak{o}/\mathfrak{p}$ induced by $\psi|_{\mathfrak{o}}$ and $\bar{u}$ is the image of u in $\mathrm{GL}_2(k)$.

   Thus $\hat{\Phi}$ vanishes on $ZK \setminus K$.

4. *The epsilon exponent is zero.* The support of $\check{f}$ is $(ZK)^{-1} = ZK$, so by the previous item

$$\zeta\big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\big) = \int_K \hat{\Phi}(u)\check{f}(u)\, d^\times u = c_1$$

is again constant in s. Since $L(\pi, s) = L(\check{\pi}, 1-s) = 1$ by corollary 5.3.2, the functional equation (5.10) reads $c_1 = \varepsilon(\pi, s, \psi)c_0$, so

$$\varepsilon(\pi, s, \psi) = \frac{c_1}{c_0}, \qquad n(\pi, \psi) = 0$$

The ε-factor is constant in s. The computation also forces $c_1 \neq 0$, since $c_0 \neq 0$ and ε is a unit; in particular the finite Fourier transform of a matrix coefficient of a cuspidal representation of $\mathrm{GL}_2(k)$ does not vanish identically on $\mathrm{GL}_2(k)$. Against a character of level zero the exponent is $n(\pi, \psi) + 2 = 2$.

5. *Unramified twisting.* For a character φ of $F^\times$ write $\varphi\pi = (\varphi \circ \det) \otimes \pi$. If φ is unramified then $\varphi = \| - \|^\sigma$ for some $\sigma \in \mathbb{C}$ with $q^{-\sigma} = \varphi(\varpi)$, so $\varphi(\det g)\|\det g\|^s = \|\det g\|^{s+\sigma}$ and
$$\zeta\big(\Phi, (\varphi \circ \det)f, s\big) = \zeta(\Phi, f, s+\sigma)$$
The coefficients of $\varphi\pi$ are exactly the $(\varphi \circ \det)f$ with $f \in C(\pi)$, and $((\varphi \circ \det)f)^\vee = (\varphi^{-1} \circ \det)\check{f}$, a coefficient of $\varphi^{-1}\check{\pi} = (\varphi\pi)^\vee$. Applying (5.10) for π at $s+\sigma$ therefore gives $\gamma(\varphi\pi, s, \psi) = \gamma(\pi, s+\sigma, \psi)$. The representation $\varphi\pi$ is cuspidal, since N lies in the kernel of $\varphi \circ \det$ and the Jacquet module is unchanged, so both γ-factors are ε-factors and
$$\varepsilon(\varphi\pi, s, \psi) = \varepsilon(\pi, s+\sigma, \psi) = \varphi(\varpi)^{n(\pi,\psi)}\varepsilon(\pi, s, \psi)$$
In particular $n(\varphi\pi, \psi) = n(\pi, \psi)$, and for the present π, where $n(\pi, \psi) = 0$, unramified twisting does not change the ε-factor at all.

6. *The constraint from the functional equation.* Since $n(\pi, \psi) = n(\check{\pi}, \psi) = 0$, the identity of corollary 5.3.2 reduces to
$$\varepsilon\big(\pi, \tfrac{1}{2}, \psi\big)\, \varepsilon\big(\check{\pi}, \tfrac{1}{2}, \psi\big) = \omega_\pi(-1)$$
The right-hand side is computable inside the finite group: $-1 \in U_F$ and $\omega_\pi|_{U_F}$ is the inflation of the central character of $\tilde{\lambda}$, so $\omega_\pi(-1)$ is the value of that central character at $-1 \in k^\times$.

7. *The remaining constant.* What the tools of this section do not give is a closed form for c_1/c_0; identifying it requires the Kirillov model, which this book does not build. Let E/F be the unramified quadratic extension and let ξ be the character of $E^\times$ of level zero attached to π by proposition 4.2.3, and put $\psi_E = \psi \circ \mathrm{Tr}_{E/F}$, of level one in E. The answer has the shape
$$\varepsilon(\pi, s, \psi) = \lambda_{E/F}(\psi)\, \frac{\tau(\xi, \psi_E)}{q}$$
where $\tau(\xi, \psi_E)$ is the Gauss sum of definition 5.1.19 formed over E and $\lambda_{E/F}(\psi)$ depends only on E/F and ψ; see [1, §25]. That the right-hand side is constant in s is the consistency check available here: ξ is ramified of level zero and $q_E = q^2$, so theorem 5.1.18 applied over E gives $\varepsilon(\xi, s, \psi_E) = q_E^{-1/2}\tau(\xi, \psi_E)$, with no dependence on s, matching $n(\pi, \psi) = 0$.

The computation is as far as elementary means go for a single cuspidal representation. A statement of a different kind becomes available once twisting is allowed, and it records how little the ε-factor remembers: past a threshold depending on π, the local constant of $\chi\pi$ depends only on χ and on the central character of π.

Theorem 5.3.6: Stability of the Local Constant for GL_2

Let (π, V) be an irreducible smooth representation of G with central character ω_π, and let $\psi \in \widehat{F}$ have level one. There is an integer $m = m(\pi) \geq 1$ with the following property. Let χ be a character of $F^\times$ of level $n \geq m$, and let $c \in F^\times$ satisfy

$$\chi(1+x) = \psi(cx), \qquad x \in \mathfrak{p}^{[n/2]+1}$$

Then, writing $\chi\pi = (\chi \circ \det) \otimes \pi$,

$$\varepsilon(\chi\pi, s, \psi) = \omega_\pi(c)^{-1}\, \varepsilon(\chi, s, \psi)^2$$

In particular, if (π', V') is a second irreducible smooth representation of G with $\omega_{\pi'} = \omega_\pi$, then $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for every character χ of $F^\times$ of sufficiently large level.

The proof is Deligne's, and it rests on machinery this book does not develop; the theorem is used here as a black box. The GL_1 analogue theorem 5.1.21 *is* proved in full above, and comparing the two shows exactly what is missing.

***Proof Key ideas*:**
At GL_1 the argument is a manipulation of Gauss sums. Theorem 5.1.18 writes $\varepsilon(\chi, s, \psi)$ explicitly in terms of $\tau(\chi, \psi)$, and proposition 5.1.20 collapses the Gauss sum of a character of level n onto a sum over the small group $U_F^{[(n+1)/2]}/U_F^{[n/2]+1}$; and a twisting character θ of level l with $2l < n$ is trivial on that group, so it comes out of the sum as the single scalar $\theta(c)^{-1}$. Every step is available because the GL_1 local constant has a closed formula.

No such formula exists at GL_2. The local constant of a cuspidal representation is not an elementary sum over $F^\times$; the worked example above produced it only as a ratio of two integrals, and even at level zero the constant had to be quoted. The substitute is an analysis of the zeta integral of $\chi\pi$ in the Kirillov model of π, in which the vectors of V are realized as functions on $F^\times$. A twist by a character of very large level makes the integrand oscillate so rapidly that the integral localizes onto a small neighbourhood, and on that neighbourhood the only surviving datum is the action of the centre. The shape of the answer is the one the GL_1 case predicts: a two-dimensional object contributes $\varepsilon(\chi, s, \psi)$ twice, and its determinant, here the central character, contributes the scalar $\omega_\pi(c)^{-1}$.

For the argument in full, see [1, §25].

What the theorem asserts is a limitation. Fix a central character; then all irreducible representations of G with that central character become indistinguishable once they are twisted deeply enough, because their local constants collapse to one expression built out of χ and ω_π. A family of local constants $\{\varepsilon(\chi\pi, s, \psi)\}_\chi$ therefore carries information about π only through the twists of bounded level, together with ω_π. Comparing two cuspidal representations through all their twists looks like an infinite amount of data; past the threshold $m(\pi)$ it is not.

Exercise 5.3.1

1. Let (π, V) be an irreducible smooth representation of G and let χ be a character of $F^\times$. Show that $\mathrm{Hom}_G(\pi, \chi\circ\det) \neq 0$ forces $\pi \cong \chi\circ\det$. Deduce that no irreducible cuspidal representation of G admits a non-zero G-homomorphism onto a one-dimensional representation.

2. Let (π, V) be an irreducible cuspidal representation of G and let χ be any character of $F^\times$. Show that $\chi\pi = (\chi \circ \det) \otimes \pi$ is irreducible and cuspidal, and deduce that $L(\chi\pi, s) = 1$ and $\gamma(\chi\pi, s, \psi) = \varepsilon(\chi\pi, s, \psi)$.
3. Let (π, V) be an irreducible smooth representation of G and let ψ have level one. Show that $n(\pi, \psi) = n(\check{\pi}, \psi)$ and that $\varepsilon(\pi, \frac{1}{2}, \psi)\varepsilon(\check{\pi}, \frac{1}{2}, \psi) = \omega_\pi(-1)$. Deduce that if $\pi \cong \check{\pi}$ and ω_π is trivial, then $\varepsilon(\pi, s, \psi) = \pm q^{(1/2-s)n(\pi,\psi)}$.
4. Let (π, V) be an irreducible smooth representation of G and let ψ have level one.
    1. Show that the level of $a\psi$ is the level of ψ minus $v_F(a)$, for $a \in F^\times$.
    2. Deduce that if ψ' has level l, then $\varepsilon(\pi, s, \psi') = \varepsilon(\pi, \frac{1}{2}, \psi')q^{(1/2-s)(n(\pi,\psi)+2(1-l))}$.
    3. Let π be a cuspidal representation of level zero as in example 5.3.5. Compute $\varepsilon(\pi, s, \varpi\psi)$ in terms of $\varepsilon(\pi, s, \psi)$ and read off the exponent attached to π by an additive character of level zero.
5. Let (π, V) be an irreducible cuspidal representation of G whose central character ω_π is ramified. Give the proof that $L(\pi, s) = 1$ that uses only proposition 5.1.8 together with the expression of $Z(\Phi, f, X)$ as a finite sum of Tate integrals obtained in the proof of proposition 5.3.1, and identify the single point at which the general argument needs ω_π to be unramified.

5.4 Epsilon Factors of Non-Cuspidal Representations

Section 5.3 settled the cuspidal half of the spectrum. What remains is the list of theorem 2.4.16: the irreducible induced representations $\iota_B^G\chi$, the one-dimensional representations $\varphi \circ \det$, and the special representations $\varphi\mathrm{St}_G$. Every coefficient of every one of these is a coefficient of some $\mathrm{Ind}_B^G\chi$, and the GL_2 zeta integral of such a coefficient unfolds into a pair of Tate integrals. The whole of this section is therefore a transcription of the GL_1 theory of section 5.1, with the two normalizations in play kept explicit throughout.

Two normalizations should be kept in view throughout, since almost every error available here is an error in one of them. The first is the shift in the GL_2 functional equation: by proposition 5.2.6 the L-function is the generator of the module of the *shifted* integrals $\zeta(\Phi, f, s + \frac{1}{2})$, which is the normalization that makes (5.10) and corollary 5.2.9 symmetric in $s \leftrightarrow 1 - s$. The generator of the unshifted module is then $L(\pi, s - \frac{1}{2})$, and nothing is lost in passing between the two: $s \mapsto s + \frac{1}{2}$ sends q^{-s} to $q^{-1/2}q^{-s}$ and so is a ring automorphism of $\mathbb{C}[q^s, q^{-s}]$ carrying one module isomorphically onto the other, generators to generators, and preserving the normalization $P(0) = 1$. The second normalization is the $\delta_B^{-1/2}$ twist of (2.10). Since $\delta_B(t) = \|t_2/t_1\|$ by proposition 2.2.8, as a character of $T = F^\times \times F^\times$ one has $\delta_B^{\pm 1/2} = \| - \|^{\mp 1/2} \otimes \| - \|^{\pm 1/2}$, and therefore

$$\iota_B^G(\chi_1 \otimes \chi_2) = \mathrm{Ind}_B^G\big(\chi_1\| - \|^{1/2} \otimes \chi_2\| - \|^{-1/2}\big), \qquad \mathrm{Ind}_B^G(\chi_1 \otimes \chi_2) = \iota_B^G\big(\chi_1\| - \|^{-1/2} \otimes \chi_2\| - \|^{1/2}\big)$$

The half-integral powers of $\| - \|$ produced by this twist are exactly what the half-integral shift $s+\frac{1}{2} \leftrightarrow \frac{3}{2}-s$ absorbs. Recall for the same purpose that $L(\chi\|-\|^a, s) = L(\chi, s+a)$ and $\zeta(\Phi, \chi\|-\|^a, s) = \zeta(\Phi, \chi, s + a)$, immediate from definition 5.1.6 and definition 5.1.9, and hence $\gamma(\chi\| - \|^a, s, \psi) = \gamma(\chi, s + a, \psi)$ and $\varepsilon(\chi\| - \|^a, s, \psi) = \varepsilon(\chi, s + a, \psi)$ by theorem 5.1.11 and definition 5.1.13, for every $a \in \mathbb{C}$ and every character χ of $F^\times$.

The first family is the irreducible induced representations. Here the answer is the one the notation predicts: the invariants of $\iota_B^G(\chi_1 \otimes \chi_2)$ are the products of the invariants of χ_1 and χ_2, with no shift left over.

Proposition 5.4.1: L-Function and Epsilon Factor of a Principal Series

Let $\chi = \chi_1 \otimes \chi_2$ be a character of T such that $\pi = \iota_B^G \chi$ is irreducible, and let $\psi \in \widehat{F}$ be non-trivial. Then

$$L(\pi, s) = L(\chi_1, s)L(\chi_2, s), \qquad \varepsilon(\pi, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$$

and consequently $\gamma(\pi, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$.

***Proof*:**
Step 1: the unfolded zeta integral. Let $\xi = \xi_1 \otimes \xi_2$ be any character of T, inflated to B trivially on N, and write $\Sigma_\xi = \mathrm{Ind}_B^G \xi$, realized on the space of smooth $\varphi\colon G \to \mathbb{C}$ with $\varphi(bg) = \xi(b)\varphi(g)$. The quotient $B\backslash G$ is compact (corollary 2.2.2), so the Duality Theorem (theorem 1.3.12) identifies $\check{\Sigma}_\xi$ with $\mathrm{Ind}_B^G \delta_B^{-1}\check{\xi}$ under the pairing $\langle \tilde{\varphi}, \varphi \rangle = \int_K \tilde{\varphi}(k)\varphi(k)\, d\mu_K(k)$; hence $C(\Sigma_\xi)$ is spanned by the functions

$$f(g) = \int_K \tilde{\varphi}(k)\varphi(kg)\, d\mu_K(k), \qquad \varphi \in \Sigma_\xi,\ \tilde{\varphi} \in \mathrm{Ind}_B^G \delta_B^{-1}\check{\xi}$$

Normalize the Haar measure μ_N on $N \cong F$ entering the decomposition $d\mu_B(tn) = d\mu_T(t)\, d\mu_N(n)$ fixed before proposition 2.2.8 to be the self-dual measure μ_ψ; this changes the left Haar measure μ_B, and hence the constant relating the two sides of (2.4), by a positive scalar and nothing else. For $\Theta \in C_c^\infty(A)$ set

$$\Psi_\Theta(t_1, t_2) = \int_F \Theta\begin{pmatrix} t_1 & y \\ 0 & t_2 \end{pmatrix} d\mu_\psi(y)$$

and for a character ξ of T and $\Psi \in C_c^\infty(F \times F)$ set

$$\Xi_\xi(\Psi, s) = \int_{F^\times}\int_{F^\times} \Psi(t_1, t_2)\, \xi_1(t_1)\|t_1\|^{s-1}\, \xi_2(t_2)\|t_2\|^s\, d^\times t_1\, d^\times t_2$$

The non-cuspidal half of the proof of proposition 5.2.6 is exactly the computation of $\zeta(\Phi, f, s)$ for such an f: substituting the pairing formula for f and replacing g by $k^{-1}g$, applying the Iwasawa decomposition $G = BK$ (proposition 2.2.1) through (2.4), writing $b = tn(x)$ and substituting $y = t_1 x$ in the N-integral produces

$$\zeta(\Phi, f, s) = c_0 \int_K \int_K \tilde{\varphi}(k)\varphi(k')\, \Xi_\xi\left(\Psi_{\Phi_{k,k'}}, s\right) d\mu_K(k)\, d\mu_K(k'), \qquad \Phi_{k,k'}(y) = \Phi(k^{-1}yk')$$

where $c_0 > 0$ depends only on the chosen Haar measures and not on ξ, Φ or f. That proof also records the two facts making the right-hand side meaningful: $\Psi_\Theta \in C_c^\infty(F \times F)$ for every $\Theta \in C_c^\infty(A)$, and all the interchanges are justified on a half plane $\mathrm{Re}(s) \gg 0$ of absolute convergence, after which both sides are rational in q^{-s} and the identity persists.

Step 2: the partial integral converts the matrix Fourier transform into a two-variable one. For $\Psi \in C_c^\infty(F \times F)$ let $\widehat{\Psi}$ denote the Fourier transform for the character $(y_1, y_2) \mapsto \psi(t_1 y_1 + t_2 y_2)$

and the measure $\mu_\psi \otimes \mu_\psi$, which is self-dual for it. The claim is

$$\Psi_{\hat{\Theta}} = \widehat{\Psi_\Theta}, \qquad \Theta \in C_c^\infty(A)$$

where $\hat{\Theta}$ is the matrix Fourier transform of definition 5.2.1. In coordinates the pairing on A is $\mathrm{tr}(xy) = x_{11}y_{11} + x_{12}y_{21} + x_{21}y_{12} + x_{22}y_{22}$, which is the standard pairing of F^4 with itself after the relabelling that swaps the two off-diagonal coordinates in one of the two arguments; a permutation of coordinates preserves self-duality, so the measure μ on A self-dual for ψ_A is the product $\mu_\psi^{\otimes 4}$ of the four self-dual measures on the entries.

Fix $t_1, t_2 \in F$. Since

$$\mathrm{tr}\left(\begin{pmatrix} t_1 & u \\ 0 & t_2 \end{pmatrix} y\right) = t_1 y_{11} + u y_{21} + t_2 y_{22}$$

the function

$$h(v) = \int_F \int_F \int_F \Theta\begin{pmatrix} y_{11} & y_{12} \\ v & y_{22} \end{pmatrix} \psi(t_1 y_{11} + t_2 y_{22})\, d\mu_\psi(y_{11})\, d\mu_\psi(y_{12})\, d\mu_\psi(y_{22})$$

lies in $C_c^\infty(F)$, and Fubini over the compact support of Θ gives $\hat{\Theta}\left(\begin{smallmatrix} t_1 & u \\ 0 & t_2 \end{smallmatrix}\right) = \hat{h}(u)$. As $\hat{h} \in C_c^\infty(F)$ by proposition 5.1.2, integrating in u against μ_ψ is an absolutely convergent integral, and by the Fourier inversion formula (definition 5.1.4)

$$\Psi_{\hat{\Theta}}(t_1, t_2) = \int_F \hat{h}(u)\, d\mu_\psi(u) = \hat{\hat{h}}(0) = h(0)$$

The value $h(0)$ is by definition $\int_F \int_F \Psi_\Theta(y_{11}, y_{22}) \psi(t_1 y_{11} + t_2 y_{22})\, d\mu_\psi(y_{11}) d\mu_\psi(y_{22}) = \widehat{\Psi_\Theta}(t_1, t_2)$, which is the claim. A scalar change of the measure used in Ψ_Θ multiplies both sides by the same scalar, so the identity is insensitive to the normalization of μ_N fixed in Step 1.

Step 3: unfolding the transformed side. Keep ξ arbitrary and let $f \in C(\Sigma_\xi)$ be as in Step 1. The pairing $\langle -, - \rangle$ is G-invariant, so

$$\check{f}(g) = f(g^{-1}) = \langle \tilde{\varphi}, \Sigma_\xi(g^{-1})\varphi \rangle = \langle \varphi, \check{\Sigma}_\xi(g)\tilde{\varphi} \rangle = \int_K \varphi(k)\tilde{\varphi}(kg)\, d\mu_K(k)$$

Thus $\check{f}$ is a coefficient of Σ_η for $\eta = \delta_B^{-1}\check{\xi}$, built from the pair $(\varphi, \tilde{\varphi})$ with the roles reversed, the outer function being φ; and φ indeed lies in $\mathrm{Ind}_B^G \delta_B^{-1}\check{\eta} = \mathrm{Ind}_B^G \xi$. Applying Step 1 to Σ_η, with test function $\hat{\Phi}$,

$$\zeta(\hat{\Phi}, \check{f}, s') = c_0 \int_K \int_K \varphi(k)\tilde{\varphi}(k')\, \Xi_\eta\big(\Psi_{(\hat{\Phi})_{k,k'}}, s'\big)\, d\mu_K(k)\, d\mu_K(k')$$

with the *same* constant c_0, since c_0 depends only on the measures. For $k, k' \in K$ the substitution $y \mapsto kyk'^{-1}$ preserves μ (its determinant as an F-linear map on A is $\det(k)^2 \det(k')^{-2}$, of absolute value one) and $\mathrm{tr}(xkyk'^{-1}) = \mathrm{tr}((k'^{-1}xk)y)$, whence $\widehat{\Phi_{k',k}} = (\hat{\Phi})_{k,k'}$. Combining this with Step 2 and relabelling $(k, k') \mapsto (k', k)$,

$$\zeta(\hat{\Phi}, \check{f}, s') = c_0 \int_K \int_K \tilde{\varphi}(k)\varphi(k')\, \Xi_\eta\big(\widehat{\Psi_{\Phi_{k,k'}}}, s'\big)\, d\mu_K(k)\, d\mu_K(k')$$

so both sides of the functional equation are integrals of the same weight $\tilde{\varphi}(k)\varphi(k')$ against the same family of functions $\Psi_{\Phi_{k,k'}} \in C_c^\infty(F \times F)$.

Step 4: Tate's functional equation in each variable. Since $\delta_B^{-1} = \|-\| \otimes \|-\|^{-1}$, the character $\eta = \delta_B^{-1}\check{\xi}$ has components $\eta_1 = \check{\xi}_1\|-\|$ and $\eta_2 = \check{\xi}_2\|-\|^{-1}$. Put $\sigma = s + \frac{1}{2}$, so that $\frac{3}{2} - s = 2 - \sigma$. Then for $\Psi \in C_c^\infty(F \times F)$,

$$\Xi_\eta(\widehat{\Psi}, 2-\sigma) = \int_{F^\times}\int_{F^\times} \widehat{\Psi}(t_1,t_2)\,\check{\xi}_1(t_1)\|t_1\|^{2-\sigma}\,\check{\xi}_2(t_2)\|t_2\|^{1-\sigma}\,d^\times t_1\,d^\times t_2$$

where the two powers of $\|-\|$ absorbed from η_1 and η_2 have moved the exponents by $+1$ and -1. Comparing with

$$\Xi_\xi(\Psi, \sigma) = \int_{F^\times}\int_{F^\times} \Psi(t_1,t_2)\,\xi_1(t_1)\|t_1\|^{\sigma-1}\,\xi_2(t_2)\|t_2\|^{\sigma}\,d^\times t_1\,d^\times t_2$$

the exponents pair up as $2-\sigma = 1-(\sigma-1)$ in the first variable and $1-\sigma$ in the second: the transformed integral is Tate's left-hand side for ξ_1 at $u_1 = \sigma - 1$ and for ξ_2 at $u_2 = \sigma$. Both Ψ and $\widehat{\Psi}$ are finite sums of pure tensors, by the argument of proposition 2.2.6 run on $F \times F$, and Fourier transform on $F \times F$ carries $\Psi_1 \otimes \Psi_2$ to $\hat{\Psi}_1 \otimes \hat{\Psi}_2$. On a pure tensor,

$$\Xi_\eta(\widehat{\Psi_1 \otimes \Psi_2}, 2-\sigma) = \zeta(\hat{\Psi}_1, \check{\xi}_1, 1-u_1)\,\zeta(\hat{\Psi}_2, \check{\xi}_2, 1-u_2)$$

so theorem 5.1.11 applied in each variable separately gives

$$\Xi_\eta(\widehat{\Psi}, 2-\sigma) = \gamma(\xi_1, \sigma-1, \psi)\,\gamma(\xi_2, \sigma, \psi)\,\Xi_\xi(\Psi, \sigma)$$

for every $\Psi \in C_c^\infty(F \times F)$, the two γ-factors being independent of Ψ. Feeding this into the two displays of Step 3 and Step 1, the common constant c_0 and the common weight $\tilde{\varphi}(k)\varphi(k')$ cancel and

$$\zeta\Big(\hat{\Phi}, \check{f}, \tfrac{3}{2} - s\Big) = \gamma\big(\xi_1, s - \tfrac{1}{2}, \psi\big)\,\gamma\big(\xi_2, s + \tfrac{1}{2}, \psi\big)\,\zeta\Big(\Phi, f, s + \tfrac{1}{2}\Big) \tag{5.11}$$

for every $\Phi \in C_c^\infty(A)$ and every $f \in C(\mathrm{Ind}_B^G \xi)$. Nothing in the derivation used irreducibility, so (5.11) holds for every character ξ of T.

Step 5: the L-factor. Still with ξ arbitrary, the proof of proposition 5.2.6 bounds the unshifted integrals: for all Φ and all $f \in C(\Sigma_\xi)$,

$$\zeta(\Phi, f, s) \in L(\xi_1, s-1)L(\xi_2, s)\,\mathbb{C}[q^s, q^{-s}]$$

since $L(\xi_1\|-\|^{-1}, s) = L(\xi_1, s-1)$. This bound is attained. Choose compact open subsets S_1, S_2 of F with $\zeta(\mathbf{1}_{S_i}, \xi_i, u) = c_i L(\xi_i, u)$ for constants $c_i > 0$: take $S_i = \mathfrak{o}$ when ξ_i is unramified, as in the proof of proposition 5.1.8, and $S_i = U_F^{n_i+1}$ with n_i the level of ξ_i when ξ_i is ramified, as in the proof of lemma 5.1.15, where $L(\xi_i, u) = 1$. Let Φ be the characteristic function of

$$\mathcal{U} = \big\{y \in A : y_{11} \in S_1,\ y_{22} \in S_2,\ y_{12} \in \varpi^{-r}\mathfrak{o},\ y_{21} \in \mathfrak{o}\big\}$$

for any $r \geq 0$; this set is compact and open, so $\Phi \in C_c^\infty(A)$, and since the matrices occurring in Ψ_Φ have vanishing $(2,1)$-entry,

$$\Psi_\Phi = \mu_\psi(\varpi^{-r}\mathfrak{o})\,\mathbf{1}_{S_1} \otimes \mathbf{1}_{S_2}$$

Being locally constant with compact support, Φ is bi-invariant under $K_1 = K \cap (1 + \varpi^M M_2(\mathfrak{o}))$ for M large: if Φ is invariant under translation by $\varpi^{M_0} M_2(\mathfrak{o})$ and supported in $\varpi^{-r_0} M_2(\mathfrak{o})$, then $M = M_0 + r_0$ works, because $k_1 y - y = (k_1 - 1)y$ lies in $\varpi^{M_0} M_2(\mathfrak{o})$ for y in the support, and k_1 preserves the lattice $\varpi^{-r_0} M_2(\mathfrak{o})$, so both values vanish off it. Shrinking K_1 further, assume also $B \cap K_1 \subseteq \ker \xi$, which is possible because ξ is smooth. Now let φ be supported on BK_1 with $\varphi(bk_1) = \xi(b)$, and $\tilde{\varphi}$ supported on BK_1 with $\tilde{\varphi}(bk_1) = \delta_B^{-1} \check{\xi}(b)$. Both are well-defined, since $bk_1 = b'k_1'$ forces $b'^{-1} b \in B \cap K_1 \subseteq \ker \xi$, and both are smooth, being right K_1-invariant with open support.

The weights and the Ξ-values now cancel exactly. Write $b^{-1}\Phi$ and $\Phi b'$ for $y \mapsto \Phi(b^{-1}y)$ and $y \mapsto \Phi(yb')$. For b, b' in B with diagonals (b_1, b_2) and (b_1', b_2'), substituting for the upper-right entry in the defining integral gives

$$\Psi_{b^{-1}\Phi}(t_1, t_2) = \|b_1\| \, \Psi_\Phi(b_1^{-1} t_1, b_2^{-1} t_2), \qquad \Psi_{\Phi b'}(t_1, t_2) = \|b_2'\|^{-1} \, \Psi_\Phi(t_1 b_1', t_2 b_2')$$

Substituting $t_i \mapsto b_i t_i$, respectively $t_i \mapsto t_i b_i'^{-1}$, in Ξ_ξ and using $\|b_1\| = \|b_2\| = \|b_1'\| = \|b_2'\| = 1$ for $b, b' \in B \cap K$ yields

$$\Xi_\xi(\Psi_{b^{-1}\Phi b'}, s) = \xi(b) \, \check{\xi}(b') \, \Xi_\xi(\Psi_\Phi, s), \qquad b, b' \in B \cap K$$

Also $K \cap BK_1 = (B \cap K)K_1$, since $k = bk_1 \in K$ forces $b = kk_1^{-1} \in B \cap K$, and δ_B is trivial on $B \cap K$. At a point $k = bk_1$, $k' = b'k_1'$ of the support the integrand of Step 1 is therefore

$$\check{\xi}(b) \, \xi(b') \, \Xi_\xi(\Psi_{\Phi_{k,k'}}, s) = \check{\xi}(b) \, \xi(b') \, \xi(b) \, \check{\xi}(b') \, \Xi_\xi(\Psi_\Phi, s) = \Xi_\xi(\Psi_\Phi, s)$$

using bi-K_1-invariance of Φ to delete k_1 and k_1'. On the pure tensor Ψ_Φ the double integral Ξ_ξ separates, so

$$\Xi_\xi(\Psi_\Phi, s) = \mu_\psi(\varpi^{-r}\mathfrak{o}) \, \zeta(\mathbf{1}_{S_1}, \xi_1, s-1) \, \zeta(\mathbf{1}_{S_2}, \xi_2, s) = c_1 c_2 \, \mu_\psi(\varpi^{-r}\mathfrak{o}) \, L(\xi_1, s-1) L(\xi_2, s)$$

and therefore

$$\zeta(\Phi, f, s) = c_0 \, \mu_K\big((B \cap K)K_1\big)^2 \, \Xi_\xi(\Psi_\Phi, s) = c \, L(\xi_1, s-1) L(\xi_2, s)$$

for a constant $c > 0$. The module of unshifted integrals of Σ_ξ is therefore exactly $L(\xi_1, s-1)L(\xi_2, s)\,\mathbb{C}[q^s, q^{-s}]$.

Step 6: specialization to $\pi = \iota_B^G \chi$ and assembly. Take $\xi_1 = \chi_1 \| - \|^{1/2}$ and $\xi_2 = \chi_2 \| - \|^{-1/2}$, so that $\Sigma_\xi = \iota_B^G \chi = \pi$, which by hypothesis is irreducible; then $C(\pi) = C(\Sigma_\xi)$ and Steps 4 and 5 apply verbatim. By Step 5 the generator of the unshifted module is

$$L(\xi_1, s-1) L(\xi_2, s) = L\big(\chi_1, s - \tfrac{1}{2}\big) \, L\big(\chi_2, s - \tfrac{1}{2}\big)$$

so the generator of the shifted module, which is $L(\pi, s)$, is obtained by replacing s by $s + \frac{1}{2}$:

$$L(\pi, s) = L(\chi_1, s) L(\chi_2, s)$$

This is the first place where the two normalizations are seen to match: the $\pm\frac{1}{2}$ carried by the $\delta_B^{-1/2}$ twist and the $+\frac{1}{2}$ carried by the Godement-Jacquet shift cancel, and no half-integral shift survives.

For the ε-factor, $\zeta(\Phi, f, s+\frac{1}{2})$ is not identically zero by part (2) of proposition 5.2.6, so the rational function $\gamma(\pi, s, \psi)$ of theorem 5.2.7 is unique; comparing (5.10) with (5.11),

$$\gamma(\pi, s, \psi) = \gamma\big(\chi_1\|-\|^{1/2}, s-\tfrac{1}{2}, \psi\big)\,\gamma\big(\chi_2\|-\|^{-1/2}, s+\tfrac{1}{2}, \psi\big) = \gamma(\chi_1, s, \psi)\,\gamma(\chi_2, s, \psi)$$

again with the two shifts cancelling. Finally, the contragredient of π is computed from theorem 1.3.12: $\check{\pi} = (\mathrm{Ind}_B^G \delta_B^{-1/2}\chi)^\vee \cong \mathrm{Ind}_B^G \delta_B^{-1}\delta_B^{1/2}\check{\chi} = \iota_B^G\check{\chi}$, so $L(\check{\pi}, 1-s) = L(\check{\chi}_1, 1-s)L(\check{\chi}_2, 1-s)$ by what has just been proved. Substituting the definitions of the two ε factors into

$$\varepsilon(\pi, s, \psi) = \gamma(\pi, s, \psi)\frac{L(\pi, s)}{L(\check{\pi}, 1-s)} = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)\frac{L(\chi_1, s)L(\chi_2, s)}{L(\check{\chi}_1, 1-s)L(\check{\chi}_2, 1-s)}$$

the four L-factors cancel in pairs against those in $\gamma(\chi_i, s, \psi) = \varepsilon(\chi_i, s, \psi)L(\check{\chi}_i, 1-s)/L(\chi_i, s)$, leaving $\varepsilon(\pi, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$, as we sought to show.

The content of the proposition is that a principal series carries no information beyond its inducing characters. Both invariants are computed from χ_1 and χ_2 by the GL_1 recipe, and neither sees anything else about π. On this part of the spectrum a correspondence preserving L and ε is therefore forced before it is constructed: $\iota_B^G(\chi_1 \otimes \chi_2)$ can only be matched with an object whose invariants factor into the GL_1 invariants of χ_1 and χ_2 in the same way, so the matching is determined by the unordered pair $\{\chi_1, \chi_2\}$ and by nothing else. That the unordered pair is the right datum is already visible on this side of the fence, since $\iota_B^G\chi \cong \iota_B^G\chi^w$ by theorem 2.4.16. The interesting behaviour is all in the reducible degenerations, which is where the remaining two families live.

Both remaining families sit inside a single reducible induced representation. Fix a character φ of $F^\times$ and recall the non-split exact sequence (2.7), in its twisted form

$$0 \to \varphi_G \to \mathrm{Ind}_B^G\varphi_T \to \varphi\mathrm{St}_G \to 0$$

where $\varphi_G = \varphi \circ \det$ and $\varphi_T = \varphi \otimes \varphi$. Written in the normalized language, $\mathrm{Ind}_B^G\varphi_T = \iota_B^G(\chi_1 \otimes \chi_2)$ with

$$\chi_1 = \varphi\|-\|^{-1/2}, \qquad \chi_2 = \varphi\|-\|^{1/2}$$

so $\chi_1\chi_2^{-1} = \|-\|^{-1}$ and the representation is reducible, exactly as lemma 2.4.15 says. Every coefficient of either constituent is a coefficient of the whole (lemma 5.2.5), so (5.11) applies to both and the γ-factor is the same for both. What is not the same is the L-factor, and the sub-representation is the one that keeps all of it.

Proposition 5.4.2: L-Function and Epsilon Factor of a One-Dimensional Representation

Let φ be a character of $F^\times$ and let $\pi = \varphi_G = \varphi\circ\det$. With $\chi_1 = \varphi\|-\|^{-1/2}$ and $\chi_2 = \varphi\|-\|^{1/2}$ as above,

$$L(\varphi_G, s) = L(\chi_1, s)L(\chi_2, s) = L\big(\varphi, s-\tfrac{1}{2}\big)L\big(\varphi, s+\tfrac{1}{2}\big)$$

and

$$\varepsilon(\varphi_G, s, \psi) = \varepsilon(\chi_1, s, \psi)\,\varepsilon(\chi_2, s, \psi)$$

That is, the one-dimensional constituent has exactly the invariants that the formula of proposition 5.4.1 would assign to the reducible $\iota_B^G(\chi_1 \otimes \chi_2)$ containing it.

***Proof*:**
Step 1: the coefficients and the zeta integral. The space $V = \mathbb{C}$ is one-dimensional and $\check{V} = \mathbb{C}$, so a matrix coefficient of φ_G is $g \mapsto \tilde{v}v\varphi(\det g)$: thus $C(\varphi_G) = \mathbb{C}\varphi_G$, and it is enough to compute $\zeta(\Phi, \varphi_G, s)$ for $\Phi \in C_c^\infty(A)$. By proposition 2.2.7 the Haar integral on G is $\int_G h(g)\, d^\times g = \int_A h(x) \|\det x\|^{-2}\, d\mu(x)$, an identity of Haar integrals which extends to any h for which either side converges absolutely. Hence, on the half plane where the integral converges,

$$\zeta(\Phi, \varphi_G, s) = \int_A \Phi(x)\, \varphi(\det x)\, \|\det x\|^{s-2}\, d\mu(x)$$

Step 2: the upper bound. By (2.7) the representation φ_G is a subrepresentation of $\mathrm{Ind}_B^G \varphi_T$, so $C(\varphi_G) \subseteq C(\mathrm{Ind}_B^G \varphi_T)$ by lemma 5.2.5 and Step 5 of the proof of proposition 5.4.1 bounds the unshifted integrals:

$$\zeta(\Phi, \varphi_G, s) \in L(\varphi, s-1) L(\varphi, s)\, \mathbb{C}[q^s, q^{-s}]$$

Step 3: the ramified case. Suppose φ is ramified of level n. Then $L(\varphi, s-1) = L(\varphi, s) = 1$ by lemma 5.1.15, so Step 2 says the module of unshifted integrals is contained in $\mathbb{C}[q^s, q^{-s}]$. It contains a unit: let Φ be the characteristic function of the compact open set $\left\{g \in K : \det g \in U_F^{n+1}\right\}$, which lies in $C_c^\infty(A)$ because G is open in A. On the support, $\varphi(\det g) = 1$ and $\|\det g\| = 1$, so $\zeta(\Phi, \varphi_G, s)$ is the positive constant $\int_{\mathrm{supp}\Phi} d^\times g$, independent of s. Hence the module is all of $\mathbb{C}[q^s, q^{-s}]$ and $L(\varphi_G, s) = 1 = L(\chi_1, s) L(\chi_2, s)$, the last equality because χ_1 and χ_2 are ramified whenever φ is.

Step 4: the unramified case. Suppose φ is unramified, and let Φ_0 be the characteristic function of $M_2(\mathfrak{o})$. For this computation rescale μ so that $\mu(M_2(\mathfrak{o})) = 1$; a rescaling multiplies every zeta integral by the same positive constant and so does not change the module they generate, which is all that is at stake here. Decompose a matrix $x \in M_2(\mathfrak{o})$ into its rows $r_1, r_2 \in \mathfrak{o}^2$; then μ is the product of the two Haar measures of total mass one on the two row spaces. Let $m = \min\{v_F(x_{11}), v_F(x_{12})\} \geq 0$ be the minimum valuation along the first row. The set of first rows with a given m is $\varpi^m \mathfrak{o}^2 \setminus \varpi^{m+1}\mathfrak{o}^2$, of measure $q^{-2m}(1 - q^{-2})$. For each such r_1 there is $k \in K$ with $r_1 k = (\varpi^m, 0)$; right translation by k preserves μ and multiplies det by the unit $\det k$, and it carries r_2 to a row that is again uniformly distributed over $\mathfrak{o}^2$. Writing $(c, d) = r_2 k$, the determinant of xk is $\varpi^m d$. Therefore, using that φ is trivial on units,

$$\int_{M_2(\mathfrak{o})} \varphi(\det x) \|\det x\|^z\, d\mu(x) = (1 - q^{-2}) \sum_{m \geq 0} \left(\varphi(\varpi) q^{-z-2}\right)^m \int_{\mathfrak{o}} \varphi(d) \|d\|^z\, d\mu_1(d)$$

where μ_1 is the measure of total mass one on $\mathfrak{o}$. Splitting $\mathfrak{o}$ into the shells $\varpi^j U_F$, of measure $q^{-j}(1 - q^{-1})$, gives $\int_{\mathfrak{o}} \varphi(d) \|d\|^z d\mu_1(d) = (1 - q^{-1}) \sum_{j \geq 0} (\varphi(\varpi) q^{-z-1})^j$. Both geometric series converge for $\mathrm{Re}(z)$ large, and summing them,

$$\int_{M_2(\mathfrak{o})} \varphi(\det x) \|\det x\|^z\, d\mu(x) = (1 - q^{-1})(1 - q^{-2})\, L(\varphi, z+1)\, L(\varphi, z+2)$$

Setting $z = s - 2$ as in Step 1, $\zeta(\Phi_0, \varphi_G, s) = (1 - q^{-1})(1 - q^{-2}) L(\varphi, s-1) L(\varphi, s)$, a positive constant times the bound of Step 2. The module of unshifted integrals is therefore exactly $L(\varphi, s-1) L(\varphi, s) \mathbb{C}[q^s, q^{-s}]$.

Step 5: assembly. In both cases the generator of the unshifted module is $L(\varphi, s-1)L(\varphi, s)$, so replacing s by $s+\frac{1}{2}$,

$$L(\varphi_G, s) = L\big(\varphi, s-\tfrac{1}{2}\big)L\big(\varphi, s+\tfrac{1}{2}\big) = L(\chi_1, s)L(\chi_2, s)$$

By (5.11) applied to $\xi = \varphi_T$, whose components are both φ, and by uniqueness of γ,

$$\gamma(\varphi_G, s, \psi) = \gamma\big(\varphi, s-\tfrac{1}{2}, \psi\big)\gamma\big(\varphi, s+\tfrac{1}{2}, \psi\big) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$$

Finally $\check{\varphi_G} = \check{\varphi} \circ \det$, so $L(\check{\varphi_G}, 1-s) = L(\check{\chi}_1, 1-s)L(\check{\chi}_2, 1-s)$ by what has just been proved applied to $\check{\varphi}$, and the computation ending the proof of proposition 5.4.1 goes through unchanged, giving $\varepsilon(\varphi_G, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$, completing the proof.

The zeta integral of the trivial coefficient is thus a GL_2 analogue of Tate's integral over $\mathfrak{o}$: the lattice $M_2(\mathfrak{o})$ replaces $\mathfrak{o}$, and the two elementary divisors of a matrix contribute one GL_1 factor each. The two factors sit at $s-\frac{1}{2}$ and $s+\frac{1}{2}$, one half-step on either side of the centre of symmetry, which is the pattern the $\delta_B^{\pm 1/2}$ twist was designed to produce. Nothing is lost on restricting from the induced representation to its one-dimensional subspace.

Passing to the quotient is a different matter. The quotient has no constant vectors to integrate, and the coefficient functionals that survive are exactly those annihilating the constants. That single linear condition is what kills one of the two poles.

Proposition 5.4.3: L-Function and Epsilon Factor of a Special Representation

Let φ be a character of $F^\times$, let $\pi = \varphi\mathrm{St}_G$, and keep $\chi_1 = \varphi\|-\|^{-1/2}$, $\chi_2 = \varphi\|-\|^{1/2}$. Then

$$L(\varphi\mathrm{St}_G, s) = L(\chi_2, s) = L\big(\varphi, s+\tfrac{1}{2}\big)$$

and

$$\varepsilon(\varphi\mathrm{St}_G, s, \psi) = \varepsilon(\chi_1, s, \psi)\,\varepsilon(\chi_2, s, \psi)\,\frac{L(\check{\chi}_2, 1-s)}{L(\chi_1, s)}$$

The correcting factor is 1 when φ is ramified, and equals $-\chi_1(\varpi)q^{-s}$ when φ is unramified.

***Proof*:**

Step 1: the γ-factor. By (2.7) the representation $\varphi\mathrm{St}_G$ is a quotient of $\Sigma = \mathrm{Ind}_B^G\varphi_T$, so $C(\varphi\mathrm{St}_G) \subseteq C(\Sigma)$ by lemma 5.2.5, and (5.11) with $\xi = \varphi_T$ together with the uniqueness of γ gives

$$\gamma(\varphi\mathrm{St}_G, s, \psi) = \gamma\big(\varphi, s-\tfrac{1}{2}, \psi\big)\gamma\big(\varphi, s+\tfrac{1}{2}, \psi\big) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$$

the same value as for φ_G.

Step 2: the ramified case. If φ is ramified then so are χ_1 and χ_2, and Step 5 of the proof of proposition 5.4.1 puts the unshifted integrals inside $L(\varphi, s-1)L(\varphi, s)\mathbb{C}[q^s, q^{-s}] = \mathbb{C}[q^s, q^{-s}]$. Writing the generator of the unshifted module as $P(q^{-s})^{-1}$ with $P(0) = 1$ as in proposition 5.2.6, that generator lies in $\mathbb{C}[q^s, q^{-s}]$ and its inverse is a polynomial with constant term 1, which forces $P = 1$ and $L(\varphi\mathrm{St}_G, s) = 1 = L(\chi_2, s)$. The correcting factor is then the ratio of two trivial L-factors, namely 1.

Step 3: the unramified case, and where the pole goes. Let φ be unramified and write $\Sigma = \mathrm{Ind}_B^G \varphi_T$ as in Step 1 of the proof of proposition 5.4.1, with $\xi_1 = \xi_2 = \varphi$. In the variable $X = q^{-s}$ the two Tate denominators are $P_{\varphi\|-\|^{-1}}(X) = 1 - \varphi(\varpi)qX$ and $P_\varphi(X) = 1 - \varphi(\varpi)X$, with distinct roots

$$X_1 = \big(\varphi(\varpi)q\big)^{-1}, \qquad X_2 = \varphi(\varpi)^{-1}$$

The claim is that no element of the module attached to $\varphi\mathrm{St}_G$ has a pole at X_1.

Let $\Psi \in C_c^\infty(F \times F)$, regard $\Xi_{\varphi_T}(\Psi, s)$ as a rational function of X, and set $\rho(\Psi) = \lim_{X\to X_1}(1 - \varphi(\varpi)qX)\Xi_{\varphi_T}(\Psi, s)$. On a pure tensor $\Psi_1 \otimes \Psi_2$ the first factor is the Tate series $Z(\Psi_1, \varphi\|-\|^{-1}, X)$; since Ψ_1 is constant near 0, its coefficients satisfy $z_m = \Psi_1(0)\mu^*(U_F)(\varphi(\varpi)q)^m$ for $m \gg 0$, so the series differs from $\Psi_1(0)\mu^*(U_F)(\varphi(\varpi)qX)^{m_0}(1 - \varphi(\varpi)qX)^{-1}$ by a Laurent polynomial, and the limit is $\Psi_1(0)\mu^*(U_F)$. The second factor $Z(\Psi_2, \varphi, X)$ has no pole at X_1, because $P_\varphi(X_1) = 1 - q^{-1} \neq 0$, and its value there is $\int_{F^\times} \Psi_2(t)\varphi(t)\|t\|^{s_1} d^\times t$ where $q^{-s_1} = X_1$. For unramified φ and t of valuation m one has $\varphi(t)\|t\|^{s_1} = (\varphi(\varpi)X_1)^m = q^{-m} = \|t\|$, so this value is $\int_F \Psi_2(t)\, d\mu(t)$ by definition 5.1.5. Summing over a decomposition of Ψ into pure tensors and using $\sum_j \Psi_1^{(j)}(0)\Psi_2^{(j)} = \Psi(0, -)$,

$$\rho(\Psi) = \mu^*(U_F) \int_F \Psi(0, t)\, d\mu(t)$$

Now specialize to $\Psi = \Psi_{\Phi_{k,k'}}$. The matrices $\left(\begin{smallmatrix} 0 & y \\ 0 & t \end{smallmatrix}\right)$ occurring here are exactly the elements of $\mathfrak{a} = \{x \in A : xe_1 = 0\}$, the left ideal of matrices with vanishing first column, and $dy\, dt$ is a Haar measure on $\mathfrak{a} \cong F^2$. Since $\mathfrak{a}$ is a left ideal, $k^{-1}\mathfrak{a} = \mathfrak{a}$, and the induced map on $\mathfrak{a}$ is $x \mapsto k^{-1}x$, whose Jacobian is $\|\det k\|^{-1} = 1$ for $k \in K$. Hence

$$\rho\big(\Psi_{\Phi_{k,k'}}\big) = \mu^*(U_F) \int_{\mathfrak{a}} \Phi(k^{-1}vk')\, dv = \mu^*(U_F) \int_{\mathfrak{a}} \Phi(vk')\, dv$$

which does not depend on k. The two K-integrations in the display of Step 1 of the proof of proposition 5.4.1 have locally constant integrands on a compact group, hence are finite linear combinations, so ρ passes through them; and since the resulting integrand no longer depends on k, the double integral factors:

$$\lim_{X\to X_1} \big(1 - \varphi(\varpi)qX\big) Z(\Phi, f, X) = c_0\mu^*(U_F) \left(\int_K \tilde{\varphi}(k)\, d\mu_K(k)\right) \left(\int_K \varphi(k') \int_{\mathfrak{a}} \Phi(vk')\, dv\, d\mu_K(k')\right)$$

The one-dimensional subspace of Σ is spanned by $\varphi_0(g) = \varphi(\det g)$, and $\varphi(\det k) = 1$ for $k \in K$ because φ is unramified, so $\langle \tilde{\varphi}, \varphi_0 \rangle = \int_K \tilde{\varphi}(k)\, d\mu_K(k)$. A coefficient of the quotient $\varphi\mathrm{St}_G$ is by definition built from a functional annihilating that subspace, that is from a $\tilde{\varphi}$ with $\int_K \tilde{\varphi} = 0$. Therefore the limit vanishes for every $f \in C(\varphi\mathrm{St}_G)$ and every Φ, and since sums and $\mathbb{C}[X, X^{-1}]$-multiples of functions regular at X_1 are regular at X_1, the generator of the module is regular there as well. Writing $L(\varphi\mathrm{St}_G, s - \frac{1}{2}) = P(X)^{-1}$ with $P(0) = 1$ and P dividing $P_{\varphi\|-\|^{-1}}P_\varphi$, the factor $1 - \varphi(\varpi)qX$ cannot divide P, so

$$L(\varphi\mathrm{St}_G, s) = 1 \qquad \text{or} \qquad L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$$

the second alternative being $P = P_\varphi$ after the shift $s \mapsto s + \frac{1}{2}$.

Step 4: the first alternative is impossible. Two preliminaries about twisting. First, let η be an unramified character of $F^\times$ and let π' be an irreducible representation of G. The coefficients of $\eta\pi'$ are the functions $g \mapsto \eta(\det g) f(g)$ with $f \in C(\pi')$, and $\eta(\det g) = \eta(\varpi)^{v_F(\det g)}$ since η is unramified, so

$$Z(\Phi, \eta f, X) = Z\big(\Phi, f, \eta(\varpi)X\big)$$

The substitution $X \mapsto \eta(\varpi)X$ is a ring automorphism of $\mathbb{C}[X, X^{-1}]$ fixing $X = 0$, so it carries the module attached to π' onto the module attached to $\eta\pi'$, generators to generators, and respects the normalization of proposition 5.2.6: writing the two generators as $P_{\pi'}(X)^{-1}$ and $P_{\eta\pi'}(X)^{-1}$ with $P(0) = 1$, this says $P_{\eta\pi'}(X) = P_{\pi'}(\eta(\varpi)X)$. In particular $L(\eta\pi', s) = 1$ if and only if $L(\pi', s) = 1$. Second, twisting commutes with the contragredient, $(\eta\pi')^\vee = \check{\eta}\,(\pi')^\vee$ directly from the definition of the twist, and $\mathrm{St}_G^\vee \cong \mathrm{St}_G$ by (2.9), so $(\varphi\mathrm{St}_G)^\vee = \check{\varphi}\mathrm{St}_G$.

Now suppose $L(\varphi\mathrm{St}_G, s) = 1$. Since $\check{\varphi}\mathrm{St}_G = \eta \cdot (\varphi\mathrm{St}_G)$ for the unramified character $\eta = \varphi^{-2}$, the first preliminary gives $L(\check{\varphi}\mathrm{St}_G, s) = 1$, and the second turns this into $L((\varphi\mathrm{St}_G)^\vee, 1 - s) = 1$. Step 1 then gives

$$\varepsilon(\varphi\mathrm{St}_G, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)\,\frac{L(\check{\chi}_1, 1 - s)L(\check{\chi}_2, 1 - s)}{L(\chi_1, s)L(\chi_2, s)}$$

By corollary 5.2.9 the left-hand side is a monomial in q^{-s}, and the two GL_1 local constants are monomials by corollary 5.1.14; so the remaining quotient must be a monomial. Write $Y = q^{-s}$ and $a_i = \chi_i(\varpi)$, both non-zero. Then $L(\chi_i, s)^{-1} = 1 - a_iY$ and $L(\check{\chi}_i, 1 - s)^{-1} = 1 - a_i^{-1}q^{-1}Y^{-1}$, so the quotient is $N(Y)/D(Y^{-1})$ with

$$N(Y) = (1 - a_1Y)(1 - a_2Y) = a_1a_2\big(Y - a_1^{-1}\big)\big(Y - a_2^{-1}\big), \qquad D(Y^{-1}) = \big(1 - a_1^{-1}q^{-1}Y^{-1}\big)\big(1 - a_2^{-1}q^{-1}Y^{-1}\big)$$

The numerator is supported in degrees 0 to 2 in Y and the denominator in degrees -2 to 0, so if $N(Y) = cY^mD(Y^{-1})$ then $m = 2$ and $m - 2 = 0$, and

$$Y^2D(Y^{-1}) = \big(Y - a_1^{-1}q^{-1}\big)\big(Y - a_2^{-1}q^{-1}\big)$$

Comparing roots forces the multisets $\{a_1^{-1}, a_2^{-1}\}$ and $\{a_1^{-1}q^{-1}, a_2^{-1}q^{-1}\}$ to agree. Matching a_1^{-1} with $a_1^{-1}q^{-1}$ gives $q = 1$, and matching a_1^{-1} with $a_2^{-1}q^{-1}$ and a_2^{-1} with $a_1^{-1}q^{-1}$ gives $a_1 = q^2a_1$, again $q^2 = 1$. Both are false, so the first alternative cannot hold and $L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$.

Step 5: the local constant. With $L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$ the same computation applied to $\check{\varphi}$ gives $L((\varphi\mathrm{St}_G)^\vee, 1 - s) = L(\check{\chi}_1, 1 - s)$, because the character playing the role of χ_2 for $\check{\varphi}$ is $\check{\varphi}\| - \|^{1/2} = \check{\chi}_1$. Hence

$$\varepsilon(\varphi\mathrm{St}_G, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)\frac{L(\chi_2, s)}{L(\check{\chi}_1, 1 - s)} = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)\frac{L(\check{\chi}_2, 1 - s)}{L(\chi_1, s)}$$

after cancelling $L(\check{\chi}_1, 1 - s)$ and $L(\chi_2, s)$. This formula also holds in the ramified case of Step 2, where every L-factor in sight is 1. Finally, when φ is unramified, put $W = \chi_1(\varpi)q^{-s}$; since $\check{\chi}_2(\varpi) = \chi_1(\varpi)^{-1}q$ one has $L(\check{\chi}_2, 1 - s)^{-1} = 1 - W^{-1}$ and $L(\chi_1, s)^{-1} = 1 - W$, so the correcting factor is

$$\frac{1 - W}{1 - W^{-1}} = -W = -\chi_1(\varpi)q^{-s}$$

as we sought to show.

So the special representation keeps the factor $L(\chi_2, s)$ and loses $L(\chi_1, s)$: of the two poles carried by the reducible induced representation, exactly the one belonging to the character $\chi_1 = \varphi\| - \|^{-1/2}$ is annihilated by the condition $\int_K \tilde{\varphi} = 0$. The mechanism behind that condition is the one part of this section that is not bookkeeping: the residue at the lost pole is computed by integrating Φ over the matrices with vanishing first column, and that integral is blind to the left K-variable, so the residue functional factors through the pairing of $\tilde{\varphi}$ against the constant function. Killing the constants kills the residue.

Three formulas are now available for the same short exact sequence, and setting them side by side says something about which of the two invariants carries the content.

5.4.4 Remark: The Invariants Along the Steinberg Sequence Keep $\chi_1 = \varphi\|-\|^{-1/2}$, $\chi_2 = \varphi\|-\|^{1/2}$, so that $\mathrm{Ind}_B^G\varphi_T = \iota_B^G(\chi_1 \otimes \chi_2)$ and (2.7) reads $0 \to \varphi_G \to \iota_B^G(\chi_1 \otimes \chi_2) \to \varphi\mathrm{St}_G \to 0$. Proposition 5.4.2 and proposition 5.4.3 give

$$\gamma(\varphi_G, s, \psi) = \gamma(\varphi\mathrm{St}_G, s, \psi) = \gamma(\chi_1, s, \psi)\gamma(\chi_2, s, \psi)$$

but

$$L(\varphi_G, s) = L(\chi_1, s)L(\chi_2, s), \qquad L(\varphi\mathrm{St}_G, s) = L(\chi_2, s)$$

The unnormalized invariant γ is therefore multiplicative along the sequence, in the sense that it is computed from the two GL_1 constituents of the inducing character and is the same for the sub and for the quotient. The L-factor is not: it is not determined by the inducing character alone, and the factor $L(\chi_1, s)$ is present for the subrepresentation and absent for the quotient.

Since $\varepsilon = \gamma \cdot L(\pi, s)/L(\check{\pi}, 1-s)$, the entire difference between the two local constants is the difference between the two L-factors:

$$\frac{\varepsilon(\varphi\mathrm{St}_G, s, \psi)}{\varepsilon(\varphi_G, s, \psi)} = \frac{L(\check{\chi}_2, 1-s)}{L(\chi_1, s)} = \begin{cases} -\chi_1(\varpi)q^{-s} & \text{if } \varphi \text{ is unramified,} \\ 1 & \text{if } \varphi \text{ is ramified} \end{cases}$$

Two consequences, in opposite directions. When φ is unramified and ψ has level one, the explicit values are $\varepsilon(\varphi_G, s, \psi) = \varphi(\varpi)^{-2}q^{2s-1}$ and $\varepsilon(\varphi\mathrm{St}_G, s, \psi) = -\varphi(\varpi)^{-1}q^{s-1/2}$, so the two constituents differ in the exponent of q as well as in the coefficient, and the symmetry of corollary 5.2.9 holds for the second with the two minus signs cancelling: $(-\varphi(\varpi)^{-1}q^{s-1/2})(-\varphi(\varpi)q^{1/2-s}) = 1 = \omega_\pi(-1)$. When φ is ramified, on the other hand, the two constituents have the *same* L-factor, namely 1, and the *same* local constant. A single pair (L, ε) therefore never separates the representations of G on its own.

What separates them is the family of all twists, under which the two invariants behave quite differently. The L-factor does move on the non-cuspidal side: if φ is ramified then $L(\varphi\mathrm{St}_G, s) = 1$, but twisting by $\check{\varphi}$ returns St_G, whose L-factor is $L(\|-\|^{1/2}, s) = (1 - q^{-s-1/2})^{-1} \neq 1$. On the cuspidal side it does not move at all: it is 1 for every cuspidal representation by proposition 5.3.1, and a twist of a cuspidal is again cuspidal, since twisting does not change the support of a matrix coefficient (theorem 3.1.3). The family of twisted L-factors therefore detects cuspidality, which is the use section 5.5 makes of it, and says nothing beyond that. The local constant collapses nowhere: it is a monomial cq^{-fs} in which both the coefficient c and the exponent f move as the twist varies, as the formulas above already show on the non-cuspidal side. This is the asymmetry that section 5.5 exploits, and it is why the characterization proved there is by ε with L along for the ride, rather than the other way round.

Everything above becomes explicit when every character in sight is unramified: the poles of the L-function can be located exactly, and the local constant is a single monomial whose coefficient is the central character evaluated at a prime element.

Example 5.4.5: The Unramified Principal Series

Let χ_1, χ_2 be unramified characters of $F^\times$, put $a_i = \chi_i(\varpi) \in \mathbb{C}^\times$, and let $\psi \in \widehat{F}$ have level one. Set $\pi = \iota_B^G(\chi_1 \otimes \chi_2)$.

Irreducibility. By lemma 2.4.15 the representation π is reducible precisely when $\chi_1\chi_2^{-1} = \|-\|^{\pm 1}$. Both sides are unramified, so this is the numerical condition $a_1/a_2 = q^{\mp 1}$, and π is irreducible as soon as $a_1 \neq qa_2$ and $a_2 \neq qa_1$. Note that $a_1 = a_2$ is allowed: the case $\chi_1 = \chi_2 = \varphi$ gives $\chi_1\chi_2^{-1} = 1$, which is not $\|-\|^{\pm 1}$ because $\|\varpi\|^{\pm 1} = q^{\mp 1} \neq 1$.

The L-factor and its poles. By definition 5.1.9 and proposition 5.4.1,

$$L(\pi, s) = \frac{1}{(1 - a_1 q^{-s})(1 - a_2 q^{-s})}$$

whose poles are the s with $q^s = a_1$ or $q^s = a_2$, each family being a single point repeated with period $2\pi i/\log q$. Corollary 5.1.10 says the two families are either disjoint or identical, and identical exactly when $\chi_1 = \chi_2$. So $L(\pi, s)$ has two disjoint families of simple poles when $\chi_1 \neq \chi_2$, and one family of double poles when $\chi_1 = \chi_2$; the latter does occur for an irreducible π, by the previous paragraph.

The local constant. Proposition 5.1.17 gives $\varepsilon(\chi_i, s, \psi) = q^{s-1/2}a_i^{-1}$, so by proposition 5.4.1

$$\varepsilon(\pi, s, \psi) = q^{s-1/2}a_1^{-1} \cdot q^{s-1/2}a_2^{-1} = (a_1a_2)^{-1}q^{2s-1}$$

which is the monomial cq^{-fs} of corollary 5.2.9 with $c = (a_1a_2)^{-1}q^{-1}$ and $f = -2$. The central character of π is $\chi_1\chi_2$: for $z = z_0 I$ in the centre the representation acts by right translation, so $\varphi(gz) = \varphi(zg) = (\delta_B^{-1/2}\chi)(z)\varphi(g)$, and $\delta_B(z) = \|z_0/z_0\| = 1$ leaves $\chi_1(z_0)\chi_2(z_0)$. Hence $\omega_\pi(\varpi) = a_1a_2$ and $\varepsilon(\pi, s, \psi) = \omega_\pi(\varpi)^{-1}q^{2s-1}$.

Two checks. First, the symmetry of corollary 5.2.9. Since $\check{\pi} = \iota_B^G(\check{\chi}_1 \otimes \check{\chi}_2)$, the same formula gives $\varepsilon(\check{\pi}, 1-s, \psi) = a_1a_2\, q^{2(1-s)-1}$, and

$$\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = (a_1a_2)^{-1}q^{2s-1} \cdot a_1a_2q^{1-2s} = 1$$

which is $\omega_\pi(-1)$, because $-1 \in U_F$ and ω_π is unramified. Second, the dependence on ψ. Replacing ψ by $\varpi\psi$, lemma 5.1.16 gives $\varepsilon(\chi_i, s, \varpi\psi) = \chi_i(\varpi)\|\varpi\|^{s-1/2}\varepsilon(\chi_i, s, \psi) = a_iq^{1/2-s} \cdot q^{s-1/2}a_i^{-1} = 1$, so $\varepsilon(\pi, s, \varpi\psi) = 1$. The monomial computed above is therefore entirely an artefact of the choice of additive character, and disappears for a suitable one.

A numerical instance. Take $\chi_1 = 1$ and $\chi_2 = \|-\|^{1/2}$, so $a_1 = 1$ and $a_2 = q^{-1/2}$. Then $a_1/a_2 = q^{1/2}$ is neither q nor q^{-1}, so π is irreducible, and

$$L(\pi, s) = \frac{1}{(1 - q^{-s})(1 - q^{-s-1/2})}, \qquad \varepsilon(\pi, s, \psi) = q^{2s-1/2}$$

with simple poles at $q^s = 1$ and at $q^s = q^{-1/2}$.

Exercise 5.4.1

1. Let $\chi = \chi_1 \otimes \chi_2$ be a character of T with $\pi = \iota_B^G\chi$ irreducible, and let η be a character of $F^\times$. Show that $\eta\pi \cong \iota_B^G(\eta\chi_1 \otimes \eta\chi_2)$ and that this representation is again irreducible, and write

down $L(\eta\pi, s)$ and $\varepsilon(\eta\pi, s, \psi)$. Evaluate both when η, χ_1, χ_2 are unramified and ψ has level one.

2. Let π be any irreducible non-cuspidal representation of G and let $a \in F^\times$. Show that

$$\varepsilon(\pi, s, a\psi) = \omega_\pi(a)\, \|a\|^{2s-1}\, \varepsilon(\pi, s, \psi)$$

Treat the three families of theorem 2.4.16 separately, and identify ω_π in each case.

3. Let $\pi = \iota_B^G(\chi_1 \otimes \chi_2)$ be irreducible with χ_1, χ_2 unramified. Show that $L(\pi, s)$ has a double pole at each of its poles if and only if $\chi_1 = \chi_2$, and that this case is not vacuous. In that case, writing $\chi_1 = \chi_2 = \varphi$, compare the three L-factors $L(\pi, s)$, $L(\varphi_G, s)$ and $L(\varphi\mathrm{St}_G, s)$ and show that no two of them agree.

4. Determine exactly which irreducible non-cuspidal representations π of G satisfy $L(\pi, s) = 1$. Combining the answer with proposition 5.3.1, explain why the L-function alone cannot decide whether an irreducible representation is cuspidal.

5. Verify the identity $\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$ of corollary 5.2.9 directly for $\pi = \varphi\mathrm{St}_G$, for an arbitrary character φ, starting from the formula of proposition 5.4.3 and using only corollary 5.1.14.

5.5 The Converse Theorem for GL_2

Everything the argument needs is now on the table: section 5.3 and section 5.4 compute $L(\pi, s)$ for every irreducible smooth representation of G, and $\varepsilon(\pi, s, \psi)$ as far as elementary means reach. This section shows that those invariants, read across all twists $\chi\pi$ rather than at π alone, determine π up to isomorphism, and records the consequence that at most one map from Galois-side parameters to representations of G can preserve them.

The first thing to ask of the invariants is that they see the division the classification is built on. Nothing in the definition of the zeta integral mentions cuspidality, yet the two families feed into it in completely different ways. The coefficients of a cuspidal representation vanish outside a set compact modulo the centre (theorem 3.1.3), so the only unbounded direction is the central one; a non-cuspidal representation is assembled from characters of the torus (theorem 2.4.16), and its zeta integral inherits their poles. The L-factor records exactly that difference, and it records it uniformly in the twist.

Lemma 5.5.1: L-functions Detect Cuspidality

Let (π, V) be an irreducible smooth representation of G. Then

$$L(\chi\pi, s) = 1 \quad \text{for every character } \chi \text{ of } F^\times$$

if and only if π is cuspidal.

***Proof*:**
Step 1: twisting preserves irreducibility, cuspidality and the three non-cuspidal families. Let χ be a character of $F^\times$ and let $\chi\pi$ denote the twist of π by χ, acting on the same space V by $\chi\pi(g) = \chi(\det g)\pi(g)$. For each $g \in G$ the operators $\chi\pi(g)$ and $\pi(g)$ differ by the non-zero scalar

$\chi(\det g)$, so a subspace of V is stable under π exactly when it is stable under $\chi\pi$, and a vector has open stabilizer for one exactly when it does for the other. The two representations therefore have the same subrepresentations, and $\chi\pi$ is smooth and irreducible.

For $n \in N$ one has $\det n = 1$, so $\chi\pi(n) = \pi(n)$: the restrictions of π and $\chi\pi$ to N are equal as representations of N. The subspace $V(N)$ spanned by the vectors $v - \pi(n)v$, $v \in V$, $n \in N$, therefore does not change, and neither does the Jacquet module $V_N = V/V(N)$. Cuspidality is the vanishing of V_N, so π is cuspidal if and only if $\chi\pi$ is. Twisting can move a cuspidal representation around inside the cuspidal family, but never out of it.

The same computation controls the induced representations. If ξ is a character of T, inflated to B, then the map $\Lambda\colon f \mapsto \big(g \mapsto \chi(\det g) f(g)\big)$ sends $\mathrm{Ind}_B^G \xi$ into $\mathrm{Ind}_B^G(\chi \cdot \xi)$, because $\Lambda(f)(bg) = \chi(\det b)\xi(b)\,\Lambda(f)(g)$. Both target actions are by right translation, so for $h \in G$

$$\Lambda\big(\chi\mathrm{Ind}_B^G\xi(h) f\big)(g) = \chi(\det g)\chi(\det h) f(gh) = \Lambda(f)(gh) = \big(\mathrm{Ind}_B^G(\chi\cdot\xi)(h)\,\Lambda f\big)(g)$$

Its inverse is multiplication by $\chi^{-1} \circ \det$, so $\chi\,\mathrm{Ind}_B^G\xi \cong \mathrm{Ind}_B^G(\chi\cdot\xi)$. Since $\chi \cdot (\delta_B^{-1/2} \otimes \sigma) = \delta_B^{-1/2} \otimes (\chi\cdot\sigma)$, the same holds for normalized induction (2.10):

$$\chi\,\iota_B^G(\mu_1 \otimes \mu_2) \cong \iota_B^G(\chi\mu_1 \otimes \chi\mu_2) \tag{5.12}$$

Directly from the same description one also has $\chi(\varphi \circ \det) = (\chi\varphi) \circ \det$, and twisting the exact sequence (2.7) attached to φ by χ gives $\chi(\varphi \cdot \mathrm{St}_G) = (\chi\varphi) \cdot \mathrm{St}_G$. So each of the three families of theorem 2.4.16 is stable under twisting, with the parameter μ_i or φ multiplied by χ.

Step 2: a cuspidal representation has trivial twisted L-factors. If π is cuspidal then $\chi\pi$ is irreducible and cuspidal for every χ, by Step 1. Applying proposition 5.3.1 to $\chi\pi$ gives $L(\chi\pi, s) = 1$ for every χ.

Step 3: a non-cuspidal representation has a twist with a pole. Suppose π is not cuspidal. By theorem 2.4.16 there are three possibilities, and in each one a single twist suffices.

If $\pi \cong \iota_B^G(\mu_1 \otimes \mu_2)$ is an irreducible normalized induction, take $\chi = \mu_1^{-1}\| - \|^{1/2}$. By (5.12) and proposition 5.4.1,

$$L(\chi\pi, s) = L(1, s)\, L(\mu_1^{-1}\mu_2, s)$$

and the first factor is $(1 - q^{-s})^{-1}$ by definition 5.1.9, so $L(\chi\pi, s) \neq 1$.

If $\pi \cong \varphi \circ \det$, take $\chi = \varphi^{-1}$, so that $\chi\pi$ is the trivial representation 1_G. By proposition 5.4.2,

$$L(1_G, s) = L(1, s)\, L(\| - \|^{-1}, s) = \frac{1}{(1 - q^{-s})(1 - q^{1-s})}$$

which is not 1. If $\pi \cong \varphi \cdot \mathrm{St}_G$, take $\chi = \varphi^{-1}$ again; then $\chi\pi \cong \mathrm{St}_G$ and proposition 5.4.3 gives $L(\mathrm{St}_G, s) = L(1, s) = (1 - q^{-s})^{-1} \neq 1$.

In every case some twist has a non-trivial L-factor. The contrapositive of Step 3 together with Step 2 gives the stated equivalence, as we sought to show.

The lemma is the first statement in which the invariants *characterize* rather than accompany. What it detects is intrinsic: by theorem 3.1.3 cuspidality is a property of the matrix coefficients themselves, and the L-factor is built out of those coefficients. The quantifier over all χ is doing real work, and cannot be dropped. A single L-factor sees only the unramified part of the data: if θ is a character with both θ and θ^2 ramified, then $\iota_B^G(\theta \otimes \check{\theta})$ is irreducible by lemma 2.4.15 and non-cuspidal, yet its L-factor is $L(\theta\|-\|^{-1/2}, s)L(\check{\theta}\|-\|^{-1/2}, s) = 1$ by proposition 5.4.1 and lemma 5.1.15, exactly as for a cuspidal representation. Twisting by $\check{\theta}\|-\|^{1/2}$ is what pushes one of the two characters governing the L-factor to an unramified one, where a pole appears; the computation is carried out in example 5.5.4.

With the two families separated, what remains is to show that inside each family the invariants pin down the representation. On the non-cuspidal side the classification supplies the parameters and section 5.4 supplies the dictionary between parameters and L-factors, so the problem becomes one of reading a multiset of characters off a family of rational functions. The three formulas of section 5.4 have a common shape: to each irreducible non-cuspidal π there is attached a multiset $\mathcal{S}(\pi)$ of characters of $F^\times$ with

$$L(\pi, s) = \prod_{\mu \in \mathcal{S}(\pi)} L(\mu, s) \tag{5.13}$$

namely

$$\mathcal{S}\big(\iota_B^G(\mu_1 \otimes \mu_2)\big) = \{\mu_1\|-\|^{-1/2},\, \mu_2\|-\|^{-1/2}\}, \qquad \mathcal{S}(\varphi \circ \det) = \{\varphi,\, \varphi\|-\|^{-1}\}, \qquad \mathcal{S}(\varphi \cdot \mathrm{St}_G) = \{\varphi\}$$

by proposition 5.4.1, proposition 5.4.2 and proposition 5.4.3 respectively. The half-integral powers record the normalization: $L(\pi, s)$ is the generator attached in proposition 5.2.6 to the unshifted integral $\zeta(\Phi, f, s)$, and the denominator $P_{\chi_1\|-\|^{-1}}P_{\chi_2}$ produced there for a subquotient of $\mathrm{Ind}_B^G(\chi_1 \otimes \chi_2)$ is exactly what (5.13) sharpens. The first multiset is well defined as an *unordered* pair because $\iota_B^G(\mu_1 \otimes \mu_2) \cong \iota_B^G(\mu_2 \otimes \mu_1)$ and no other coincidences occur (lemma 2.4.15, theorem 2.4.16). By Step 1 of the proof above, $\mathcal{S}(\chi\pi) = \chi\mathcal{S}(\pi) = \{\chi\mu \big| \mu \in \mathcal{S}(\pi)\}$ for every character χ.

Theorem 5.5.2: Converse Theorem for GL_2

Let π and π' be irreducible smooth representations of G with the same central character, $\omega_\pi = \omega_{\pi'}$, and let $\psi \in \widehat{F}$, the group of characters of the additive group of F, have level one. Suppose that

$$L(\chi\pi, s) = L(\chi\pi', s) \qquad \text{and} \qquad \varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$$

for every character χ of $F^\times$. Then $\pi \cong \pi'$.

Fixing ψ of level one costs nothing. It is the normalization under which the explicit local constants and the stability theorem are stated, and the exercises below show that the hypothesis holds for one non-trivial additive character exactly when it holds for every other.

Steps 1 and 2 of the argument below are complete proofs. Step 3 quotes two inputs, and neither of them is new here. The first is the stability theorem of theorem 5.3.6, stated but not proved in section 5.3. The second is the Kirillov model, the same machinery that section 5.3 had to quote for the explicit local constant of a cuspidal representation; it is the one construction of this chapter that the book does not build. Everything the book does build is used in full, the environment is marked accordingly, and both quoted inputs are named again at the point of use.

***Proof Key ideas*:**
Step 1: the two representations lie in the same family. The hypothesis includes $L(\chi\pi, s) = L(\chi\pi', s)$ for every χ, so the condition of lemma 5.5.1 holds for π exactly when it holds for π'. Hence π and π' are either both cuspidal or both non-cuspidal.

Step 2: the non-cuspidal case. Assume both are non-cuspidal, and let $\mathcal{U}$ denote the group of unramified characters of $F^\times$. The claim is that the multiset $\mathcal{S}(\pi)$ is determined by the family $\{L(\chi\pi, s)\}_\chi$.

Write $L(\pi, s) = P_\pi(q^{-s})^{-1}$ with $P_\pi \in \mathbb{C}[X]$ and $P_\pi(0) = 1$; this normalization determines P_π uniquely by proposition 5.2.6, so two irreducible representations have equal L-factors precisely when they have equal polynomials P. Combining (5.13) with $\mathcal{S}(\chi\pi) = \chi\mathcal{S}(\pi)$ and with the two-case formula of definition 5.1.9 and lemma 5.1.15,

$$P_{\chi\pi}(X) = \prod_{\substack{\mu\in\mathcal{S}(\pi)\\ \chi\mu \text{ unramified}}} \big(1 - \chi\mu(\varpi)X\big) \tag{5.14}$$

the product being taken with multiplicity. A character μ satisfies "$\chi\mu$ unramified" exactly when χ lies in the coset $\mu^{-1}\mathcal{U}$, so the product in (5.14) runs over the elements of $\mathcal{S}(\pi)$ lying in $\chi^{-1}\mathcal{U}$.

Fix a coset $\mathcal{C}$ of $\mathcal{U}$ in the character group of $F^\times$ and choose χ with $\chi^{-1}\mathcal{U} = \mathcal{C}$. The polynomial $P_{\chi\pi}$ then has degree equal to the number of elements of $\mathcal{S}(\pi)$ lying in $\mathcal{C}$, counted with multiplicity, and its roots are the numbers $\chi\mu(\varpi)^{-1}$ for those μ. An unramified character is trivial on U_F and $F^\times$ is generated by ϖ together with U_F, so such a character is determined by its value at ϖ; equivalently, distinct unramified characters have disjoint pole sets (corollary 5.1.10). The multiset $\{\chi\mu | \mu \in \mathcal{S}(\pi) \cap \mathcal{C}\}$ is therefore read off from $P_{\chi\pi}$, and multiplying back by χ^{-1} recovers $\mathcal{S}(\pi) \cap \mathcal{C}$, again with multiplicity. Every element of $\mathcal{S}(\pi)$ lies in exactly one coset, so running the recovery over all cosets produces the whole of $\mathcal{S}(\pi)$. The recovery uses nothing but the family $\{L(\chi\pi, s)\}_\chi$, and it is carried out by the same choices of χ for π and for π'; the hypothesis therefore gives

$$\mathcal{S}(\pi) = \mathcal{S}(\pi')$$

It remains to see that $\mathcal{S}$ separates the three families and determines the representation inside each. If $\mathcal{S}(\pi)$ has one element, then π is neither of the first two types, since those have two; so $\pi \cong \varphi \cdot \mathrm{St}_G$ with $\mathcal{S}(\pi) = \{\varphi\}$, and φ is the unique element of $\mathcal{S}(\pi)$. The same applies to π', and $\pi \cong \varphi \cdot \mathrm{St}_G \cong \pi'$.

If $\mathcal{S}(\pi) = \{\nu_1, \nu_2\}$ has two elements, the ratio $\nu_1\nu_2^{-1}$ decides between the two remaining types. For $\pi \cong \varphi \circ \det$ the ratio is $\| - \|^{\pm 1}$. For $\pi \cong \iota_B^G(\mu_1 \otimes \mu_2)$ irreducible the ratio is $\mu_1\mu_2^{-1}$, and lemma 2.4.15 forbids exactly that value: the normalized induction $\iota_B^G(\mu_1 \otimes \mu_2)$ is reducible if and only if $\mu_1\mu_2^{-1} = \| - \|^{\pm 1}$. So the ratio, which is a function of the multiset alone, tells the two types apart. In the first case, relabelling so that $\nu_1\nu_2^{-1} = \| - \|$ gives $\varphi = \nu_1$ and $\pi \cong \varphi \circ \det \cong \pi'$. In the second, $\mu_i = \nu_i \| - \|^{1/2}$ recovers the inducing characters and $\pi \cong \iota_B^G(\mu_1 \otimes \mu_2) \cong \pi'$, the order of the two being immaterial because $\iota_B^G(\mu_1 \otimes \mu_2) \cong \iota_B^G(\mu_2 \otimes \mu_1)$ (lemma 2.4.15). This settles the non-cuspidal case; note that neither the ε-factors nor the equality of central characters was used.

Step 3: the cuspidal case. Assume both are cuspidal. Then $L(\chi\pi, s) = L(\chi\pi', s) = 1$ for every χ by lemma 5.5.1, so the L-hypothesis is vacuous and all the information sits in the local constants. By corollary 5.3.2 the hypothesis is equivalent to $\gamma(\chi\pi, s, \psi) = \gamma(\chi\pi', s, \psi)$ for every χ.

Two consequences are immediate. First, each $\varepsilon(\chi\pi, s, \psi)$ is a monomial $c\,q^{-ms}$ with $c \in \mathbb{C}^\times$ and $m \in \mathbb{Z}$ by corollary 5.2.9, and that exponent is what definition 5.3.4 records; the hypothesis therefore forces the epsilon exponents of $\chi\pi$ and $\chi\pi'$ to agree for every χ, together with the leading constants. Second, the deeply ramified twists carry no information at all: by theorem 5.3.6 there is an integer n_π such that for every χ of level at least n_π the constant $\varepsilon(\chi\pi, s, \psi)$ depends only on χ, ψ and the central character ω_π, and likewise for π' with some $n_{\pi'}$. Since $\omega_\pi = \omega_{\pi'}$, the two sides of the hypothesis agree automatically for every χ of level at least $\max(n_\pi, n_{\pi'})$. The whole content of the hypothesis is thus concentrated in the twists of bounded level, of which there are finitely many up to unramified twist.

The remaining implication, that the local constants of those finitely many levels of twist determine a cuspidal representation, is the one step of this chapter that the machinery built here does not reach. The standard argument realizes π on the mirabolic subgroup in its Kirillov model, a space of functions on $F^\times$ carrying an explicit action of B, and recovers that model from the γ-factors of twists by Mellin inversion; in the language of theorem 3.5.5, what is being shown is that the twisted local constants determine the G-conjugacy class of the cuspidal type $(\mathfrak{A}, J, \Lambda)$. The Kirillov model is precisely the construction that section 5.3 had to quote rather than build, and it is quoted again here; see [1, §25]. Granting that input, $\pi \cong \pi'$ in the cuspidal case as well, and the theorem follows.

The theorem is an injectivity statement and nothing more. It says that the assignment

$$\pi \longmapsto \Big(\omega_\pi,\ \big\{L(\chi\pi, s)\big\}_\chi,\ \big\{\varepsilon(\chi\pi, s, \psi)\big\}_\chi\Big)$$

is injective on isomorphism classes of irreducible smooth representations of G. It says nothing about which families of rational functions arise: there is no claim here that a prescribed system of L- and ε-factors is realized by some representation. Existence is a separate problem, solved by construction rather than by comparison. The proof also distributes the work very unevenly. In the non-cuspidal case the L-factors alone suffice and the local constants are never consulted; in the cuspidal case the L-factors are all equal to 1 and the local constants carry the entire burden.

The role of the quantifier over χ deserves to be stated plainly. Matching invariants for the trivial twist alone is far too weak. The L-factor of a principal series induced from two ramified characters equals 1, exactly as for a cuspidal representation, so the untwisted L-factor cannot even see the division of lemma 5.5.1; and the untwisted ε-factor is a single monomial, which is very little data with which to distinguish an infinite family. The mechanism by which twisting extracts more is visible in (5.14): the L-factor is blind to ramified characters, and a twist by μ^{-1} moves a ramified μ to the trivial character, where the L-factor acquires a pole. Running over all χ interrogates every character of $F^\times$ in turn. Example 5.5.4 makes the failure of the untwisted invariants concrete: it produces two non-isomorphic irreducible representations agreeing in central character, in L-factor and in ε-factor.

The immediate payoff is a uniqueness statement about correspondences, and it is purely formal: once the theorem is in hand, nothing about the Galois side is needed to conclude that a map preserving these invariants is unique if it exists at all.

Corollary 5.5.3: Uniqueness of an Invariant-Preserving Correspondence

Let $\psi \in \widehat{F}$ have level one. Let $\mathcal{P}$ be a set, and suppose that each $\rho \in \mathcal{P}$ is equipped with a character ω_ρ of $F^\times$ and, for each character χ of $F^\times$, with functions $L(\chi \otimes \rho, s)$ and $\varepsilon(\chi \otimes \rho, s, \psi)$. Let

$$\Pi, \Pi' \colon \mathcal{P} \longrightarrow \{\text{isomorphism classes of irreducible smooth representations of } G\}$$

be two maps such that, for every $\rho \in \mathcal{P}$ and every character χ of $F^\times$,

$$\omega_{\Pi(\rho)} = \omega_\rho, \qquad L(\chi\,\Pi(\rho), s) = L(\chi \otimes \rho, s), \qquad \varepsilon(\chi\,\Pi(\rho), s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi)$$

and such that Π' satisfies the same three conditions. Then $\Pi = \Pi'$.

***Proof*:**
Fix $\rho \in \mathcal{P}$ and put $\pi = \Pi(\rho)$, $\pi' = \Pi'(\rho)$. Both are irreducible smooth representations of G, and the first condition applied to each map gives

$$\omega_\pi = \omega_\rho = \omega_{\pi'}$$

For every character χ of $F^\times$ the second condition gives $L(\chi\pi, s) = L(\chi \otimes \rho, s) = L(\chi\pi', s)$, and the third gives $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi) = \varepsilon(\chi\pi', s, \psi)$. The hypotheses of theorem 5.5.2 are met, so $\pi \cong \pi'$. Since ρ was arbitrary and both maps take values in isomorphism classes, $\Pi = \Pi'$, completing the proof.

The set $\mathcal{P}$ is left abstract on purpose: the argument uses only that its elements come with a character of $F^\times$ and a system of twisted invariants, never what they are. Later, in chapter 6, that data is constructed for two-dimensional Deligne representations of the Weil group, and chapter 7 applies the corollary with $\mathcal{P}$ taken to be that set of parameters, which is where the word "the" in "the local Langlands correspondence" comes from. One point of bookkeeping is needed there. The invariants of this chapter are attached to the unshifted integral $\zeta(\Phi, f, s)$ of proposition 5.2.6, whereas the factors usually matched against Galois-side parameters are the shifted ones $L(\pi, s + \frac{1}{2})$ and $\varepsilon(\pi, s + \frac{1}{2}, \psi)$; the two conventions must not be mixed. Nothing in the corollary depends on which is chosen, since it compares two maps into the same set of representations and a shift applied on both sides at once changes neither its hypotheses nor its conclusion. What the corollary does not do is produce the map. Existence requires building, for each parameter, a representation of G with the prescribed invariants, and that construction proceeds family by family: characters of the torus for the reducible parameters, admissible pairs and the cuspidal types of theorem 3.5.5 for the irreducible ones. The corollary guarantees that there is at most one answer, not that there is one.

The promised example makes the necessity of the quantifier concrete. It exhibits two non-isomorphic irreducible representations whose untwisted invariants agree in every respect, and locates the ramified twist at which they part.

Example 5.5.4: Two Principal Series with Equal Untwisted Invariants

Assume $q > 3$, fix $\psi \in \widehat{F}$ of level one, and let η be the unramified character $\eta(x) = (-1)^{v_F(x)}$ of $F^\times$, so that $\eta^2 = 1$, $\eta \neq 1$ and $\eta(\varpi) = -1$.

Choose a character θ of $F^\times$ of level $n \geq 0$ such that both θ and θ^2 are ramified. Such a θ exists: take θ trivial on U_F^1 with $\theta|_{U_F}$ of order at least 3, which is possible because $U_F/U_F^1 \cong k^\times$ is cyclic of order $q-1 \geq 3$; then $\theta^2|_{U_F} \neq 1$. Put

$$\pi = \iota_B^G(\theta \otimes \check{\theta}), \qquad \pi' = \eta\pi \cong \iota_B^G(\eta\theta \otimes \eta\check{\theta})$$

the second isomorphism being (5.12).

Both are irreducible: the relevant ratio is $\theta\check{\theta}^{-1} = \theta^2$ for π and $(\eta\theta)(\eta\check{\theta})^{-1} = \theta^2$ for π', and θ^2 is ramified, hence not equal to $\| - \|^{\pm 1}$, so lemma 2.4.15 applies. They have the same central character: $\delta_B(\mathrm{diag}(a,a)) = \|a/a\| = 1$ by proposition 2.2.8, so the normalizing factor $\delta_B^{-1/2}$ contributes nothing on Z and the central character is the inducing character of T evaluated at $\mathrm{diag}(a,a)$. This gives $\omega_\pi = \theta\check{\theta} = 1$ and $\omega_{\pi'} = \eta^2\theta\check{\theta} = 1$.

They are not isomorphic. By theorem 2.4.16 an isomorphism would force $\{\eta\theta, \eta\check{\theta}\} = \{\theta, \check{\theta}\}$. If $\eta\theta = \theta$ then $\eta = 1$, which is false; if $\eta\theta = \check{\theta}$ then $\theta^2 = \eta^{-1} = \eta$, which is false because θ^2 is ramified and η is not.

Their L-factors agree. Write $\nu_1 = \theta\| - \|^{-1/2}$ and $\nu_2 = \check{\theta}\| - \|^{-1/2}$, so that $\mathcal{S}(\pi) = \{\nu_1, \nu_2\}$ and $\mathcal{S}(\pi') = \{\eta\nu_1, \eta\nu_2\}$ by proposition 5.4.1. All four of these restrict on U_F to $\theta|_{U_F}^{\pm 1}$, because $\| - \|^{-1/2}$ and η are unramified, hence are ramified of level n; so lemma 5.1.15 and (5.13) give

$$L(\pi, s) = L(\nu_1, s)L(\nu_2, s) = 1 = L(\eta\nu_1, s)L(\eta\nu_2, s) = L(\pi', s)$$

Their ε-factors agree as well. Let μ be any ramified character of level n and let $c \in F^\times$ with $v_F(c) = -n$. Since η is trivial on U_F, the value $\check{\eta}(cx) = \eta(\varpi)^n$ is constant as x runs over U_F/U_F^{n+1}, so the Gauss sum of definition 5.1.19 satisfies

$$\tau(\eta\mu, \psi) = \sum_{x \in U_F/U_F^{n+1}} \check{\eta}(cx)\check{\mu}(cx)\psi(cx) = \eta(\varpi)^n\, \tau(\mu, \psi)$$

The character $\eta\mu$ again has level n, so theorem 5.1.18 turns this into $\varepsilon(\eta\mu, s, \psi) = \eta(\varpi)^n \varepsilon(\mu, s, \psi)$. Applying this to $\mu = \nu_1$ and to $\mu = \nu_2$ and multiplying, proposition 5.4.1 gives

$$\varepsilon(\pi', s, \psi) = \varepsilon(\eta\nu_1, s, \psi)\varepsilon(\eta\nu_2, s, \psi) = \eta(\varpi)^{2n}\varepsilon(\nu_1, s, \psi)\varepsilon(\nu_2, s, \psi) = \varepsilon(\pi, s, \psi)$$

because $\eta(\varpi)^2 = 1$.

So π and π' are non-isomorphic irreducible representations with the same central character, the same L-factor and the same ε-factor. The twist that separates them is $\chi = \nu_1^{-1} = \check{\theta}\| - \|^{1/2}$, which is ramified because $\check{\theta}$ is and $\| - \|^{1/2}$ is not. Indeed (5.12) gives $\chi\pi \cong \iota_B^G(\| - \|^{1/2} \otimes \check{\theta}^2\| - \|^{1/2})$ and $\chi\pi' \cong \iota_B^G(\eta\| - \|^{1/2} \otimes \eta\check{\theta}^2\| - \|^{1/2})$, so $\mathcal{S}(\chi\pi) = \{1, \check{\theta}^2\}$ and $\mathcal{S}(\chi\pi') = \{\eta, \eta\check{\theta}^2\}$; both are irreducible by the ratio test above, and since θ^2 is ramified while 1 and η are not, (5.13) together with definition 5.1.9 and lemma 5.1.15 gives

$$L(\chi\pi, s) = L(1, s)L(\check{\theta}^2, s) = \frac{1}{1 - q^{-s}}, \qquad L(\chi\pi', s) = L(\eta, s)L(\eta\check{\theta}^2, s) = \frac{1}{1 - \eta(\varpi)q^{-s}} = \frac{1}{1 + q^{-s}}$$

These are different, so the hypothesis of theorem 5.5.2 fails at $\chi = \check{\theta}\| - \|^{1/2}$, exactly as it must.

Exercise 5.5.1

1. Step 1 of the proof of lemma 5.5.1 shows that twisting preserves cuspidality by comparing Jacquet modules. Give the argument instead through matrix coefficients: show that $C(\chi\pi) = \{(\chi \circ \det)f \big| f \in C(\pi)\}$ for any smooth representation π of G and any character χ of $F^\times$, and deduce from definition 3.1.1 and theorem 3.1.3 that, for irreducible π, the representation π is cuspidal if and only if $\chi\pi$ is.

2. Let φ, φ' be characters of $F^\times$, let $\pi = \varphi \circ \det$ and let $\pi' = \varphi' \cdot \mathrm{St}_G$. Show that $L(\chi\pi, s) = L(\chi\pi', s)$ cannot hold for every character χ of $F^\times$. Show also that if $\varphi = \varphi'$ is ramified then the untwisted L-factors do agree, so that a twist is needed to separate these two families.

3. Let μ_1 be an unramified and μ_2 a ramified character of $F^\times$, and put $\pi = \iota_B^G(\mu_1 \otimes \mu_2)$.
    1. Show that π is irreducible.
    2. Determine the set of characters χ for which $L(\chi\pi, s) \neq 1$, and show it is a disjoint union of two cosets of the group $\mathcal{U}$ of unramified characters.
    3. Recover μ_1 and μ_2 explicitly from the family $\{L(\chi\pi, s)\}_\chi$.

4. In the situation of example 5.5.4, show that $L(\chi\pi, s) = L(\chi\pi', s)$ and $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for *every* unramified character χ of $F^\times$. Conclude that no unramified twist detects the difference between π and π', so that the separating twist found in the example had to be ramified.

5. The statement of theorem 5.5.2 fixes a $\psi \in \widehat{F}$ of level one. Show that the hypothesis does not in fact depend on that choice: if π, π' are irreducible with $\omega_\pi = \omega_{\pi'}$ and $\varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi\pi', s, \psi)$ for every χ, then the same equalities hold with ψ replaced by any non-trivial $\psi' \in \widehat{F}$, and conversely. Deduce that the theorem applies as soon as its hypothesis holds for one non-trivial additive character. Recall from section 5.1 that every such ψ' has the form $a\psi$ for a unique $a \in F^\times$, and use lemma 5.3.3.

6

Representation of Galois Theory: Weil Groups

We now cross the bridge to the other side of the correspondence: the Galois side. The absolute Galois group $\mathrm{Gal}(\overline{F}/F)$ is the key object of study, but its profinite topology is too coarse to see the complex representations that match the smooth representations of $\mathrm{GL}_2(F)$. The goal of this chapter is to refine the Galois group into the *Weil Group* $\mathcal{W}_F$ and to develop its representation theory to a point where it matches the complexity of the $\mathrm{GL}_2(F)$ side.

We begin by defining the Weil group and its topology. We then review Local Class Field Theory (Section 6.2), which essentially constitutes the Langlands correspondence for $n = 1$. We define L-functions and ε-factors for representations of $\mathcal{W}_F$, parallel to those defined for G in the previous chapter. Finally, we introduce *Deligne Representations* (Section 6.4), which incorporate a monodromy operator and allows us to generalize from complex representations to choosing any algebraically closed field of characteristic different from the residual characteristic of F. This modification is necessary to account for the difference between the semisimple Galois representations and the non-semisimple (Special) representations of G, and to align the complex theory with ℓ-adic geometry.

6.1 Weil Group and Smooth Representations

We start by recalling some features of the Galois theory of F. Throughout, let p means the characteristic of the residue class field $\mathbf{k} = \mathfrak{o}/\mathfrak{p}$. Choose a separable algebraic closure $\overline{F}$ of F. We set $\Omega_F = \mathrm{Gal}(\overline{F}/F)$, and view it as a profinite group with its natural (Krull) topology. Thus

$$\Omega_F = \varprojlim \mathrm{Gal}(E/F),$$

where E/F ranges over finite Galois extensions with $E \subset \overline{F}$. If K/F is a finite field extension, with $K \subset \overline{F}$, denote by Ω_K the open subgroup $\mathrm{Gal}(\overline{F}/K)$ of Ω_F. If $\widetilde{F}/F$ is some other separable algebraic closure of F, an F-isomorphism $\widetilde{F} \cong \overline{F}$ induces an isomorphism $\mathrm{Gal}(\widetilde{F}/F) \cong \Omega_F$. This does depend on the choice of isomorphism $\widetilde{F} \cong \overline{F}$. Fortunately, it is only up to inner automorphism, so in particular, it induces a *canonical* bijection between the sets of equivalence classes of irreducible smooth representations of the two groups $\mathrm{Gal}(\widetilde{F}/F)$, Ω_F.

Fixing an $\overline{F}/F$, consider extensions E/F where $E \subset \overline{F}$. Then the field F admits a unique unramified extension F_m/F of degree m, for each $m \geqslant 1$. Let F_∞ denote the composite of all these fields. Then F_∞/F is the unique maximal unramified extension of F. The extension F_m/F is Galois and $\mathrm{Gal}(F_m/F)$ is cyclic. An F-automorphism of F_m is determined by its action on the residue field $\mathbf{k}_{F_m} \cong \mathbb{F}_{q^m}$ (see [10, section 1.6]). In particular, there is a unique element φ_m of $\mathrm{Gal}(F_m/F)$ which acts on $\mathbf{k}_{F_m}$ as $x \mapsto x^q$. Set Φ_m to be the map for $\mathrm{Gal}(F_m/F)$ corresponding to the Frobenius map φ_m for the extension of finite fields. The map $\Phi_m \mapsto 1$ gives a canonical isomorphism $\mathrm{Gal}(F_m/F) \to \mathbb{Z}/m\mathbb{Z}$. Taking the limit over m, we get a canonical isomorphism of topological groups

$$\mathrm{Gal}(F_\infty/F) \cong \varprojlim \mathbb{Z}/m\mathbb{Z} = \widehat{\mathbb{Z}}$$

and a unique element $\Phi_F \in \mathrm{Gal}(F_\infty/F)$ which acts on F_m as Φ_m, for all m. Note that $\langle \Phi_F \rangle \cong \mathbb{Z}$ and hence cannot generate the entire $\mathrm{Gal}(F_\infty/F)$, however it is *dense* (on every finite group the restriction generates the entire group).

Definition 6.1.1: Geometric Frobenius

The element Φ_F is the *geometric Frobenius substitution* on F_∞. The map φ_F for the limit of galois groups of finite fields is the *arithmetic Frobenius substitution.*

An element of Ω_F is called a *geometric Frobenius element* (over F) if its image in $\mathrm{Gal}(F_\infty/F)$ is Φ_F. The Chinese Remainder Theorem gives a canonical isomorphism of topological groups

$$\widehat{\mathbb{Z}} \cong \prod_\ell \mathbb{Z}_\ell,$$

where ℓ ranges over all prime numbers and $\mathbb{Z}_\ell$ is the (additive) group of ℓ-adic integers.

Let $\mathcal{I}_F = \mathrm{Gal}(\overline{F}/F_\infty)$ be the *inertia group* of F. This fits in the exact sequence of topological groups

$$1 \to \mathcal{I}_F \to \Omega_F \to \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \cong \widehat{\mathbb{Z}} \to 0$$

Now extend to the maximal tamely ramified extensions.For each integer $n \geqslant 1$, $p \nmid n$, the field F_∞ has a unique extension E_n/F_∞ of degree n. If ϖ is a prime element of F, the extension E_n is generated over F_∞ by an n-th root of ϖ. The extension E_n/F_∞ is cyclic, and the Galois group $\mathrm{Gal}(E_n/F_\infty)$ admits a canonical description using Kummer theory (presented in [12, section 18.4]). Choose $\alpha \in E_n$ such that $\alpha^n = \varpi$. The map

$$\begin{aligned} \mathrm{Gal}(E_n/F_\infty) &\longrightarrow \mu_n, \\ \sigma &\longmapsto \sigma(\alpha)/\alpha, \end{aligned} \tag{6.1}$$

gives a canonical isomorphism of $\mathrm{Gal}(E_n/F_\infty)$ with the group μ_n of n-th roots of unity in F_∞. The composite E_∞ of all extensions E_n/F_∞ is the maximal tamely ramified extension of F. Taking the limit of these isomorphisms gives topological isomorphisms

$$\mathrm{Gal}(E_\infty/F_\infty) \cong \varprojlim \mu_n \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$$

In the limit, n ranges over all positive integers not divisible by p; in the product, ℓ ranges over all prime numbers other than p.

Finally, let $\mathcal{P}_F = \mathrm{Gal}(\overline{F}/E_\infty)$ be the *wild inertia group*, completing the tower

$$\overline{F}/E_\infty/F_\infty/F$$

Any finite extension of E_∞ has degree a power of p, so the profinite group $\mathcal{P}_F$ is a pro p-group. It is the unique pro p-Sylow subgroup of $\mathcal{I}_F$, and hence it is normal, giving the isomorphism $\mathcal{I}_F/\mathcal{P}_F \cong \prod_{\ell\neq p} \mathbb{Z}_\ell$ which is uniquely determined up to multiplication by an element of $\prod \mathbb{Z}_\ell^\times$.

The group $\mathrm{Gal}(F_\infty/F)$ acts on $\mathrm{Gal}(E_\infty/F_\infty)$ by conjugation. An isomorphism transfers this action to one on the group $\prod_{\ell\neq p}\mathbb{Z}_\ell$. This result is boxed for future reference:

Proposition 6.1.2: Tame Inertia Action

If $t_0 : \mathcal{I}_F/\mathcal{P}_F \to \prod_{\ell\neq p}\mathbb{Z}_\ell$ is a topological isomorphism, then

$$t_0(\Phi_F\sigma\Phi_F^{-1}) = q^{-1}t_0(\sigma), \quad \sigma\in\mathcal{I}_F/\mathcal{P}_F$$

***Proof*:**
The tame quotient is described by roots of unity, and the formula is Frobenius acting on them.

The field E_∞ is obtained from F_∞ by adjoining the m-th roots of a prime element for all m prime to p, so restriction gives a canonical isomorphism

$$\mathcal{I}_F/\mathcal{P}_F \xrightarrow{\sim} \varprojlim_{(m,p)=1} \mu_m(\overline{k})$$

under which $\sigma\in\mathcal{I}_F/\mathcal{P}_F$ corresponds to the compatible system $\left(\sigma(\varpi^{1/m})/\varpi^{1/m}\right)_m$ of roots of unity. Identifying $\varprojlim_m \mu_m(\overline{k}) \cong \prod_{\ell\neq p}\mathbb{Z}_\ell$ is the isomorphism t_0.

Now compute the conjugate. Since Φ_F is a geometric Frobenius it acts on $\overline{k}$, hence on each $\mu_m(\overline{k})$, by $\zeta\mapsto\zeta^{q^{-1}}$, and it fixes ϖ. Therefore

$$\frac{\Phi_F\sigma\Phi_F^{-1}(\varpi^{1/m})}{\varpi^{1/m}} = \Phi_F\left(\frac{\sigma(\varpi^{1/m})}{\varpi^{1/m}}\right) = \left(\frac{\sigma(\varpi^{1/m})}{\varpi^{1/m}}\right)^{q^{-1}}$$

for every m prime to p. Passing to the limit and transporting along t_0, raising to the power q^{-1} becomes multiplication by q^{-1} in $\prod_{\ell\neq p}\mathbb{Z}_\ell$, which is the stated identity, completing the proof.

Let ${}_a\mathcal{W}_F$ be defined by:

$${}_a\mathcal{W}_F = \{\sigma\in\Omega_F : \sigma|_{F_\infty}\in\langle\Phi_F\rangle\} = \mathrm{res}^{-1}(\langle\Phi_F\rangle)\subseteq\Omega_F,$$

where $\mathrm{res}:\Omega_F\to\mathrm{Gal}(F_\infty/F)$ is the restriction map. As any $\varphi\in\mathcal{I}_F$ restricts to the identity, $\mathcal{I}_F\subseteq{}_a\mathcal{W}_F$. For If $G_E = \mathrm{Gal}(E/F)$ for a finite F, Then the restriction to E is surjective, and hence ${}_a\mathcal{W}_F$ is *dense* in $\mathrm{Gal}(\overline{F}/F)$.

Next, the map:

$$\mathrm{res} : {}_a\mathcal{W}_F \to \mathrm{Gal}(F_\infty/F) \qquad \sigma\mapsto\sigma|_{F_\infty}$$

is by construction surjective onto $\langle \Phi_F \rangle$ with kernel $\mathcal{I}_F$ forming a short exact sequence:

$$\begin{tikzcd}
1 \arrow[r] & \mathcal{I}_F \arrow[r] & G_F \arrow[r] & \hat{Z} \cong \mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \arrow[r] & {} \\
1 \arrow[r] & \mathcal{I}_F \arrow[r] \arrow[u, equal] & {}_a\mathcal{W}_F \arrow[r] \arrow[u, hook] & \mathbb{Z} \arrow[r] \arrow[u, hook] & {}
\end{tikzcd}$$

Notice the difference between these two quotients:

$${}_a\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z} \quad \text{(Generated by Frobenius)} \qquad \text{and} \qquad \mathrm{Gal}(F_\infty/F) \cong \widehat{\mathbb{Z}} \quad \text{(Profinite completion)}$$

Inertia		Total Group		Unramified
$\mathcal{I}_F$	$\subset$	Ω_F (Galois)	$\twoheadrightarrow$	$\widehat{\mathbb{Z}}$
$\vert$		$\cup$		$\cup$
$\mathcal{I}_F$	$\subset$	${}_a\mathcal{W}_F$ (Weil)	$\twoheadrightarrow$	$\mathbb{Z}$

For a concrete element in Ω_F that is not in ${}_a\mathcal{W}_F$, assume char$F \neq 2$ so that 2 is invertible in F (or else pick 3). Let α be the element that is $1/2$ mod p^n for each n and $p \neq 2$. Let $\sigma \in \Omega_F$ associate to this element and notice that $\sigma^2|_{F_\infty} = \Phi_F$, showing that it cannot be in ${}_a\mathcal{W}_F$.

Now, ${}_a\mathcal{W}_F$ is a profinite group, hence not "enough" representations to correspond to irreducible representations mentioned earlier. A better topology is needed to carry the arithmetic information of ${}_a\mathcal{W}_F$, that is a locally profinite topology:

Definition 6.1.3: Weil Group

The Weil group $\mathcal{W}_F$ of F (relative to $\bar{F}/F$) is the topological group, with underlying abstract group $a\mathcal{W}_F$, so that

1. $\mathcal{I}_F$ is an open subgroup of $\mathcal{W}_F$, and
2. the topology on $\mathcal{I}_F$, as subspace of $\mathcal{W}_F$, coincides with its natural topology as $\mathrm{Gal}(\bar{F}/F_\infty) \subset \Omega_F$.

As $\mathcal{I}_F$ is open, $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ must be discrete, and as it is infinite it must be that $\mathcal{W}_F$ is no longer compact, and hence it is *locally profinite* (recall in a profinite group a set is open if and only if it is finite index). The definition of $\mathcal{W}_F$ does depend on the choice of $\overline{F}/F$ up to inner automorphism of Ω_F.

Lemma 6.1.4: Injection is Continuous

The identity map $\iota_F : \mathcal{W}_F \to {}_a\mathcal{W}_F \subset \Omega_F$ is a continuous injection.

***Proof*:**
Injectivity is immediate, since ι_F is the identity on underlying sets: $\mathcal{W}_F$ and ${}_a\mathcal{W}_F$ have the same elements and differ only in their topology.

For continuity, it is enough to check that the preimage of a basic open set of Ω_F is open in $\mathcal{W}_F$. The topology of Ω_F has a basis of cosets of the subgroups $\mathrm{Gal}(\overline{F}/L)$ for L/F finite, and each such subgroup meets $\mathcal{W}_F$ in a set that is open for the Weil topology: by construction $\mathcal{I}_F$ is open in $\mathcal{W}_F$ and carries its natural profinite topology, so the open subgroups of $\mathcal{I}_F$ remain open in $\mathcal{W}_F$. Hence every Ω_F-open set pulls back to a $\mathcal{W}_F$-open set, so ι_F is continuous.

The map is not a homeomorphism: the Weil topology is strictly finer, since $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ is discrete for it while the corresponding quotient of Ω_F is $\widehat{\mathbb{Z}}$, which is not, as we sought to show.

Let $\nu_F : \mathcal{W}_F \to \mathbb{Z}$ be the canonical map taking a geometric Frobenius element to 1 and define:

$$\|x\| = q^{-v_F(x)}, \quad x \in \mathcal{W}_F$$

Proposition 6.1.5: Weil subgroup and Finite Extensions

1. Let E/F be a finite extension, $E \subset \overline{F}$.
   (a) The group $\mathcal{W}_F$ has a unique subgroup $\mathcal{W}_F^E$ such that
   $$\iota_F(\mathcal{W}_F^E) = {}_a\mathcal{W}_F \cap \Omega_E$$
   (b) The subgroup $\mathcal{W}_F^E$ is open and of finite index in $\mathcal{W}_F$; it is normal in $\mathcal{W}_F$ if and only if E/F is Galois.
   (c) The canonical map $\mathcal{W}_F^E \backslash \mathcal{W}_F \to \Omega_E \backslash \Omega_F$ is a bijection.
   (d) The canonical map $\iota_E : \mathcal{W}_E \to \Omega_E$ induces a topological isomorphism $\mathcal{W}_E \cong \mathcal{W}_F^E$.
2. The map $E/F \mapsto \mathcal{W}_F^E$ is a bijection between the set of finite extensions E of F inside $\overline{F}$ and the set of open subgroups of $\mathcal{W}_F$ of finite index.

***Proof*:**
Throughout, ι_F is the continuous injection of lemma 6.1.4, and we identify $\mathcal{W}_F$ with its image when convenient.

(1)(a). Uniqueness is forced because ι_F is injective, so $\mathcal{W}_F^E$ must be the preimage $\iota_F^{-1}({}_a\mathcal{W}_F \cap \Omega_E)$; that preimage is a subgroup because both factors are.

(1)(b). The subgroup Ω_E is open of finite index $[E : F]$ in Ω_F by infinite Galois theory, so its preimage is open of index at most $[E : F]$ in $\mathcal{W}_F$ by continuity. The index is exactly $[E : F]$ because $\mathcal{W}_F$ surjects onto $\Omega_E \backslash \Omega_F$, which is (c). Normality transfers along the preimage, and $\Omega_E \trianglelefteq \Omega_F$ holds if and only if E/F is Galois.

(1)(c). The map is injective by construction. For surjectivity, note that $\mathcal{W}_F$ is dense in Ω_F, since it contains $\mathcal{I}_F = \mathrm{Gal}(\overline{F}/F_\infty)$ and a Frobenius lift, and Ω_E is open; a dense subgroup meets every coset of an open subgroup, so every class in $\Omega_E \backslash \Omega_F$ has a representative in $\mathcal{W}_F$.

(1)(d). Both $\mathcal{W}_E$ and $\mathcal{W}_F^E$ have the same underlying set, namely the elements of $\mathcal{W}_F$ fixing E, and both carry the topology in which the inertia subgroup $\mathcal{I}_E$ is open with its profinite topology and the quotient by it is discrete. So the identity is a topological isomorphism.

(2). Injectivity is Galois theory: E is recovered from $\mathcal{W}_F^E$ as the fixed field of the closure of its image in Ω_F. For surjectivity, let $H \leq \mathcal{W}_F$ be open of finite index. Its closure $\overline{H}$ in Ω_F is open of finite index, hence equals Ω_E for a unique finite E/F, and then $H = \mathcal{W}_F^E$ because both are the full preimage of Ω_E: an open finite-index subgroup of $\mathcal{W}_F$ contains an open subgroup of $\mathcal{I}_F$ and is therefore determined by its closure, completing the proof.

Henceforth $\mathcal{W}_E$ is identified with the subgroup $\mathcal{W}_F^E$ of $\mathcal{W}_F$.

Attention now turns to the smooth representations of $\mathcal{W}_F$ themselves. Note that as Ω_F is profinite, smooth representations must be semi-simple, while $\mathcal{W}_F$ being locally profinite (as it admits a discrete quotient $\mathbb{Z}$) must admit non-semisimple representations. As will be seen, the irreducible representations of $\mathcal{W}_F$ are closely related to the irreducible representations of Ω_F , but the reducible ones exhibit a greater variety, namely the principal sedries shall be associated to them, and these are "hidden" from the point of view of Ω_F. Just as in the previous chapters, the focus is on *complex* representations. P-adic representations are treated in section 6.4.2, where the ℓ-adic theory enters.

Lemma 6.1.6: Irreducible Representations of Weil Group is Finite

Let (ρ, V) be an irreducible representation of $\mathcal{W}_F$. Then V is finite dimensional.

***Proof*:**
let $v \in V$. As ρ is smooth it is fixed by a compact open subgroup $K \subseteq \mathcal{W}_F$. Let $\mathcal{J} = K \cap \mathcal{I}_F$ so that v is fixed by an open subgroup of $\mathcal{I}_F$. As $\mathcal{I}_F$ has the same topology as the subspace topology of $\mathcal{I}_F \subseteq \Omega_F$ and $\mathcal{J}$ is open, it corresponds to a finite extension E/F. By taking the galois closure of E/F, we get a smaller group that is still in the stabilizer of v, hence we may assume E/F is galois hence $\mathcal{J}$ is normal in $\mathcal{W}_F$. It is an exercise in classical representation theory to show that if ρ is irreducible and $N \trianglelefteq G$ is normal, if N fixes a point v, N acts trivially on all of V. Hence, $\mathcal{J} \subseteq \ker \rho$.

Now, as $\mathcal{J} \subseteq \mathcal{I}_K$ is an open subgroup, $\mathcal{I}_K/\mathcal{J}$ is finite. Let $\Phi \in \mathcal{W}_F$ be a Frobenius element (an element of the coset $1 + \mathcal{I}_F$ in the map $\mathcal{W}_F \xrightarrow{\text{res}} \langle \Phi_F \rangle \cong \mathbb{Z}$). As both $\mathcal{I}_K$ and $\mathcal{J}$ are normal in $\mathcal{W}_F$ (they are both associated to a galois group of an extension of F), Φ acts by conjugation on $\mathcal{I}_K/\mathcal{J}$, and as this group is finite some Φ^d acts trivially on this group. Thus:

$$\Phi^d \cdot x \cdot \Phi^{-d} = xj$$

for appropriate $j \in \mathcal{J}$. Applying ρ:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x)\rho(j)$$

As $\mathcal{J} \subseteq \ker \rho$:

$$\rho(\Phi^d)\rho(x)\rho(\Phi^{-d}) = \rho(x) \cdot I = \rho(x)$$

Moving terms around:

$$\rho(\Phi^d)\rho(x) = \rho(x)\rho(\Phi^d)$$

Thus the map $\rho(\Phi^d)$ is in the center of $\rho(\mathcal{W}_F) \subseteq \mathrm{Aut}(V)$, and as ρ is irreducible this means $\rho(\Phi^d)$ acts like scalar multiplication.

Using this, I claim that for the v that was fixed at the very beginning of the proof, the space

$$U = \{\rho(g) \cdot v : g \in \mathcal{W}_F\}$$

is finite dimensional. By definition of $\mathcal{W}_F$, $g = \Phi^k t$ for $t \in \mathcal{I}_F$. By taking the finite $\mathcal{I}_F/\mathcal{J}$ we have coset representatives $r_1, ..., r_n$ so that:

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha)\rho(j) \cdot v$$

where $j \in \mathcal{J}$ is any element. As $\mathcal{J} \subseteq \ker \rho$ we simplify to

$$\rho(g) \cdot v = \rho(\Phi^k)\rho(r_\alpha) \cdot v$$

Next, write $\Phi^k = \Phi^{ad+c}$ where $0 \leq c < d$ so that we can use the fact that $\rho(\Phi^d)$ is scalar multiplication to get

$$\rho(\Phi^d) = \lambda^a \rho(\Phi^c)$$

But now, we get that there finitely many possible vectors that are used to span U

$$\rho(\Phi^c)\rho(r_\alpha) \cdot v \qquad 0 \leq c < d, \text{ and } 0 < \alpha < n$$

showing it is finite. Certainly this space is $\mathcal{W}_F$-stable, and as V is irreducible it must be that $U = V$, as we sought to show.

Hence the representations of $\mathcal{W}_F$ are interest will be finite! Let ρ be a finite representation of $\mathcal{W}_F$. Then the restriction to $\mathcal{I}_F$ (a profinite group) will be semi-simple by an easy generalization of Maschke's theorem (proposition 1.2.9), and $\rho(\mathcal{I}_F)$ is finite. Therefore, all the complexity comes from the Frobenius elements of $\mathcal{W}_F$.

Proposition 6.1.7: Irreducible in Galois, then in Weil

Let $\mathcal{W}_F$ be the Weil group, $\iota_F : \mathcal{W}_F \to {}_a\mathcal{W}_F$ the identity map (a continuous but not homeomorphic bijection), (ρ, V) a smooth representation of Ω_F, the absolute galois group of F so that $\rho \circ \iota_F$ can be thought of as the restriction to $\mathcal{W}_F$. Then:

1. if ρ is irreducible (on Ω_F), then $\rho \circ \iota_F$ is irreducible (on $\mathcal{W}_F$)
2. If ρ_1, ρ_2 are irreducible on Ω_F, then $\rho_1 \cong \rho_2$ if and only if $\rho_1 \circ \iota_F \cong \rho_2 \circ \iota_F$.

***Proof*:**

1. Take $\ker \rho \subseteq \Omega_F$. This is an open subset. Let $K = \ker \rho$. Then gK is open for all $g \in \Omega_F$. As $\mathcal{W}_F$ is dense, $gK \cap \mathcal{W}_F \neq \emptyset$. Then $gk = w$ for some $k \in K$ and $w \in \mathcal{W}_F$, in particular:

   $$g = wk^{-1} \qquad w \in \mathcal{W}_F, \ k^{-1} \in K$$

   As this is true for all $g \in \Omega_F$:

   $$\Omega_F = \ker(\rho) \cdot \mathcal{W}_F$$

   Taking ρ of both sides we get:

   $$\rho(\Omega_F) = \rho(\mathcal{W}_F)$$

   Thus, if there are no nontrivial $\rho(\Omega_F)$-invariant subspaces, there are no nontrivial $\rho(\mathcal{W}_F)$-invariant subspaces.

2. If $\rho_1 \to \rho_2$ is an Ω_F-isomorphism, restriction certainly makes it a $\mathcal{W}_F$-isomorphism, and the equality $\rho(\Omega_F) = \rho(\mathcal{W}_F)$ proven above shows the other way is true too, completing the proof.

The converse is true *up to tensoring* with an unramified character:

Definition 6.1.8: Unramified Character

Let χ be a (complex) character on Ω_F or $\mathcal{W}_F$. Then χ is *unramified* if it is trivial on $\mathcal{I}_F$ ($\chi|_{\mathcal{I}_F}$ is trivial).

If ρ is a smooth, finite-dimensional representation of Ω_F or $\mathcal{W}_F$, the map $x \mapsto \det(\rho(x))$ is a character, $\det \rho$.

Proposition 6.1.9: Irreducible and Finite in Weil, then in Galois

Let τ be an irreducible smooth representation of $\mathcal{W}_F$. The following are equivalent:

1. the group $\tau(\mathcal{W}_F)$ is finite;
2. $\tau \cong \rho \circ \iota_F$, for some irreducible smooth representation ρ of Ω_F;
3. the character $\det \tau$ has finite order.

For any irreducible smooth representation τ of $\mathcal{W}_F$, there is an unramified character χ of $\mathcal{W}_F$ such that $\chi \otimes \tau$ satisfies (1) to (3).

***Proof*:**

(1) $\Rightarrow$ (2) We need to find the appropriate representation ρ of Ω_F. Let $\Phi \in \mathcal{W}_F$ be a Frobenius element. Then $\tau(\Phi)$ has finite order, say d. Since $\hat{\mathbb{Z}}/d\hat{\mathbb{Z}} \cong \mathbb{Z}/d\mathbb{Z}$, for $a \in \hat{\mathbb{Z}}$, there exists an integer $\overline{a}$ such that $\overline{a} \equiv a \bmod d\hat{\mathbb{Z}}$ up to $d\mathbb{Z}$. Now, as shown earlier any $w \in \Omega_F$ can be written as

$$w = \Phi^a \sigma \qquad a \in \hat{\mathbb{Z}},\ \sigma \in \mathcal{I}_F$$

Define ρ of Ω_F as :

$$\rho(w) = \rho(\Phi^a \sigma) = \tau(\Phi)^{\overline{a}} \tau(\sigma)$$

Certainly $\tau = \rho \circ \iota_F$ and ρ is irreducible

(2) $\Rightarrow$ (3) Let τ be an irreducible representation that extends to ρ so that $\det \tau = (\det \rho)|_{\mathcal{W}_F}$. Let $\chi = \det \rho$. Then it is a *character*:

$$\chi : \Omega_F \to \mathbb{C}^*$$

As χ is smooth, it's kernel K is open and as Ω_F is compact Ω_F/K is finite Thus, χ factors though Ω_F/K which as finite order, hence $\chi^n = \mathrm{id}$ for appropriate n, hence $\tau^n = 1$.

(3) $\Rightarrow$ (1) Assume $\det \tau$ has finite order. First, as $\mathcal{I}_F$ is compact, $\tau(\mathcal{I}_F)$ is finite, hence some order of the conjugation by $\tau(\varphi)$ acts trivially, say $\tau(\varphi)^k$. Trivially, it commutes with $\tau(\Phi)$, and hence with all of $\mathcal{W}_F$, and so by Schur's lemma $\tau(\Phi)^k = \lambda I$.

Now, letting $n = \dim \tau$ (which exists as τ is irreducible by lemma 6.1.6), $\det(\tau(\Phi)^k) = \det(\lambda I) = \lambda^n I$. As $\det \tau$ has finite order (say m):

$$\det(\tau(\Phi)^k)^m = \lambda^{nm} = 1$$

Hence λ is a root of unity. Hence $\tau(\Phi)^k = \lambda I$ has finite order. As $\mathcal{W}_F = \langle\Phi\rangle \cdot \mathcal{I}_F$ and the τ of both groups on the right hand side has finite order, $\tau(\mathcal{W}_F)$ has finite order

For the last part, notice the proof that $\tau(\Phi)^k$ did not rely on any assumption, but on the compactness of $\mathcal{I}_F$. Say $\tau(\Phi)^k = cI$. Using this c, pick a $z_0 \in \mathbb{C}$ such that $(z_0^n = c$. The define:

$$\chi : \mathcal{W}_F \to \mathbb{C}^* \qquad \chi(\Phi^n \mathcal{I}_F) = z_0^n$$

This is certainly a smooth character and unramified by construction. Then letting $\tau' = \chi^{-1} \otimes \tau$ we get:

$$\tau'(\Phi)^k = (\chi^{-1}(\Phi)\tau(\Phi))^k = \chi(\Phi)^{-k}\tau(\Phi)^k = c^{-1} \cdot (c \cdot I) = I$$

Hence τ' has finite order, and like in the proof of $(3) \Rightarrow (1)$ as $\mathcal{W}_F = \langle\Phi\rangle \cdot \mathcal{I}_F$ and τ' is finite om both pieces, $\tau'(\mathcal{W}_F)$ is finite, satisfying the first condition and completing the proof.

This result is as good as it can get, that is there exist irreducible representations τ of $\mathcal{W}_F$ where $\tau(\mathcal{W}_F)$ has infinite order:

Example 6.1.10: Irreducible in Weil But Not Defined in Galois Group

Let $\chi : \mathcal{W}_F \to \mathbb{C}^*$, $\omega(w) = |w| = q^{-\nu_F(w)}$. This is certainly a smooth one-dimensional irreducible representation, and $\chi(\mathcal{W}_F) = \mathbb{Z}$. This map simply does not exist over Ω_F (the image must be compact).

Next, relate the representations of $\mathcal{W}_F$ and $\mathcal{W}_E$ where E/F is a finite-separable/galois extension where $E \subseteq \overline{F}$.

Definition 6.1.11: Induction On Weil Representation

Let E/F be a finite separable extension where $E \subseteq \overline{F}$. Then write:

$$\mathrm{Ind}_{E/F} := \mathrm{Ind}_{\mathcal{W}_E}^{\mathcal{W}_F}$$

for the functor of smooth induction from $\mathcal{W}_E$ to $\mathcal{W}_F$, and:

$$\mathrm{Res}_{E/F} := \mathrm{Res}_{\mathcal{W}_E}^{\mathcal{W}_F}$$

From now on, if ρ is a smooth representation of $\mathcal{W}_F$, $\rho|_E = \mathrm{Res}_{E/F}\rho = \rho|_{\mathcal{W}_E}$. Recall lemma 1.2.37, for ease of reference the following translates this lemma with the groups of interest:

Lemma 6.1.12: Weil Representation of Finite Extension

Let E/F be a finite separable field extension, $E \subset \bar{F}$.

1. Let ρ be a smooth representation of W_F; then ρ is semisimple if and only if ρ_E is semisimple.
2. Let τ be a smooth representation of W_E; then τ is semisimple if and only if $\mathrm{Ind}_{E/F}\tau$ is semisimple.

***Proof*:**
Directly apply lemma 1.2.37.

Definition 6.1.13: Class of Semisimple Weil Representations

Let F be a local field. For each integer $n \geq 1$, let

$$\mathcal{G}_n^{\mathrm{ss}}(F)$$

denote the set of isomorphism classes of semisimple smooth representations of $\mathcal{W}_F$ of dimension n. Let

$$\mathcal{G}_n^0(F)$$

denote the set of isomorphism classes of *irreducible* smooth representations of W_F of dimension n.

If E/F is a finite extension where $E \subseteq \overline{F}$. Then lemma 6.1.12 gives the induction and restriction maps

$$\begin{aligned}\mathrm{Ind}_{E/F} &: \mathcal{G}_n^{\mathrm{ss}}(E) \to \mathcal{G}_{nd}^{\mathrm{ss}}(F),\\ \mathrm{Res}_{E/F} &: \mathcal{G}_n^{\mathrm{ss}}(F) \to \mathcal{G}_n^{\mathrm{ss}}(E),\end{aligned}$$

where $d = [E : F]$.

It is worth recognising that choices were made in the definition of the Weil group (the field closure, the Frobenius element) , but W_F is, colloquially speaking, well-defined up to inner automorphism of Ω_F. The sets $\mathcal{G}_n^0(F)$, $\mathcal{G}_n^{\mathrm{ss}}(F)$ are thus well-defined and independent of all choices (and so are the above two maps). In particular, given two separable field extensions E/F, E'/F' of finite degree d and an isomorphism $\alpha : E \to E'$ such that $\alpha(F) = F'$. There is a well-defined bijection $\mathcal{G}_n^{\mathrm{ss}}(F) \to \mathcal{G}_n^{\mathrm{ss}}(F')$ and the diagram commutes

$$\begin{tikzcd}
\mathcal{G}_n^{\mathrm{ss}}(E) \arrow[r] \arrow[d, "\mathrm{Ind}_{E/F}"] & \mathcal{G}_n^{\mathrm{ss}}(E') \arrow[d, "\mathrm{Ind}_{E'/F'}"] \\
\mathcal{G}_{nd}^{\mathrm{ss}}(F) \arrow[r] & \mathcal{G}_{nd}^{\mathrm{ss}}(F')
\end{tikzcd}$$

Similarly for restriction. The induction and restriction map are defined for any finite separable extension E/F, independent of any choice of F-embedding of E in $\bar{F}$.

Proposition 6.1.14: Semisimple Weil Representation

Let (ρ, V) be a smooth representation of $\mathcal{W}_F$ of finite dimension, and let $\Phi \in \mathcal{W}_F$ be a Frobenius element. The following are equivalent:

(1) the representation ρ is semisimple;

(2) the automorphism $\rho(\Phi) \in \mathrm{Aut}_{\mathbb{C}}(V)$ is semisimple;

(3) the automorphism $\rho(\Psi) \in \mathrm{Aut}_{\mathbb{C}}(V)$ is semisimple, for every element Ψ of $\mathcal{W}_F$.

Proof:
(3) $\Rightarrow$ (2) is obvious, so let's prove (2) $\Rightarrow$ (1) and then (1) $\Rightarrow$ (3)

(2) $\Rightarrow$ (1) Recall a semisimple operator means it is diagonalizable. As $\rho(\Phi)$ is semisimple, so is $\rho(\Phi)^d$ for any d. Pick d so that $\rho(\Phi)^d$ commutes with $\rho(\mathcal{I}_F)$ (as it is finite, such a d exists). Then as Φ^d and $\mathcal{I}_F$ commutes, $\rho|_H$ where $H = \langle \Phi^d, \mathcal{I}_F \rangle$ is semisimple. Now, G/H is finite and $\rho|_H$ is semisimple, hence by lemma 6.1.12 ρ is semisimple

(1) $\Rightarrow$ (3) Let ρ be semisimple and pick any $\Psi \in \mathcal{W}_F$. If $\Psi \in \mathcal{I}_F$, it $\rho(\Psi)$ semisimple as $\mathcal{I}_F$ is profinite, so assume $\Psi \notin \mathcal{I}_F$. Now, let $n = \nu_F(\Psi)$ which is nonzero as $\Psi \notin \mathcal{I}_F$. Choose an unramified degree n extension E/F (such an extension exists, see [10, existence of unramified extensions]) and consider $\mathcal{W}_E$ which by proposition 6.1.5 is an open subgroup of $\mathcal{W}_F$ of finite index and consists of elements whose valuation is divisible by n. Certainly Ψ belongs in this group as:

$$\nu_E(\Psi) = \frac{1}{n}\nu_F(\Psi) = \frac{1}{n} \cdot n = 1$$

and $\mathcal{I}_E = \mathcal{I}_F$ (certainly the bijective as E/F is unramified). For the generation notice that:

$$1 \to \mathcal{I}_E \to \mathcal{W}_E \xrightarrow{\nu_F} \langle \Psi \rangle \to 1$$

And hence $\mathcal{W}_E = \langle \Psi, \mathcal{I}_E \rangle$. As ρ is assumed to be semisimple and E/F is finite, by lemma 6.1.6 $\rho|_E$ is semisimple. As we want to show $\rho(\Psi)$ is semisimple, we may focus on E, or equivalently we may as well assume $E = F$. As we want to prove it is semisimple, we need to prove each block-diagonal is diagonalizable, so we may further reduce to assuming ρ is irreducible. But now, by proposition 6.1.9, any irreducible representation can be twisted by an unramified character χ to get a finite image. By untwisting ($c = \chi(\Psi)^{-1}$), we get:

$$\rho(\Psi) = c\varphi$$

where φ has finite order. But then by regular Maschke's theorem $\rho(\Psi)$ is semisimple, as we sought to show.

6.2 Weil Group and Local Class Field Theory

Let $\mathcal{W}_F^{\mathrm{der}}$ denote the closure of the commutator subgroup of $\mathcal{W}_F$ and let

$$\mathcal{W}_F^{\mathrm{ab}} = \mathcal{W}_F / \mathcal{W}_F^{\mathrm{der}}$$

The group $\mathcal{W}_F^{\mathrm{ab}}$ is a locally profinite abelian group (defined independently of the choices made in the definition of $\mathcal{W}_F$). The following summarizes the core result of CFT:

Theorem 6.2.1: Local Class Field Theory

There is a canonical continuous group homomorphism

$$\mathbf{a}_F \colon \mathcal{W}_F \longrightarrow F^\times$$

Then:

1. The map $\mathbf{a}_F$ induces a topological isomorphism $\mathcal{W}_F^{\mathrm{ab}} \cong F^\times$.
2. An element $x \in \mathcal{W}_F$ is a geometric Frobenius if and only if $\mathbf{a}_F(x)$ is a prime element of F.
3. $\mathbf{a}_F(\mathcal{I}_F) = U_F$ and $\mathbf{a}_F(\mathcal{P}_F) = U_F^1$.
4. If E/F is a finite separable extension, the diagram

$$\begin{array}{ccc} \mathcal{W}_E & \xrightarrow{\mathbf{a}_E} & E^\times \\ \big\downarrow & & \big\downarrow{\scriptstyle N_{E/F}} \\ \mathcal{W}_F & \xrightarrow{\mathbf{a}_F} & F^\times \end{array}$$

   commutes.
5. Let $\alpha \colon F \to F'$ be an isomorphism of fields. The map α induces an isomorphism $\alpha \colon \mathcal{W}_F^{\mathrm{ab}} \to \mathcal{W}_{F'}^{\mathrm{ab}}$, and the diagram

$$\begin{array}{ccc} \mathcal{W}_F^{\mathrm{ab}} & \xrightarrow{\alpha} & \mathcal{W}_{F'}^{\mathrm{ab}} \\ {\scriptstyle \mathbf{a}_F}\big\downarrow & & \big\downarrow{\scriptstyle \mathbf{a}_{F'}} \\ F^\times & \xrightarrow{\alpha} & F'^\times \end{array}$$

   commutes.

The map $\mathbf{a}_F$ is the Artin Reciprocity Map.

***Proof*:**
This is the main theorem of local class field theory (see [10, the local reciprocity map])

Theorem 6.2.2: Local Class Theory: Norm

1. The map $E/F \mapsto N_{E/F}(E^\times)$ gives a bijection between the set of finite abelian extensions E/F, $E \subset \bar{F}$, and the set of open subgroups of $F^\times$ of finite index.
2. The map $\mathbf{a}_F$ induces an isomorphism $\mathrm{Gal}(E/F) \cong F^\times/N_{E/F}(E^\times)$, for any finite abelian extension E/F.
3. Let E/F be a finite separable extension. If E_1/F is the maximal abelian sub-extension of E/F, then $N_{E/F}(E^\times) = N_{E_1/F}(E_1^\times)$.

***Proof*:**
This is the main theorem of local class field theory (see [10, the local reciprocity map])

There is also the need for "lifting" a representation given E/F. To that end, let G be some group and H a subgroup of G of finite index. Let $D(G)$ be the commutator subgroup of G (denoted $[G, G]$ in [10] and [5] but abbreviated to $D(G)$ since it is all "derivation"). Choose a section $t\colon G/H \to G$ of the quotient map $G \to G/H$. For $x \in G/H$ and $g \in G$, define an element $h_{x,g} \in H$ by $gt(x) = t(gx)h_{x,g}$. Then define the map:

Definition 6.2.3: Verlagerung Map

$$\begin{aligned}\mathrm{ver}_{G/H}\colon G/D(G) &\longrightarrow H/D(H),\\ g\,D(G) &\longmapsto \prod_{x \in G/H} h_{x,g}\,D(H)\end{aligned}$$

The map $\mathrm{ver}_{G/H}$, called the *transfer* or *Verlagerung*, and is a homomorphism of abelian groups.

If G is locally compact, write G^{der} for the closure of $D(G)$ in G and $G^{\mathrm{ab}} = G/G^{\mathrm{der}}$. If H is a closed subgroup of G of finite index, the transfer then induces a continuous homomorphism $G^{\mathrm{ab}} \to H^{\mathrm{ab}}$, still denoted $\mathrm{ver}_{G/H}$. In particular, if E/F is a finite separable field extension, the transfer gives a map $\mathrm{ver}_{E/F}\colon \mathcal{W}_F^{\mathrm{ab}} \to \mathcal{W}_E^{\mathrm{ab}}$.

Theorem 6.2.4: Transfer Theorem

If E/F is a finite separable field extension, the diagram

$$\begin{tikzcd} \mathcal{W}_E^{\mathrm{ab}} \arrow[r, "\mathbf{a}_E"] & E^\times \\ \mathcal{W}_F^{\mathrm{ab}} \arrow[u, "\mathrm{ver}_{E/F}"] \arrow[r, "\mathbf{a}_F"'] & F^\times \arrow[u] \end{tikzcd}$$

commutes, where the map $F^\times \to E^\times$ is inclusion.

The compatibility of the transfer with the reciprocity map is a theorem of local class field theory, and this book's prerequisite *Class Field Theory* [10] proves it in full; by the de-duplication convention of the series the statement is used here and the proof is cited rather than repeated. What matters downstream is the shape of the diagram: it says that restricting a character of $F^\times$ along the inclusion into $E^\times$ corresponds, on the Weil side, to composing with the transfer, and that is exactly the input proposition 6.2.5 needs.

This links to representations as follows. As $\mathcal{W}^{\text{ab}}$ is abelian, all representations are characters. By theorem 6.2.1, there is an isomorphism:

$$\operatorname{Hom}_{\mathbf{Top}}(F^*, \mathbb{C}) \xrightarrow{\sim} \operatorname{Hom}_{\mathbf{Top}}(\mathcal{W}_F^{\text{ab}}, \mathbb{C}) \qquad \chi \mapsto \chi \circ \mathbf{a}_F$$

The isomorphism preserves unramified characters, and maps tamely ramified characters of F^* to characters of $\mathcal{W}_F$ trivial on $\mathcal{P}_F$. Thus one may simply refer to characters of $\mathcal{W}_F$ without composing with the Artin map.

An important character to consider is $\det \rho$ where $\rho \in \mathcal{G}_d^{ss}(F)$ viewed as a character on F^*. It shall often come up as it natural reduces an n-representation to a character.

Let K/F is a separable extension of finite degree d and 1_K denotes the trivial character of $\mathcal{W}_K$. Then its regular representation is:

$$R_{K/F} = \operatorname{Ind}_{K/F} 1_K \in \mathcal{G}_d^{\text{ss}}(F),$$

and its determinant

$$\varkappa_{K/F} = \det R_{K/F}$$

Thus, $\varkappa_{K/F}$ is a character of $F^\times$ satisfying $\varkappa_{K/F}^2 = 1$ (as $\varkappa_{K/F} \in \{1, -1\}$). It arises in the following context:

Proposition 6.2.5: Determinant of Inflation

Let K/F be a separable extension of degree d, let $\rho \in \mathcal{G}_m^{\text{ss}}(K)$ and put $\sigma = \operatorname{Ind}_{K/F} \rho \in \mathcal{G}_{md}^{\text{ss}}(F)$. Then

$$\det \sigma = \varkappa_{K/F}^m \otimes \det \rho|_{F^\times}$$

Lemma 6.2.6: Determinant of an Induced Representation

Let G be a group, let H be a subgroup of G of finite index, and ρ a finite dimensional representation of H. Set $\sigma = \operatorname{Ind}_H^G \rho$. Then

$$\det \sigma = \det(R_{G/H})^{\dim \rho} \otimes (\det \rho \circ \operatorname{ver}_{G/H})$$

where $R_{G/H} = \operatorname{Ind}_H^G 1_H$.

***Proof*:**

Fix coset representatives $t(x)$, $x \in G/H$, and realize $\operatorname{Ind}_H^G \rho$ on $\bigoplus_x t(x) \otimes W$ where W is the space of ρ. For $g \in G$ the action permutes the summands according to $x \mapsto gx$ and acts on the summand indexed by x through $\rho(h_{x,g})$, where $h_{x,g} \in H$ is the element defined by $g\,t(x) = t(gx)h_{x,g}$, exactly as in definition 6.2.3.

The matrix of $\sigma(g)$ in this decomposition is therefore a block permutation matrix whose non-zero blocks are the $\rho(h_{x,g})$. Its determinant splits into two contributions. The first is the sign of the permutation of the blocks, each counted $\dim\rho$ times, which is precisely the determinant of the permutation representation $R_{G/H}$ raised to the power $\dim\rho$. The second is the product of the determinants of the blocks,

$$\prod_{x\in G/H}\det\rho(h_{x,g}) = \det\rho\Big(\prod_{x\in G/H} h_{x,g}\Big) = \det\rho\big(\mathrm{ver}_{G/H}(g)\big)$$

where the first equality uses that $\det\rho$ is a homomorphism into an abelian group, so the order of the factors is immaterial, and the second is the definition of the transfer. Multiplying the two contributions gives the stated formula, as we sought to show.

***Proof*:**
(of proposition 6.2.5)

Combining lemma 6.2.6 with the Transfer Theorem (theorem 6.2.4): the lemma gives $\det\sigma = \varkappa_{K/F}^m \otimes (\det\rho\circ\mathrm{ver}_{K/F})$ with $m = \dim\rho$, and the transfer theorem identifies $\det\rho\circ\mathrm{ver}_{K/F}$ with the restriction of $\det\rho$ to $F^\times$ along the inclusion $F^\times \hookrightarrow K^\times$, completing the proof.

6.3 L-functions and Epsilon-Factors

We now turn to the construction of the fundamental invariants, L-functions and ε-factors, attached to finite-dimensional, semisimple, smooth representations of the Weil group $\mathcal{W}_F$. This theory generalizes the Tate theory for characters of $F^\times$ (which corresponds to the 1-dimensional case via Class Field Theory), but requires significantly more machinery to establish existence in higher dimensions.

Take the Artin reciprocity map $\mathbf{a}_F : \mathcal{W}_F \to F^\times$ and identify characters of $F^\times$ with 1-dimensional representations of $\mathcal{W}_F$. For such a character χ, we have already defined the local L-function $L(\chi, s)$ and the local constant $\varepsilon(\chi, s, \psi)$ in section 5.1. We translate these to $\mathcal{W}_F$ by setting:

$$L(\chi\circ\mathbf{a}_F, s) = L(\chi, s), \qquad \varepsilon(\chi\circ\mathbf{a}_F, s, \psi) = \varepsilon(\chi, s, \psi)$$

The objective is to extend these definitions to $\sigma \in \mathcal{G}_n^{\mathrm{ss}}(F)$.

The extension of the L-function is straightforward. For an irreducible representation $\sigma \in \mathcal{G}_n^0(F)$, the definition divides based on dimension. If $n = 1$, it is given by the formula above. If $n \geq 2$:

$$L(\sigma, s) = 1$$

We then extend this to semisimple representations $\sigma = \bigoplus_i \sigma_i$ by additivity (multiplicativity of the function):

$$L(\sigma, s) = \prod_i L(\sigma_i, s)$$

6.3.1 Artin L-functions There is a more uniform description. Let (σ, V) be a representation of $\mathcal{W}_F$. The space $V^{\mathcal{I}_F}$ of inertia-fixed vectors carries a representation of $\mathcal{W}_F/\mathcal{I}_F \cong \mathbb{Z}$ (generated by a geometric Frobenius Φ). The Artin L-function is defined as:

$$L(\sigma, s) = \det(1 - \sigma|_{\mathcal{I}_F}(\Phi)q^{-s})^{-1}$$

This matches the definition given above: if $\dim(\sigma) \geq 2$ and σ is irreducible, the inertia-fixed space $V^{\mathcal{I}_F}$ is usually zero (unless σ is unramified, which implies dimension 1).

The definition of the local constant is far more involved. It is specified axiomatically by its behavior under induction.

Theorem 6.3.2: Existence of the Langlands-Deligne Local Constant

Let $\psi \in \widehat{F}, \psi \neq 1$. Let E/F range over finite extensions inside $\bar{F}$, and let $\psi_E = \psi \circ \mathrm{tr}_{E/F}$. There exists a unique family of functions

$$\mathcal{G}^{\mathrm{ss}}(E) \to \mathbb{C}[q^s, q^{-s}]^\times, \qquad \rho \mapsto \varepsilon(\rho, s, \psi_E),$$

satisfying the following properties:

1. **Dimension 1:** If χ is a character of $E^\times$, then $\varepsilon(\chi \circ \mathbf{a}_E, s, \psi_E) = \varepsilon(\chi, s, \psi_E)$.
2. **Additivity:** If $\rho_1, \rho_2 \in \mathcal{G}^{\mathrm{ss}}(E)$, then
$$\varepsilon(\rho_1 \oplus \rho_2, s, \psi_E) = \varepsilon(\rho_1, s, \psi_E)\varepsilon(\rho_2, s, \psi_E)$$
3. **Inductivity:** If $\rho \in \mathcal{G}^{\mathrm{ss}}_n(E)$ and $E \supset K \supset F$, then
$$\varepsilon(\mathrm{Ind}_{E/K}\rho, s, \psi_K) = \varepsilon(\rho, s, \psi_E)\left(\frac{\varepsilon(R_{E/K}, s, \psi_K)}{\varepsilon(1_E, s, \psi_E)}\right)^n, \tag{6.2}$$
where $R_{E/K} = \mathrm{Ind}_{E/K}1_E$ is the regular representation.

The goal for the rest of this section is to establish this theorem. The uniqueness half is local and is proved here in full; the existence half rests on one global input, isolated as theorem 6.3.7 and cited there.

***Proof Key ideas*:**
Uniqueness. Properties (1) to (3) say precisely that $\rho \mapsto \varepsilon(\rho, s, \psi)$ is an induction constant in the sense of definition 6.3.4, whose boundary is the division $(E, \chi) \mapsto \varepsilon(\chi, s, \psi_E)$ fixed by property (1). By lemma 6.3.5 an induction constant is determined by its boundary, so at most one such family exists. This half uses only Brauer induction and is complete.

Existence. By lemma 6.3.6 it suffices to produce, for each finite Galois L/F, an induction constant on $\mathrm{Gal}(L/F)$ with the prescribed boundary, that is, to show the division $D^{\psi}_{L/F}$ is pre-inductive. That is theorem 6.3.7, and it is the step requiring global methods; lemma 6.3.8 supplies the local input it needs. Granting it, the constants glue over the tower of finite Galois extensions and the family exists.

The dependence on ψ is then removed by proposition 6.3.3(2), which shows that changing ψ multiplies ε by an explicit factor, so a family constructed for one $\psi \neq 1$ determines one for every other.

Before turning to the machinery, here are the properties the rest of the book uses.

Proposition 6.3.3: Properties of the Local Constant

Let $\psi \in \widehat{F}, \psi \neq 1$, and let $\rho \in \mathcal{G}^{\mathrm{ss}}(F)$.

1. There is an integer $n(\rho, \psi)$ such that
$$\varepsilon(\rho, s, \psi) = q^{n(\rho,\psi)(1/2-s)}\varepsilon(\rho, 1/2, \psi)$$
2. Let $a \in F^\times$. Then:
$$\varepsilon(\rho, s, a\psi) = \det \rho(a)\|a\|^{\dim(\rho)(s-1/2)}\varepsilon(\rho, s, \psi)$$
Consequently, $n(\rho, a\psi) = n(\rho, \psi) + v_F(a) \dim \rho$.
3. The local constants satisfy the functional equation
$$\varepsilon(\rho, s, \psi)\varepsilon(\check{\rho}, 1-s, \psi) = \det \rho(-1)$$
4. There is an integer n_ρ such that, if χ is a character of $F^\times$ of level $k \geq n_\rho$, then
$$\varepsilon(\chi \otimes \rho, s, \psi) = \det \rho(c(\chi))^{-1}\varepsilon(\chi, s, \psi)^{\dim \rho},$$
for any $c(\chi) \in F^\times$ such that $\chi(1 + x) = \psi(c(\chi)x)$ for $x \in \mathfrak{p}^{[k/2]+1}$.

***Proof*:**
Granting theorem 6.3.2, every assertion reduces to the one-dimensional case already settled in section 5.1. The reduction is the same each time and worth stating once. By the Brauer induction theorem, as used in lemma 6.3.5, the class of ρ minus $\dim \rho$ copies of the trivial representation is a $\mathbb{Z}$-linear combination of terms $\mathrm{Ind}_{H_i}^{H} \chi_i - 1_{H_i}$ with each χ_i a character. Since ε is additive (axiom (2)) and inductive in degree zero (axiom (3)), any identity that holds for characters and is multiplicative in the appropriate sense propagates from those generators to ρ.

1. By theorem 6.3.2 the value $\varepsilon(\rho, s, \psi)$ lies in $\mathbb{C}[q^s, q^{-s}]^\times$. The units of that ring are exactly the monomials cq^{-ms} with $c \in \mathbb{C}^\times$ and $m \in \mathbb{Z}$, so writing the monomial in the form $q^{n(1/2-s)}$ times its value at $s = 1/2$ defines the integer $n(\rho, \psi)$ and gives the stated shape.
2. For a character χ this is lemma 5.1.16, which gives $\varepsilon(\chi, s, a\psi) = \chi(a)\|a\|^{s-1/2}\varepsilon(\chi, s, \psi)$. Both sides of the asserted identity are multiplicative under direct sums, and the correction factor $\det \rho(a)\|a\|^{\dim \rho(s-1/2)}$ is multiplicative in ρ because both det and dim are additive on direct sums; the induction step is compatible because replacing ψ by $a\psi$ replaces ψ_E by $(a\psi)_E = a\psi_E$ for $a \in F^\times$. So the character case propagates, and specializing to $\dim \rho = 1$ recovers the familiar form. The consequence for $n(\rho, a\psi)$ follows by comparing the exponents of q on the two sides, using $\|a\| = q^{-v_F(a)}$.

3. For a character this is corollary 5.1.14, which reads $\varepsilon(\chi, s, \psi)\varepsilon(\check{\chi}, 1-s, \psi) = \chi(-1)$. Again both sides are multiplicative under direct sums, and $\det\rho(-1)$ is multiplicative in ρ, so the identity propagates from characters to all of $\mathcal{G}^{\mathrm{ss}}(F)$.

4. This is the Galois-side stability statement, and for a character it is theorem 5.1.21: if θ has level l and χ has level $k > 2l$ then $\varepsilon(\theta\chi, s, \psi) = \theta(c(\chi))^{-1}\varepsilon(\chi, s, \psi)$. Take n_ρ larger than twice the level of every character occurring in a Brauer decomposition of ρ, which is possible because that decomposition is finite; lemma 6.3.8 is what guarantees such a bound exists uniformly over the intermediate fields. For χ of level $k \geq n_\rho$ the character case applies term by term, and multiplying up gives $\varepsilon(\chi \otimes \rho, s, \psi) = \det\rho(c(\chi))^{-1}\varepsilon(\chi, s, \psi)^{\dim\rho}$, completing the proof.

The proof of the existence theorem itself requires more induction machinery, which is developed next in the abstract before being applied to Galois groups.

Let G be a profinite group. Let K_0G be the Grothendieck group of G, the free abelian group generated by isomorphism classes of irreducible smooth representations. Define $\mathcal{K}_0G = \bigoplus_H K_0H$, where H ranges over open subgroups of G. Let $\Gamma(G) = \bigoplus_H \mathrm{Hom}(H, \mathbb{C}^\times)$ be the set of pairs (H, χ) of open subgroups and characters.

Definition 6.3.4: Induction Constant and Division

Let A be an abelian group.

1. An *induction constant* on G is a function $\mathcal{F} : \mathcal{K}_0G \to A$ such that:
   (a) For each open H, $\mathcal{F}|_{K_0H}$ is a homomorphism.
   (b) If $H \subset J$ are open and ρ is a representation of H of dimension 0 (virtual), then $\mathcal{F}(J, \mathrm{Ind}_H^J\rho) = \mathcal{F}(H, \rho)$.
2. A *division* on G is a function $D : \Gamma(G) \to A$.

An induction constant defines a division $\partial\mathcal{F}$ by restriction; $\partial\mathcal{F}$ is called the *boundary* of $\mathcal{F}$. Conversely, a division D is *pre-inductive* if $D = \partial\mathcal{F}$ for some induction constant.

Lemma 6.3.5: Uniqueness of Induction Constants

Let $\mathcal{F}$ be an induction constant on G. The division $\partial\mathcal{F}$ determines $\mathcal{F}$ uniquely.

***Proof*:**
It suffices to treat the case where G is finite. Let ρ be an irreducible representation of $H \subset G$ of dimension $m \geq 1$. By the Brauer Induction Theorem, there exist characters χ_i of subgroups H_i and integers n_i such that

$$[\rho] - m[1_H] = \sum_i n_i(\mathrm{Ind}_{H_i}^H[\chi_i] - [1_{H_i}]) \tag{6.3}$$

Writing A multiplicatively, axiom (b) of the definition implies:

$$\mathcal{F}(H, \rho) = \mathcal{F}(H, 1_H)^m \prod_i \left(\frac{\mathcal{F}(H_i, \chi_i)}{\mathcal{F}(H_i, 1_{H_i})}\right)^{n_i}$$

All terms on the RHS are values of the division $\partial\mathcal{F}$.

Furthermore, one can reduce to finite groups.

Lemma 6.3.6: Reduction to Finite Quotients

If $\mathcal{H}$ is a family of open normal subgroups such that $G \cong \varprojlim G/H$, and a division D restricts to a pre-inductive division on every quotient G/H, then D is pre-inductive on G.

***Proof*:**
For each $H \in \mathcal{H}$ let $\mathcal{F}_H$ be an induction constant on G/H with $\partial\mathcal{F}_H = D|_{G/H}$; by lemma 6.3.5 it is unique. Uniqueness gives compatibility: if $H' \subseteq H$ then inflating along $G/H' \to G/H$ carries $\mathcal{F}_{H'}$ to an induction constant on G/H with the same boundary, hence equal to $\mathcal{F}_H$.

Every smooth representation of G is trivial on some $H \in \mathcal{H}$, because G is the limit of the quotients and the representation is smooth; so $\mathcal{K}_0 G$ is the colimit of the groups $\mathcal{K}_0(G/H)$ along inflation. The compatible family $(\mathcal{F}_H)$ therefore defines a single function $\mathcal{F}$ on $\mathcal{K}_0 G$. Both axioms of definition 6.3.4 are checked one level at a time and hold on each G/H, so they hold for $\mathcal{F}$; and $\partial\mathcal{F} = D$ because that holds at every level, completing the proof.

This applies to the Galois group of a finite Galois extension L/F. Let $G = \mathrm{Gal}(L/F)$. Via Class Field Theory, $\Gamma(G)$ corresponds to pairs (E, χ) where $F \subset E \subset L$ and χ is a character of $E^\times$ trivial on $N_{L/E}(L^\times)$.

Theorem 6.3.7: Existence of Pre-Inductive Division

Let L/F be a finite Galois extension and $G = \mathrm{Gal}(L/F)$. There exists $\psi \in \widehat{F}, \psi \neq 1$, such that the division

$$D^{\psi}_{L/F} : (E, \chi) \mapsto \varepsilon(\chi, s, \psi_E)$$

is pre-inductive on G.

This is the one step of the construction that leaves the local world, and it is the reason theorem 6.3.2 is hard. The local constants of characters are defined by Tate's local functional equation, which knows nothing about how the fields E sit inside one another; producing an induction constant whose boundary is $D^{\psi}_{L/F}$ means finding a coherent system across all of them, and the known route is to realize L/F as the completion of a global extension and to use the functional equation of a global Hecke L-function, whose ε-factor is by construction a product over all places. The global machinery is outside this book's scope; the argument is sketched at the end of this section and the details are cited. See [1, §29-§30].

Before turning to it, a technical local lemma on levels is needed.

Lemma 6.3.8: Level Separation

Let L/F be a finite Galois extension. There exists an integer $n_{L/F} \geq 1$ such that if α is a character of $F^\times$ of level $n_{L/F}$ and $(E, \chi) \in \Gamma(G)$, the level of $\alpha \circ N_{E/F}$ is greater than twice the level of χ.

***Proof*:**
For each intermediate field E, there is a level $\ell(E)$ such that $U_E^{\ell(E)} \subset N_{L/E}(L^\times)$. If $(E, \chi) \in \Gamma(G)$, the level of χ is bounded by $\ell(E)$. Set $m(E) = 2\ell(E) + 1$. Choose $n(E)$ such that $U_F^{n(E)} \subset N_{E/F}(U_E^{m(E)})$. The maximum of these $n(E)$ suffices.

Assuming theorem 6.3.7, the main results follow, and the restriction on ψ can be removed at the same time. First, the restriction on ψ is removed. By lemma 6.3.5, the division $\varepsilon(\chi, s, \psi_E)$ defines an induction constant $\varepsilon(\rho, s, \psi_E)$ on Ω_F. For $a \in F^\times$, consider the function

$$(E, \rho) \mapsto \det \rho(a) \|a\|^{(s-1/2)\dim \rho_E}$$

This is clearly an induction constant. Its boundary is the division $(E, \chi) \mapsto \chi(a)\|a\|_E^{s-1/2}$. Recall from lemma 5.1.16 that $\varepsilon(\chi, s, a\psi_E) = \chi(a)\|a\|_E^{s-1/2}\varepsilon(\chi, s, \psi_E)$. Since the product of pre-inductive divisions is pre-inductive, the division $(E, \chi) \mapsto \varepsilon(\chi, s, a\psi_E)$ is pre-inductive. Thus the theorem holds for all ψ.

Properties (1) and (2) of proposition 6.3.3 now hold for Galois representations. The extension to the Weil group uses the unramified twist.

Proposition 6.3.9: Unramified Twist of Epsilon Factor

Let $(\Omega_E, \rho) \in \mathcal{K}_0\Omega_F$, and let φ be an unramified character of $F^\times$ of finite order. Let $\varphi(\varpi) = q^{-s(\varphi)}$. Then $\varepsilon(\varphi_E \otimes \rho, s, \psi_E) = \varepsilon(\rho, s + s(\varphi), \psi_E)$.

***Proof*:**
Both sides define induction constants (as functions of ρ) with the same boundary on 1-dimensional characters. By uniqueness, they are equal.

Next, define the following:

Definition 6.3.10: Langlands Constant

Let E/F be a finite extension, let 1_E be the trivial character of $\mathcal{W}_E$ and set $R_{E/F} = \mathrm{Ind}_{E/F} 1_E$. Define the *Langlands Constant*:

$$\lambda_{E/F}(s, \psi) = \frac{\varepsilon(R_{E/F}, s, \psi)}{\varepsilon(1_E, s, \psi_E)}$$

Corollary 6.3.11: Langlands Constant Property

The function $\lambda_{E/F}(s, \psi)$ is constant in s.

***Proof*:**
Using the proposition above, $\lambda_{E/F}(s + s(\varphi), \psi) = \lambda_{E/F}(s, \psi)$. Since φ can be any character of finite order, this function is periodic with dense periods, and thus must be constant in s. We write $\lambda_{E/F}(\psi)$.

Now let ρ be an irreducible representation of $\mathcal{W}_E$. There is an unramified character φ for which $\varphi_E \otimes \rho$ factors through a representation ρ_0 of the Galois group Ω_E (see proposition 6.1.9). Define:

$$\varepsilon(\rho, s, \psi_E) = \varepsilon(\rho_0, s - s(\varphi), \psi_E)$$

This definition is well-defined and satisfies the inductive properties.

The remaining properties of proposition 6.3.3 follow.

- **Functional Equation (3):** The maps $\tau \mapsto \varepsilon(\tau, s, \psi)\varepsilon(\check{\tau}, 1-s, \psi)$ and $\tau \mapsto \det \tau(-1)$ are both induction constants. Their boundaries are equal by the functional equation for GL(1) (corollary 5.1.14). Thus the maps are equal.
- **Stability (4):** It suffices to consider ρ irreducible. Use the Brauer relation $[\rho] - m[1_F] = \sum(\mathrm{Ind}\varphi_i - 1)$, which gives

$$\frac{\varepsilon(\chi \otimes \rho, s, \psi)}{\varepsilon(\chi, s, \psi)^m} = \prod \frac{\varepsilon(\chi_{E_i}\varphi_i, s, \psi_{E_i})}{\varepsilon(\chi_{E_i}, s, \psi_{E_i})}$$

  If χ has sufficiently large level (using lemma 6.3.8), the Stability Theorem for GL(1) (theorem 5.1.21) applies to each term in the product. The RHS becomes $\prod \varphi_i(c(\chi))^{-1}$. Taking the determinant of the Brauer relation yields $\det \rho = \prod \varphi_i \circ \mathrm{ver}$, which implies the result.

What remains is the global input. The following sketches the proof of theorem 6.3.7. This requires global methods (see [10] for more details). Let K be a global field and $\mathbb{A}_K$ its adele ring. For a representation ρ of Ω_K, the global Artin L-function is defined as the product of local L-functions $L(\rho, s) = \prod_v L(\rho_v, s)$. This satisfies a global functional equation $L(\rho, s) = \varepsilon(\rho, s)L(\check{\rho}, 1-s)$, where the global epsilon factor is the product of local factors $\varepsilon(\rho, s) = \prod_v \varepsilon(\rho_v, s, \psi_v)$. Critically, $\varepsilon(\mathrm{Ind}_{L/K}\tau, s) = \varepsilon(\tau, s)$.

Lemma 6.3.12: Properties of Global L-functions

1. Let L/K be a finite extension, $L \subset \bar{K}$, and let τ be a smooth representation of $\Omega_L = \mathrm{Gal}(\bar{K}/L)$ of finite dimension. Then

$$L(\mathrm{Ind}_{L/K}\tau, s) = L(\tau, s)$$

2. Let ρ be a finite-dimensional smooth representation of Ω_K. The product defining the global L-function converges in some half-plane and admits analytic continuation to a meromorphic function on the whole s-plane. It satisfies a functional equation

$$L(\rho, s) = \varepsilon(\rho, s)L(\check{\rho}, 1-s), \tag{6.4}$$

   for a uniquely determined exponential function $\varepsilon(\rho, s)$ having neither zeros nor poles.

3. In the situation of (1),

$$\varepsilon(\mathrm{Ind}_{L/K}\tau, s) = \varepsilon(\tau, s) \tag{6.5}$$

***Proof*:**
Part (1) is given by an elementary argument, based on the definitions of the local factors (which align with the definition of Artin L-functions). Part (2) then follows from the Brauer Induction Theorem (as in lemma 6.3.5) and the known case for 1-dimensional characters (Hecke L-functions). Part (3) is an immediate consequence of (1) and the functional equation (2).

Lemma 6.3.13: Global Realization

There exists a finite Galois extension $\mathcal{L}/\mathcal{F}$ of global fields and a place v_0 of $\mathcal{F}$ such that:

1. The completion $\mathcal{F}_{v_0} \cong F$.
2. There is a unique place u_0 of $\mathcal{L}$ above v_0, and $\mathcal{L}_{u_0} \cong L$ (the local field).
3. $\mathcal{F}$ has no real embeddings.

***Proof*:**
This follows from the Weak Approximation Theorem. If $\mathrm{char} F = 0$, we approximate local polynomials to ensure the local behavior at v_0 and full decomposition at real places.

Define local integers n_v for places $v \neq v_0$ based on the ramification of $\mathcal{L}/\mathcal{F}$.

Lemma 6.3.14: Existence of Global Character

By the Grunwald-Wang Theorem, there exists a global character α of the idele class group $C_{\mathcal{F}}$ such that:

1. $\alpha_{v_0} = 1$, and
2. the level l_v of α_v satisfies $l_v \geq n_v$, for every non-Archimedean $v \neq v_0$.

Finally:

***Proof*:**
(of theorem 6.3.7 and theorem 6.3.2)
Let Ψ be a global additive character. Set $\psi = \Psi_{v_0}$. For any (E, χ) in the local Galois group $\mathrm{Gal}(L/F)$, we lift it to the global setting (using lemma 6.3.13). We analyze the product formula for the global ε-factor:

$$\varepsilon(\chi\alpha_E, s) = \prod_w \varepsilon((\chi\alpha_E)_w, s, \Psi_w)$$

We analyze the factors at the different places w:

- At w_0 (the unique place above v_0): We constructed α such that $\alpha_{v_0} = 1$. Thus, this factor is simply $\varepsilon(\chi, s, \psi_E)$, which is the local term we wish to define.
- At $w \neq w_0$: The level of α is sufficiently high (by construction in lemma 6.3.14). We apply the 1-dimensional Stability Theorem (theorem 5.1.21) to substitute these factors with terms depending only on α_E, effectively removing χ.

We now rearrange the global functional equation. By lemma 6.3.12(3), the global epsilon factor is inductive: $\varepsilon(\mathrm{Ind}(\chi\alpha)_E) = \varepsilon((\chi\alpha)_E)$. We can therefore express the unknown local term $\varepsilon(\chi, s, \psi_E)$ as a quotient of terms that are known to be inductive (global epsilon factors, character values, and local constants of α):

$$\varepsilon(\chi, s, \psi_E) = \frac{\varepsilon(\mathrm{Ind}(\chi\alpha_E))}{\prod_{w\neq w_0} \varepsilon((\chi\alpha_E)_w, s, \Psi_w)}$$

Since the right-hand side is built from inductive terms, the local division $D^{\psi}_{L/F}$ is pre-inductive. By lemma 6.3.5 and lemma 6.3.6, this uniquely defines the induction constant for the local field, proving the existence theorem.

6.4 Deligne Representations

We now encounter an interesting friction in the theory due to our choice to work with *complex* representations. Our goal is to relate representations of $G = \mathrm{GL}_2(F)$ to representations of the Weil group $\mathcal{W}_F$. On the group side, we have encountered representations like the Steinberg representation St_G, which is an irreducible quotient of a reducible principal series. On the Galois side, however, representations of $\mathcal{W}_F$ are often constructed from geometry (for example étale cohomology), naturally appearing as representations on $\overline{\mathbb{Q}}_\ell$-vector spaces rather than $\mathbb{C}$.

These ℓ-adic representations exhibit behaviour, specifically regarding continuity and image size, that does not align with the smooth complex representations we have studied so far. To resolve this, and to ensure our theory is as independent of the coefficient field as possible, we introduce *Deligne Representations*. These objects decouple the "semisimple part" of the action from the "monodromy" (nilpotent) part, allowing us to classify representations over $\mathbb{C}$ that correspond to continuous representations over $\overline{\mathbb{Q}}_\ell$.

6.4.1 Definition and Algebraic Structure

A Deligne representation enriches a smooth representation with a nilpotent operator that tracks how the norm scales under conjugation.

Definition 6.4.1: Deligne Representation

A *Deligne representation* of $\mathcal{W}_F$ is a triple $(\rho, V, \mathfrak{n})$, where:

1. (ρ, V) is a finite-dimensional, smooth representation of $\mathcal{W}_F$ on a complex vector space V.
2. $\mathfrak{n} \in \mathrm{End}_{\mathbb{C}}(V)$ is a nilpotent endomorphism.
3. The operators satisfy the compatibility relation:

$$\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}, \quad \forall w \in \mathcal{W}_F \tag{6.6}$$

A Deligne representation is called *semisimple* if the underlying smooth representation ρ is semisimple. Write $\mathcal{G}_n(F)$ for the set of equivalence classes of n-dimensional, semisimple Deligne representations.

Note that one can identify the "ordinary" smooth representations $\mathcal{G}_n^{\mathrm{ss}}(F)$ with the subset of triples where $\mathfrak{n} = 0$.

Example 6.4.2: The Special Representation $Sp(n)$

Let $V = \mathbb{C}^n$ with basis $v_0, \dots, v_{n-1}$. Let $\mathfrak{n}$ be the standard principal nilpotent element (Jordan block): $\mathfrak{n}v_{n-1} = 0$ and $\mathfrak{n}v_i = v_{i+1}$. Define a smooth representation $\rho_0(w)v_i = \|w\|^i v_i$, and set $\rho(w) = \|w\|^{(1-n)/2}\rho_0(w)$. The triple $(\rho, \mathbb{C}^n, \mathfrak{n})$ is a semisimple Deligne representation denoted $Sp(n)$ (not to be confused with the spin group!)

The standard operations on representations must be adapted to the operator $\mathfrak{n}$. These definitions are chosen to mirror the behaviour of ℓ-adic representations (as will see in section 6.4.2).

Definition 6.4.3: Operations on Deligne Representations

Let $\sigma_1 = (\rho_1, V_1, \mathfrak{n}_1)$ and $\sigma_2 = (\rho_2, V_2, \mathfrak{n}_2)$ be Deligne representations.

1. **Direct Sum:** $\sigma_1 \oplus \sigma_2 = (\rho_1 \oplus \rho_2, V_1 \oplus V_2, \mathfrak{n}_1 \oplus \mathfrak{n}_2)$.
2. **Tensor Product:** $\sigma_1 \otimes \sigma_2 = (\rho_1 \otimes \rho_2, V_1 \otimes V_2, \mathfrak{n}_1 \otimes 1 + 1 \otimes \mathfrak{n}_2)$.
3. **Dual:** $\check{\sigma}_1 = (\check{\rho}_1, \check{V}_1, -\check{\mathfrak{n}}_1)$.

6.4.4 Note: Classification of the Indecomposables The special representations account for all of the indecomposable semisimple Deligne representations: every one is of the form $\rho \otimes Sp(n)$ for some $n \geq 1$ and some $\rho \in \mathcal{G}_m^0(F)$. The mechanism is that the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ makes $\mathfrak{n}$ shift the $\|-\|$-grading by one step, so a semisimple ρ forces V to break into $\mathfrak{n}$-strings, and indecomposability leaves exactly one.

6.4.2 Relation with ℓ-adic Representations

We now motivate the definition above. Let ℓ be a prime with $\ell \neq p$. In nature (arithmetic geometry), representations of $\mathcal{W}_F$ act on vector spaces over $\overline{\mathbb{Q}}_\ell$. These are continuous in the ℓ-adic topology, but almost never smooth (the image of inertia is usually infinite).

Fix a continuous surjection $t_\ell : \mathcal{I}_F \to \mathbb{Z}_\ell$ (this is unique up to a unit in $\mathbb{Z}_\ell^\times$). It satisfies the conjugation relation $t_\ell(gxg^{-1}) = \|g\|t_\ell(x)$.

Theorem 6.4.5: Grothendieck's Monodromy Theorem

Let (σ, V) be a finite-dimensional, continuous representation of $\mathcal{W}_F$ over $\overline{\mathbb{Q}}_\ell$. There exists a unique nilpotent endomorphism $\mathfrak{n}_\sigma \in \mathrm{End}(V)$ such that, for x in some open subgroup of $\mathcal{I}_F$:

$$\sigma(x) = \exp(t_\ell(x)\mathfrak{n}_\sigma)$$

***Proof*:**
Uniqueness is straightforward. For existence, we view σ as a homomorphism into $\mathrm{GL}_d(\overline{\mathbb{Q}}_\ell)$. The kernel of t_ℓ has pro-order prime to ℓ, so its image must be finite (and eventually trivial in the pro-ℓ subgroups of GL_d). Thus σ factors through t_ℓ on an open subgroup. The image of inertia must satisfy $\sigma(\Phi h \Phi^{-1})^q = \sigma(h)$. This forces the eigenvalues of $\sigma(h)$ to be roots of unity that are permuted by the q-power map. Since the image is in a pro-ℓ group, we use the following lemma:

Lemma 6.4.6: Integrality of Roots

If $\alpha \in \overline{\mathbb{Q}}_\ell$ is a root of unity such that $(\alpha - 1)/\ell^2$ is integral, then $\alpha = 1$.

***Proof*:**
If α has order ℓ^r, the valuation of $\alpha - 1$ is $1/(\ell^r - \ell^{r-1})$, which is too small. If the order is prime to ℓ, $\alpha - 1$ is a unit.

Using the matrix logarithm (which converges on the pro-ℓ group), we define $\mathfrak{n} = t_\ell(h_0)^{-1} \log \sigma(h_0)$ for a suitable h_0.

This theorem strips away the non-smooth part of the representation. Define a new representation σ_Φ by:

$$\sigma_\Phi(\Phi^a x) = \sigma(\Phi^a x) \exp(-t_\ell(x)\mathfrak{n}_\sigma), \quad a \in \mathbb{Z}, x \in \mathcal{I}_F$$

This representation σ_Φ is trivial on an open subgroup of inertia, hence *smooth*. The triple $(\sigma_\Phi, V, \mathfrak{n}_\sigma)$ is a Deligne representation over $\overline{\mathbb{Q}}_\ell$.

Theorem 6.4.7: Equivalence of Categories (ℓ-adic to Deligne)

Let $\mathrm{Rep}^f_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F)$ be the category of finite-dimensional continuous ℓ-adic representations. The functor

$$(\sigma, V) \longmapsto (\sigma_\Phi, V, \mathfrak{n}_\sigma)$$

is an equivalence of categories $\mathrm{Rep}^f_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F) \xrightarrow{\sim} \text{D-Rep}_{\overline{\mathbb{Q}}_\ell}(\mathcal{W}_F)$.

***Proof*:**
We sketched the construction of the object. The inverse functor is given by $\tau(g) = \tau_\Phi(g)\exp(t_\ell(x)\mathfrak{n})$. The check that this is well-defined and functorial relies on the conjugation relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$, which corresponds exactly to the relation satisfied by the Monodromy operator.

Finally, the notion of semisimplicity needs clarifying.

Proposition 6.4.8: Semisimplicity of ℓ-adic Representations

Let (σ, V) be a continuous ℓ-adic representation. The following are equivalent:

1. The associated Deligne representation is semisimple (i.e., σ_Φ is semisimple).
2. There is a Frobenius element Ψ such that $\sigma(\Psi)$ is semisimple.
3. $\sigma(g)$ is semisimple for every $g \in \mathcal{W}_F \setminus \mathcal{I}_F$.

***Proof*:**
Write $(\rho, V, \mathfrak{n})$ for the Deligne representation attached to σ by theorem 6.4.7, so that $\sigma(g) = \rho(g)\exp(t_\ell(x)\mathfrak{n})$ in the notation of that theorem, and $\rho = \sigma_\Phi$.

(1) implies (3). If σ_Φ is semisimple, take $g \in \mathcal{W}_F \setminus \mathcal{I}_F$. Then $\sigma(g) = \rho(g)u$ with u unipotent commuting with $\rho(g)$ up to the conjugation relation; the multiplicative Jordan decomposition of $\sigma(g)$ has semisimple part $\rho(g)$ and unipotent part u. The relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ with $\|g\| \neq 1$, which holds because g is outside inertia, forces $\mathfrak{n}$ to act as zero on the generalized eigenspaces where the two parts would interact, so $u = 1$ and $\sigma(g) = \rho(g)$ is semisimple.

(3) implies (2). Immediate: a Frobenius element lies in $\mathcal{W}_F \setminus \mathcal{I}_F$.

(2) implies (1). Suppose $\sigma(\Psi)$ is semisimple for some Frobenius Ψ. Since $\sigma(\Psi) = \rho(\Psi)\exp(t_\ell(x)\mathfrak{n})$ and the two factors are the semisimple and unipotent parts of a Jordan decomposition, uniqueness of that decomposition gives $\rho(\Psi) = \sigma(\Psi)$ semisimple. The representation ρ is smooth, so it is trivial on an open subgroup of inertia and is determined by $\rho(\Psi)$ together with a finite-image representation of inertia; a finite-image representation in characteristic zero is semisimple, so ρ is semisimple, completing the proof.

Such representations are called Φ-semisimple.

Corollary 6.4.9: Bijection of Classes

The isomorphism classes of n-dimensional Φ-semisimple representations over $\overline{\mathbb{Q}}_\ell$ are in canonical bijection with isomorphism classes of n-dimensional semisimple Deligne representations over $\mathbb{C}$ (once an isomorphism $\mathbb{C} \cong \overline{\mathbb{Q}}_\ell$ is chosen).

***Proof*:**
Theorem 6.4.7 already gives an equivalence between continuous ℓ-adic representations of $\mathcal{W}_F$ and Deligne representations over $\overline{\mathbb{Q}}_\ell$, and it preserves dimension since the underlying space is unchanged. By proposition 6.4.8 that equivalence matches the Φ-semisimple objects on one side with the semisimple Deligne representations on the other, so it restricts to a bijection on those classes.

It remains to transport from $\overline{\mathbb{Q}}_\ell$ to $\mathbb{C}$. A Deligne representation is a triple $(\rho, V, \mathfrak{n})$ with ρ smooth, so ρ has open kernel and its matrix entries generate a finite extension of $\mathbb{Q}$; the datum is therefore defined over a field embeddable in both $\mathbb{C}$ and $\overline{\mathbb{Q}}_\ell$, and a choice of isomorphism $\mathbb{C} \cong \overline{\mathbb{Q}}_\ell$ carries triples to triples, preserving semisimplicity and dimension. Hence the classes correspond, as we sought to show.

The bijection depends on the chosen isomorphism, which is exactly the ambiguity that theorem 7.4.2 removes by twisting.

6.4.3 L-functions and Epsilon-factors

We extend the definitions from section 6.3 to Deligne representations. The presence of the operator $\mathfrak{n}$ modifies the L-function by restricting to the kernel of the monodromy.

Let $\sigma = (\rho, V, \mathfrak{n})$ be a semisimple Deligne representation. Let $V_{\mathfrak{n}} = \ker(\mathfrak{n})$. This subspace is invariant under ρ (due to the relation (6.6)). We define:

$$L(\sigma, s) = L(\rho|_{V_{\mathfrak{n}}}, s) \tag{6.7}$$

For the epsilon factor, we incorporate the full structure and the dual:

$$\varepsilon(\sigma, s, \psi) = \varepsilon(\rho, s, \psi)\frac{L(\check{\rho}, 1-s)}{L(\rho, s)}\frac{L(\rho, s)}{L(\check{\rho}|_{\check{V}_{\check{\mathfrak{n}}}}, 1-s)} \tag{6.8}$$

This simplifies to $\varepsilon(\sigma, s, \psi) = \varepsilon(\rho, s, \psi)\frac{L(\check{\rho},1-s)}{L(\check{\sigma},1-s)}$.

Example 6.4.10: L-function of $Sp(2)$

Consider $\sigma = Sp(2)$. Here $\rho = \|\cdot\|^{-1/2} \oplus \|\cdot\|^{1/2}$. The kernel of $\mathfrak{n}$ corresponds to the subrepresentation $\|\cdot\|^{1/2}$. Let $\zeta_F(s) = (1 - q^{-s})^{-1}$. Then:

$$L(Sp(2), s) = L(\|\cdot\|^{1/2}, s) = \zeta_F(s + 1/2)$$

In contrast, $L(\rho, s) = \zeta_F(s-1/2)\zeta_F(s+1/2)$. The Monodromy operator "kills" the pole at $s = 1/2$. Calculating the epsilon factor (assuming ψ has level 1):

$$\varepsilon(Sp(2), s, \psi) = -q^{s-1/2}\frac{\zeta_F(s + 1/2)}{\zeta_F(3/2 - s)}$$

This calculation perfectly matches the local constant for the Steinberg representation of $\mathrm{GL}_2(F)$.

7

The Langlands Correspondence

We have now assembled all the necessary components to state and prove the local Langlands correspondence for $\mathrm{GL}_2(F)$. This correspondence establishes a canonical bijection between two-dimensional representations of the Weil group $\mathcal{W}_F$ and irreducible smooth representations of $G = \mathrm{GL}_2(F)$. It is determined by the preservation of arithmetic invariants, specifically, L-functions and ε-factors, which we explored in chapter 5.

In this chapter, we first state the general correspondence. We then handle the "reducible" cases (where the Galois representation is a sum of characters) which correspond to the Principal Series representations. Then, we look at the irreducible cases. For odd residual characteristic ($p \neq 2$), we establish the correspondence explicitly using the method of Admissible Pairs from chapter 4. The dyadic case ($p = 2$) is significantly more involved and is reserved for future chapters, though the existence theorem stated here covers it.

7.1 Statement of the Correspondence

Recall the notations for our categories of representations.

Definition 7.1.1: Representation Sets

Let F be a non-Archimedean local field.

1. Let $\mathcal{G}_2(F)$ denote the set of equivalence classes of 2-dimensional, semisimple, Deligne representations of the Weil group $\mathcal{W}_F$.
2. Let $\mathcal{A}_2(F)$ denote the set of equivalence classes of irreducible smooth representations of $G = \mathrm{GL}_2(F)$.

As usual, a character χ of $F^\times$ is identified via local class field theory with a character of $\mathcal{W}_F$, denoted $\chi \circ a_F$ or simply χ.

Theorem 7.1.2: The Local Langlands Correspondence

Let $\psi \in \widehat{F}$ be a non-trivial additive character ($\psi \neq 1$). There exists a unique surjective map:

$$\boldsymbol{\pi}_F : \mathcal{G}_2(F) \to \mathcal{A}_2(F)$$

such that for all $\rho \in \mathcal{G}_2(F)$ and all characters χ of $F^\times$:

$$L(\chi \otimes \boldsymbol{\pi}_F(\rho), s) = L(\chi \otimes \rho, s), \tag{7.1}$$

$$\varepsilon(\chi \otimes \boldsymbol{\pi}_F(\rho), s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi) \tag{7.2}$$

Furthermore:

1. The map $\boldsymbol{\pi}_F$ is a bijection.
2. The defining relations hold for *all* $\psi \neq 1$.
3. The central character matches the determinant: $\omega_{\boldsymbol{\pi}_F(\rho)} = \det \rho$.
4. The correspondence commutes with twisting: $\boldsymbol{\pi}_F(\chi \otimes \rho) = \chi \boldsymbol{\pi}_F(\rho)$.
5. The correspondence commutes with contragredients: $\boldsymbol{\pi}_F(\check{\rho}) = \boldsymbol{\pi}_F(\rho)^\vee$.

***Proof*:**
Step 1: uniqueness. Suppose $\boldsymbol{\pi}_F$ and $\boldsymbol{\pi}'_F$ both satisfy the two displayed relations. Fix $\rho \in \mathcal{G}_2(F)$ and put $\pi = \boldsymbol{\pi}_F(\rho)$, $\pi' = \boldsymbol{\pi}'_F(\rho)$. Both are irreducible smooth representations of G, and for every character χ of $F^\times$ the relations give

$$L(\chi\pi, s) = L(\chi \otimes \rho, s) = L(\chi\pi', s), \qquad \varepsilon(\chi\pi, s, \psi) = \varepsilon(\chi \otimes \rho, s, \psi) = \varepsilon(\chi\pi', s, \psi)$$

Taking χ trivial in the third listed property gives $\omega_\pi = \det \rho = \omega_{\pi'}$, so π and π' have the same central character. Theorem 5.5.2 then gives $\pi \cong \pi'$. This is the step that was unavailable until the converse theorem was proved, and it is the whole content of uniqueness.

Step 2: the two halves are separately determined. By lemma 5.5.1 a representation $\pi \in \mathcal{A}_2(F)$ is cuspidal if and only if $L(\chi\pi, s) = 1$ for every χ, and the corresponding statement on the Galois side characterizes $\mathcal{G}_2^0(F)$. Since the defining relations equate the two families of L-functions, any map satisfying them restricts to maps $\mathcal{G}_2^1(F) \to \mathcal{A}_2^1(F)$ and $\mathcal{G}_2^0(F) \to \mathcal{A}_2^0(F)$.

Step 3: existence and bijectivity on the reducible half. This is theorem 7.2.1, which constructs a bijection $\mathcal{G}_2^1(F) \to \mathcal{A}_2^1(F)$ preserving both invariants under all twists.

Step 4: existence and bijectivity on the irreducible half. For $p \neq 2$ this is theorem 7.3.2, which uses the parametrization of both sides by admissible pairs together with the rectifier of definition 7.3.1. For $p = 2$ the construction is outside the scope of this book; see [1, §34-§35]. The statement of the present theorem is made for all residue characteristics, and only its proof in the dyadic case is deferred.

Step 5: the listed properties. Given the map, (1) is Steps 3 and 4 combined with Step 2. For (3), the central character and the determinant are both recovered from the ε-factors of twists by the same recipe, so they agree; this is checked case by case in the two construction theorems. Property (4) holds because twisting acts compatibly on both sides in each of the three reducible families and, in the cuspidal case, because the parametrization by admissible pairs is twist-equivariant. Property (5) follows from (4) and the functional equations $\varepsilon(\pi, s, \psi)\varepsilon(\check{\pi}, 1-s, \psi) = \omega_\pi(-1)$ of corollary 5.2.9 and $\varepsilon(\rho, s, \psi)\varepsilon(\check{\rho}, 1-s, \psi) = \det\rho(-1)$ of proposition 6.3.3, which match under (3). For (2), lemma 5.3.3 and proposition 6.3.3 give the dependence of each side on the additive character as $\omega_\pi(a)\|a\|^{2(s-1/2)}$ and $\det\rho(a)\|a\|^{\dim\rho(s-1/2)}$ respectively; since $\dim\rho = 2$ and $\omega_\pi = \det\rho$ by (3), the two agree, so the relations hold for one $\psi \neq 1$ if and only if they hold for all, completing the proof.

The correspondence respects the natural partition of each side into a reducible and an irreducible part.

- *Galois side:* $\mathcal{G}_2(F) = \mathcal{G}_2^1(F) \sqcup \mathcal{G}_2^0(F)$, where $\mathcal{G}_2^1(F)$ consists of the triples $(\rho, V, \mathfrak{n})$ whose underlying smooth representation ρ is *reducible*, hence a sum of two characters, and $\mathcal{G}_2^0(F)$ of those with ρ irreducible. In the second case $\mathfrak{n} = 0$ automatically: the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ makes $\ker\mathfrak{n}$ a $\mathcal{W}_F$-subspace, so irreducibility forces $\mathfrak{n}$ to be zero or invertible, and a nilpotent endomorphism cannot be invertible.
- GL_2 *side:* $\mathcal{A}_2(F) = \mathcal{A}_2^1(F) \sqcup \mathcal{A}_2^0(F)$, the non-cuspidal and the cuspidal representations. By theorem 2.4.16 the first consists of the irreducible $\iota_B^G\chi$, the one-dimensional $\varphi \circ \det$, and the special representations $\varphi\mathrm{St}_G$.

By lemma 5.5.1, $\pi \in \mathcal{A}_2^0(F)$ if and only if $L(\chi\pi, s) = 1$ for every character χ of $F^\times$, and the same criterion on the Galois side characterizes $\mathcal{G}_2^0(F)$. Any map satisfying the conditions of theorem 7.1.2 therefore carries $\mathcal{G}_2^1(F)$ into $\mathcal{A}_2^1(F)$ and $\mathcal{G}_2^0(F)$ into $\mathcal{A}_2^0(F)$, and the two halves can be treated separately.

Both parts of the reducible side are needed, and the nilpotent is what separates them. Fix a character φ of $F^\times$ and put $\chi_1 = \varphi\|-\|^{-1/2}$, $\chi_2 = \varphi\|-\|^{1/2}$, so that $\chi_1\chi_2^{-1} = \|-\|^{-1}$ and $\rho = \chi_1 \oplus \chi_2$ is reducible with $\iota_B^G(\chi_1 \otimes \chi_2)$ reducible as well. There are exactly two triples in $\mathcal{G}_2(F)$ with this underlying ρ, namely $\mathfrak{n} = 0$ and $\mathfrak{n} \neq 0$, the latter being $\varphi \otimes Sp(2)$; and there are exactly two irreducible representations of G built from that induced representation, namely $\varphi \circ \det$ and $\varphi\mathrm{St}_G$. The invariants computed in section 5.4 match them up: by proposition 5.4.2 the one-dimensional representation carries the *full* product $L(\chi_1, s)L(\chi_2, s)$, which is what ρ with $\mathfrak{n} = 0$ has, while by proposition 5.4.3 the special representation carries only $L(\chi_2, s)$, which is what the nilpotent cuts the ρ-side L-factor down to. So the one-dimensional representations do lie in the image, paired with $\mathfrak{n} = 0$, and it is the special representations that need $\mathfrak{n} \neq 0$.

7.2 The Reducible Correspondence

The reducible half is settled entirely by the computations of section 5.4, because on both sides the objects are built from a pair of characters of $F^\times$ and the invariants are built from the GL_1 invariants of that pair. What has to be got right is the bookkeeping at the reducible locus, where two Galois-side triples and two group-side representations share the same pair of characters.

Theorem 7.2.1: Correspondence for $\mathcal{G}_2^1(F)$

There is a unique bijection $\boldsymbol{\pi}_1 \colon \mathcal{G}_2^1(F) \to \mathcal{A}_2^1(F)$ preserving L-functions and ε-factors under all twists. Writing an element of $\mathcal{G}_2^1(F)$ as $(\chi_1 \oplus \chi_2, \mathfrak{n})$, it is given by:

1. if $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$, then necessarily $\mathfrak{n} = 0$ and $\boldsymbol{\pi}_1$ sends it to the irreducible principal series $\iota_B^G(\chi_1 \otimes \chi_2)$;
2. if $\chi_1\chi_2^{-1} = \|-\|^{\mp 1}$, so that $\{\chi_1, \chi_2\} = \{\varphi\|-\|^{-1/2}, \varphi\|-\|^{1/2}\}$ for a character φ of $F^\times$, and $\mathfrak{n} = 0$, then $\boldsymbol{\pi}_1$ sends it to the one-dimensional representation $\varphi \circ \det$;
3. in the same situation with $\mathfrak{n} \neq 0$, that is for $\varphi \otimes Sp(2)$, then $\boldsymbol{\pi}_1$ sends it to the special representation $\varphi\mathrm{St}_G$.

***Proof*:**
Step 1: the three cases are exhaustive and disjoint on both sides. An element of $\mathcal{G}_2^1(F)$ has $\rho = \chi_1 \oplus \chi_2$ with χ_1, χ_2 characters of $\mathcal{W}_F$, identified with characters of $F^\times$ by local class field theory (theorem 6.2.1). If $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$ then $\mathfrak{n} = 0$: the relation $\rho(w)\mathfrak{n}\rho(w)^{-1} = \|w\|\mathfrak{n}$ says that $\mathfrak{n}$ intertwines ρ with $\rho \otimes \|-\|$, and for a sum of two characters that forces $\{\chi_1, \chi_2\}$ and $\{\chi_1\|-\|, \chi_2\|-\|\}$ to meet, which is exactly the excluded relation. In the remaining case the two values $\mathfrak{n} = 0$ and $\mathfrak{n} \neq 0$ give two distinct classes, and rescaling shows all non-zero $\mathfrak{n}$ give the same class. On the other side, theorem 2.4.16 lists $\mathcal{A}_2^1(F)$ as the irreducible $\iota_B^G\chi$, the one-dimensional $\varphi \circ \det$, and the special $\varphi\mathrm{St}_G$, and by lemma 2.4.15 the first family is irreducible precisely when $\chi_1\chi_2^{-1} \neq \|-\|^{\pm 1}$. The three cases therefore biject with the three families, once each case is shown to preserve the invariants.

Step 2: case (1). By proposition 5.4.1,

$$L\big(\iota_B^G(\chi_1 \otimes \chi_2), s\big) = L(\chi_1, s)L(\chi_2, s), \qquad \varepsilon\big(\iota_B^G(\chi_1 \otimes \chi_2), s, \psi\big) = \varepsilon(\chi_1, s, \psi)\,\varepsilon(\chi_2, s, \psi)$$

On the Galois side $\rho = \chi_1 \oplus \chi_2$ with $\mathfrak{n} = 0$ has, by additivity of the Artin L-function and of the local constant (proposition 6.3.3), exactly the same two products. Twisting by a character χ replaces χ_i by $\chi\chi_i$ on both sides simultaneously, so the agreement persists under all twists.

Step 3: case (2). Take $\chi_1 = \varphi\|-\|^{-1/2}$ and $\chi_2 = \varphi\|-\|^{1/2}$. By proposition 5.4.2 the one-dimensional representation $\varphi \circ \det$ has $L(\varphi \circ \det, s) = L(\chi_1, s)L(\chi_2, s)$ and $\varepsilon(\varphi \circ \det, s, \psi) = \varepsilon(\chi_1, s, \psi)\varepsilon(\chi_2, s, \psi)$, which is again the invariant of $\chi_1 \oplus \chi_2$ with $\mathfrak{n} = 0$. This is the case the naive count overlooks: the reducible locus carries a one-dimensional representation as well as a special one, and it is the one-dimensional representation that pairs with the vanishing nilpotent.

Step 4: case (3). For a Deligne representation the L-factor is computed on the subspace where both inertia and $\mathfrak{n}$ act trivially, so passing from $\mathfrak{n} = 0$ to $\mathfrak{n} \neq 0$ deletes the factor belonging to χ_1 and leaves $L(\chi_2, s)$. By proposition 5.4.3 that is precisely $L(\varphi\mathrm{St}_G, s)$, and the same proposition supplies the matching correcting factor in the ε-factor. So $\varphi \otimes Sp(2)$ corresponds to $\varphi\mathrm{St}_G$.

Step 5: uniqueness. Any two maps satisfying the conditions agree by theorem 5.5.2: their values at a given ρ are irreducible smooth representations of G with the same twisted L- and ε-factors, hence isomorphic. Since the three cases exhaust both sides and each is a bijection onto its family, $\boldsymbol{\pi}_1$ is a bijection, as we sought to show.

The asymmetry between $\chi_1\chi_2^{-1} = \|-\|$ and $\chi_1\chi_2^{-1} = \|-\|^{-1}$ is only apparent: the two are exchanged by swapping χ_1 and χ_2, which does not change the isomorphism class of $\chi_1 \oplus \chi_2$, and by lemma 2.4.15 it does not change $\iota_B^G \chi$ either.

7.3 The Tame Correspondence ($p \neq 2$)

We now turn to the heart of the matter: matching irreducible representations of $\mathcal{W}_F$ ($\mathcal{G}_2^0(F)$) with supercuspidal representations of G ($\mathcal{A}_2^0(F)$).

When the residual characteristic p is odd, we can use the classification via *Admissible Pairs* from chapter 4. Recall that $P_2(F)$ is the set of isomorphism classes of admissible pairs $(E/F, \xi)$, where E/F is a quadratic extension and ξ is a character of $E^\times$ not factoring through the norm.

We have two "naive" maps from $P_2(F)$:

1. *To Galois:* $(E/F, \xi) \mapsto \rho_\xi = \mathrm{Ind}_{E/F}\xi \in \mathcal{G}_2^0(F)$.
2. *To GL(2):* $(E/F, \xi) \mapsto \pi_\xi$, the cuspidal representation constructed via compact induction in chapter 4.

If $p \neq 2$, both maps are bijections. However, the map $\rho_\xi \mapsto \pi_\xi$ is *not* the Langlands correspondence. There is a subtle mismatch in the central characters.

$$\det(\rho_\xi) = \kappa_{E/F} \cdot (\xi|_{F^\times}) \quad \text{but} \quad \omega_{\pi_\xi} = \xi|_{F^\times}$$

Here $\kappa_{E/F}$ is the quadratic character of $F^\times$ associated to E/F by local class field theory. To fix this, we must twist by a "rectifier" character.

Definition 7.3.1: Rectifier Character Δ_ξ

For an admissible pair $(E/F, \xi)$, define a character Δ_ξ of $E^\times$ as follows:

1. If E/F is unramified, Δ_ξ is the unique unramified character of $E^\times$ of order 2.
2. If E/F is totally ramified, let $\lambda_{E/F}(\psi)$ be the Langlands constant defined in definition 6.3.10. Let α be an element relating ξ to the additive character ψ_E. Then Δ_ξ is the unique character such that:

$$\Delta_\xi|_{U_E^1} = 1, \quad \Delta_\xi|_{F^\times} = \kappa_{E/F}, \quad \Delta_\xi(\varpi_E) = \kappa_{E/F}(\zeta)\lambda_{E/F}(\psi)^n$$

   where n is the level of ξ and ζ is a specific root of unity derived from α.

Crucially, Δ_ξ depends only on the isomorphism class of the pair.

Theorem 7.3.2: The Tame Correspondence

Let $p \neq 2$. The Langlands correspondence $\boldsymbol{\pi}_F : \mathcal{G}_2^0(F) \to \mathcal{A}_2^0(F)$ is given by:

$$\boldsymbol{\pi}_F(\rho_\xi) = \pi_{\xi'} \qquad \text{where } \xi' = \xi \cdot \Delta_\xi$$

The bijectivity of the map costs nothing, and the central characters can be matched outright. What needs machinery this book does not build is the ε-factor identity, for the reason recorded in section 5.3: the explicit local constant of a cuspidal representation is available only through the Kirillov model. The proof is therefore given in full for everything except that one comparison, which is cited.

***Proof Key ideas*:**
Step 1: bijectivity. For $p \neq 2$ the two maps out of $P_2(F)$ are bijections onto $\mathcal{G}_2^0(F)$ and $\mathcal{A}_2^0(F)$ respectively, the first by the theory of induced Weil representations and the second by theorem 4.3.1. The rectifier attaches to each admissible pair $(E/F, \xi)$ the pair $(E/F, \xi\Delta_\xi)$, and since Δ_ξ depends only on the isomorphism class of the pair and is invertible, this is a bijection of $P_2(F)$ onto itself. The composite $\rho_\xi \mapsto \pi_{\xi\Delta_\xi}$ is therefore a bijection $\mathcal{G}_2^0(F) \to \mathcal{A}_2^0(F)$.

Step 2: central characters, in full. On the Galois side, the determinant of an induced representation is $\det(\mathrm{Ind}_{E/F}\xi) = \kappa_{E/F} \cdot (\xi|_{F^\times})$, where $\kappa_{E/F}$ is the quadratic character of $F^\times$ attached to E/F by local class field theory (theorem 6.2.1). On the group side $\omega_{\pi_{\xi'}} = \xi'|_{F^\times}$, and by construction $\Delta_\xi|_{F^\times} = \kappa_{E/F}$ (definition 7.3.1), so

$$\omega_{\pi_{\xi\Delta_\xi}} = (\xi\Delta_\xi)|_{F^\times} = (\xi|_{F^\times})\,\kappa_{E/F} = \det(\rho_\xi)$$

which is property (3) of theorem 7.1.2. This also shows why the naive map $\rho_\xi \mapsto \pi_\xi$ cannot be the correspondence: it fails this identity by exactly the factor $\kappa_{E/F}$.

Step 3: the ε-factor comparison, cited. Inductivity of the local constant in degree zero (proposition 6.3.3) gives, for $\rho_\xi = \mathrm{Ind}_{E/F}\xi$,

$$\varepsilon(\rho_\xi, s, \psi) = \lambda_{E/F}(\psi)\,\varepsilon(\xi, s, \psi_E)$$

with $\lambda_{E/F}(\psi)$ the Langlands constant of definition 6.3.10. The corresponding computation on the group side expresses $\varepsilon(\pi_{\xi'}, s, \psi)$ in terms of $\varepsilon(\xi', s, \psi_E)$, and the definition of Δ_ξ is arranged so that the discrepancy between the two is exactly $\lambda_{E/F}(\psi)$. That group-side computation requires the Kirillov model, which is the deferral already taken in section 5.3; see [1, §34] for it. Granting it, the twisted ε-factors agree for all twists, and the L-factors agree trivially since both sides are 1 by proposition 5.3.1.

7.4 The ℓ-adic Correspondence

So far, we have worked with representations over $\mathbb{C}$. However, in arithmetic geometry (e.g., the cohomology of modular curves), representations naturally arise over $\overline{\mathbb{Q}}_\ell$, where $\ell \neq p$.

There always exists a (choice of) isomorphism $\iota : \mathbb{C} \cong \overline{\mathbb{Q}}_\ell$, but using it directly is problematic because the correspondence depends on $\sqrt{q}$. The automorphism group $\mathrm{Aut}(\mathbb{C})$ acts on the sets $\mathcal{G}_2(F)$ and $\mathcal{A}_2(F)$, but the correspondence $\boldsymbol{\pi}_F$ is *not* $\mathrm{Aut}(\mathbb{C})$-equivariant. Specifically, if $\sigma \in \mathrm{Aut}(\mathbb{C})$, generally $\boldsymbol{\pi}_F(\sigma\rho) \neq \sigma\boldsymbol{\pi}_F(\rho)$.

To fix this, we introduce a normalization.

Definition 7.4.1: Normalized Galois Representation

For $\rho \in \mathcal{G}_2(F)$, define the twist:

$$\tilde{\rho} = \rho \otimes \| - \|^{-1/2}$$

(Note: This is a twist by a character of the Weil group).

Theorem 7.4.2: The ℓ-adic Correspondence

Define a modified correspondence $\Pi_F : \mathcal{G}_2(F) \to \mathcal{A}_2(F)$ by:

$$\Pi_F(\rho) = \boldsymbol{\pi}_F(\tilde{\rho}) = \boldsymbol{\pi}_F(\rho \otimes \| - \|^{-1/2})$$

This map Π_F commutes with the action of $\mathrm{Aut}(\mathbb{C})$. Consequently, for any $\ell \neq p$ and any field isomorphism $\iota\colon \mathbb{C} \to \overline{\mathbb{Q}}_\ell$, transporting Π_F along ι gives a bijection, independent of the choice of ι, between the 2-dimensional Frobenius-semisimple Deligne representations of $\mathcal{W}_F$ over $\overline{\mathbb{Q}}_\ell$ and the irreducible smooth $\overline{\mathbb{Q}}_\ell$-representations of $\mathrm{GL}_2(F)$.

***Proof Key ideas*:**
The obstruction is a square root of q and nothing else, and it is worth seeing exactly where it enters. By corollary 5.2.9 the local constant of $\pi \in \mathcal{A}_2(F)$ has the form $\varepsilon(\pi, s, \psi) = \varepsilon(\pi, \frac{1}{2}, \psi)\, q^{(1/2-s)n(\pi,\psi)}$, and the value at the centre of symmetry $s = \frac{1}{2}$ is the one the correspondence pins down. Evaluating the GL_2 functional equation (5.10) there involves $\zeta(\Phi, f, s + \frac{1}{2})$ at $s = \frac{1}{2}$, so half-integral powers of q appear; a field automorphism σ of $\mathbb{C}$ does not act on $q^{1/2}$ in any way determined by its action on q, since $\mathbb{C}$ has no distinguished square root. Hence $\boldsymbol{\pi}_F$ cannot be $\mathrm{Aut}(\mathbb{C})$-equivariant.

Twisting by $\| - \|^{-1/2}$ shifts the variable by $\frac{1}{2}$ and so moves the point of evaluation from $s = \frac{1}{2}$ to $s = 0$, where only integral powers of q occur. The transported relations then involve only quantities on which $\mathrm{Aut}(\mathbb{C})$ acts through its action on roots of unity and on q, and the equivariance $\Pi_F(\sigma\rho) = \sigma\Pi_F(\rho)$ follows by comparing the two sides of the defining relations after applying σ. Given equivariance, transporting along any ι produces the same bijection, since two choices of ι differ by an element of $\mathrm{Aut}(\mathbb{C})$.

The verification that the transported relations are preserved under σ is a computation with the explicit local constants of both sides, and on the group side those are exactly what this book quotes rather than derives (section 5.3); it is therefore stated here and not carried out. See [1, §35].

7.5 Summary

We have established the Local Langlands Correspondence for $\mathrm{GL}_2(F)$:

7.5.1 The Correspondence Maps

- *Principal Series* $\pi(\chi_1, \chi_2) \longleftrightarrow \chi_1 \oplus \chi_2$ (Reducible, decomposable).
- *Special Representations* $\chi\mathrm{St}_G \longleftrightarrow \chi \otimes \mathrm{Sp}(2)$ (Reducible, indecomposable Deligne rep).
- *Supercuspidal Representations* $\pi \longleftrightarrow \rho$ (Irreducible).

For the supercuspidal case with $p \neq 2$:

1. Take an admissible pair $(E/F, \xi)$.
2. Construct the Weil representation $\rho = \mathrm{Ind}_{E/F}\xi$.
3. Construct the rectifying character Δ_ξ.
4. The matching GL(2) representation is the compactly induced representation $\pi_{\xi\Delta_\xi}$ constructed in chapter 4.

This satisfies the preservation of L-functions (which are 1 for cuspidals) and ε-factors (which encode the arithmetic information, including Gauss sums).

A

The Projectivity Theorem

Throughout the text, we have focused on the construction and classification of representations. In this appendix, we establish a categorical property of the cuspidal representations. The goal of this appendix is to prove that irreducible cuspidal representations are *projective objects* in the category of smooth representations with a fixed central character. This result highlights the "isolated" nature of cuspidal representations: they cannot be deformed or connected to other representations. It also provides a rigorous underpinning for some of the homological algebra used implicitly in the main text. We prove this using the Schur Orthogonality Relations for square-integrable representations.

The main result is:

Theorem A.0.1: Projectivity Theorem

Let G be a unimodular locally profinite group satisfying the countability hypothesis, with centre Z. Assume that any character $\chi : Z \to \mathbb{R}_+^\times$ extends to a character $G \to \mathbb{R}_+^\times$.

Let (π, V) be an irreducible γ-cuspidal representation of G. Let (τ, U) be a smooth representation of G admitting ω_π as central character. Let $f : U \to V$ be a surjective G-homomorphism. Then there exists a G-homomorphism $\varphi : V \to U$ such that $f \circ \varphi = 1_V$ (i.e., π is projective in this category).

The proof requires the orthogonality relations for square-integrable representations. The group G/Z is locally profinite and unimodular. Let $d\dot{g}$ be a Haar measure on G/Z.

Theorem A.0.2: Schur's Orthogonality Relation

Let (π, V) be an irreducible γ-cuspidal representation. For $v_1, v_2 \in V$ and $\tilde{v}_1, \tilde{v}_2 \in \check{V}$,

$$\int_{G/Z} \langle \check{\pi}(g)\tilde{v}_1, v_1\rangle\langle \tilde{v}_2, \pi(g)v_2\rangle \, d\dot{g} = d(\pi)^{-1}\langle \tilde{v}_1, v_2\rangle\langle \tilde{v}_2, v_1\rangle,$$

for a constant $d(\pi) > 0$ depending only on π and the measure $d\dot{g}$.

***Proof*:**
Since π is γ-cuspidal, the integrand has compact support in G/Z, so the integral converges. Fixing $\tilde{v}_1$ and v_2, the integral defines a G-invariant pairing $\check{V} \times V \to \mathbb{C}$, given by a homomorphism $\Theta : \check{V} \to \check{V}$. Since π is admissible and irreducible, Schur's Lemma implies Θ is a scalar. Thus the integral is equal to $c(\tilde{v}_1, v_2)\langle \tilde{v}_2, v_1\rangle$. The bilinear form $c(-,-)$ is again G-invariant, hence proportional to the evaluation pairing. This gives the formula. To show $d(\pi) > 0$, one twists to make ω_π unitary, then defines a G-invariant Hermitian form using the integral. Schur's Lemma implies this form is unique up to a scalar, which must be positive definite.

The constant $d(\pi)$ is given a name as it generalizes the notion of dimension:

Definition A.0.3: Formal Degree

The constant $d(\pi)$ is called the *formal degree* of π.

***Proof*:**

of theorem A.0.1

Let χ be a character of $F^\times$. Let $\mathcal{H}_\chi(G)$ be the space of locally constant functions $f : G \to \mathbb{C}$, compactly supported modulo Z, such that $f(zg) = \chi(z)^{-1}f(g)$. This is an algebra under convolution. Let $\omega = \omega_\pi$. Since π is γ-cuspidal, matrix coefficients of π lie in $\mathcal{H}_\omega(G)$. Let $u \in U$ such that $f(u) \neq 0$. The map $H_\omega(G) \to V$ given by $\varphi \mapsto f(\tau(\varphi)u)$ is surjective (using the irreducibility of π). It suffices to show this map splits. We choose $\tilde{v}_0 \in \check{V}$ such that $d(\pi)^{-1}\langle \tilde{v}_0, f(u)\rangle = 1$. The function $\varphi_0(g) = \langle \check{\pi}(g)\tilde{v}_0, f(u)\rangle$ lies in $\mathcal{H}_\omega(G)$. Define $\Phi : V \to H_\omega(G)$ by $v \mapsto \varphi_v$, where $\varphi_v(g) = \langle \check{\pi}(g)\tilde{v}_0, v\rangle$. The composite map $V \to H_\omega(G) \to V$ sends $w \in V$ to

$$\pi(\varphi_w)f(u) = \int_{G/Z} \langle \check{\pi}(g)\tilde{v}_0, w\rangle \pi(g)f(u) \, d\dot{g}$$

By Schur's orthogonality relations, this equals $\langle \tilde{v}_0, f(u)\rangle d(\pi)^{-1}w = w$. Thus the map splits.

B

What Next?

The correspondence proved here is one case of a programme, and it is a small one: a single group, GL_2, over a single kind of field. What follows sketches the directions that open from it, and where in the series each is taken up.

Larger groups

The obvious next move is to replace GL_2 by GL_n, and then by a general reductive group. Almost every mechanism used here survives, but each becomes harder in a specific way: parabolic induction runs over more parabolic subgroups than the single Borel, so the classification acquires the Bernstein-Zelevinsky combinatorics. Compact induction still produces supercuspidals, but the chain orders of section 3.3 become the much richer theory of Bruhat-Tits buildings, and admissible pairs are replaced by the types of Bushnell-Kutzko. The series takes this up in *Smooth Representations of p-adic Groups*, which generalizes the worked case of this book. For the archimedean analogue, where $F = \mathbb{R}$ or $\mathbb{C}$ and smooth representations give way to $(\mathfrak{g}, K)$-modules, see *Real Reductive Groups*.

The global correspondence

The local statement exists to be assembled. Given a global field K, the local correspondences at all places v glue into a statement about automorphic representations of $\mathrm{GL}_2(\mathbb{A}_K)$, and the invariants of chapter 5 become the local factors of a global L-function. The mechanism that makes the gluing meaningful is the trace formula. The series treats this in *Automorphic Forms and Representations*, with the trace formula itself in a volume of its own.

Worth noticing on the way out: the global argument already appeared here, once. The proof of theorem 6.3.7, the one step of chapter 6 that this book quotes rather than proves, runs through the functional equation of a global Hecke L-function. The local theory could not close that gap by itself, which is a fair preview of how the two levels depend on each other.

Modularity

The case that motivated much of the subject is the correspondence between elliptic curves over $\mathbb{Q}$ and modular forms. An elliptic curve produces a compatible system of ℓ-adic Galois representations, and section 6.4.2 is where those enter the picture here: Grothendieck's monodromy theorem converts them into the Deligne representations of chapter 6, and the local correspondence then attaches to each place a representation of $\mathrm{GL}_2(\mathbb{Q}_p)$. Modularity is the statement that the resulting object is automorphic. The series develops the curve side in *Elliptic Curves* [3] and the form side in *Modular Forms*.

The dyadic case, and p-adic coefficients

Two gaps left open in this book are themselves research-shaped. The correspondence for $p = 2$ is genuine mathematics rather than an exercise in bookkeeping: the tame parametrization of chapter 4 needs quadratic extensions to be tamely ramified, which fails at $p = 2$, and repairing it takes the wildly ramified theory. Separately, this book works with complex coefficients throughout. Replacing $\mathbb{C}$ by $\overline{\mathbb{Q}}_p$ produces the p-adic local Langlands correspondence, a different subject with a different shape, taken up in *p-adic Local Langlands*.

External reading

Bushnell and Henniart, *The Local Langlands Conjecture for GL(2)* [1], is the source this book follows and the place to go for everything deferred here, in particular the dyadic case and the explicit local constants. Bump, *Automorphic Forms and Representations*, is the standard bridge from the local theory to the global one. For the Galois side in its own right, Tate's *Number Theoretic Background* in the Corvallis proceedings remains the compact reference for Weil-Deligne representations and local constants.

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